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Preface to the Nineth Edition 2027

Hello dear students.

This is the 9th edition. Our book is now 20 years old, but is even growing younger, as we are keen to update the information within in accordance with the latest studies in our field. You will find in the current edition, in an appealing form, full of illustrations.

Thanks to the enthusiasm and perseverance of Obstetrics and Gynecology the editorial board, and the valuable contribution of the faculty members of the department, in Cairo University, this edition in its final version met my expectations, and I hope it will meet yours as well.

This book is a valuable addition to our department's efforts to upgrade our education standards to meet and hopefully surpass the most recent education advances worldwide.

This book will be a valuable learning tool for both our undergraduate students as a requirement, as well as for our postgraduate students as a baseline for further reading sources.

I hope you make good use of this rich source of medical information both in your undergrad education and in your practice later on in your career.

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Part I

Obstetrics

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01 ANATOMY AND PHYSIOLOGY OF PREGNANCY

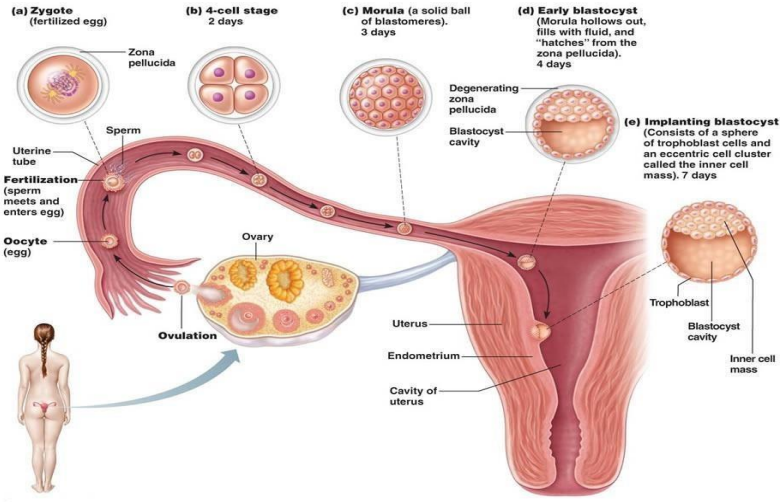


Fig 1-1: Fertilization and Implantation

FERTILIZATION

It is fusion of male and female gametes. It occurs in the distal 1/3 of the Fallopian tube, within the first 24 hours after ovulation.

IMPLANTATION

Implantation is imbedding or nidation of fertilized ovum in the uterine mucosa.

IMPLANTATION BED (THE DECIDUA):

- The decidua is the secretory endometrium that was formed under estrogen (E) and progesterone (P) during the 2nd half of the menstrual cycle. It consists of endometrial gland full of glycogen secretion.

The decidual thickness is 10 mm or more.

THE PLACENTA

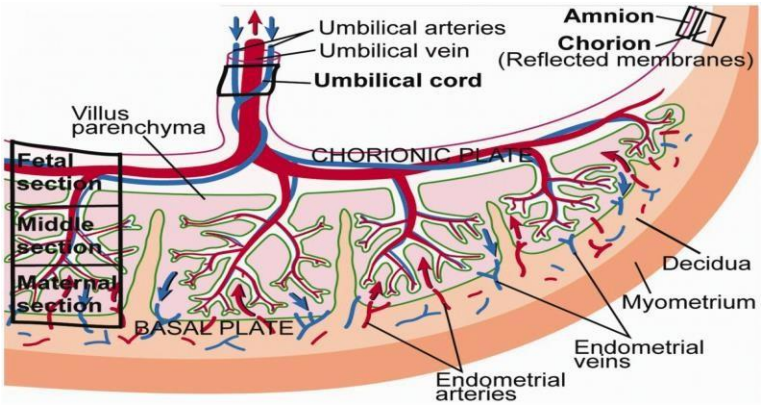


Fig 1-2: The Placenta

ANATOMY AND HISTOLOGY:

- **The placenta** represents the interface between the mother and the fetus. It is formed from the chorion frondosum and the decidua basalis.
- **Morphology:** the placenta is discoid in shape, about 20 cm in diameter, 1 inch in thickness at the center, 0.5 inch at the periphery and about 500 gm in weight. It has 2 surfaces:
  1. *Fetal surface:* Smooth, glistening and is covered by amnion which is reflected on the umbilical cord which is inserted at or near the center of this surface.
  2. *Maternal surface:* Dull greyish red in color, divided into 10-38 lobules (called cotyledons).
- **Attachment:** the placenta is normally attached to the upper uterine segment, 60% in the posterior wall, and 40% in the anterior wall.

PLACENTAL PHYSIOLOGY:

The human placenta has three main functions, namely transfer of nutrients, production of a group of hormones and enzymes (endocrine and metabolic function), and an immunological role.

A) TRANSFER FUNCTION: can occur through many mechanisms: ***Simple diffusion, Facilitated diffusion, Active transport or Pinocytosis.***

B) ENDOCRINE FUNCTION:

1. ***Human chorionic gonadotrophin (hCG):***

- It is a *glycoprotein hormone* produced by the *syncytiotrophoblast*, consisting of an □-subunit (similar to all anterior pituitary hormones) and a β-subunit which is specific to hCG.
- hCG main function is *maintenance of corpus luteum function* that supports early pregnancy till
  the development of the placenta. Its level nearly doubles every 48 hours in the first 6-8 weeks of pregnancy and correlates with the integrity of the pregnancy and its duration.

2. ***Human placental lactogen (hPL):***

- HPL is a large *protein hormone* secreted from the *syncytiotrophoblast.* HPL starts to appear since early pregnancy and *steadily increase*s till before labor.

3. ***Estrogen:***

- It is a *steroid hormone* secreted form the *fetoplacental unit* (placenta and fetus).
- Functions of estrogens:
  a. Hyperplasia of uterine muscles and increased uterine blood flow.

b. Stimulation of the development of the breast duct system.
c. Stimulation of prolactin synthesis.
d. Inhibits pituitary gonadotropin secretion leading to inhibition of ovulation.
e. Stimulates oxytocin receptor formation and increases uterine excitability in late pregnancy preparing for labor.

4. ***Progesterone (P):***

- P is a *steroid hormone* secreted by the *syncytiotrophoblast*. It progressively increases throughout pregnancy then declines during the last week before labor due to placental aging.
- Functions:
  a. Induces hypertrophy of the myometrium.
  b. Maintains secretory activity of the endometrium making it suitable for implantation.
  c. Inhibits uterine activity throughout pregnancy preventing it from expelling the conceptus.
  d. May have a role in immunological materno-fetal acceptance.
  e. Responsible for uterine softness (edema).
  f. Stimulates the development of the alveolar system of the breast.

C) IMMUNOLOGICAL FUNCTION:

The placenta appears to be an important factor in maternal acceptance of the fetus.



Fig 1-3: Normal placenta

ABNORMALITIES OF THE PLACENTA

1. ***Succenturiate placenta:*** The placenta is made of a large part and a small accessory part which is separate, and is connected by membranes. The accessory part may be retained in the uterus leading to postpartum hemorrhage and in this case, we find a circular gap in the membranes from which vessels pass to the edge of the main placenta.
2. ***Abnormal adhesion of the placenta*** to the myometrium (placenta accreta, increta and percreta).
3. ***Abnormal encroachment on the lower uterine segment:*** placenta previa.

THE UMBILICAL CORD

The Umbilical cord is developed from the ventral (connecting) stalk. At term, it measures about 50 cm in length and 1-2 cm in diameter.



Fig 1-4: The umbilical Cord

ABNORMALITIES OF THE UMBILICAL CORD

1. **Abnormalities of insertion:**
   a. Marginal insertion in the placenta (battledore), occurs in nearly 8% of pregnancies.
   b. Insertion In the membranes (velamentous insertion), occurs in nearly 2% of pregnancies.
2. **Knots of the cord:**
   a. False Knots: Localized collection of Wharton's jelly containing loop of umbilical vessels.
   b. True Knots (1%): Occurs when the fetus passes through a loop formed by abnormally long cord. If tight, asphyxia results from interference of circulation.

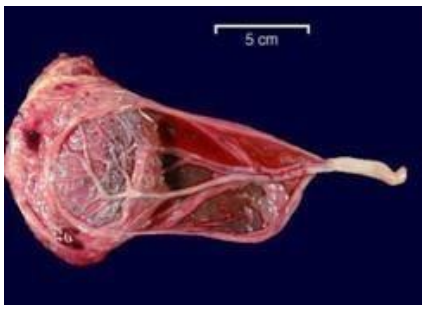


Fig 1-5: Velamentous cord insertion

Abnormal length:

   c. Too long (> 60 cm): Predispose to cord prolapse, true knots and coiling loops of the cord around the neck of the fetus.
   d. Too short (< 35 cm): this may be true or apparent (due to coiling around neck). It predisposes to prolonged 2nd stage of labor due to delay descent.

3. **Vasa previa:**

When placental vessels are below the presenting part. This occurs in cases of velamentous insertion of the cord especially when the placenta is low lying. This condition increases the risk of rupture leading to dangerous fetal

hemorrhage.





Fig 1-7: False Knot of the cord

Maternal Physiologic changes during Pregnancy

Maternal physiological changes help adaptation of the woman's body to pregnancy. Understanding these changes is essential for:

- Discriminating between normal pregnancy symptoms and those related to pathological conditions.
- Understanding the effect of pregnancy on pre-existing diseases e.g. diabetes and heart diseases.

1. GENITAL CHANGES

A) The Body of the Uterus:

- **The Height:** The height increases from 7.5 cm to reach 35 cm at term, due to:
  - Estrogen and progesterone leading to myometrial hypertrophy and hyperplasia.
  - The contents of the uterus i.e. fetus, placenta and amniotic fluid etc…
- **The Weight:** The weight increases from 50 gm to reach 1000 gm at term.
- **The Shape:** The shape of the uterus is globular till 14 weeks then it becomes pyriform.
- **The uterine ligaments:** Show hypertrophy.
- **Dextro-rotation:** The uterus is tilted and twisted to the right in 80% of cases.
- **Braxton-Hicks's contractions:** Irregular, painless contractions that help placental circulation.

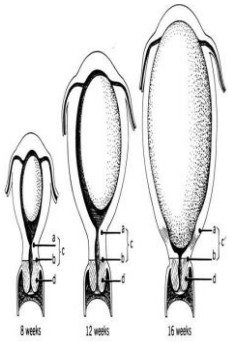


Fig 1-8: LUS

B) The Lower Uterine Segment (LUS):

- The LUS is formed from the isthmus.
- It starts to be formed from the 20th week, to reach 10 cm at full term.
- Obstetric significance of the LUS:

  1. Site of lower segment cesarean section (LSCS)

2. Site of rupture in cases of obstructed labor.
3. Site of implantation in cases of placenta previa.

a. anatomical internal os
b. histological internal os
c. isthmus
c'. Former isthmus
d. cervix

DIFFERENCES BETWEEN THE UPPER AND LOWER UTERINE SEGMENTS:

Upper uterine segment (UUS)	Lower uterine segment (LUS)
▪ Active during labor ▪ Contracts and retracts to become shorter & thicker. ▪ Thick wall: It consists of 3 muscle layers: – *Outer longitudinal* – *Middle oblique (for hemostasis)* – *Inner circular.* ▪ Covered by adherent peritoneum. ▪ The membranes are firmly attached.	▪ Less active during labor ▪ dilates and stretches to be longer and thinner ▪ Thin wall: It consists of 3 muscle layers: – *Outer longitudinal* – *Middle oblique layer is defective* – *Inner circular* ▪ Covered by loose peritoneum. ▪ The membranes are loosely attached.

C) The Cervix:

- Edema and congestion; it becomes soft "Goodell's sign" and bluish "Chadwick sign".
- Mucus plug, which consists of the cervical mucus closing the cervical canal.
- Hormonal erosion may occur; it disappears spontaneously 3-6 months after labor.
- Cervical ripening mainly at the end of pregnancy due to edema and decreased collagen.

D) The Vulva: shows increased vascularity and varicosities may develop.

E) The Vagina: shows increased vascularity which makes it soft, moist and bluish.

F) The Ovary: shows increased vascularity and increased size. One of the ovaries contains the corpus luteum, which degenerates after the 10th week.

2. BREAST CHANGES:

Due to the effect of estrogen (E) progesterone (P) & prolactin (PRL)

- Increased size and vascularity (warm, tense and tender).
- Increased pigmentation of the nipple and areola.
- 2ry areola appears (light pigmentation around the 1ry areola).
- Montgomery tubercles appear on the areola (dilated sebaceous glands).
- Colostrum-like fluid is expressed at the end of the 4th month.

3. SKIN CHANGES:

A. Pigmentation: Melanocyte Stimulating Hormone (MSH + E):

- Linea nigra: Pigmentation of the linea alba, more marked below the umbilicus.
- Chloasma gravidarum: Butterfly pigmentation of the face

B. Striae gravidarum "stretch marks":

These are pink lines in the flanks due to rupture



Fig 1-9: Breast changes



Fig 1-10: Linea nigra



Fig 1:11 Striae gravidarum

of the subcutaneous elastic fibers as a result of stretch of the abdominal wall during pregnancy. After labor, the color turns white due to fibrosis "striae albicans".

C. Loss of hair: is a common complaint that occurs during pregnancy.

4. METABOLIC CHANGES:

A. Protein metabolism: (anabolic state)

Tendency to *nitrogen retention*, with *increased binding globulins* produced by the liver.

B. Carbohydrates metabolism: (diabetogenic state)

- *Progressive increase in insulin requirements* due to insulin antagonism and resistance by;

1. Placental hormones; human placental lactogen (HPL), estrogen (E), and progesterone (P).

2. Production of insulinase enzyme by the placenta.

C. **Fat metabolism:** Increased lipolysis to deliver more free fatty aids to the fetus.

D. **Minerals:** Increased requirements of iron, calcium, phosphorus and iodides.

E. **Water:** Salt and water retention occurs due to the effect of progesterone.

F. **Weight:** The average total weight gain is 9-14 kg, most of it occurs in the 3rd trimester.

5. URINARY CHANGES:

A. Dilatation of the ureters: due to:

- *Atony* of the ureteric muscles caused by progesterone and relaxin.
- Ureteric atony and pressure of gravid uterus (more on the right side) lead to *stasis* of urine that predispose to *urinary tract infection (UTI).*

B. Frequency of micturition: due to pressure of the pregnant uterus on the urinary bladder in the 1st trimester and again with engagement of the head late in pregnancy in last few weeks.

6. GASTRO-INTESTINAL CHANGES:

- Emesis gravidarum "morning sickness", and sometimes with increased salivation (ptyalism).
- Heart burn "reflux esophagitis" caused by relaxation of the cardiac sphincter due to the effect of progesterone and relaxin.
- Constipation due to reduced gut motility caused by progesterone.
- Relaxation of the gall bladder with a tendency to stone formation.

7. CARDIOVASCULAR CHANGES:

A. The Blood Volume:

- Total blood volume increases almost 40% by the end of 32nd week of pregnancy.
- Plasma volume increases more than RBCs volume increase leading to hemodilution (physiological anemia, hyperdynamic circulation, and functional murmurs).

B. The Cardiac Output (CO):

- The CO increases to a peak at the 32nd week of pregnancy (about 40% increases) which remains elevated till the end of pregnancy. Increased CO is due to a both increased stroke volume (the main reason) and increased in the heart rate (15%) towards the end of pregnancy.

C. The blood pressure (BP):

- A normal BP in general has a mean of **120/80**. During pregnancy a **slight**

drop in the BP occurs in the 2nd trimester due to opening of arterio-venous shunts at the placenta.

- Hypertension (HTN) is diagnosed in pregnancy when the BP is **140/90** or more, on two occasions, 6 hours or more apart.
- Supine hypotension syndrome: in 2nd half of pregnancy, maternal hypotension occurs in the supine position due to pressure of the pregnant uterus on the inferior vena cava. This is due to decreased venous return and cardiac output.

D. Cardiac changes:

- Displacement of the apex of heart in late pregnancy, to the 4th intercostal space outside the midclavicular line, due to elevation of the heart by the fundus of the growing uterus.
- Functional murmurs are detected and are usually systolic.

E. Venous stasis:

Pressure of the uterus on the pelvic veins leads to ankle edema, varicose veins and piles.

F. Blood changes:

Increased fibrinogen (up to 600 mg %) and WBCs (leukocytosis up to 12,000/mm^3).

8. RESPIRATORY CHANGES:

- Hyperventilation occurs, as progesterone stimulates the respiratory center.
- Limitation of movement of the diaphragm in late pregnancy causes dyspnea.

9. SKELETAL CHANGES:

- Increased lumbar lordosis may cause low back pain.
- Relaxation of pelvic joints and ligaments due to progesterone and relaxin may cause pelvic pains.

10. ENDOCRINAL CHANGES

- ***The Pituitary gland:*** The anterior pituitary increases in size and activity.
- ***The Thyroid gland:*** increases in size and activity (physiological goiter). Total T4 is increased but the free T4 is within normal levels.
- ***The Parathyroid gland:*** increases in size and activity to regulate calcium metabolism.
- ***The Adrenal gland:*** increases in size and activity. Cortisol is increased but the main increase is in the bound portion

02 DIAGNOSIS OF PREGNANCY, PRENATAL Diagnosis and Antenatal Care

Diagnosis of pregnancy

Diagnosis of pregnancy is based on the **triad** of:

- Delayed Cycle (missed period)
- Positive Pregnancy test (urine or serum).
- Pelvic Ultrasound (TAS/TVS) showing intrauterine pregnancy.

A. DELAYED MENSTRUAL CYCLE (missed period):

A missed period is the *earliest symptom* of pregnancy.

- *Cause:* persistent corpus luteum **(CL)** function.
- *Mechanism:* **hCG** produced by the proliferating trophoblasts stimulates **CL** for continuous production of both estrogen **(E)** and progesterone **(P),** thus preventing hormone withdrawal and endometrial shedding that normally occur in absence of a pregnancy.



Fig 2:1 positive urine pregnancy test ELISA

- *Associated symptoms in 1st trimester:* as nausea, with or without vomiting, frequency of micturition, fatigue, weakness and lack of concentration, breast enlargement.
- *Associated symptoms in 2nd trimester:* gradual abdominal enlargement and perception of fetal movements (quickening).

B. POSITIVE PREGNANCY TESTS:

Laboratory tests for diagnosis of pregnancy are based on the detection of **hCG** in blood or urine samples of pregnant women.

- **Serum β-hCG tests**
  - It is the *earliest* available method for accurate diagnosis of pregnancy.
  - The quantitative test may be positive even before the missed period (levels > 50 mIU/ml).
  - β-hCG levels *rise sharply* in the first 8 weeks of pregnancy, nearly doubling every 48hrs.
  - Levels usually > 1500 mIU/ml by end of the 5th week (calculated

from the LMP).

- **Urine ELISA tests:** are simple, cheap, accurate, and readily available. A positive result will be detectable in the first 48 hours of a missed period.

N.B.: Serum B-HCG test is superior to urine pregnancy test in the early diagnosis of normal pregnancies, and in the follow up of abnormal pregnancies since it is a quantitative test of high sensitivity.

c. ULTRASOUND DIAGNOSIS OF PREGNANCY:

TAS and TVS are the mainstay in the diagnosis of an **intrauterine** pregnancy. US is safe, cheap, readily available, and easy to interpret.

- **First trimester US:**
  - 5-6 weeks: Gestational sac < 2.5 cm with no embryonic echoes.
  - 7-9 weeks: Embryo with positive heart pulsations (GS ≥ 2.5cm)
  - Measurements of fetal crown to rump length **(CRL)** give the most accurate dating of pregnancy, and hence the most acceptable expected delivery date (EDD), +ve pulsations if CRL ≥ 1cm.
- **Second trimester US:**
  - Confirmation of fetal life (FHR)
  - Fetal biometry for estimation of gestational age, fetal weight, and growth patterns.
    - Biparietal diameter (BPD)
    - Abdominal circumference (AC)
    - Femur length (FL)



GS 5 wks



GS + YS 6 wks



GS +YS + Embryo 7 wks



Embryo 8 wks + CRL

Fig 2-2: TVS in diagnosis of early pregnancy

N.B.: Calculation of the EDD (Naegle's formula)

- Pregnancy duration is 280 days from first day of LMP.
- Pregnancy duration = 9 month ± 2 weeks.
- Calculation of EDD: LMP + 9 months + 7 days.
- e.g.: If LMP at **01/01/2025**, then EDD will be at **08/10/2025**.

ANTENATAL CARE (ANC)

Antenatal care (ANC) is a program of preventive obstetrics.

OBJECTIVES OF ANC:

1. Early detection and, possibly prevention, of complications specific to pregnancy, as preeclampsia, eclampsia, and obstetric hemorrhage.
2. Detection and management of any medical disorder complicating pregnancy as anemia, diabetes mellitus, cardiac, renal, or endocrine disorders.
3. Detection of complications which may affect labor as disproportion and malpresentations.
4. Education of the patient and her family about pregnancy, labor and delivery, the hygiene and diet in pregnancy, and the warning or alarming symptoms that necessitate consultation.
5. Finally, classification of patients into normal or high risk to put the plan of proper management.

THE PRECONCEPTION VISIT

The ideal first visit should be at a preconception clinic where health education and risk assessment can be directed towards the planned pregnancy.

Advice can be given regarding the avoidance of harmful and teratogenic factors, ensuring an optimal dietary intake and control of chronic medical disorders, if present, in order to allow pregnancy to be started in the optimum conditions.

THE FIRST ANC VISIT

- **Aim:** Thorough history taking and clinical examination to **identify important risk factors:**
- **History:**
  - Menstrual History: To identify LMP, calculate gestational age, and the EDD (Naegle's formula).
  - Obstetric History: Previous pregnancies provide important clues to potential problems in the current one.
  - Medical History: Medical disorders exacerbated by pregnancy e.g. hypertension, diabetes, heart disease…etc.
  - Surgical History: e.g. uterine surgery as myomectomy, previous CS.
  - Family History: e.g. diabetes, twins, familial disorders.

- **General examination:** Pulse, temperature and B.P., pallor…etc.
- **Abdominal Examination**: abdominal masses, enlarged liver or spleen, hernias, etc.
- **Fundal level assessment:** in relation to pregnancy duration.
- **Vaginal examination:** done only, if necessary, e.g.: suspected pelvic masses, vaginal bleeding, etc...
- **Routine laboratory tests:**
  - Blood group and Rh typing, to identify RH negative patients.
  - Complete blood picture: for Hb%, WBCs, and platelets.
  - Blood sugar level: random blood glucose, or fasting and 2 hrs postprandial levels.
  - Complete urine analysis.
  - Other tests as: hepatitis B and C if necessary.

RETURN ANC VISITS

Monthly visits are required in the first 6 months, twice weekly visits in 7th and 8th months, then weekly visits in the 9th month until delivery. In each visit important data include;

- Warning symptoms, e.g. Bleeding or regular menstrual like colicky pains, persistent vomiting, sudden escape of liquor amnii, severe persistent headache, blurring of vision, marked swelling of the lower limbs.
- Weight gain: the average weight gain during pregnancy is 9- 14 Kg. Excessive weight gain many denote occult edema (developing preeclampsia) while inadequate weight gain may reflect nutritional deficit or fetal growth restriction).
- Examination for LL edema: ankle edema is acceptable in late 2nd and 3rd trimesters.
- Blood pressure measurements: to detect early gestational hypertension or preeclampsia.
- Fundal level: Should be measured and recorded at each visit after 20 weeks.
- Fetal heart sounds, heard by the Sonicaid Duplex instrument, or by Pinard stethoscope.
- Ultrasonography whenever needed to ensure gestational age, to exclude major fetal anomalies, to assure normal fetal growth, to assess fetal wellbeing, to evaluate placental location and amniotic fluid volume at various pregnancy trimesters.
- Tetanus toxoid to be given in 5th and 6th months in PHCs of rural areas.

INSTRUCTIONS TO THE PATIENT:

- **Exercise:** Mild to moderate exercise, as walking, and regular daily house

work are allowed.

- **Sleep and rest:** Proper night sleep (8 hrs), and adequate periods of rest are advisable.
- **Care of teeth:** To avoid dental caries caused by increased acidity, and septic foci.
- **Bowel habit:** Avoiding constipation by fresh vegetables and drinking water.
- **Clothes:** Avoid tight and too heavy uncomfortable clothing.
- **Breasts:** Regular hygiene by daily washes throughout pregnancy. In the last few weeks, massage of the nipples using lubricant creams might reduce the incidence of cracking. Retracted nipple is withdrawn by the thumb and finger using a lubricant.
- **Sexual intercourse:** Is avoided if there is bleeding or suspected rupture of the membranes.
- **Smoking:** Should be avoided as it may result in placental insufficiency, delivery of small babies, sudden infant death, or premature labor.
- **Travelling:** Only comfortable travelling is allowed. However, travelling should be avoided in the last month.

NUTRITION IN PREGNANCY

Nutritional Requirements of the pregnant mother should include**:**

- Caloric requirements average 2300 Kcal/day.
- Protein: 80-100 gm/day, Calcium: 1-1.5 gm/day, Iron: 30-60 mg/day.
- Vitamins and minerals: Especially B, C, D, K.
- Folic acid is important for cell division and replication. In the first trimester, a dose of 400 µg/day has been shown to effectively reduce the risk of neural tube defects.
- Salt restriction, is advisable in cases with marked edema or tendency to hypertension.

Effect of severe Malnutrition on the mother:

1. Loss of weight, anemia, and lowered resistance against infection.
2. Decalcification of bones, caries of teeth.
3. Affection of lactation.

Effect of Malnutrition on the fetus:

1. Low birth weight infants.
2. Higher incidence of rickets and anemia, in severe cases.

Vaccination (immunization) in pregnancy:

- Live attenuated vaccines are contraindicated.
- The vaccines for the following diseases may be given if needed,

preferably after the 1st trimester:

- Tetanus, poliomyelitis, rabies, influenza, cholera and typhoid (as they are killed vaccines).
- Passive immunization with Immunoglobulins against hepatitis A and B may be given.

DRUG INTAKE DURING PREGNANCY

Drug categories during pregnancy according to FDA classification:

- ***Category A:*** Safe during pregnancy. Examples are folic acid & levothyroxine.
- ***Category B:*** Risky in animal, no enough data on humans. Examples are penicillin.
- ***Category C:*** Risk in human cannot be ruled out. Examples are corticosteroids.
- ***Category D:*** Risky in human pregnancy, but the benefits may outweigh the risks. Examples are tetracyclines.
- ***Category X:*** Contraindicated in pregnancy, as they are teratogenic. If really needed, termination of pregnancy is indicated. Examples are anticancer chemotherapy.

PLACE OF DELIVERY:

The place of delivery is chosen according to the risk assessment of the case performed by the end of the ANC program. High risk cases should be delivered at hospitals equipped for care of obstetric emergencies including C.S., blood transfusion, maternal ICU service, and neonatal ICU.

POSTNATAL CARE:

The mother is still at risk for complications during the immediate and late postpartum period. (See later maternal care during puerperium).

PRENATAL DIAGNOSIS OF FETAL ANOMALIES

Prenatal diagnosis entails in utero diagnosis of fetal **congenital** or **chromosomal** abnormalities, to facilitate dealing with such problems appropriately during pregnancy or immediately after delivery. Prenatal diagnosis can be achieved in early pregnancy through:

- ***Screening tests:*** as *maternal blood sampling* and *level 2 ultrasonography*
- ***Diagnostic tests:*** as *fetal sampling* and level *3 ultrasonography*.

FETAL STRUCTURAL CONGENITAL ANOMALIES:

These describe abnormalities or malformation in organ development as occurs in:

- *Neural tube defects (NTDS)* as anencephaly, spina-bifida, hydrocephalus, etc…
- *Abdominal wall defects* (exomphalos) and congenital *diaphragmatic hernia*.

- *Cardiac anomalies* as ASD, VSD, and transposition of great vessels
- *Renal anomalies* as renal agenesis, unilateral or pelvic kidney and obstructive uropathy.
- *Tumors* as sacro-coccygeal teratoma, congenital goiter, …
- *Lymphatic channel obstruction* as cystic hygromas, and many others.

FETAL CHROMOSOMAL ANOMALIES:

- ***Aneuploidy*** occurs when the chromosome **number** is not an exact multiple of **23**. Most common aneuploidy are **trisomy 21** (Down syndrome), trisomy **18** (Edward syndrome) and trisomy **13** (Pateau syndrome).
- ***Nondisjunction*** is an error in **meiosis** that results in a gamete with one chromosome too few, or too many. They increase with advancing maternal age. Fertilization of the gamete with one extra chromosome results in a conception with **47** chromosomes. Nondisjunction may involve sex chromosomes resulting in abnormalities such as 47 XXY, 47 XYY, etc…

N.B.: Maternal age **above 35** years is a risk factor for a fetus with ***nondisjunction*** chromosomal defect, **aneuploidy**, and some structural **congenital anomalies**. The incidence increases with advancing maternal age.

HIGH RISK CASES FOR PRENATAL SCREENING AND DIAGNOSIS:

1. Maternal **age** more than **35** years (increased chromosomal and structural anomalies) Cases with a history of **familial** genetic disease, past **obstetric** history of previous babies with congenital or chromosomal anomalies, and cases with previous recurrent pregnancy loss.
2. History of exposure to **teratogenic drugs** in early pregnancy.
3. Exposure to certain **infections** in the pre-conception period or during early pregnancy.

TYPES OF PRENATAL TESTS:

SCREENING TESTS:

Screening tests should be *cheap, safe, non-invasive*, and *readily available*, with a reasonable diagnostic accuracy & high sensitivity. The aim of screening is to select a subgroup which will benefit from more complicated expensive diagnostic procedures. Prenatal screening includes:

- Maternal *serum biochemical markers* tests (double marker DMT & triple marker TMT)
- US (level 2) screening for nuchal translucency (NT).
- New techniques as *cell-free fetal DNA* (cfFDNA).

DIAGNOSTIC TESTS:

Diagnostic tests have a high degree of accuracy with low high specificity, they are generally more complicated, more expensive, and sometimes more invasive, that is why they are reserved to high-risk cases with positive

screening tests only and not to general low risk population. Prenatal diagnosis includes:

- *Chorionic villus sampling* (CVS)
- *Amniocentesis* for chromosomal anomalies.
- *Fetal blood sampling as cordocentesis.*
- New techniques as *cell-free fetal DNA* (cfFDNA) are considered diagnostic in some cases.
- *Advanced ultrasonography* (level 3) for fetal anatomy survey to diagnose various types of structural congenital anomalies.

SCREENING TESTS FOR FETAL CONGENITAL ABNORMALITIES

1. MATERNAL SERUM BIOCHEMICAL MARKERS:

A maternal **serum** blood sample, may allow testing for different **markers** which **separately**

or in **combination**, may point to an increased risk of having a child with a chromosomal defect.

a) ***AFP:*** Alpha Fetoprotein is glycoprotein produced by the yolk sac, fetal liver and GIT. Levels are **increased** in cases with open neural tube defects **(NTDs)** and **Turner** syndrome, while, levels are abnormally **decreased** in cases with trisomy 21 (Down syndrome).

b) ***HCG:*** human chorionic gonadotropin levels correlate with gestational age in normal pregnancies. Levels are abnormally **increased** in cases with trisomy 21.

c) ***PAPP-A:*** Pregnancy associated plasma protein A is a placental protein that shows abnormally **decreased** levels in trisomy 21 fetuses.

d) ***UE3:*** Unconjugated Oestriol levels are **decreased** in association with trisomy 21 fetuses.

THE DOUBLE MARKER TEST (DMT):

Combining maternal **serum markers** (hCG & PAPP-A) together with **US** measurement of the nuchal translucency **(NT)** at the fetal neck in one test performed at **11-13** weeks gestations.

- **Down syndrome** fetuses usually show **high** hCG, **low** PAPP-A levels, together with increased NT > 3.0 mm at 12 weeks gestation. A high-risk score calls for further diagnostic tests.

THE TRIPLE MARKER TEST (TMT):

The combination of **MSAFP, hCG,** and **UE3**, in one test performed in the second trimester (15-16 weeks gestation) yields good information about the risk of trisomies.

Down syndrome risk is high when the test shows **high** hCG levels with **low** MSAFP and **UE3** levels. The test is less commonly used nowadays in favor for the double marker test which is both more accurate and provides and earlier

first trimesteric diagnosis.

2.ULTRASOUND SCREENING:

- **US MEASUREMENT OF NUCHAL TRANSLUCENCY (NT)**:

  Measurement of skin fold thickness of the fetal neck at 11 -13 weeks gestation shows good correlation with the presence of fetal chromosomal abnormalities. NT thickness **> 3.0 mm** weeks should call for more invasive diagnostic work up to exclude Down syndrome.

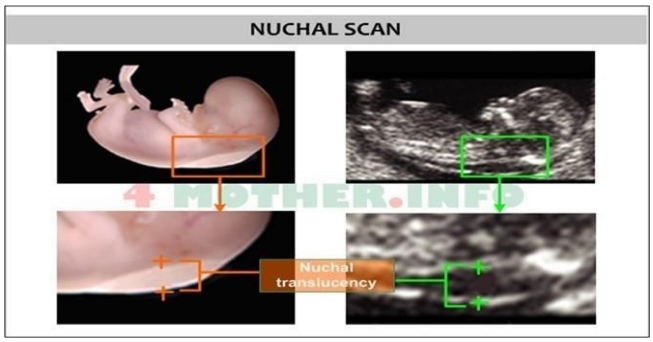


Fig 2-3: Nuchal translucency

- **Adjusting maternal age:** The risk of Down's syndrome increases with **maternal age**. The individual risk for each woman may be calculated by taking her age-related risk and then adjusting this on the measurements obtained for the NT, hCG and PAPP-A.
- **Adding soft markers:** Improving the accuracy of the results of the screening tests for Down's

  syndrome can be refined by adding further soft markers such as measuring the **nasal bone**.

3.CELL FREE FETAL DNA (cfFDNA) for screening and diagnosis:

Fragments of **fetal DNA** are released in minute amounts by the fetus into maternal circulation with increasing concentration from 10 weeks gestation onwards. By amplifying cfFDNA using **PCR** it is possible to check the fetal genotype, define the sex, and exclude many chromosomal abnormalities.

The test although being a novel promising **non-invasive diagnostic** technique with high accuracy yet its use is limited by an expensive test, not readily available in many centers, and

is still of **limited** capacity in diagnosing only 5-12 chromosomes compared to CVS and Amniocentesis which give a complete accurate genetic diagnosis of all 23 pairs of chromosomes in human pregnancies.

DIAGNOSTIC TESTS FOR FETAL CONGENITAL ABNORMALITIES

1. ULTRASOUND DIAGNOSIS OF FETAL CONGENITAL ANOMALIES:

a. First Trimester: by end of first trimester some important anomalies can be diagnosed as anencephaly, cystic hygromas, and some skeletal

abnormalities.

b. **Second Trimester Anomaly scan:** in the 2nd trimester between **18-22** weeks a detailed fetal anatomy survey is usually performed for diagnosis of important major fetal congenital anomalies as; NTDs, most of the skeletal, cardiac, renal and GIT anomalies, in addition to diaphragmatic and ventral hernias. This may allow proper decision making for continuation of pregnancy or planning the place and time of delivery for better fetal outcome.

2. CHORIONIC VILLOUS SAMPLING (CVS):

CVS procedure aims at taking a sample of the rapidly dividing **trophoblastic** cells as early as **11** weeks, performed with US guidance:

- Transvaginal: by passing a fine catheter through the cervix into the placenta.
- Transabdominal: by passing a needle through abdominal wall and myometrium.

- Cells obtained are subjected to **karyotyping** for genetic diagnosis of chromosomal defects.

3. Overall miscarriage risk for a CVS procedure is around **2%**

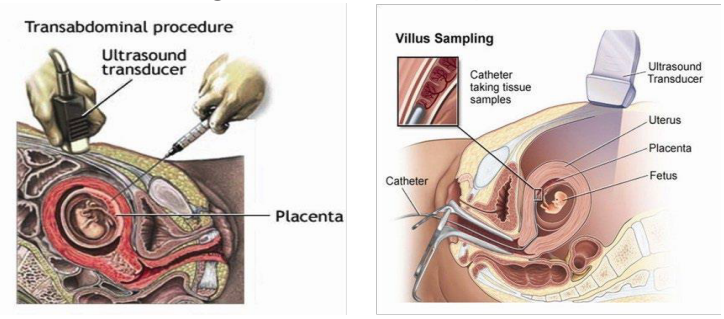


Fig 2-4: Chorionic villus sampling CVS Prenatal Diagnosis of Fetal Congenital Abnormalities

4. **AMNIOCENTESIS:** Amniotic fluid contains amniocytes and fibroblasts shed from fetal membranes, skin, and GIT. In amniocentesis a sample (15-20 ml) of amniotic fluid is obtained via a **fine needle** 18-22 gauge introduced through the **abdominal wall** and myometrium under **ultrasound** guidance.

- *Timing of amniocentesis:* **14-16** weeks gestation to obtain adequate fluid sample.
- Fetal cells obtained will be subjected to **Karyotyping**, fetal sexing, **DNA** analysis using polymerase chain reaction (PCR).
- *Risk of the procedure:* **1%** risk of **miscarriage**, very small risk of infection, rupture of membranes, accidental hemorrhage, and feto-maternal transfusion with the risk of sensitization in cases of RH negative mothers.
- **N.B.:** The advantage of *CVS* over *amniocentesis* is that it is performed earlier allowing earlier diagnosis and possible interventions. However, in later

gestational ages, amniocentesis is preferred due to less risk of abortion.

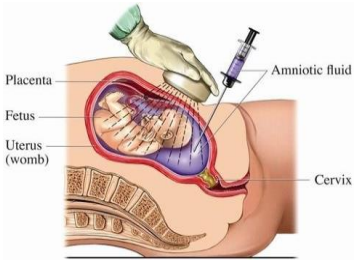


Fig 2-5: Amniocentesis

5. CORDOCENTESIS:

- *Cordocentesis* entails obtaining fetal blood sample.
- *Indications:* The most common reason is suspected severe fetal anemia such as in cases of RH incompatibility with availability of immediate intra-uterine transfusion with packed RBCs.
- *Technique:* A needle is passed under ultrasound guidance through the abdominal wall and myometrium and, into the umbilical cord at the point where it inserts into the placenta.
- *Timing:* Cordocentesis can be performed from about 20 weeks' gestation. The risk of miscarriage varies with indication and position of the placenta.

PREIMPLANTATION GENETIC DIAGNOSIS (PGD)

Early in human gestation, a single cell at the 8-cell stage, or a dozen cells at the blastocyst stage, can be removed without subsequent damage to the fetus. These cells provide sufficient DNA for PCR or fluorescent in situ hybridization for aneuploidy.

Preimplantation biopsy for PGD involves participation in an IVF program and is currently available for a limited number of genetic conditions.

03 BLEEDING IN EARLY PREGNANCY

Causes of bleeding in Early pregnancy:

- ***Abortion.***
- ***Ectopic pregnancy.***
- ***Gestational trophoblastic disease.***
- ***Associated gynecologic conditions (cervical erosion, polypi or cancer).***

SPONTANEOUS ABORTION (MISCARRIAGE)

DEFINITION: Termination of pregnancy before viability of the fetus (in Egypt, before 24 weeks).

INCIDENCE: Spontaneous abortion occurs in 15-20% of pregnancies, mostly in early 1st trimester.

ETIOLOGY:

1. Genetic or chromosomal abnormalities in > 50% of cases.
2. Accidental trauma.
3. Hormonal as luteal phase defect.
4. Infections e.g. acute maternal fever.
5. Drugs: some teratogenic drugs may cause fetal demise in utero.
6. Psychological factors or idiopathic.

CLINICAL TYPES OF SPONTANEOUS ABORTION:

- Threatened abortion.
- Inevitable abortion.
- Incomplete abortion.
- Complete abortion.
- Missed abortion.
- Septic abortion.

THREATENED ABORTION

Clinical picture:

a. Symptoms &signs of pregnancy are present according to its duration.
b. Minimal vaginal bleeding.
c. Mild suprapubic colicky pain may be present.
d. Closed cervix.

Investigations:

a. Serial quantitative βHCG estimation to detect adequate doubling.
b. Ultrasonography (abdominal or vaginal) is the main diagnostic tool to confirm fetal condition.

Treatment:

a. Physical & mental rest for 1week after the last attack.
b. Hormonal support e.g. natural progesterone (oral-vaginal-IM) is beneficial in cases with luteal phase defect.
c. Follow up to ensure clinical improvement & continuation of fetal growth.

Prognosis: 50% of cases will continue pregnancy and 50% will end in inevitable or missed abortion.

INEVITABLE ABORTION

There is no chance for pregnancy to continue.

Clinical Picture:

a. Symptoms &signs of pregnancy are present according to its duration.
b. Moderate to severe vaginal bleeding which may cause hypovolemic shock. **NB:**
ROM <24 weeks is considered a sign for inevitability.
c. Marked suprapubic colicky pain similar to labor pains is usually present.
d. PV examination: **Cervix is usually dilated**. Conception products may leave the cavity and felt distending the cervical canal (cervical abortion).

Investigations:

a. Ultrasonography (abdominal or vaginal) to detect the exact gestational age to decide route of termination.
b. Blood group and others to detect general condition especially CBC.

Treatment:

Anti-shock measures (if needed) and termination of pregnancy:

a. **Anti-shock measures**: rest – sedation – I.V crystalloids infusion (glucose-saline) through two wide bored I.V cannula till blood transfusion is available. Close monitoring of vital signs and urine output.
b. **Termination of pregnancy**: according to size of uterus:
<12weeks surgical vaginal evacuation by curettage or suction is done under Anesthesia.
>12weeks: according to severity of bleeding.
- Mild bleeding: medical evacuation by prostaglandins as oxytocin in not effective in early pregnancy due to decrease receptors.

-Severe bleeding: abdominal hysterotomy. The scar in uterus is weaker than C.S scar as it is an upper segment scar.

INCOMPLETE ABORTION

Part of the conception products are expelled from the uterus.

Clinical picture:

a. Symptoms & signs of pregnancy according to its duration.
b. Bleeding & pain: The same as inevitable abortion with passage of parts of conception products.
c. PV examination: **Cervix is opened** & retained parts may be felt coming out.

Investigations:

Ultrasonography to detect retained products (hyper-echoic irregular tissues inside the uterine cavity).

Treatment:

a. Anti-shock measures (see before).
b. Surgical evacuation.

COMPLETE ABORTION

All conception products are expelled from the uterus.
Bleeding and pain cease, and ultrasound confirm empty uterus.

MISSED ABORTION

The term missed abortion is used when death of an embryo (or fetus) occurs but is retained in the uterus.

Clinical picture:

a. The condition may be **symptomless** at the start but after sometime vaginal brownish discharge and recurrent bleeding may start to occur, with loss of symptoms of pregnancy.
b. PV examination will reveal a smaller than expected uterus with closed internal os.
c. Infection or coagulation problems may develop with hypofibrinogenaemia and DIC if pregnancy has been prolonged for several weeks, especially in the 2nd trimester.

Investigations:

a. **US** examination is diagnostic revealing intrauterine fetal demise or anembryonic sac (blighted ovum).
b. Inadequate doubling or decreasing hcg levels may be an earlier evidence.

Treatment:

Termination of pregnancy: according to size of uterus:

<12weeks surgical vaginal induction by curettage or suction is done under Anesthesia.

>12weeks:

- Medical Induction by prostaglandins as oxytocin in not effective in early pregnancy due to decrease receptors.
- Abdominal hysterotomy: rarely needed when medical induction failed or complicated by severe bleeding.
- Treatment of complications as DIC.

N.B: Following abortion, Anti D immunoglobulin I.M injection should be given to RH – ve mothers to guard against maternal sensitization.

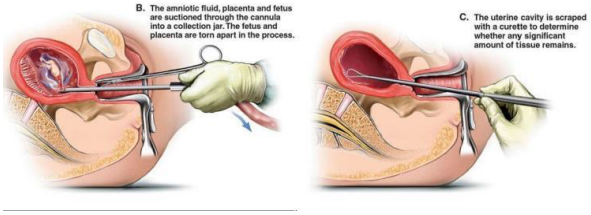


Fig 3-1: Surgical evacuation by suction and curettage

SEPTIC ABORTION

Infection could be superimposed on any type of abortion (usually on top of incomplete abortion).

Routes of infection:

a. *Exogenous*: from instruments and personnel e.g. staphylococci, hemolytic streptococci, Clostridium Welchii.
b. *Endogenous*: from bacteria flora in the genital tract as E. coli, bacteroides, non-hemolytic streptococci.
c. *Autogenous*: from septic focus in the body e.g. tonsils (blood born infection) as staphylococci and streptococci.

Clinical picture:

a. Clinical picture of abortion (amenorrhea followed by bleeding and abdominal colics).
b. Clinical picture of infection as:
   Generally: Fever headache, malaise, anorexia or may be septic shock.
   Locally: Lower abdominal pain, tender uterus, and offensive vaginal discharge.

Investigations:

a. U.S. to detect presence of intrauterine contents.
b. Blood picture: leukocytosis with shift to the left denotes infection.
c. Blood culture, Kidney functions, liver functions and coagulation profile.
d. Culture and sensitivity for a swab from vagina and cervix is of limited value due to polymicrobial nature of infection.

Treatment:

a. Isolate the patient, Fowler's position.
b. Antishock measures may be needed as warmth, sedation, fluid replacement and Blood transfusion may be needed.
c. Parenteral broad-spectrum antibiotics.
d. Obstetrical management:
   - If uterus is < 12 weeks: vaginal surgical induction under good umbrella of antibiotics. There is high incidence of uterine perforation with development of septic peritonitis.
   - If uterus is >12 weeks: medical Induction could be done. If failed, or severe bleeding, abdominal hysterotomy or rarely hysterectomy is

done if the uterus is markedly infected.

Complications:

a. Septic shock.
b. Acute renal & liver failure.
c. DIC.

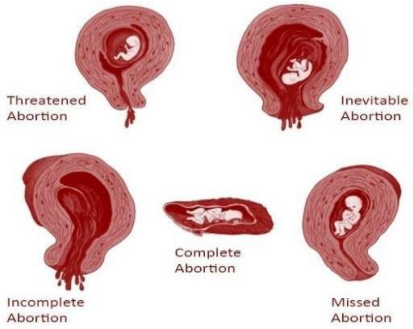


Fig 3-2: Clinical types of spontaneous abortion

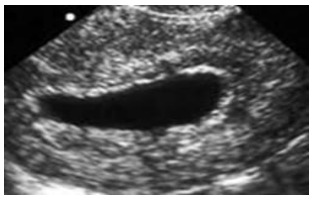


Fig 3-3: An-embryonic sac



Fig 3-4: Missed abortion

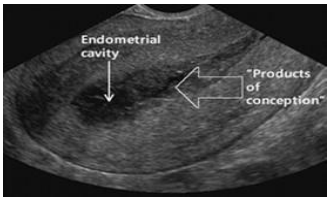


Fig 3-5: Incomplete abortion

RECURRENT PREGNANCY LOSS (HABITUAL ABORTION)

Definition: Three or more successive spontaneous abortions.

Incidence: 1-2%

Causes:

1. **Immunologic disorders are the commonest cause of early RPL:** Antiphospholipid syndrome: auto immune disorder with elevated level of antibodies as lupus anticoagulant, anticardiolipin antibodies or others. These antibodies lead to thrombosis of placental vessels. RPL, fetal growth restriction or intrauterine fetal death may occur. These patients are also liable to hypertension and deep venous thrombosis.
2. **Genetic or chromosomal abnormalities of the fertilized ovum**: Parents may show balanced translocation which is responsible for recurrence.
3. **Local causes in the uterus:**
   - *Congenital*: septate uterus (1st trimesteric).
   - *Traumatic:* Cervical incompetence (2nd trimesteric).
   - *Neoplastic:* Submucous fibroid especially fundal.
4. **General and chronic diseases** e.g. heart diseases, lung diseases, hypertension, diabetes or renal impairment.
5. **Hormonal as luteal phase defect** or thyroid dysfunction.
6. Drugs as alcoholism, morphine, heroine or excessive smoking.
7. Chronic severe psychological disorders may contribute.

Diagnosis: Evaluation may start after the second loss rather than waiting for a third abortion

History:

- Increased maternal age and family consanguinity may suggest genetic or chromosomal cause.
- Chronic medical diseases and endocrine disorders; as hypertension, diabetes or thyroid disorders.
- Early abortions are commonly due to genetic/chromosomal cause or luteal phase defect.
- Late 1st and 2nd trimester abortions may suggest anti- phospholipid syndrome/ thrombophilia (there is loss of fetal heart activity after appearance in ultrasound) or cervical incompetence (see later).

Examination: May detect general disease as hypertension, heart diseases or thyroid enlargement. Local examination to exclude local uterine causes as fibroid.

Investigations:

a. Vaginal ultrasound: to detect uterine anomalies, fibroids or signs suggestive of incompetent cervix during pregnancy (short cervix with funneling).
b. Hysteroscopy: for evaluation of abnormalities within the endometrial cavity
c. Chromosomal karyotyping for the couple: to exclude inherited disorders
d. Thyroid function, blood sugar, HbA1C, etc….
e. Screening for anti-phospholipid syndrome: Lupus anticoagulant, anticardiolipin and β2 glycoprotein.
f. Screening for other causes of thrombophilia: as Antithrombin III, Protein-S and Protein-C deficiency.

Treatment:

A) In-between pregnancies:

- Treatment of the cause e.g. myomectomy or hysteroscopic excision of the septum.
- Control any general disease as diabetes, hypertension or thyroid dysfunction.
- **Folic acid** started before pregnancy and throughout 1st trimester

B) During pregnancy:

- Progesterone supplementation in cases with luteal phase defect till the end of 1st trimester.
- Low dose aspirin (LDA) 75-81 mg/day and Low molecular weight Heparin in cases of anti-phospholipid syndrome and thrombophilia from very early beginning of pregnancy.
- Surgical treatment: for incompetent os (see later).

Cervical (Isthmic) Incompetence

Definition: Inability of cervix to retain conception past the 1st trimester.
Incidence: about 0.5-1% of pregnancies.
Etiology:

- Traumatic injury of cervix e.g. excessive forcible dilatation during D&C, lacerations during previous delivery (spontaneous or by forceps), high amputation of cervix or conization.
- Congenital: very rare, may associate bicornuate or septate uterus.

Diagnosis:

- Past history of trauma.
- Abortions are usually mid trimesteric with decreasing duration of pregnancies. Abortions are usually easy, preceded by R.O.M & associated with minimal pain & bleeding.
- Aborted fetus appears normal with no congenital anomalies.
- During pregnancy: Ultrasound reveals funneling of membranes through cervical canal.
  Cervix is short <2.5cm.

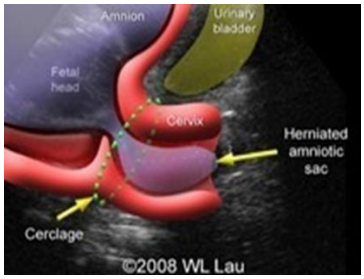


Fig 3-6: Protrusion of membrane TVS



Fig 3-7: Short cervical canal by

Treatment: Cerclage operation during pregnancy, more rest & tocolytic supplementation is also needed.

- **Cervical cerclage procedures** describe the placement of a circumferential suture into the cervix, encircling the upper cervical canal at the cervico-vaginal junction.
- It is generally performed as an **elective prophylactic** procedure, under general or regional anesthesia, at **11- 14** weeks gestation.
- Cerclage suture is **removed** as an office procedure, at 37weeks, or at onset of **labor** pains to avoid traumatic cervical lacerations, or at the end of a **cesarean** delivery.

Types of Cerclage Operations:

1. **Vaginal Cerclage procedures:** are the most commonly performed procedures including:
   - ***McDonald's procedure*:** Four bites of a non-absorbable tape (Mersilene tape) are taken as high as possible in the cervix externally like a purse string. This is by far the **most popular** procedure being rapidly and easily performed with excellent results.

- ***Shirodkar's procedure:*** Mersilene tape or silk suture is inserted under the vaginal mucosa surrounding the cervix at a level as near as possible to the internal Os (**after upward dissection of urinary bladder**). It is lengthier and more associated with more blood loss. It is mainly indicated when there are deep cervical lacerations or failed McDonald's.

2. **Abdominal Cerclage Procedure**: In **rare** occasions, an abdominal approach for cerclage via laparotomy may be recommended in cases where vaginal approach is difficult with distorted cervical anatomy. The patient is delivered by **CS** and cerclage suture is left in situ for subsequent pregnancies.

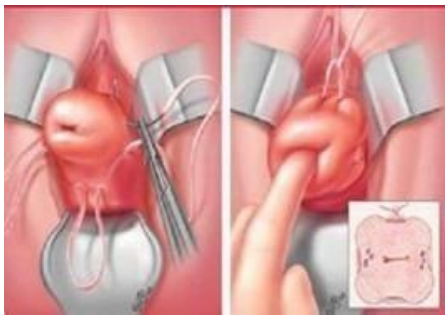


Fig 3-8: McDonald's cerclage

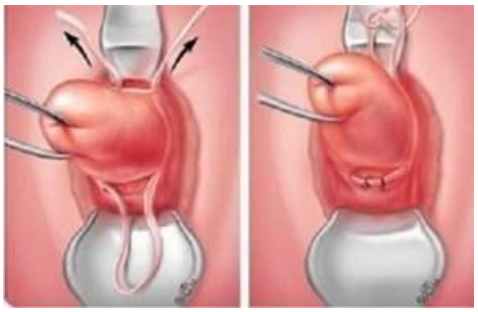


Fig 3-9: Shirodkar's procedure

ECTOPIC PREGNANCY

Definition: Implantation of the fertilized ovum outside the normal uterine cavity.

Incidence: 0.5-2% of all pregnancies may be ectopic according to associated risk factors present.

Sites of ectopic pregnancy:

1. **Fallopian tube:** is the most common site, mostly in ampullary portion, less common in isthmic portion or interstitial part of the tube.
2. **Other sites**:
   i. Ovary.
   ii. Advanced abdominal pregnancy.
   iii. Rudimentary horn.
   iv. Cervix.
   v. CS scar.

TUBAL ECTOPIC

Etiology and Risk factors:

1. **Previous ectopic gestation:** recurrence rates for an ectopic pregnancy may reach up to **30%.**
2. **Previous PID e.g gonococcal, chlamydial or puerperal.**
3. **Endometriosis:** leading to peritubal adhesions causing tubal kinking and stenosis.
4. **Pelvic and tubal surgery:** leading to tubal scarring or pelvic adhesions.
5. **IUD:** IUD prevents intrauterine pregnancy not extra-uterine.

6. **Progesterone effect:** results in alteration of tubal motility thus interfering with oocyte and sperm transfer. PRG effect may be encountered in progesterone only contraception.
7. **ICSI** procedure is a known risk factor possibly due to multiple ovulation and high PRG support.

Pathology:

- **Uterus**: It is slightly enlarged up to 8 weeks and soft (hormonal effect). Decidual is present without chorionic villi (Arias- Stella reaction).
- **Tube:** commonest in ampullary portion, less commonly in isthmus and rarely in interstitial part.

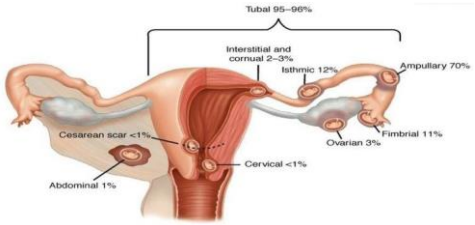


Fig 3-10: Sites of ectopic pregnancy

Results of tubal ectopic:

Due to defective decidua, tube cannot with stand proliferation of chorionic villi with growth of the embryo. Disturbance of pregnancy and damage of the tube occur as early as 8- 10 weeks. This may result in any of the following:

- ***Tubal mole*:** Recurrent attacks of slight bleeding occur, and the dead ovum is surrounded by a blood clot forming a mole in the tubal lumen.
- ***Tubal abortion:*** the ovum passes through the abdominal ostium. Abortion may be complete or incomplete with persistent bleeding. This may cause hematosalpinx, pelvic hematocele**,** or diffuse intraperitoneal hemorrhage.
- ***Tubal rupture*** results in pelvic hematocele and diffuse intraperitoneal hemorrhage.

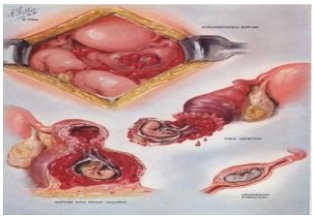


Fig 3-11: Pathology and fate of tubal ectopic gestation

Clinical Picture: Many clinical presentations can occur:

[I] Disturbed ectopic pregnancy, subacute type (hemodynamically stable)

Symptoms:

- Short period of amenorrhea.
- Pain is the **most prominent symptom**. It may be aching pain, stabbing pain or colicky pain. Shoulder pain occurs in massive internal hemorrhage with accumulated blood under the diaphragm irritating the

phrenic nerve.
- Vaginal bleeding: Usually slight due to shedding of decidua after death of the embryo. It follows pain.
- Syncope (fainting attacks): Nausea and vomiting due to peritoneal irritation by blood.

Signs:

General examination:
- Temperature: Low, normal or slightly elevated due to absorption of blood.
- Pulse: Rapid.
- B.P.: falls in proportion of internal bleeding.

Abdominal examination:
- Lower abdominal tenderness and rebound tenderness due to peritoneal irritation.
- Shifting dullness could be present.

Vaginal examination:
- Marked tenderness on moving the cervix on the other side of ectopic.
- Uterus is slightly enlarged and an adnexal mass could be felt.

[II] Disturbed ectopic pregnancy with severe intra peritoneal hemorrhage, acute type (hemodynamically unstable):

The patient is shocked with evidence of severe intra peritoneal hemorrhage. It is an emergency case and urgent surgical interference is mandatory.

[III] Undisturbed tubal pregnancy:

No symptoms or mild pelvic pain is present. The patient is discovered accidently during antenatal care.

Specific investigations:

The most important are combined U.S. and estimation of β-hCG.

- **TVS**:
  - Diagnostic findings are empty uterus with adnexal mass and pelvic fluid collection.
  - Problems: the minimal limit for vaginal U.S. to visualize early intrauterine pregnancy is about 5 weeks, so empty uterus alone by U.S. can't exclude earlier uterine pregnancy. Also, adnexal mass is not necessarily an ectopic pregnancy; it could be a corpus luteum cyst.
- **Serum βhCG level**:
  - In lower levels, pregnancy is either ectopic pregnancy or early intrauterine pregnancy. Repeat βhCG level after 48 hours (doubling time). If the level is doubled with an increase by at least 60 % from the previous level, it is mostly normally growing intrauterine pregnancy that will be detected later on by US. Improper doubling indicates unhealthy ectopic pregnancy.
- **Discrimination zone: combine B-HCG with US.**
  - In TVS, a gestational sac should be seen intrauterine with a B-HCG level of

1500 IU/ML., if this level is reached while the uterus is empty, this confirms an ectopic pregnancy.

- In TAS, the discrimination level is 3500 IU/ ML.

- **Laparoscopy:**

It could be used to confirm the diagnosis in doubtful cases (hemodynamically stable cases).

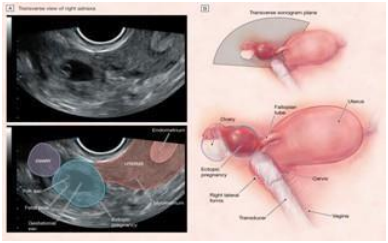


Fig 3-12: TVS right Unruptured tubal ectopic ruptured pregnancy

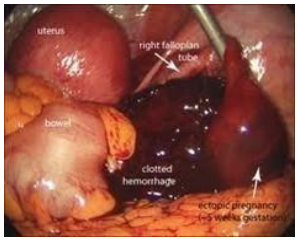


Fig 3-13: laparoscopy right tubal ectopic

Management of tubal pregnancy:

- Blood transfusion and antishock measures may be needed in acute cases.
- AntiD should be given to RH–ve cases after management.

(A) Surgical treatment:

Either by laparoscopy or open laparotomy:

- Conservative surgical treatment**: Salpingostomy** with linear incision in the tube to remove the ectopic and this incision will heal spontaneously by 2ry intention. It could be done in young patients to preserve the tube if not markedly damaged. Recurrence rate is very high on the same side.
- Unilateral **salpingectomy**.

(B) Medical treatment by methotrexate injections:

- Mode of action: antifolinic acid that helps autolysis of chorionic villi.
- Indicated in undisturbed early cases with small ectopic < 3cm, no heart activity and hCG is <6000mIU/ml.
- CBC, liver and kidney functions should be checked before receiving injection.
- Single IM injection regimen; 1mg/kg body weight.
- Repeat serum B-HCG at days **4 & 7** from 1st injection to insure regression by at least **15%** between return days

N.B: *if HCG levels fail to decline properly another IM dose of MTX may be given and if levels still fail to decline properly laparoscopic intervention should be considered.*

- Subsequent pregnancy should be delayed for 3months after methotrexate.

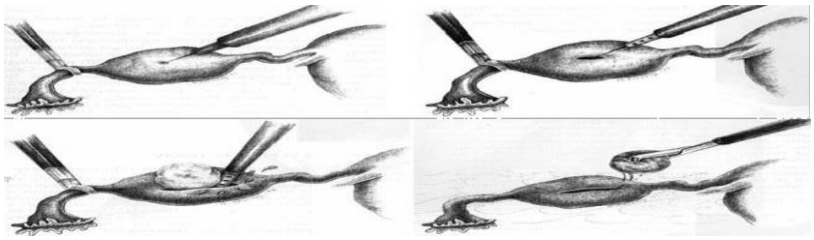


Fig 3-14: Laparoscopic Salpingostomy for undisturbed tubal ectopic pregnancy

Other Types of Ectopic Pregnancy

Ovarian Ectopic Pregnancy:

- Ovarian ectopic produces similar symptoms and signs to tubal ectopic with slightly less prominent pain. Diagnosis is difficult without laparoscopy which can be used for treatment.
- Aim of treatment is to remove ectopic pregnancy while preserving reasonable ovarian tissue.
- IM Methotrexate if pre-requisites apply.
- Oophorectomy is sometimes mandatory to prevent or stop severe hemorrhage.

Advanced Abdominal Pregnancy:

- Pregnancies rarely implant within the abdominal cavity (e.g.; on omentum, bowel, or visceral peritoneum), where it may sometimes proceed to 2nd or rarely third trimester.
- Diagnosis is usually only made at laparotomy where removing the placenta poses a great risk, being attached to important vital organs and sometimes major vessels. Any attempt at removal may cause massive internal hemorrhage.
- In most cases it is best to leave the placenta attached allowing for spontaneous autolysis.

Pregnancy in a Rudimentary Horn:

- It ruptures at about 4th and 5th months (later than tubal ectopic due to presence of myometrium, although poorly developed).
- Clinical picture is that of ectopic but its diagnosis is difficult due to later disturbance and usually associated with marked internal hemorrhage.

Cervical Ectopic Pregnancy:

- It usually presents with severe vaginal bleeding rather than pain. Attempts to remove products of conception from the cervix might fail leading to profuse bleeding. US will show contents filling the cervical canal below the internal cervical os with empty endometrial cavity.
- IM injection of methotrexate and arterial embolization can be used if bleeding is mild, and sac is small.

- Hysterectomy may be needed in cervical ectopic with life threatening vaginal bleeding.

CS Scar Pregnancy:

- The presence of a gestational sac within a previous CS scar is being reported with increasing frequency. Diagnosis is done mainly by transvaginal ultrasonography.
- IM methotrexate or ultrasound guided feticide by a special needle are all treatment options.
- In some cases, surgical intervention by hysterotomy may be necessary especially if deep invasion of scar has occurred.

N.B. Pregnancy of Unknown Location (PUL)

This term is used to describe a possible ectopic pregnancy (abnormal B-HCG doubling and absent intrauterine gestational sac), without reassuring evidence about being extra uterine. The term is used until a final diagnosis is made which could be either; a normal intrauterine pregnancy, an abnormal intrauterine pregnancy (abortion), or an ectopic pregnancy.



Fig 3-15: tubal ectopic pregnancy



Fig 3-16: ovarian ectopic



Fig 3-17: US picture of cervical ectopic



Fig 3-18: Rudimentary horn ectopic

GESTATIONAL TROPHOBLASTIC DISEASE (GTD)

Gestational Trophoblastic Disease (GTD) is the term used to describe a spectrum of cellular proliferations arising from the placental trophoblasts.

Clinico-pathological forms:

- **Benign GTD**: Vesicular (Hydatidiform) Mole. This may be complete and partial.
- **Malignant GTD:**
  - **Non metastatic:** Invasive Mole
  - **Metastatic:** Choriocarcinoma

VESICULAR (HYDATIDIFORM) MOLE

Molar pregnancies (hydatidiform mole) are rare. It may reach up to 1:125 pregnancies in some Asian countries. Molar pregnancies represent the **benign** form of **GTD** and are either complete or partial moles.

Risk factors:

- Age: increased risk 5-10 times with extremes of age (<20years &>35years).
- Dietary factors: increased risk with low intake of vitamin A and animal fat.
- The risk of a second molar pregnancy is **doubled,** while that of a third one may reach **33%**.

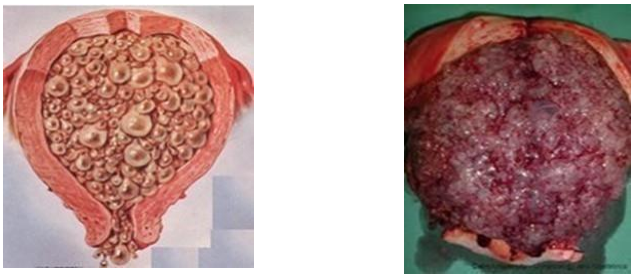


Fig 3-19: vesicular mole

Pathology:

A. **Complete mole:**

A complete mole originates from **fertilization** of an **empty ovum.** **Duplication** of the **DNA** of **one** sperm resulting in a **46 XX** karyotype.

Gross Morphology:

The uterine cavity will be distended with **grape like** thin-wall **vesicles** that vary in size from few millimeters to a few centimeters in diameter (distended chorionic villi).

-Hydatidiform changes are **generalized** with **absence** of **amniotic sac** or **fetal tissue**.

Histological features:

- Diffuse **proliferation** of the trophoblastic cells.
- Marked **edema** and enlargement of the villous stroma (hydropic villi).
- Disappearance of the villous fetal blood vessels.

B. **Partial mole:**

A partial mole originates from fertilization of a normal ovum by 2 sperms leading to an embryo with triploidy (69 chromosomes). Many spontaneous abortions may represent undiagnosed partial moles.

Gross Morphology:

- Hydatidiform changes are only **focal**, with **presence** of **amniotic sac** or **fetal tissues**.
- The **placenta** shows large areas with molar change and malformed fetus.

Histologic Features:

- Focal hydropic villi with only focal trophoblastic proliferation.

- Presence of umbilical cord, amniotic membrane, or fetal tissues.

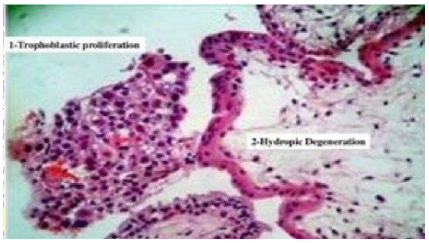


Fig 3-20: Histology of complete mole

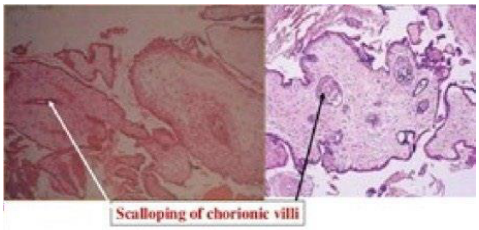


Fig 3-21: Histology of partial mole

c. Associated ovarian cysts:

- Functional ovarian **theca lutein cysts** commonly occur in **25-60 %** of **complete moles**, as a form of ovarian hyperstimulation in response to the huge amounts of hCG secreted by proliferating trophoblastic cells. They are much less commonly associated with partial moles.
- Cysts are usually bilateral, varying in size (few mm up to 10 cm in diameter), with a yellowish color and a smooth surface.
- ***Fate and complications:***
  These *theca lutein cysts* will undergo spontaneous regression after molar evacuation.



Fig 3-22: Theca lutein cysts NE appearance



Fig 3-23: Theca lutein cysts US picture

Clinical picture of molar pregnancy:

- Recurrent early pregnancy bleeding is the chief symptom.
- Other symptoms include:
  - Severe *nausea and vomiting* in early pregnancy (due to high hCG levels).
  - Characteristic **prune juice** discharge or may be vesicles expulsion.
  - *Pain* is uncommon. It may be due to separation of vesicles and hemorrhage with starting expulsion.
- *General Examination:*
  - *Pallor* due to anemia or *Hypovolemic shock;* in cases of sudden severe bleeding.
  - *Signs of preeclampsia (PE)* are rarely encountered in cases reaching the 2nd trimester.
- *Abdominal Examination:*
  - *Fundal level:* larger than calculated gestational age due to rapidly proliferating vesicles

- *Consistency:* uterus is **soft** and **doughy** due to absence of amniotic fluid.
- *Fetal parts* and *fetal pulsations* are **absent** in complete moles.

- *Vaginal Examination:*
  - *The cervix* is soft in consistency.
  - *Enlarged ovaries* with theca lutein cysts may be palpable in the pouch of Douglas.
  - *Passage of vesicles* in association with vaginal bleeding is diagnostic of the condition.

Diagnostic investigations:

- **Ultrasonography** is the **gold standard** in diagnosis of molar pregnancy;
  Complete mole: shows a characteristic picture of **snowstorm** appearance due to presence of vesicles surrounded by blood in absence of fetal echoes or amniotic sac. *Ovarian theca lutein cyst s*are usually present in association with complete moles
  Partial Mole: An **embryo** usually in the first trimester, commonly associated with multiple fetal congenital anomalies and vesicular **hydropic** changes within the **placenta**.
- *Serum β-hCG:* Abnormally high serumβ-hCG levels, reaching >100,000mIU/ml in the first 8 weeks gestation. Also, a baseline for follow up after treatment.
- Metastatic workup for possibility of malignant changes e.g. CT chest and brain or others.



Fig 3-24: US showing snowstorm appearance characteristic of a complete HDFM

Complications of molar pregnancy:

- Severe nausea & vomiting (Hyperemesis gravidarum): due to abnormally elevated hCG levels.
- *Severe bleeding* when starting expulsion which may lead to hypovolemic shock
- *Early onset pregnancy induced hypertension* (Gestational HTN or preeclampsia PE)
- *Hyperthyroidism*: in7% of cases may occur due to thyroid stimulating effect of hCG.
- *Trophoblastic embolization* to the lungs causing RDS
- *Complications of ovarian cysts* (torsion, hemorrhage, rupture…)

- *Risk of developing an invasive Mole* (**20%** in a complete mole, and **4%** in a partial moles)
- *Risk of developing Choriocarcinoma* (**5%** of cases)

Management of molar pregnancy:

Suction evacuation:

- This is the gold standard technique for evacuating molar pregnancies
- Vaginal PGLE1(Misoprostol) may be used to ripen the cervix for easier dilatation
- IV Ecbolics are used to minimize blood loss and enhance uterine contractility.
- Gentle curettage of the uterine cavity is performed after suction to ensure complete molar evacuation with no residual trophoblastic tissue or

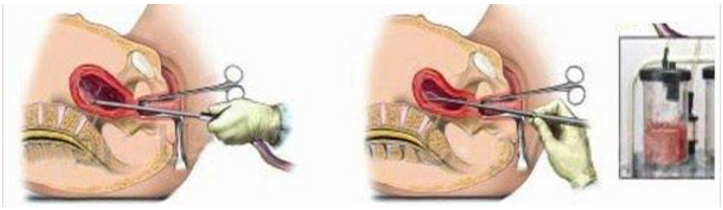


Fig 3-25: Suction evacuation and curettage

Hysterectomy:

Abdominal hysterectomy has a limited place in the management of hydatiform mole. It is indicated only in elderly patients above 40 years with completed family.

Follow up:

- **Repeated serum B-hCG** assays made at 48 hours post-evacuation, then weekly until results are negative for 3 successive weeks, then monthly for 1 year.
- **Contraception for 1 year** after evacuation of molar pregnancy is recommended and best achieved by combined oral contraceptive pills.
- **Persistent trophoblastic disease (PTD): PTD** is the term used to describe cases with incomplete persistently higher than expected **β-hCG** levels after evacuation (abnormal post-molar regression curves). Methotrexate chemotherapy is usually given.

Malignant Gestational Trophoblastic Diseases (GTD)

- These tumors include: Invasive mole and Choriocarcinoma.
- They are rarely occurring trophoblastic tumors with an incidence of nearly **1/20.000** pregnancies.
- **They** are characterized by their aggressive invasion into the myometrium, and their tendency to metastasize.
- **More than 50 % follows molar pregnancy. Also, they may develop after abortion and rarely after full term pregnancy.**

Clinical classification:

- Non-Metastatic: disease confined to the uterus.
- Metastatic: Blood born spread to the lung (80%), vagina (30%), brain (10%), and liver (10%).
- Metastatic GTN are further divided in to **low** risk and **high-risk** cases according to;

Low Risk Metastatic GTN	High Risk Metastatic GTN
Occurs < 4 months after a previous pregnancy	*Occurs > 4 months after a previous pregnancy*
No previous chemotherapy treatment No spread to brain or liver Good prognosis	*Previous chemotherapy treatment Spread to brain or liver Poor prognosis*

Histopathological classification:

- **Invasive Mole:**
  - These tumors have a tendency for **deep myometrial penetration** yet with a **limited** ability for metastases (more like locally malignant tumors).
  - They can occur after complete moles (20%), or partial moles (4%).
  - **Grossly:** deep myometrial invasion with focal areas of necrosis and hemorrhage
  - **Microscopically:** deep myometrial penetration by trophoblastic cells and whole villi.
- **Gestational Choriocarcinoma:**
  - A highly malignant trophoblastic tumor usually associated with **early** blood borne **metastases**, mostly to the lungs and the vagina.
  - **Grossly:** rapidly growing dark **red friable mass** invading both myometrium and blood vessels, causing hemorrhage and necrosis.
  - **Microscopically:** columns and sheets of trophoblastic cells invading the myometrial muscles and blood vessels. There is complete **loss** of the **villous pattern.**

FIGO anatomical staging:

- Stage I: Disease confined to the **uterus**
- Stage II: Outside uterus but limited to **genital tract** structures.
- Stage III: Extends to the **lungs**, with or without known genital tract involvement
- Stage IV: **All** other metastatic sites e.g. liver or brain.

Diagnosis of Malignant GTD:

- Persistent **vaginal bleeding following pregnancy event; mostly molar pregnancy, abortion or rarely persistent bleeding after delivery.**
- Abnormal **high hCG** levels that either rise, plateau, or fail to decrease after a recently terminated pregnancy.
- *TVS/TAS:* May or may not show evidence of uterine lesion. Theca lutein

cysts may be present.

- **Pelvic MRI** is useful in evaluation myometrial **invasion** if present
- **Other metastatic sites may be detected by chest and abdomen CT** scan or MRI.

Management:

Chemotherapy: It is the **main line of treatment**, either single agent as **methotrexate** (in low-risk cases) or by multiple agents as MTX, Actinomycin-D, and Cyclophosphamide (MAC) (in high-risk cases). Follow up protocol by **weekly** serial hCG levels until negative then **monthly** for 1 year.

Hysterectomy: May be offered to elderly multiparous patients with no desire for future fertility to decrease incidence of recurrence and to decrease dose and toxicity of chemotherapy.

It is also indicated in cases with severe vaginal bleeding and resistant to chemotherapy.

04 ANTEPARTUM HEMORRHAGE

ANTEPARTUM HEMORRHAGE

Antepartum hemorrhage **(APH)** is the term used to describe bleeding occurring after fetal **viability** till **delivery** (i.e.; bleeding between **24-40** weeks).

Bleeding in **APH** is mostly **maternal** in origin, and when severe may directly affect maternal vital signs and general condition before affecting fetal wellbeing. Less commonly bleeding may be **fetal** in origin (vasa previa) rapidly affecting fetal wellbeing before significant maternal affection.

CAUSES AND DIFFERENTIAL DIAGNOSIS OF APH

1. **OBSTETRIC CAUSES:** (detected by ultrasound examination)
   - *Placental abruption:* bleeding from separation of a normally implanted placenta.
   - *Placenta Previa:* bleeding from separation of an abnormally low implanted placenta.
   - *Vasa Previa:* bleeding from the umbilical cord (fetal origin).
2. **GYNECOLOGICAL CAUSES:** (detected by speculum examination)
   - *Vaginal trauma or lesion;* as bleeding after sexual intercourse
   - *Benign (erosion or polyp) or malignant cervical conditions.*
3. **GENERAL OR SYSTEMIC CAUSES:** (detected from medical history of the patient)
   - *Anticoagulant therapy:* using drugs as Heparin, Clexane, and oral anticoagulants.
   - *Idiopathic thrombocytopenic purpura (ITP).*
   - *Hypertension (HTN)* only if severe uncontrolled.

N.B.: *Vaginal bleeding should be always differentiated from;*

- Urinary tract bleeding; as bleeding in association with severe UTI or renal stones.
- Rectal bleeding; as in cases of bleeding piles.

PLACENTAL ABRUPTION

Placental abruption describes bleeding due to **separation** of a normally implanted placenta. It may complicate up to **1%** of all pregnancies, and is considered the most common cause for antepartum and intrapartum hemorrhage. In 2/3 of cases, abruption is classified as severe based on associated maternal, fetal, and neonatal morbidity.

Bleeding from placental abruption, also known as **accidental hemorrhage,** significantly affects fetal wellbeing and is an important cause for intrauterine fetal

death (IUFD), preterm labor (PTL), intrapartum hemorrhage (IPH), intrapartum fetal bradycardia and distress, and may be associated with a serious maternal coagulopathy and DIC.

RISK FACTORS:

1. **Abruption i**n a previous pregnancy.
2. **Maternal hypertension** and severe preeclampsia (PE).
3. **Sudden ROM:** in cases with polyhydramnios due to associated rapid decompression.
4. **Traumatic abruption;** may occur due to external trauma as car accidents, fall from height or direct hit on the abdomen.
5. **Smoking:** increases the risk significantly relevant to the number of cigarette packets.
6. **Thrombophilias,** Factor V Leiden and prothrombin gene mutations, have been associated with IUGR, PE, IUFD, placental infarcts, and placental abruption.

TYPES OF BLEEDING IN PLACENTAL ABRUPTION:

1. **Revealed type:** where blood dissects its way between the placental membranes to the internal os, to be revealed externally through the vagina (this is the commonest type).
2. **Concealed type:** blood remains entrapped forming a retro-placental hematoma with no clinically detectable external vaginal bleeding (least common and most serious type). Mostly occurs with severe PE, associated with severe pain and hard tender uterus and fetal distress.
3. **Mixed type:** where bleeding is partially concealed and partially revealed (the 2nd most com-mon type).

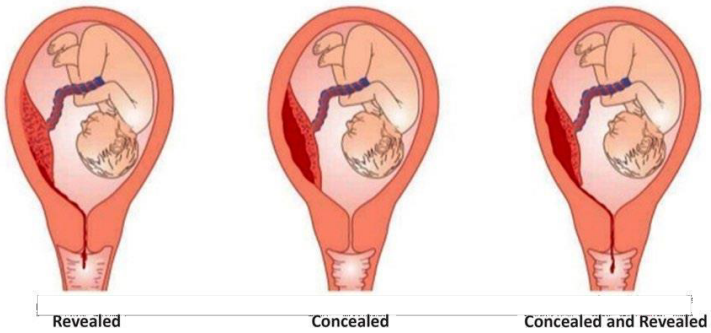


Fig 4-1: Clinical types of placental abruption

CLINICAL PICTURE:

- ***Mild placental separation and bleeding:*** does not affect the vital signs or the general condition.
- ***Severe placental separation and bleeding:*** is usually associated with **hemorrhagic shock.**
  - The degree of shock may not correlate with the amount of external

bleeding in cases with combined or concealed hemorrhage.

- Hypotension may be masked by the essentially elevated BP in cases with pre-eclampsia (PE)
- DIC may occur in cases with concealed hemorrhage with large retroplacental hematoma.

ABDOMINAL EXAMINATION:

- *Uterine tenderness* is usually present with severity related to the degree of abruption and amount of retroplacental bleeding.
- A *rising fundal level* may be noticed in cases with an accumulating retroplacental hematoma in cases with concealed accidental hemorrhage.
- **N.B.: Vaginal examination** could be performed ONLY after EXCLUSION of PL PRV by US (as DD being the other common cause of APH)

DIAGNOSIS:

- TAS is a gold standard in identifying placental site and hence **exclusion** of PL PRV. It also detects a large retroplacental **hematoma** in cases with combined or concealed HE.
- **Laboratory tests:** CBC, liver & kidney function tests, in addition to coagulation profile.
- **Assessment of fetal wellbeing** through FHR tracing, CTG, Doppler US.

PATHOLOGY AND COMPLICATIONS OF SEVERE PALCENTAL ABRUPTION:

1. **Hemorrhagic shock** which might be complicated by acute renal tubular necrosis.
2. **Sheehan syndrome** (acute pituitary necrosis) may also occur as a result of hemorrhagic shock.
3. **DIC** due to thromboplastin release from the damaged muscle fibers and consumption of clotting factors in the retroplacental hematoma (hypofibrinogenemia)
4. **Couvelaire's uterus:** in cases of concealed HGE when entrapped blood in a retroplacental hematoma dissects its way within myometrium to reach the serosal surface leading to bruises like lesions with failure of uterine contractions and atonic postpartum hemorrhage.
5. **Fetal Complications:** include fetal distress, severe IUGR, IUFD, preterm delivery with high neonatal morbidity and mortality especially in cases of severe concealed accidental hemorrhage.

MANAGEMENT:

Placental abruption can be managed either **actively** by immediate termination of pregnancy, or
expectantly by conservative prolongation of pregnancy according to:

- Severity of bleeding (whether external or internal) and its effect on maternal & fetal conditions.
- Gestational age at the time of APH.
- Onset of true labor pains.

A) ACTIVE MANAGEMENT by IMMEDIATE DELIVERY in cases with;

1. *Severe bleeding* compromising maternal or fetal lives, **regardless** gestational age.
2. *Gestational age > 36 weeks* **regardless** amount of bleeding.
3. *Spontaneous onset of labor pains,* regardless amount of bleeding and gestational age.

B) EXPECTANT MANAGEMENT to DELAY DELIVERY in preterm pregnancies < 36 weeks to allow more time to achieve better lung maturity whenever:

1. *Bleeding is mild* not affecting maternal or fetal well-being.
2. *Labor pains* are absent.

Expectant management is continued until any indication for active management arises (lung maturity achieved, onset of labor pains, or an attack of severe APH).

- Vaginal delivery is attempted in cases with mild to moderate bleeding whenever possible.
- CS is indicated whenever bleeding is severe affecting maternal or fetal wellbeing, or in presence of any other obstetric indication for CS (Prior CS, CPD, Breech presentation, etc…)

N.B.: Non sensitized **Rh-ve** mothers will deserve an **Anti-D immunoglobulin** injection during their expectant period of management or if bleeding occurs ≥ 3 weeks from a previous attack.

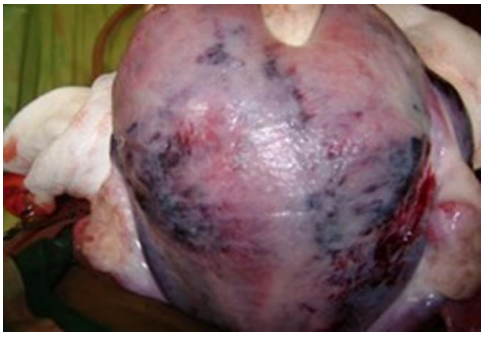


Fig 4-2: Couvelaire's uterus in concealed accidental hemorrhage

PLACENTA PREVIA

Placenta previa **(PL PRV)** describes **implantation** of the placenta partially or totally on the
lower uterine segment **(LUS)**. It is classified according to its relation to the internal cervical os into **total (central), partial**, and **marginal** PL PRV. The nearer the placental margin to the internal os, the more will be the liability to earlier and more severe type of **APH**.

PL PRV is the 2nd **most common** cause of **APH,** being preceded only by placental abruption.

PL PRV may sometimes **adhere to** and/or **invade** the myometrium especially at the site of a previous CS scar, leading to an abnormally invasive placenta **(AIP),** one of the most life-threatening obstetric complications encountered nowadays especially with the increasing rates of CS.

INCIDENCE:

PL PRV with all its types may be encountered in around **0.5%** of **all** pregnancies.

The incidence of PL PRV has been showing a ***progressive rise*** in the last 2 decades due to increasing maternal age and increasing incidence of repeated deliveries by CS.

PREDISPOSING AND RISK FACTORS:

1. **Previous placenta previa,** increases the risk of a future PL PRV in a next pregnancy.
2. **Previous cesarean section,** risk increases proportionately with the number of previous cesarean sections, especially the risk of abnormally **invading placenta.**
3. **Increasing** maternal **age and parity.**
4. **Increased placental size** (as in **GDM, twins,** & **multifetal** pregnancies) is an important risk factors for a PL PRV.

MECHANISM OF BLEEDING IN PL PRV:

Bleeding in PL PRV occurs due to **unavoidable separation** of the placenta as a result of stretching of the LUS while the pregnancy is ongoing. Stretching of the LUS may occur either;

- **Gradually** at 3rd trimester (causing mild recurrent APH), **or**
- **Suddenly** with the onset of labor pains and uterine contractions causing rapid separation with sudden onset of severe bleeding.

PLACENTAL MIGRATION:

- Some patients will have a degree of encroachment of their lower placental margin on the LUS as revealed by US examination done **< 24weeks**

gestation. With the development of the LUS, a relative upward **placental migration** occurs, with almost **90%** of these resolving by the third trimester.

CLASSIFICATION AND TYPES OF PL PRV:

PL PRV is classified according to the relationship of the placenta to the internal cervical os into;

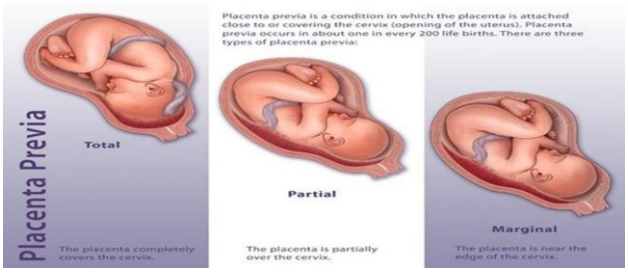


Fig4-3: Classification of the types of PL PRV

Grade I: (lateralis) lower edge of placenta is more than 2cm away from internal os.
Grade II: (marginalis), lower edge of placenta reaching the internal os.
Grade III (incomplete centralis) placenta partially covers the cervix.
Grade IV (complete centralis) placenta completely covers the cervix.

SYMPTOMS & SIGNS:

- **No Symptoms:** 1/3 of cases of PL PRV may be asymptomatic till the onset of labor.
- **Vaginal Bleeding:** painless causeless recurrent **vaginal bleeding** is the most common clinical presentation in the majority of cases (APH).
- **General Examination:**
  - *Mild APH:* no significant signs
  - *Severe APH:* signs of acute massive blood loss including; pallor, tachycardia, tachypnea, rising pulse, falling BP, oliguria, and loss of consciousness (hemorrhagic shock).
- **Abdominal Examination:** shows no specific signs, however **malpresentations** especially breech and transverse lie are more frequently encountered as the low-lying placenta will usually interfere with adaptation of the presenting part through the pelvic inlet.
- **Vaginal Examination (PV):** should not be attempted for diagnosis as it may provoke placental separation with consequent severe uncontrollable bleeding.

DIAGNOSIS:

A) **TRANSABDOMINAL US (TAS):** this is the **gold standard** for diagnosis of PL PRV. It is also accurate in classifying PL PRV into complete, partial, or marginal according to its relation to the internal cervical os. If the lower

placental margin is > 2.0 cm away from the internal os it may be then considered a low-lying placenta (lateral PL PRV) which carries less risk for bleeding and separation, and does not contraindicate vaginal delivery.

B) **TRANSVAGINAL US (TVS):** is more reliable than TAS in measuring the distance between lower placental margin and internal os to differentiate a marginal from lateral PL PRV when in doubt.

C) ***Maternal laboratory test:*** CBC, liver & kidney function test, blood glucose level, PT, PC, INR.

D) ***Tests for fetal wellbeing:*** NST, BPP, UA Doppler velocimetry studies, amniotic fluid index.

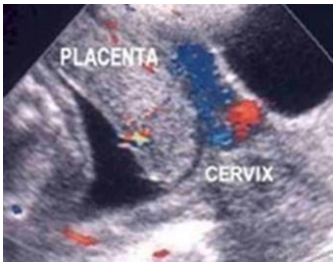


Fig 4-4: US picture of a partial PL PRV

MATERNAL AND FETAL RISKS AND COMPLICATIONS IN PL PRV:

1. Severe **APH** leading to hypovolemic **shock** with all its complications including **DIC.**
2. Severe postpartum hemorrhage **(PPH):** due to increased vascularity of the LUS and its defective retraction, together with possible retained placental fragments and uterine atony.
3. Increased risk of **CS hysterectomy** due to uncontrolled bleeding from dilated placental bed on the LUS which has a less contractile and retractile power than the upper uterine segment.
4. Increased risk of **abnormally invasive placenta** especially in presence of previous CS scar, with all its associated complications (see later).
5. **Fetal complications** primarily increased perinatal morbidity and mortality due to prematurity, IUGR, and massive blood loss.

MANAGEMENT OF PL PRV:

All cases of PL PRV (total, partial & marginal) will be delivered by CS. The timing of the procedure will depend upon many variables including;

- The *severity of APH* if present, and its effect on maternal and/or fetal wellbeing.
- The *gestational age* at time of diagnosis (term or preterm).
- Presence or absence of *uterine contractions* (onset of labor pains).

A) **ACTIVE MANAGEMENT BY IMMEDIATE CS**:

1. *Severe bleeding* compromising maternal or fetal lives, **regardless** gestational age.
2. *Gestational age > 36 weeks* **regardless** amount of bleeding.
3. *Spontaneous onset of labor pains,* regardless amount of bleeding and gestational age

- ***Precautions during performing CS for PL PRV:***
  - Immediate **resuscitation**, anti-shock measures and **blood transfusion** before surgery.
  - Liberal use of **ecbolics** (Oxytocin, Ergometrin, & PGL) to avoid uterine atony.
  - Consider the need for bilateral **uterine artery ligation** or **CS hysterectomy** if required.

B) **EXPECTANT MANAGEMENT to DELAY DELIVERY**:

In preterm pregnancies < 36 weeks to allow more time to achieve better lung maturity whenever:

1. *Bleeding is mild* not affecting maternal or fetal well-being.
2. *Labor pains* are absent.

- ***Prerequisites for expectant management:*** Hospitalization, bed rest, blood bank reservation, and biweekly evaluation of maternal and fetal wellbeing performed on regular basis and continued until any indication for active management arises (best achieved lung maturity, or onset of labor pains, or an attack of severe vaginal bleeding, or non-reassuring tests of fetal wellbeing).

C) **Cases with Lateral PL PRV** (lower edge 2 cm or more above the internal os) can be offered a trial vaginal delivery at onset of labor pains whenever bleeding is minimal.

- **NB:** Unsensitized Rh-ve patients should receive anti-D Ig during the expectant management.

Placenta Accreta Spectrum (Abnormally Invasive Placenta) (Placenta Accreta, Increta, & Percreta)

This is a **rare** but a serious, and may be **fatal**, condition in which the placenta **adheres** to or **invades** the myometrium. It occurs mostly in association with PL PRV due to the poor decidual reaction in the LUS especially at the site of a previous CS scar.

TYPES OF ADHERENT PLACENTA:

- **Accreta;** the placenta is adherent to the myometrium (most common 80%).
- **Increta:** placenta invading the myometrium short of serosa.
- **Percreta:** placenta perforating the myometrium reaching the serosa, and

may invading bladder wall as well.

DIAGNOSIS DURING PREGNANCY:

- Diagnosis can be achieved by second level **expert US** (2D or 3D US and color Doppler).
- US reveals absence of a plane of cleavage between placental and myometrium denoting adherence with presence of blood lacunae in cases of invasion.
- Diagnosis can be also achieved by **MRI** however with no significant benefit over expert US

DIAGNOSIS DURING LABOR:

Some cases of focal placenta accreta will be discovered during labor after delivery of the fetus when the placenta is totally or partially retained leading to severe atonic postpartum hemorrhage PPH (see retained placenta).

MANAGEMENT:

The majority of cases will require CS hysterectomy (classical CS with total hysterectomy) as a lifesaving procedure. Ovaries are preserved during surgery (*see retained placenta)*.

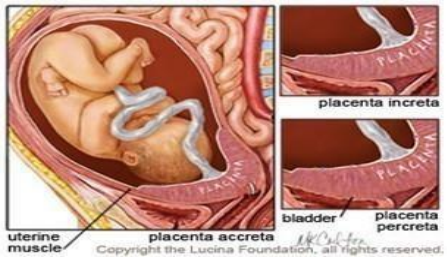


Fig 4-5: Types of adherent placenta

RUPTURE VASA PREVIA (FETAL BLEEDING)

Vasa previa is a very rare condition in which the umbilical **cord** vessels **cross over** the internal os (as in velamentous insertion of the cord, or succenturiate lobe of the placenta). During labor, progressive cervical dilatation may lead to separation and rupture of these vessels leading to fetal bleeding with associated severe fetal distress, and ultimately intrapartum **fetal death** unless the condition is urgently recognized and dealt with properly by an **emergency CS.**

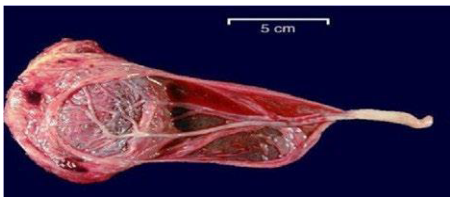


Fig 4-6: Velamentous cord insertion with vasa previa

05 MEDICAL COMPLICATIONS

HYPERTENSIVE DISORDERS OF PREGNANCY

During **pregnancy** both systolic and diastolic blood pressures (BP) show a **constant decline** starting in the late first trimester, followed by a **gradual rise** to pre-pregnancy levels by the end of the 3rd trimester.

Hypertension (HTN) during pregnancy is diagnosed with a **systolic** BP **140** mm Hg or **more** and/or **diastolic** BP of **90** mm Hg or **more**, measured at **2** occasions at least **4** hours apart in the **sitting** position.

Hypertensive disorders in pregnancy, especially **preeclampsia** and **eclampsia** are considered worldwide a leading cause for maternal **mortality**, with mortality mostly due to **CNS** hemorrhage.

Classification of Hypertensive Disorders during Pregnancy
▪ ***Gestational hypertension:*** *hypertension alone first occurring > 20*
▪ *weeks gestation* ***Preeclampsia*** *: hypertension + Proteinuria first*
▪ *occurring > 20 weeks gestation* ***Eclampsia*** *: preeclampsia + seizures*
▪ ***Chronic hypertension*** *: hypertension alone diagnosed before pregnancy < 20 weeks gestation*
▪ ***Superimposed PE*** *: Preeclampsia developing on top of chronic HTN*

GESTATIONAL HYPERTENSION

Gestational hypertension **(G-HTN)** is diagnosed when sustained **hypertension without proteinuria** occurs **> 20 weeks** gestation.

G-HTN is not usually associated with specific **symptoms** apart from mild recurrent headaches encountered in some patients.

Laboratory **tests** show no alterations in kidney or liver function tests, with absent proteinuria on repeated urine analysis.

Preeclampsia **(PE)** should be always and repeatedly ruled out, since cases with G-HTN may develop PE later in pregnancy.

MANAGEMENT:

- **Conservative** outpatient management with close observation on repeated antenatal care visits and repeated urine analysis to rule out PE.
- **Oral antihypertensive drugs**; are only indicated with increasing BP levels ≥ 160/ 100.
- **Fetal surveillance** for proper evaluation of fetal wellbeing including daily fetal movement count (DFMC), non-stress test (NST) and repeated US (see fetal surveillance).

PREECLAMPSIA

Preeclampsia **(PE)** is a **pregnancy specific** syndrome of **hypertension** and **proteinuria** occurring > 20 weeks gestation. It may complicate **3-7%** of all pregnancies, especially cases with high risk factors. **Eclampsia** describes PE with grand mal **seizures.**

DIAGNOSTIC CRITERIA OF PE:

a) **Sustained** elevation of the **BP** to **140/90** mm Hg or **more** on **two** occasions **4** hours apart.

b) **Proteinuria:** > 0.3 g in a 24 hours urine collection, or >1+ on a dipstick test, or a protein (albumin) /creatinine ratio of > 0.3 (**A/C > 0.3)**

In absence of proteinuria, **PE** can be also diagnosed if **new onset** hypertension **> 20** weeks

is associated with a recent **organ dysfunction** manifested by one or more of the following:

- Thrombocytopenia (platelets < 100.000)
- Elevated serum liver transaminases (AST & ALT > twice the norms)
- Rising serum creatinine, CNS symptoms, DIC and pulmonary edema.

PE is generally associated with the development of rapid onset generalized **edema**, especially facial and hand edema, but they are **not diagnostic** of PE.

RISK FACTORS:

1. **Primigravidas:** PE occurs **8 times** more frequently in PGs.
2. Multifetal pregnancy **(MFP)** and hydatidiform mole **(HDFM)** with abnormal placentation.
3. Extremes of **age** especially the very young primigravidas and the very elderly ones.
4. Diabetes mellitus **(DM)** and marked **obesity.**
5. Chronic hypertension **(C-HTN)** and chronic **renal disease** with poor organ perfusion.
6. History of **Previous** PE or Eclampsia or **Family history** of PE or Eclampsia.
7. Systemic Lupus Erythematosus **(SLE)** due to small vessel disease.

PATHOPHYSIOLOGY and THEORIES:

PE is called a disease of **theories** because genetic, immunologic, vascular, hormonal, nutritional, and behavioral factors have all been proposed as theories for its occurrence. Because the disease starts to **resolve** shortly after delivery, a lot of attention has been payed to the role of the **placenta** and **uteroplacental-fetal interface** in the development of the disease.

During a normal pregnancy, cytotrophoblastic **invasion** of intra-

myometrial spiral vessels renders these vessels **wider** and **less responsive** to vasoconstrictor substances, leading to a state of **decreased** vascular resistance and **increased** uteroplacental blood flow.

Inadequate cytotrophoblastic **invasion** of the intra-myometrial spiral arterioles will result in **failure** to establish the normal **low resistance** uteroplacental circulation with a resultant placental **ischemia** with involvement of secondary **mediators** responsible for;

A) **Intravascular Volume Constriction;** capillary ischemia will lead to endothelial injury and loss of intravascular fluid volume into the interstitial space causing **edema**.

B) **Decreased Organ Perfusion;** kidneys, liver, brain, & heart causing;

- Decreased renal clearance of metabolites and proteinuria.
- Sub-capsular liver hemorrhage and necrosis.
- Cerebral edema, petechial hemorrhage, and thrombi.
- Retinal hemorrhage.
- Vascular endothelial damage leading to platelet thrombosis and DIC in severe cases.

CLINICAL AND LABORATORY MANIFESTATIONS:

PE is a disease of **signs** rather than symptoms, with diagnosis based on **BP** measurements together with **laboratory** investigations including **urine** and **blood** samples. It is often a **progressive** disease and its severity is graded both by the severity of hypertension and the degree to which other organ systems are affected. PE is **graded** into mild (non-severe) and severe PE.

CRITERIA FOR MILD (NON-SEVERE) PE:

- Sustained **hypertension > 140/90** mm Hg, but **< 160/ 100mmHg.**
- **Proteinuria**: > 0.3 g in 24 hours, or dip stick test 1-2 +, or Protein / creatinine ratio > 0.3.
- Mild non-specific **symptoms** and **signs:** as recurrent **headaches**, rapid **weight** gain and **bloating,** with pitting lower limb **edema**.
- Normal or mild changes in CBC, platelet count, kidney and liver functions.

CRITERIA FOR SEVERE PE
– ***Severe Hypertension:*** *sustained systolic BP* ***160*** *or* ***more*** *and/or* – *diastolic BP* ***100*** *or* ***more Renal Insufficiency:*** *serum Creatinine* ***> 1.1*** *mg/dl or doubling of baseline values* ***Elevated Liver Enzymes:*** *AST &* – *ALT increased* ***> twice*** *their normal values* ***Thrombocytopenia:*** *Platelet* – *count* ***< 100.000*** */uL.* – ***Epigastric pain*** *or right upper quadrant pain: due to stretch of liver capsule* = ***Cerebral*** *or* ***visual disturbances Pulmonary edema*** – ***Oliguria < 750 ml/24 hours***

- ***HELLP symdrome:*** *Hemolysis, elevated liver enzymes, low platelet count.*

MANAGEMENT OF PREECLAMPSIA:

The only definitive cure for PE is **termination** of pregnancy. Once the placenta is delivered a rapid decline in most of the signs and symptoms of PE will occur.

The goal of management of PE will be to **prevent** maternal complications of severe PE while minimizing the neonatal hazards of **prematurity**.

The **decision** for **termination** of pregnancy and the route of delivery will be strongly affected by the **severity** and progression of PE, the **degree** of fetal compromise if present, and the **gestational age**.

GUIDELINES FOR MANAGEMENT:

a) **Mild** cases with **NON** severe **NON** progressive PE with **NO** evidence of **fetal** compromise, will be managed **conservatively**, not to be delivered unless gestational age is **38** weeks or older.

b) Cases with **Severe** PE in which the **fetus** is **not** compromised, can be offered an **expectant management** (anti-hypertensive ttt and follow up) till reaching **36 weeks**, whenever possible, to ensure adequate lung maturity.

c) Cases with **Severe** or **Progressive** PE, with associated **fetal compromise**, or with **HELLP** syndrome should be immediately **delivered** after **stabilization**.

MEASURES FOR STABILIZATION OF A PATIENT WITH SEVERE PE:

- **Hospitalization** is necessary for severe PE until stabilization is achieved or termination of pregnancy is indicated.
- **Antihypertensive therapy:** (see later)
  - Mild non-progressive disease does not need antihypertensive therapy.
  - Cases with **severe** or **progressive** PE will receive treatment to control the rising BP and avoid cerebral hemorrhage. Therapy is immediately started for control of BP, and continues till delivery.
- **Delivery** is indicated whenever pregnancy reaches **36** weeks, or **earlier** if signs and symptoms of worsening disease develop, or if there is evidence of fetal compromise.
- **Initial stabilization** in cases with **preterm severe PE** is achieved through hospitalization and;
  - *Antihypertensive Therapy*: ***Labetalol*** and ***hydralazine*** as drugs of choice (see later)
  - *Seizure Prophylaxis:* ***Mg SO4*** *IV* as loading and maintenance doses (see later)
  - *Enhancement of lung maturity: IM* ***Betamethasone*** *or* ***Dexamethasone*** *24 mg in 24 hrs* for minimizing risks of prematurity.

- **Termination of pregnancy** can be achieved by either Induction of labor or Elective CS.

ANTIHYPERTENSIVE DRUGS IN PE/ECLAMPSIA:

Antihypertensive medications are indicated in **all** cases with **severe** PE (BP ≥ 160/100)**.** The goal for therapy is to **control** the BP, and **avoid** complications of cerebral hemorrhage.

BP should be lowered gradually (not too rapid) to levels **not lower** than **140/90** mm Hg to avoid a sudden decrease in uteroplacental perfusion that may cause acute fetal distress.

1. **Labetalol:** is an alpha and beta receptor inhibitor that is currently the most popularly used medication for safe and efficient control of BP in cases with **severe PE** (dose: 20 mg oral tablets 1-2 tablets daily).
2. **Methyldopa:** is an alpha receptor inhibitor that has long been the drug of choice in the management of **PE**. It has been lately reported to be associated with IUGR and postpartum depression (dose: 250 mg orally 2-6 tablets daily).
3. **Hydralazine:** is an efficient drug that acts directly on the blood vessels leading to rapid vasodilatation and lowering the BP. It is used in cases with severe and progressive PE and impending eclampsia; however, an IV bolus dose may be associated with a rapid drop in BP which is sometimes hazardous to the fetus as it may compromise placental perfusion.
4. **Calcium Channel Blockers:** act as direct vasodilators and are effective in cases with progressive severe PE especially in preterm PE as they also have a tocolytic effect. They are not contraindicated in association with Mg SO_4, but may be associated with some undesirable side effects as headaches and tachycardia. Dosage: Epilat 10 mg oral tablets twice daily till control.
5. **Magnesium Sulphate (Mg SO_4):** this drug has long been used both IV & IM mainly for seizure prophylaxis and control (see later).

PREVENTION AND PREDICTION OF PE:

- ***Serum biomarkers*** as pregnancy associated specific proteins (PAPP-A), inhibin A , have been proposed, and promising results are now appearing.
- ***Uterine Artery Doppler ultrasound*** done in each trimester may show **high** resistance flow with **diastolic notching** in women at risk for PE later in pregnancy.
- ***Uric acid level elevation*** *is seen in patients who may develop PE later on.*
- ***Low Dose Aspirin (LDA)*** 75-150 mg **daily** given early in pregnancy has been shown to minimize the risk for recurrent PE and IUGR in

high-risk cases with a previous history of the disease. LDA acts by inhibiting platelet aggregation and inhibition of thromboxane A2(vasoconstrictor PGL) without impairing synthesis of prostacyclin (vasodilator PGL).

ECLAMPSIA

Eclampsia is the occurrence of **seizures** (convulsions or fits) in a woman with **PE** that cannot be attributed to any other underlying cause or pre-existing disease. It is considered a true obstetric emergency that needs an expert team for proper management.

INCIDENCE AND ETIOLOGY:

Eclampsia may complicate **1-2%** of cases of **PE**. The exact etiology is **unknown** however **severe diffuse cerebral vasospasm** may be the chief underlying factor that causes poor cerebral **perfusion** and cerebral **edema.**

TIMING and CLINICAL PICTURE:

- **Antepartum:** during pregnancy (65%).
- **Intrapartum:** during delivery (20%).
- **Postpartum:** after delivery in early puerperium (15%).

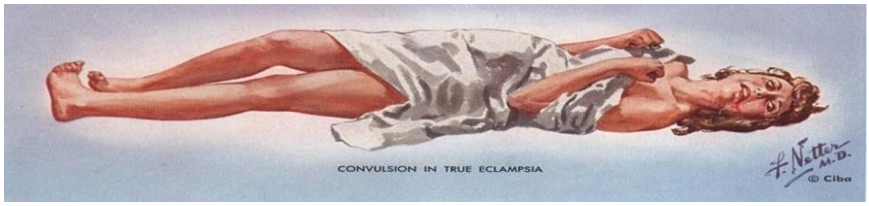


Fig 5-1: Eclamptic fit

The patient with **severe PE** will experience;

- Sudden unexplained **tonic-clonic seizures** that may persist for **1-2 minutes**.
- During the fit the **tongue** may be bitten and **cyanosis** may occur.
- Seizures are followed by a short period of **coma** followed by **recovery.**
- Convulsions may **recur** without preceding signs or symptoms.

DIFFERENTIATION FROM EPILEPSY:

Epileptic seizures differ from eclamptic fits by having a **history** antedating pregnancy, being preceded by **aura**, and lacking associated **hypertension** and proteinuria.

MANAGEMENT OF ECLAMPSIA:

1. **Protect** the patient's **tongue** (mouth gag) and **ensure airway.**
2. **Stop convulsions:**
   - **$MgSO_4$** bolus IV dose 5 g followed by **maintenance** with IV infusion 2 g/hr (see later).

- **Diazepam** 5-10 mg slowly IV to control convulsions in absence of MgSO4

3. **Lower diastolic BP** < 100 mm Hg with IV hydralazine and or labetalol IV.
4. **Prompt delivery** irrespective gestational age with either **induction of labor** or **CS** according to obstetric indications and fetal condition.
5. **Postpartum care:** MgSO4 should be continued for 24 hr to guard against postpartum seizures.

Mg SO4 THERAPY AND TOXICITY:

- **Mode of action:**
  - Anticonvulsant action through inhibition of neuromuscular transmission.
  - CNS depressant with a sedative effect.
  - Tocolytic with a mild antihypertensive effect.
- **Dose and administration:**
  - Loading dose: 5 g as an IV dose on 100 ml fluid over 15-20 min
  - Maintenance dose: 1-2 g/hr as an IV infusion on 500 ml fluid
- **Therapeutic serum Mg level:** should be kept < 8.0 mg/dL (N: 4.5-9.5 ml/dL)
- **Precautions to prevent toxicity:** ensure the following before giving next dose
  - Preserved patellar (knee jerk) reflex (first to be lost in cases of toxicity).
  - Urine output (> 30 ml/hr).
  - Normal respiratory rate (> 16/min).
- **Antidote in case of toxicity:** Calcium Gluconate 10%, 10 ml by IV route over 3-5 minutes.

GASTROINTESTINAL DISORDERS DURING PREGNANCY

NAUSEA AND VOMITING OF PREGNANCY (NVP)

NVP is a very common condition occurring in up to **80%** of pregnant women during the **first** trimester being most pronounced between **7-11** weeks. The condition is usually tolerable and is mostly expressed as morning sickness with few vomiting episodes throughout the day.

It is a **self-limited** and starts to **decline** spontaneously by the end of the 1st trimester.

- **Etiology:** may be due to the first exposure to **hCG** and its severity correlates with hCG levels.
- **Management:**
  - Reassurance: as the condition is usually self-limiting.
  - Instructions: to eat small frequent protein rich meals and avoid recumbent position after meals, as well to avoid known triggers to nausea as; strong odors, coffee & fatty foods.
  - Antiemetic therapy: Pyridoxine (vitamin B6), and anti-emetics of first line (as Doxylamine).

HYPEREMESIS GRAVIDARUM (HEG)

HEG is defined as **severe persistent NVP** of the **1st trimester** that **affects** the maternal general condition**,** and is associated with **ketosis** and **loss of > 5%** of total **body weight**.

HEG may complicate **1-2 %** of **all pregnancies,** being more prevalent and severe in primigravida **(PGDA)**, multifetal pregnancy **(MFP)**, and gestational trophoblastic disease **(GTD)**.

- **Etiology**: exact etiology is unknown but acceptable theories include;
  - *Hormonal changes* as high hCG and E2 levels.
  - *Psychological abnormalities.*
- **Clinical and Biochemical Changes in Severe HEG:**
  - *Signs of Dehydration:* as hypotension, oliguria, dry tongue, and decreased skin turgor.
  - *Electrolyte imbalance:* decreased Na & K levels (due to excessive loss).
  - *Ketonuria* with depleted urinary chlorides (due to starvation ketosis).
  - *Hemoconcentration* and *decreased BUN* (decreased plasma volume).
  - Slightly *elevated liver enzymes* (AST & ALT) and decreased liver glycogen.
- **Complications:**
  - *Dehydration,* hypotension, pre-renal failure and liver failure in severe cases.
  - *Malory Weiss syndrome;* hematemesis due to vascular tears and rupture with vomiting.
  - *Wernick's Encephalopathy:* is a rare complication due to **degenerative** brain changes related to **vitamin B1** (Thiamine) deficiency. It is manifested by delirium, ataxia and nystagmus. **MRI** reveals petechial hemorrhage in Wernick's area in the brain.

- **Management:**

A) **Mild cases:** treatment as NVP in addition to stronger antiemetic drugs.

B) **Severe cases:** hospitalization, IV fluids, and NPO are mandatory;

- IV normal saline and lactated Ringer's solution are important to correct dehydration and electrolyte imbalance, and avoid prerenal failure.
- Liver supports and correct ketosis and Ketonuria.
- Thiamine (vitamin B1) 100 mg IV diluted in saline infusion.
- Antiemetic drugs are used (most are class B category) including;
  - *Metoclopramide* (Primperan) IM or suppository (Dopamine receptor block).

- *Meclizine* (Navidoxine) IM, or suppository (histamine receptor block).
- *Ondansetron* (Zofran) IV infusion, or IM (serotonin receptor block).

- The few patients who do not respond to the above-mentioned medical therapy may require **nasogastric feeding** or total parenteral nutrition **(TPN).**
- Induction of abortion has a very limited place in management and is rarely resorted to.

DIABETES MELLITUS WITH PREGNANCY

Diabetes mellitus **(DM)** is a disorder in carbohydrate metabolism characterized by hyperglycemia and glucosuria. Prolonged disease will be associated with microangiopathy.

TYPES OF DIABETES MELLITUS IN PREGNANCY

- **Gestational Diabetes Mellitus (GDM):**
  - **GDM** describes carbohydrate intolerance having its **onset** during pregnancy.
  - It is mostly due to the **anti-insulin placental hormones effect HPL**, placental **insulinase**, cortisol and progesterone **(P).**
- **Overt Diabetes Mellitus (ODM):**

- **ODM** describes DM that is known to be present **before** the onset of pregnancy. It may be

- ***Insulin Dependent (type 1):*** juvenile onset, caused by pancreatic islet cell deficiency.
- ***Non-insulin dependent (type 2):*** adult onset, caused by insulin resistance.

RISK FACTORS FOR GDM:

1. Marked Obesity (class 2 or more with BMI > 35).
2. Maternal age > 35 years.
3. Fetal macrosomia in a previous pregnancy (fetal weight > 4.5 kg).
4. Unexplained IUFD, repeated fetal congenital anomaly, and recurrent pregnancy loss.

SCREENING and DIAGNOSIS OF DM IN PREGNANCY:

Done For **All** pregnant women between **24-28** weeks (maximum hormonal anti-insulin effect)

- **Screening Test:** oral glucose tolerance tests:

A) 50g 1-hour oral glucose tolerance test (OGTT);

- Normal values **< 140** mg/dl: denoting a **negative** screening test for GDM.
- Abnormal values **≥ 140** mg/dl: require repeat screening with a **100-g** 1hour **OGTT.**

- **Diagnostic Tests:**
  - Fasting blood sugar (FBS) **> 95** mg/dl.
  - 100-g 1 hour **OGTT** values **> 200 mg/dl** in presence of **FBS** > 95 mg/dl.

COMPLICATIONS OF DIABETES DURING PREGNANCY:

A) Maternal complications

1. Difficult glycemic control due to the anti-insulin effect of placental hormones.
2. Increased incidence of diabetic ketoacidosis (DKA) and preeclampsia (PE).
3. Increased incidence of abortion, PTL, CS, instrumental deliveries, and maternal birth injuries.
4. Increased chances for chorioamnionitis, puerperal sepsis and wound infection.

B) Fetal complications:

1.Fetal congenital anomalies **(FCA):**

- Most common are VSD and open NTD.
- Most specific is caudal regression syndrome (sacral agenesis)

2.Fetal **macrosomia** (weight > 4.5 kg):

Maternal hyperglycemia leads to fetal **hyperglycemia** and **hyperinsulinemia**, resulting in enhanced glycogen synthesis, lipogenesis, and fetal protein synthesis.

3.Fetal **birth injuries** during instrumental delivery and **shoulder** dystocia due to macrosomia.

4.Sudden IUFD in cases with hypoglycemic coma**.**

C) Neonatal complications:

1. **Hypoglycemia:** due to persistent hyperinsulinemia 2ry to hyperglycemia.
2. **Hypocalcemia:** due to failure of neonate to increase parathyroid hormone synthesis.
3. **Polycythemia:** due to elevated erythropoietin 2ry to relative intrauterine hypoxia.
4. **Hyperbilirubinemia:** due to breakdown of excessive neonatal RBCs.
5. **Respiratory distress syndrome (RDS):** caused by delayed pulmonary surfactant production.

MANAGEMENT OF DM DURING PREGNANCY:

In general, diabetic pregnancies are at high risk for poor maternal and fetal outcome unless good glycemic control is achieved starting from early pregnancy till delivery and puerperium.

Management of a diabetic pregnancy entails three important measures to ensure best outcome;

- Proper glycemic control; including diet, exercise, and insulin.
- Fetal surveillance and exclusion of FCA, polyhydramnios and macrosomia.
- Proper timing and preferable route for delivery (NVD or CS).

A) ANTEPARTUM GLYCEMIC CONTROL:

- **Diet Control:** supplying 30 Kcal/kg/day offers good control in 80% of cases of **GDM.**
- **Home** blood glucose **monitoring:** to keep FBS **< 90** mg/dl, **and** 1h after meal **< 140** mg/dl.
- **Oral Hypoglycemic Drugs (OHD):** are contraindicated because of concerns of crossing the placenta and causing fetal adverse effects. Only **Metformin** can be safely given.
- **Insulin Therapy:** Insulin is a large molecule that does not cross the placenta and can be safely used to maintain normal blood glucose levels.
- **Titrating Insulin Doses:** Total daily insulin units = body weight in Kg x 1. Doses are divided 2/3 in the morning and 1/3 at the evening (NPH and regular insulin 50/50).
- **Glycosylated hemoglobin (Hb A1C):** It reflects glycemic control over the past 3months (N: 4.5_6.5).

B) ANTEPARTUM FETAL SURVEILLANCE:

1. **Screening for Fetal Anomalies:**
   - Cases of uncontrolled **Overt DM** may suffer hyperglycemia early in pregnancy and are thus at a higher risk for development of FCA especially NTDs, Cardiac anomalies, and Caudal Regression Syndrome (rare but specific).
   - Cases with **GDM** develop hyperglycemia in the 2nd half of pregnancy and are thus more prone to fetal macrosomia and polyhydramnios.
   - Screening for fetal abnormalities include;
     - First trimester **US** scan for NTDs, anencephaly and nuchal translucency (NT)
     - **US** fetal **anomaly**-scan at 18-22 weeks to assess for structural anomalies as cardiac anomalies and caudal regression.
2. **US Assessment of Fetal Growth and Amniotic fluid volume: Macrosomia** and **Polyhydramnios** are commonly associated with poor diabetic control.
3. **Fetal Surveillance: DFMC, NST, BPP, Doppler (see fetal surveillance)**

C) TIMING DELIVERY IN DIABETIC PREGNANCIES:

- In cases with **good** glycemic control and **no growth disorders** or

polyhydramnios, pregnancy can be prolonged even up to **40** weeks and **vaginal delivery** may be allowed.

- In cases with **poor** glycemic control associated with **growth disorders** and or polyhydramnios, pregnancy should be **terminated** whenever fetal lung maturity is ensured (>36 weeks), or whenever fetal surveillance points to increased risk of IUFD.
- **Corticosteroids** prophylaxis given at 28-32 weeks to enhance lung maturity and minimize neonatal complications (IM Dexamethasone 24 mg/24 hours on 4 divided doses).

INDICATIONS FOR CS IN DIABETIC PREGNANCIES

- Fetal **macrosomia** (weight **≥ 4.5** Kg) for prevention of obstructed labor, shoulder dystocia, and fetal birth injuries.
- **Obstetric** indications as Breech, Transverse lie, placenta previa, contracted pelvis, etc…

CARE OF THE NEWLY BORN INFANT

- Infant of diabetic mother **(IDM)** is more liable to the development of hyaline membrane disease leading to respiratory distress syndrome (RDS).
- 25% glucose I.V. solution given after delivery may be needed to combat the possibility of neonatal hypoglycemia due to overactive fetal pancreas**.**

CARE IN PUERPERIUM

- Reduction of insulin dose to prevent hypoglycemia.
- Proper Antibiotic coverage to avoid puerperal sepsis.

ANEMIA DURING PREGNANCY

PHYSIOLOGIC ANEMIA OF PREGNANCY:

During pregnancy both the plasma volume and red cell mass show a significant increase to meet pregnancy demands. However, the **increase** in plasma volume exceeds the increase in the red cell mass leading to a state of **hemodilution**, usually referred to as physiologic anemia of pregnancy. These changes are more pronounced by the end of the **2nd** trimester.

By the end of **3rd** trimester plasma volume **decreases** gradually with a relative **increase** in the red cell mass leading to a state of mild **hemoconcentration** with spontaneous elevation of Hb & HTC levels, with correction of physiologic anemia.

DEFINITION AND PREVALENCE:

Anemia is an extremely **common** condition in pregnancy and postpartum period. In its severe forms it carries number of health risks to both the mother and the child. It is estimated that about **50%** of women world-wide will suffer some degree of anemia during pregnancy.

During pregnancy anemia is **diagnosed** with hemoglobin (Hb) levels **< 11** gm/dl in the 1st and 3rd trimesters, and **< 10.5** gm/dl in the 2nd trimester.

Maternal **symptoms** and **signs** are usually non-specific, but can include: fatigue, pallor, dyspnea, palpitations and dizziness. They are all only prominent in **severe** types of anemia.

Anemia is **classified** according to its **severity** as being:

- *Mild: Hb 10-10.9 g/dl*
- *Moderate: Hb 7-10 g/dl*
- *Severe: Hb < 7 g/dl*
- *Decompensated: Hb < 4 g/dl*

CAUSES OF ANEMIA IN GENERAL

- In the simplest of terms, anemia results from either; impaired RBCs production, increased RBCs destruction, or excessive blood loss.
- **Iron deficiency** anemia is by far the **commonest** type of anemia encountered during pregnancy mainly due to increasing iron demands, improper dietary intake and inadequate supplementation during pregnancy.
- Anemia due to **blood loss** is the **2nd most** frequently encountered cause in association with;
  - Early pregnancy bleeding; as with abortion, ectopic and molar pregnancies.
  - Antepartum hemorrhage (APH); as with placental abruption and placenta previa.
  - Intrapartum hemorrhage (IPH) and Postpartum hemorrhage (PPH).
- Other types may be present as: **Megaloblastic** anemia, **Hemolytic** anemia, **thalassemia.**

EFFECTS OF ANEMIA ON PREGNANCY OUTCOME:

- **Mild** and **moderate** anemia do not carry really maternal or fetal risks or hazards.
- **Women** with **severe** anemia are at **higher risk** for preterm labor (PTL), postpartum hemorrhage (PPH) and puerperal sepsis.
- **Fetuses** of women with **severe** anemia are at an increased risk of IUGR, LBW, PTL, and susceptibility to neonatal sepsis.

DIAGNOSIS OF IDA:

Complete blood count (CBC) showing **microcytic** hypochromic anemia with low Hb & HTC levels is the mainstay in diagnosis. In addition, the following will be confirmatory;

- Low serum **ferritin <** 30 ng/ml, and low serum **iron** levels.
- Increased total serum iron-binding capacity **TIBG.**

MANAGEMENT OF IDA:

A)Prevention:

- Proper **diet** rich in iron and vitamin C.
- Iron **supplementation** using oral iron preparations.

B) Treatment:

- ***Oral iron therapy***: in cases of mild anemia, hemoglobin rises about 0.5 g per week.
- ***Parenteral iron therapy:*** in cases with moderate anemia.
- ***Blood transfusion*** (Packed cells are preferred): in cases of decompensated anemia or cases with acute blood loss, antepartum, intrapartum, or postpartum.

N.B.: During Puerperium:

- Adequate rest, iron and folate therapy for at least 3 months.
- Any infection should be promptly treated.

CARDIAC DISEASE DURING PREGNANCY

Cardiac disease may be associated with **0.5-1%** of pregnancies. **Rheumatic** heart **valve** diseases account for most of cases in the developing countries. In developed countries, there is more predominance of **congenital** heart disease.

CARDIAC CHANGES DURING PREGNANCY: (see physiologic changes in pregnancy**)**

ASSESSMENT OF A PATIENT'S CARDIAC CONDITION: New-York Heart-Association (NYHA) classification:

Class I: Organic heart disease without limitation of house hold activity

Class II: Limitation of activity (dyspnea and chest pain) at ordinary house-hold duties.

Class III: Limitation of activity at less than ordinary house-hold duties.

Class IV: Dyspnea at rest

N.B.: Pregnancy usually makes the classification worse by one grade.

MANAGEMENT DURING PREGNANCY:

- **Frequency:** More frequent ANC than usual, attended by both obstetricians and cardiologists.
- **Examination:**
  - Cardiac assessment and early detection of heart failure.
  - Recognition of risk factors (anemia, infection, hypertension)
- **Advice:**
  - Bed rest; 9 hours by night, 2 hours by day.
  - Guard against excessive weight gain and risk factors.
  - Dental care and use of an umbrella of antibiotics when there is tooth

extraction.

- **Indications for Digitalis:**
  - If already on this regimen before pregnancy has started.
  - In cases of Class II or more or impending heart failure.
- **Hospitalization:**
  - Cases of Class II at 24 -32 weeks of pregnancy and one week before expected delivery date.
  - Cases of Class III and IV are admitted earlier and for longer periods.

MANAGEMENT DURING LABOR:

- *Proper pain relief* to minimize anxiety and tachycardia. Epidural anesthesia may sometimes cause systemic hypotension.
- *Straining is prohibited*, to minimize venous return and decrease heart rate.
- Delivery in the *Semi-sitting position* with adequate oxygenation.
- *Digitalis* in cases at high risk for heart failure.
- *Shorten the second stage* by low-forceps when necessary.
- *Cesarean section* is done when obstetrically indicated.
- *Antibiotic cover* to guard against SBE.
- *Lasix* may be given at third stage; there is increased venous return after placental delivery.

N.B.: Ergometrine is NOT given in third stage, as VC causes sudden increase in venous return, predisposing to heart failure.

- Breast-feeding is contraindicated ONLY if there is heart failure.
- Proper selection of a method of contraception, for adequate pregnancy spacing.

INDUCTION OF ABORTION:

In the past, first trimester termination of pregnancy was indicated in cases of Class III and IV heart classification, or history of heart failure in a previous pregnancy.

In the past two decades induction of abortion is **rarely indicated** after introduction of recent surgical treatment of heart lesions, and availability of proper medical control of chronic cases.

Apart from some cases with pulmonary hypertension and right to left shunt, termination of pregnancy is not indicated in cardiac cases.

URINARY TRACT INFECTIONS DURING PREGNANCY (UTI)

ASYMPTOMATIC BACTERIURIA:

This is the **commonest UTI** in pregnancy affecting almost **6%** of all pregnant women.

The condition is usually **diagnosed** during **screening** for pregnant women, between 12-16 weeks gestation, when a *fresh mid-stream* urine sample is positive in absence of UTI symptoms. If untreated up to **30-45 %** of cases may develop **acute pyelonephritis**.

- **Laboratory Diagnosis:** positive **urine culture** from a fresh midstream sample showing **> 100.000** colony forming unit (CFU) of a single organism (usually being E. coli).
- **Treatment:** single agent outpatient oral antibiotic according to culture and sensitivity. Empirical ampicillin (against E. coli) 500 mg/12 hours for 1 week may be given until results of culture and sensitivity are available.
- **Follow up:** repeat midstream urine examination 3 weeks after treatment to ensure cure.

ACUTE CYSTITIS

This is an acute UTI **localized** to the bladder **without (fever) systemic** findings. If not properly treated it may progress to **acute pyelonephritis**.

- **Symptoms:** urgency, frequency, and burning micturition
- **Laboratory Diagnosis:** positive **urine culture** showing **> 100.000** CFU of a single organism
- **Treatment:** single agent outpatient oral antibiotic according to culture and sensitivity.
- **Follow up:** repeat midstream urine examination after 3 weeks to ensure eradication of infection

ACUTE PYELONEPHRITIS

This is an **acute UTI** involving the **upper** urinary tract with **(fever) systemic** findings that usually manifest in the 2nd or 3rd trimesters with pronounced symptoms of acute **cystitis** followed by systemic manifestations as nausea and vomiting in association with **rigors**, **chills**, and **loin pain.**

- **Signs**: high fever, tachycardia, and tender loin (right is more common due to dextro-rotation of the uterus with more pressure on the right ureter)
- **Significance:** if not early detected and adequately managed acute pyelonephritis may lead to serious maternal and fetal complications as:
  - Sepsis, anemia, and pulmonary dysfunction.
  - PROM and PTL with marked fetal morbidity and/or mortality.
- **Laboratory Diagnosis:** positive **urine culture** showing **> 100.000 CFU** of a single organism.
- **Management:**
  - Hospitalization is necessary with IV fluid infusion, antipyretics and pain killers.
  - 3rd generation Cephalosporins (Ceftriaxone) may be started immediately until results of culture and sensitivity are available.
  - IV antibiotics according to culture and sensitivity.
  - Fetal surveillance to assure fetal wellbeing.

HEPATIC DISORDERS DURING PREGNANCY

INTRAHEPATIC CHOLESTASIS OF PREGNANCY (ICP)

ICP is an uncommon condition that may affect **1-4%** of all pregnancies. It is characterized by **cholestasis** and **pruritus** in the **second half** of pregnancy **without** other major liver dysfunction.

ICP usually runs a **benign course** with no serious maternal hepatic sequela but has a tendency for **recurrence** in subsequent pregnancies, and is sometimes associated with **meconium** stained amniotic fluid and rarely fetal demise **(IUFD).**

- **Etiology:** exact etiology is **unknown** and may be multifactorial; however, *Gene mutation* in the MDR3 gene is associated in 15% of cases.
- **Diagnosis:**
  - *Symptoms:* itching without rash or abdominal pain, is the main symptom, being more intense on the palms and soles
  - *Signs:* Jaundice may be rarely observed.
  - *Laboratory tests:* elevated levels of serum bile acids with normal or mildly elevated AST, ALT & bilirubin.
- **Management:**
  - *Symptomatic treatment:* with cold baths, emollients, and anti-histaminic drugs.
  - *Medical treatment:* with oral ***ursodeoxycholic acid*** tablets is effective in ameliorating pruritus, and reducing serum levels of bile acids and bilirubin.
  - *Postpartum follow up:* rapid improvement occurs with normalization of laboratory tests.
  - *Fetal Surveillance:* Intrapartum surveillance for fetal wellbeing is important and in some cases timing delivery > 36 weeks may be indicated if treatment fails to control the condition.

ACUTE FATTY LIVER OF PREGNANCY (AFLP)

AFLP is a very **rare** but an extremely **serious** condition that may affect 1:14000 of all pregnancies in the **3rd trimester**. It is associated with diffuse micro-vesicular **fatty infiltration** of the liver that may lead to **hepatic failure**.

- **Etiology:** Unknown, but may be related to an inborn **error** of **metabolism** leading to enzyme deficiency resulting in intrahepatic accumulation of fatty acids.
- **Symptoms & Signs** usually include;
  - *Nausea and vomiting, abdominal pain, and jaundice.*
  - *Hypoglycemia* is frequently present and may be severe.
  - *Hypertension* and *preeclampsia (PE)* may be present in up to 50% of cases.

- *Coagulopathy* may develop in late cases with *hepatic and renal failure.*
- *Hematemesis may occur in association with DIC.*

- **Diagnosis:**
  - Increased PT & PTT, hyperbilirubinemia and hypoglycemia.
  - Elevation of ALT & AST, BUN, and serum creatinine.
  - Liver biopsy is diagnostic, but not usually performed for hazards of bleeding.
- **Treatment:**
  - Prompt **delivery** with intensive **supportive care** is indicated once diagnosis confirmed.
  - Management in **ICU** unit with specialized medical team.
  - The neonate should be tested for enzyme deficiency.

EPILEPSY IN PFREGNANCY

Epilepsy may affect **0.5%** of women in childbearing age. They may be either;

- **Partial Seizures:** do not involve both hemispheres, with no loss of consciousness.
- **Generalized Seizures:** involve both hemispheres associated with *Tonic-clonic convulsions:* duration lasting up to several minutes formerly known as grand-mal.

COURSE DURING PREGNANCY:

- In almost **75%** of cases the condition will remain **controlled** on the same dose of anticonvulsant therapy.
- In around **25%** the condition may **worsen** due to higher hepatic microsomal activity during pregnancy leading to lower blood level of anticonvulsant drugs with seizures occurring more frequently.

MANAGEMENT:

- Extra folic acid supplementation (800 µg daily) before and during early pregnancy.
- Monitoring anticonvulsant blood level in each trimester to adjust the dose - Mono drug therapy with the lowest possible dose controlling fits.

Effect of Anticonvulsant Drugs on the **Fetus** and the **Newborn**

- Increased incidence of fetal congenital anomalies (FCA) 3-7%
- Increased incidence of cerebral palsy (CP), seizures, and mental retardation (MR)
- Maternal use of phenytoin drug products may lead to neonatal deficiency of vitamin K dependent clotting factors (factors II, VII, IX, and X).

VENOUS THROMBOEMBOLISM IN PREGNANCY

INTRODUCTION AND INCIDENCE:

Pregnancy is a known hypercoagulable state, with a five-fold risk of venous thromboembolism. 25% of untreated venous thrombosis may develop to pulmonary embolism (PE), which may be fatal.

PHYSIOLOGIC CHANGES DURING PREGNANCY:

Pregnancy increases the risk for thrombosis due to the changes in the thrombotic and fibrinolytic activities that normally occurs during pregnancy, such as:

1. Increase in the levels of coagulation factors VII, VIII, IX, and X.
2. Increased fibrinogen levels.
3. Increased platelet activation.
4. Decreased protein-S and antithrombin III concentrations.
5. Venous stasis in the lower limbs due to pressure by the gravid uterus.

THROMBOPHILIAS

Some women are predisposed to thrombosis through changes in their coagulation fibrinolytic systems, which may be either inherited or acquired.

A) Acquired thrombophilia

This is mostly associated with the antiphospholipid syndrome **(APS).**

- **APS** is the combination of lupus anticoagulant **(LAC)** with or without, anticardiolipin anti-bodies **(ACA)**, with a history of **recurrent miscarriage** and or **thrombosis**.
- **APS** may or may not be associated with other autoantibody disorders as systemic lupus erythematosus (SLE).

B) Inherited thrombophilia

- Protein-C, protein-S, and Antithrombin III deficiency.

RISK FACTORS FOR DVT/PE DURING PREGNANCY:

1. Maternal age > 35 years.
2. Pre-pregnancy weight > 80 kg (obesity).
3. Previous DVT (hypercoagulability).
4. Pre-existing Thrombophilia (hypercoagulability).
5. Severe varicose veins VVs (stasis + vasculitis).
6. Prolonged bed rest (stasis).
7. Multifetal pregnancies (stasis).
8. Severe pre-eclampsia (hemoconcentration).
9. Cesarean section delivery (pelvic surgery).
10. Sepsis, especially pelvic (hypercoagulability + stasis + vasculitis).

DEEP VEIN THROMBOSIS (DVT)

CLINICAL PRESENTATION:

DVT usually presents clinically with ***pain in the calf muscles*** associated with varying degrees of ***redness***, ***hotness***, or ***swelling***, especially ***unilateral edema***.

Pressure on the sole of the leg during extension results in severe pain in calf muscles.

N.B.: Bilateral edema of the lower limbs is a common finding during pregnancy that may confuse or delay the diagnosis.

INVESTIGATIONS:

- **Color Doppler Ultrasound:**
  - Allows assessment of the deep veins between the knee and the iliac veins and also allows for the dynamic assessment of femoral and iliac veins.
  - It is currently the preferred first-line method for investigating a suspected case of DVT during pregnancy being an accurate, non-invasive procedure.
- **Venography:**
  - Allows excellent visualization of veins both below and above the knee.
  - It is an invasive procedure, requiring the injection of a contrast medium and the use of Xray. This method is not used during pregnancy.

MANAGEMENT:

A) Prevention OF DVT:

- Women with a history of DVT occurring during a previous pregnancy,
- Women with a history of DVT occurring in the non-pregnant state should be screened for thrombophilia,
- Women with artificial heart valves and cases of APS with recurrent DVT, or cases with a history of previous PE

All of the previous three types are given anticoagulant in a prophylactic dose during subsequent pregnancies till the end of the puerperium.

B) **Established cases of DVT:** Complete rest and immobilization for the first 24 hours, together with anticoagulant in therapeutic doses.

N.B.: in cases of DVT avoid dehydration by allowing excessive oral fluid intake or IV fluid infusion

ANTICOAGULANT THERAPY

1. *Heparin and LMWH (Clexane):*

- *Advantages:* they are the preferred line of treatment due to:
  1. They do not cross the placenta, and therefore are not teratogenic.
  2. Their effect can be stopped within hours by withholding further doses,

therefore can be used safely before and after delivery without increased risk for hemorrhage.

- *Mode of action:* Heparin prolongs the activated partial thromboplastin time (APTT), while calcium heparins affect factor X activity.
- *Side effects: on prolonged use:* prolonged use > 6 months may be associated with thrombocytopenia and higher risk of osteoporosis.
- *The major drawback* in the use of parenteral anticoagulant therapy is the need for *daily* repeated S.C. or I.V. injections, which is troublesome for most patients.

2. Oral anticoagulants (warfarin):

- *Advantage:* oral intake which is convenient for long use, and easily monitored and adjusted by INR measurement. It is the drug of choice that can be used safely in the second trimester and during lactation.
- *Mode of action:* Warfarin prolongs the prothrombin time (PT).
- *Side effects:* The drug crosses the placenta and can cause limb and facial defects in the first trimester, and fetal intracerebral hemorrhage in the third trimesters.

06 RH ISOIMMUNIZATION

It is an immunologic disorder characterized by excessive hemolysis of fetal RBCs by anti-bodies that pass through the placenta from maternal blood.

Erythroblastosis fetalis (also called hemolytic disease of the newborn), caused by Rh isoimmunization, is an important cause of infant morbidity and mortality.

THE RHESUS FACTOR:

The Rhesus (Rh) factor is a complex antigen that is present on the surface of RBCs. It consists of 3 pairs of genes Cc, Dd, Ee, the most important of which is the D. The D gene is dominant therefore a Rh-positive individual may be homozygous (DD) or heterozygous (Dd) and an Rh-negative individual has a (dd) genotype. The C and E antigens may also lead to incompatibility, but they are quite rare.

ETIOLOGY:

Rh-negative females develop anti-Rh antibodies if they were subjected to:

1. Blood transfusion with a Rh-positive blood. The immune system responds by the production of antibodies against the D antigen.
2. Married to Rh-positive male and get pregnant in Rh-positive baby. At time of delivery or (also disturbance of ectopic pregnancy, antepartum hemorrhage, amniocentesis, or external cephalic version) feto-maternal hemorrhage occurs (passage of fetal RBCs to maternal circulation). These fetal Rh-positive RBCs stimulate the immune system to produce antibodies against the Rh-positive antigen.

INCIDENCE:

85% of the Caucasian population is Rh-positive while 15% is Rh-negative. However, the incidence of Erythroblastosis fetalis is less than 1% due to:

1. A Rh-negative female may get married to a Rh-negative male.
2. A Rh-negative female may get married to a Rh-positive heterozygous (Dd) male. The baby has a 50% chance of being Rh-negative.
3. ABO incompatibility between the mother and her fetus results in hemolysis of transfused fetal cells into the maternal circulation before they can induce Rh antibody response.
4. Immunologic variation in the response to Rh-positive antigen. Some Rh-negative women are non-responders.

5. The first baby is usually unaffected because on the first exposure to Rh antigen, during the first pregnancy, the immune system responds by producing IgM antibodies (first immune response) that have a large molecular weight so cannot cross the placenta. While on the second exposure, the immune system responds by producing IgG antibodies (second immune response) that have a small molecular weight thus cross the placenta affecting the second baby. It is to be noted that the first baby can also be affected if the mother had previous Rh-positive blood transfusion or previous abortion.

CLINICAL TYPES:

In all types, hemolysis of fetal Rh-positive RBCs occurs due to the passage of anti Rh antibodies from maternal to fetal circulation in a previously sensitized pregnant woman.

I. Congenital hemolytic anemia: (Mildest form)

- Hemolysis results in fetal anemia that develops 2 weeks after birth.
- Due to faster rate of erythropoiesis, erythroblasts (immature nucleated RBCs) are produced, hence, the name Erythroblastosis fetalis.

II. Icterus gravis neonatorum: (Commonest and moderate form)

- The baby is delivered anemic but never jaundiced at birth as the placenta carries on the function of conjugating fetal bilirubin.
- Hepatosplenomegaly is usually present due to faster erythropoiesis.
- Jaundice develops within 48 hours after birth due to the inability of the fetal liver to conjugate the high amount of bilirubin produced by the hemolyzed RBCs.
- **Kernicterus** develops when fetal bilirubin level exceeds 20 mg%. Bilirubin crosses the blood brain barrier and becomes deposited in the basal ganglia of the brain stem, a serious condition that may lead to neonatal death or mental retardation.

III. Hydrops fetalis: (Severest form)

- Intrauterine fetal death usually occurs due to severe hemolytic anemia that leads to heart failure. If the baby was born alive, it dies within few hours.
- The fetus shows generalized edema, pleural effusion and ascites due to heart failure.
- Hepatosplenomegaly.
- The placenta is large and edematous.
- If ultrasound is done during pregnancy the fetus shows the "Buddha" attitude due to flexion of the thighs and abdominal distension and shows a halo around the skull due to scalp edema.

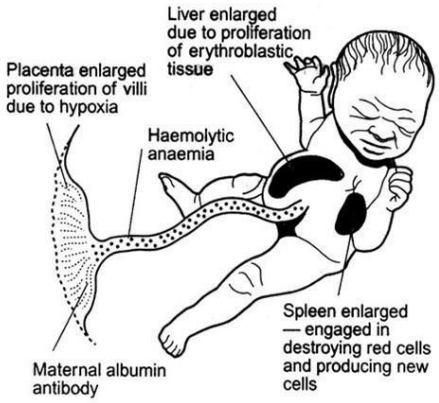


Fig 6-1: Hydrops fetalis

DIAGNOSIS DURING PREGNANCY:

Rh group should be checked in all pregnant women at the first antenatal visit and if Rh-negative:

- Ask about previous blood transfusion, fetal anemia, neonatal jaundice or IUFD.
- Determine the Rh group of the husband and if positive proceed for:
  - **Indirect Coombs' test:** to detect the presence and the titer of anti-Rh antibodies (Anti-D IgG). If <1/16 repeat the test every 4 weeks. If the titer > 1/16 amniocentesis is indicated.
  - **Amniocentesis:** determine the amount of bilirubin in the amniotic fluid that reflects the degree of hemolysis in fetal blood.
  - **Cordocentesis:** more accurate but more hazardous.
  - **Ultrasonography:** May show fetal hepatosplenomegaly, and generalized edema.
  - **Doppler US:** To assess middle cerebral artery (MCA) peak systolic velocity (PSV) which increases proportionate to the degree of fetal anemia

DIAGNOSIS AFTER LABOR:

Cord Blood is obtained for Rh grouping and if positive:

- Assess hemoglobin level to detect fetal anemia (normal 18 gm%), and Assess serum bilirubin to detect jaundice (normal = 2 mg%).
- **Direct Coombs' test:** detects the antibodies absorbed to the RBCs (sensitization).

PROPHYLAXIS AGAINST MATERNAL SENSITIZATION AND ERYTHROBLASTOSIS FETALIS:

1. Rh-negative females should never receive Rh-positive blood transfusion.
2. Anti-D immunoglobulin should be given to all Rh-negative **non-sensitized** (negative indirect Coomb's test) females married to Rh-positive males in the following conditions:
   - After delivery of an Rh-positive baby to destroy fetal RBCs before they

initiate maternal immune response (300 µg Anti-D I.M. within 72 hours of delivery).

- At time of abortion, disturbance of ectopic pregnancy, antepartum hemorrhage, amniocentesis, or external cephalic version (50-100 µg Anti-D I.M.).
- Recently, 300 µg of Anti-D immunoglobulin I.M. are given at 28 weeks of pregnancy is recommended to prevent sensitization during pregnancy.

N.B.: In already sensitized women (with Anti-D antibodies in their blood) as detected by positive indirect comb's test or who have a previously affected baby, Anti-D prophylaxis is contraindicated.

MANAGEMENT DURING PREGNANCY:

1. Intrauterine blood transfusion:

This is indicated if the fetus is severely affected before 36 weeks gestation. Rh-negative group O blood is injected into the peritoneal cavity of the fetus or into the umbilical vein. It is done under ultrasonographic guidance and local anesthesia.

2. Termination of pregnancy:

It is indicated if the fetus is severely affected after 36 weeks gestation. Induction of labor or cesarean section is done when lung maturity is achieved.

07 NORMAL LABOR

ANATOMY OF THE FEMALE PELVIS

The female pelvis is formed of two innominate bones, the sacrum, and the coccyx

- The **innominate bones** are joined together anteriorly at the symphysis pubis **(SP)** and are attached posteriorly each to one side of the sacrum at the right and left sacro-iliac joints.
- The **coccyx** is joined with the sacrum above at the sacro-coccygeal joint.
- The **pelvic brim** is an imaginary line separating the pelvic cavity into the **false** pelvis above, and the **true** pelvis below. The false pelvis has no obstetric importance.
- **The true pelvis:** is further divided into an **inlet**, a **cavity** and an **outlet**.

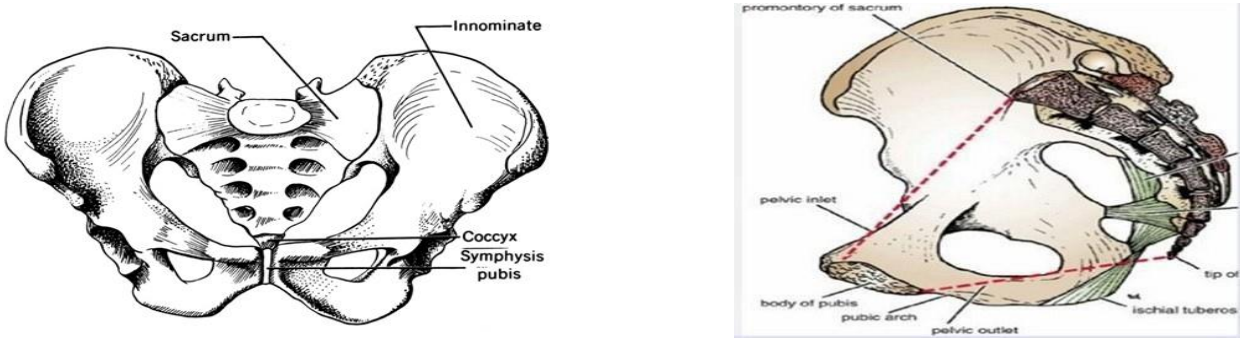


Fig 7-1: Bones of female pelvis and planes of pelvic inlet and outlet

THE PELVIC INLET (Brim)

The inlet is **oval** in shape with largest diameters being the transverse and oblique diameters.

Its plane makes an **inclination** angle 55° with the horizon in the standing position.

Boundaries of the Pelvic Inlet (plane of the pelvic inlet):

- **Posteriorly:** the promontory of the sacrum, the alae of the sacrum and the sacro-iliac joint.
- **On each side:** the iliopectineal line of the innominate bone, the iliopectineal eminence and the upper border of the superior pubic ramus.
- **Anteriorly:** the pubic tubercle, upper border of the pubic bone and upper border of the SP.

A)ANTERO-POSTERIOR DIAMETERS (APD):

- *The Anatomical APD or the True Conjugate **(11 cm)**:* from the tip of the sacral promontory to the upper border of the symphysis pubis (SP).
- *The Obstetric conjugate **(10.5 cm)**:* from the sacral promontory to the most bulging point on the back of the SP. It is 0.5 cm shorter than the true anatomical conjugate.
- *The Diagonal conjugate: **(12.5 cm):*** from the sacral promontory to the lower border of the symphysis pubis. It is the diameter that can be measured clinically by PV examination.

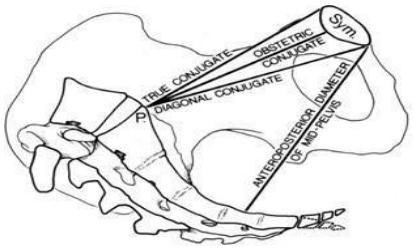


Fig 7-2: AP diameters of the pelvis

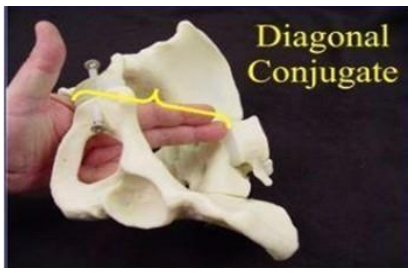


Fig 7-3: Diagonal conjugate

B) TRANSVERSE DIAMETERS (TD):

- *Anatomical TD **(13 cm):*** between the farthest points on the iliopectineal lines. It is nearer to the sacral promontory (not used by the fetus).
- *Obstetrical TD **(12 cm):*** it bisects the true conjugate and is slightly shorter than the anatomical (It is of obstetric importance).

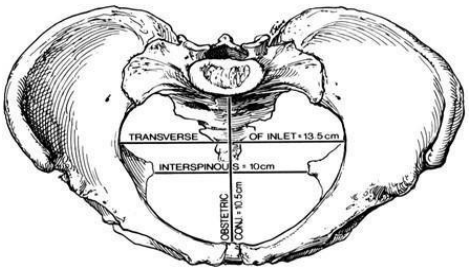


Fig 7-4: Transverse Diameters of the inlet

C) OBLIQUE DIAMETERS OF THE INLET (OD):

- ***R**ight **O**blique **D**iameter **(ROD) = 12** cm:* from **right** sacroiliac joint to **left** iliopectineal eminence.
- ***L**eft **O**blique **D**iameters **(LOD) = 12** cm:* from **left** sacroiliac joint to **right** iliopectineal eminence. It is slightly shorter than ROD being encroached upon by the descending pelvic colon.

N.B.: The ROD and LOD are the usual diameters for engagement.

- ***S**acro-**C**otyloid **D**iameter **SCD = 9.5 cm**:* extends from the promontory of the sacrum to the right and left iliopectineal eminence. The right SCD ends at the right eminence and vice versa.

THE PELVIC CAVITY

The pelvic cavity is an **imaginary segment** bounded **above** by the pelvic brim, **below** by the plane of least pelvic dimensions, **anteriorly** by the symphysis pubis and **posteriorly** by the sacrum.

- The plane of the pelvic cavity (**plane of the greatest pelvic dimensions**) lies in its widest part that extends from the center of the back of the symphysis pubis to the junction between the 2nd and 3rd pieces of sacrum.
- It is rounded in shape with all diameters equal 12.5 cm.
- It has a short anterior wall (symphysis pubis) and a long posterior wall (sacrum).
- The posterior wall (sacrum) is concave and this serves to increase the capacity of the pelvis.
- Internal rotation of fetal head occurs when the BPD lies in this plane.

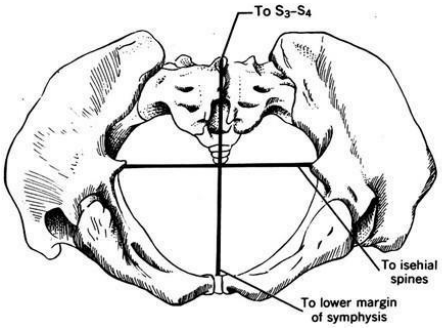


Fig 7-5: The pelvic cavity

THE PELVIC OUTLET

The pelvic outlet is a lozenge shaped area made of **2 triangles** (the anterior and posterior triangles), in **2 planes** (the anterior and posterior sagittal planes)

- **Anatomical outlet:** is a **lozenge** shaped area the boundaries of which are;
  - *Anteriorly:* The lower border of the SP.
  - *Laterally:* The pubic arch, ischial tuberosities, the sacrotuberous & sacrospinous ligaments.
  - *Posteriorly:* The tip of the coccyx.
- **Obstetrical outlet:** is a **wedge-shaped** segment the boundaries of which are;
  - *Above (the roof):* the plane of the **least pelvic dimensions** (plane of the ischial spines).
  - *Below (the floor):* the anatomical outlet.
  - *Laterally:* the part of the ischium between the ischial spine and the ischial tuberosity.
  - *Anteriorly:* the lower border of the SP.

- *Posteriorly:* the coccyx. N.B.: the coccyx moved backwards during 2nd stage of labor.

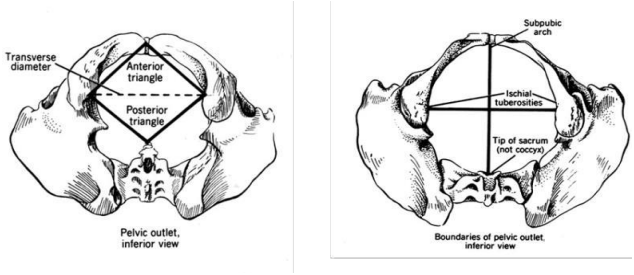


Fig 7-6: Pelvic outlet Inferior view

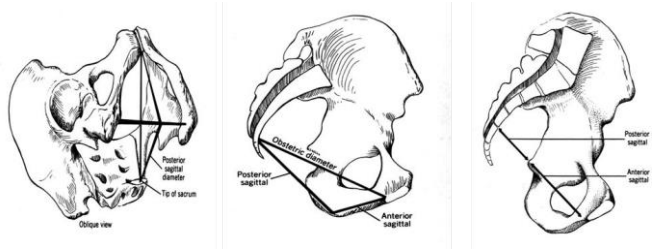


Fig 7-7 obstetric outlet Inferior view

- **Transverse Diameters:**
  - *Bituberous diameter:* ***11*** *cm*, between the inner aspects of the 2 ischial **tuberosities**.
  - *Bispinous diameter:* ***10.5*** *cm*, between the tips of the 2 ischial **spines**.
- **Anteroposterior Diameters:**
  - *Anatomical APD:* ***11 cm****,* from the lower border of the **SP** to the tip of the **coccyx.**
  - *Obstetric APD:* ***13 cm,*** from the lower border of the **SP** to the tip of the **sacrum.**

THE SUBPUBIC ANGLE:

It is normally between 90° and 100° while in the male it may be reduced to 70°

NB. A narrow subpubic angle will get the head pushed posterior necessitating generous episiotomy.

THE OBSTETRIC AXIS:

- It is an imaginary line representing the way passed by the head during labor.
- It is J-shaped passing *downwards* and *backwards* along the axis of the inlet, till the level of the ischial spines, where it passes *downwards* and *forwards* along the axis of the pelvic outlet.

IMPORTANCE OF THE LEVEL OF THE ISCHIAL SPINES:

- It is the level of the levator ani muscles, which is attached to the ischial

spines.
- It is the level at which lies the external os of the cervix and the vaginal vault.
- It is the level of the plane of the least pelvic dimensions.
- The obstetric axis changes its direction at this level.
- The head is considered engaged if the vault is felt at or below the level of the ischial spines.
- Forceps should not be applied when the fetal head is above this level.
- Anesthetic agent for pudendal nerve block is injected at this level.

ANATOMY OF THE FETAL SKULL

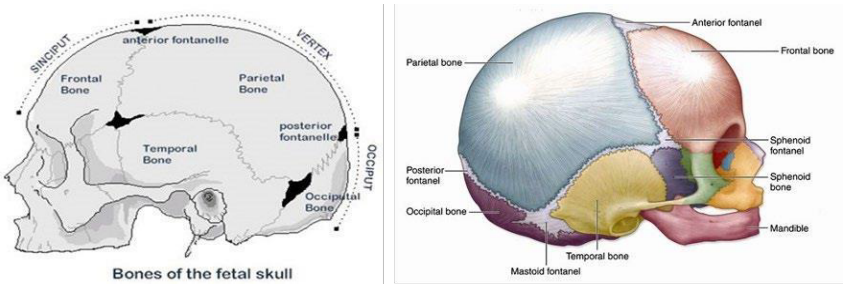


Fig 7-8: Different parts of fetal skull

The fetal skull consists of 3 parts separated by sutures and fontanelles:

1. **Base:** from the chin to the foramen magnum.
2. **Face**: from the chin to the root of the nose and supraorbital margins.
3. **Vault:** consists of three regions.

A) **Brow:** from the root of the nose and supraorbital margins to the anterior fontanelle (bregma) and coronal sutures. It includes the 2 frontal bones separated by the frontal suture.

B) **Vertex:** from the bregma and coronal sutures to the posterior fontanel and lambdoid suture. It consists of 2 parietal bones separated by the sagittal suture. It is bounded laterally by the parietal eminences.

C) **Occiput:** from the posterior fontanelle and lambdoid suture to the foramen magnum.

DIAMETERS OF THE FETAL SKULL

A. Longitudinal Diameters (LD):

1. ***Suboccipito-bregmatic (SOB) "9.5 cm":*** from below the occipital protuberance to the center of the anterior fontanelle (bregma). *The engaging diameter when the head is fully flexed.*
2. ***Suboccipito-frontal (SOF) "10 cm":*** from below the occipital protuberance to the anterior end of the anterior fontanelle. It is the diameter that extends the vulva in vaginal delivery.

3. ***Occipito-frontal (OF) "11.5 cm":*** from the occipital protuberance to the anterior end of the anterior fontanelle. *The engaging diameter when the head is deflexed (occipito-posterior).*
4. ***Submento-bregmatic (SMB) "9.5 cm":*** from the junction of the chin and neck to the center of the bregma. *The engaging diameter when the head is fully extended (Face).*
5. ***Submento-vertical (SMV) "11.5 cm":*** from the junction of the chin and neck to the vertical point (point on the sagittal suture midway between anterior and posterior fontanelles).
6. ***Mento-vertical (MV) "13.5 cm":*** from the tip of the chin to the vertical point. It is the engaging diameter when the head is incompletely extended (brow presentation). It is larger than any pelvic diameter hence engagement will not normally occur in an average sized head.

B. Transverse Diameters (TVD):

1. ***Biparietal diameter (BPD) "9.5 cm":*** Between 2 parietal eminences (*widest transverse diameter).*
2. ***Bitemporal diameter (BTD) "8cm":*** Between the anterior ends of the temporal sutures.

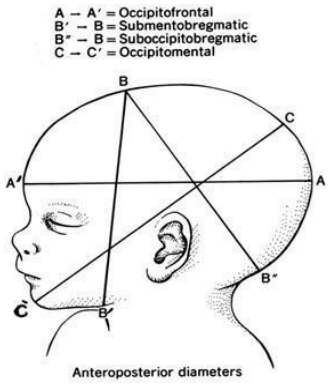


Fig 7-9: LD of fetal skull

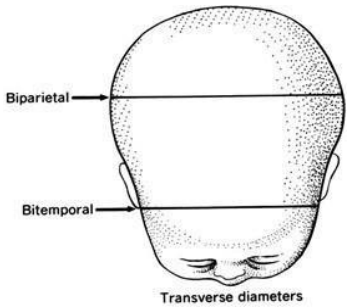


Fig 7-10: TVD of fetal skull

OBSTETRICAL DEFINITIONS

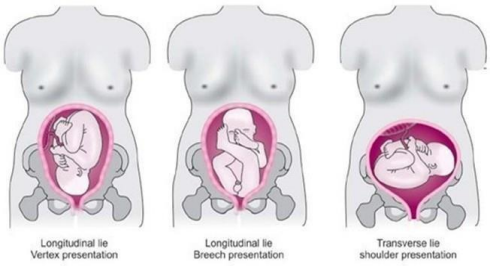


Fig 7-11: fetal lie & Presentation

- **FETAL PRESENTATION:** It is the lowest part of the fetus related to the pelvic brim and first felt by vaginal examination. It may be cephalic, breech, or shoulder presentation:

A) **Cephalic Presentation:** The most commonly encountered presentation as it easily allows for better adaptation of the fetus to the pyriform shape of the uterus with the broader buttocks and thighs lying at the wider fundus. Cephalic presentations include:
   - ***Vertex (95%)***: the commonest in full term pregnancies at the onset of labor.
   - ***Face (0.5%) and Brow (0.1%)***.

B) **Breech (3.5%):** The **buttocks** with or without the limbs form the presenting part.

C) **Shoulder (0.5%):** The **scapula** with or without the arms form the presenting part.

- **MALPRESENTATION:** It is any presentation other than the vertex including face, brow, breech, shoulder, cord and complex presentation.
- **FETAL LIE:** The relation between the longitudinal axis of the fetus to that of the mother. Fetal lie may be either **longitudinal** or **transverse**.
- **FETAL ATTITUDE:** The relation of the fetal parts to each other (flexion or extension attitude).
- **FETAL POSITION:** It is the relation of the back of the fetus to the **right** or **left** side of the mother and whether it is directed **anteriorly** or **posteriorly**.
- **FETAL DENOMINATOR:** A bony landmark on the presenting part used to denote the position either anterior or posterior, whether to the right or to the left in relation to the spine of the mother.
  - The **Occiput** in vertex presentation, the **Mentum** (chin) in face presentation.
  - The **Sacrum** in breech presentation, and the **Scapula** in shoulder presentation.

NOTES OF CLINICAL IMPORTANCE

- Occipito anterior positions **OAP** are more common than Occipito posterior positions **OPP** because the concavity of the front of the fetus fits into the convexity of the vertebral column (Lumbar lordosis) of the back of the mother.
- The right oblique diameter of the inlet is slightly longer than the left oblique (due to encroachment of the descending pelvic colon on the latter), this results in more prevalence of LOA over ROA, and ROP over LOP.
- OP positions are considered a malposition not a malpresentation.

STATIONS OF THE PELVIS

- **Station-3, -2, -1**: describes a head which is not yet engaged.
- **Station 0**: The head is **engaged,** with the **BPD passing** the plane of the pelvic inlet and the **vault** felt on PV examination **at** the **ischial spines.**

- **Station +1, +2, +3:** The BPD is **below the** level of ischial spines

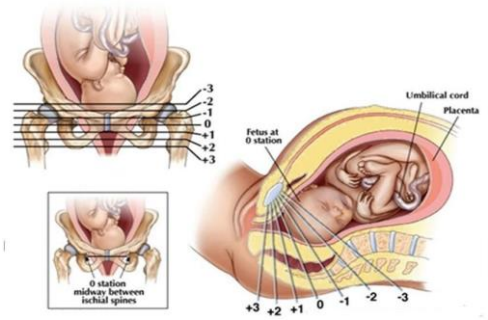


Fig 7-12: Stations of head descent

ENGAGEMENT:

Engagement describes passage of the widest **transverse** diameter of the presenting part through the plane of the pelvic inlet (the **BPD** in case of the head and the bitrochanteric **BTD** in case of breech).

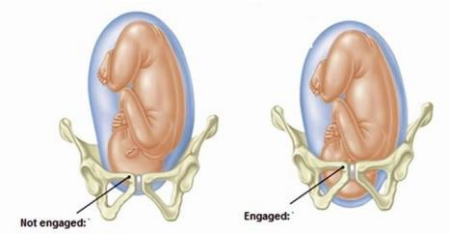


Fig 7-13: Engagement of fetal head

Abdominally: only 2/ 5 of the head is felt. The engaged head can't be grasped by the fingers of the right hand by the first pelvic grip but it can be palpated by the 2nd pelvic grip.

- Vaginally: the vault is felt at or below the level of the ischial spines on PV examination.
- In a primigravida, engagement of the fetal head may occur as early as 36 weeks due to the strong tonicity of the abdominal and uterine musculature.
- In multigravidas, engagement will occur only during the first stage of labor due to the down forces of contractions and retractions induced by true labor pains.

CAUSES OF DELAYED HEAD ENGAGEMENT IN A PRIMIGRAVIDA

MATERNAL CAUSES	FETAL CAUSES

1. *Contracted pelvis.* 2. *Placenta previa.* 3. *Pelvic tumors.* 4. *Full bladder or rectum.* 5. *Weak abdominal muscles.*	1. *OP position (commonest cause).* 2. *Malpresentations (face, brow, breech).* 3. *Multiple pregnancy.* 4. *Fetal Macrosomia (large baby).* 5. *Prematurity & polyhydramnios.* 6. *Fetal head anomalies (hydrocephalus and anencephaly).*

PHYSIOLOGY AND STAGES OF LABOR

DEFINITIONS AND DURATION OF LABOR:

- **Labor** is a process in which regular uterine contractions result in timely cervical effacement and dilatation, resulting in delivery of the fetus and expulsion of the placenta.
- **Normal labor** is a process of delivery of a single living cephalic vertex term baby, through the birth canal, within 4- 24 hours, without complications to the mother or the new born.
- **Duration of Normal labor** varies from one patient to another, however;
  - In a *primigravida* duration is usually within 12-18 hours.
  - In a *multigravida* duration is usually shorter being nearly 6-9 hours.

ONSET OF LABOR:

The exact cause for onset of labor is **unknown**, theories and hypothesis include:

1. ***Progesterone (PRG) withdrawal:*** onset of labor is preceded by sharp decline in PRG levels.
2. ***PGL Increase:*** PGL induce ripening of cervix and can initiate labor pains.
3. ***Placental Oxytocinase decrease:*** allow pituitary gland oxytocin to act freely on myometrium.
4. ***Uterine distension:*** as in cases of polyhydramnios and multifetal pregnancy (MFP) is commonly associated with PTL.
5. ***Increased fetal Cortisol:*** towards end of 3rd trimester may accelerate physiologic changes.

CHARACTERISTIC OF TRUE LABOR PAINS:

- Start in the **UUS** near the fundus as **low** amplitude **short** duration involuntary, infrequent pains.
- Gradually **increase** in frequency, amplitude and duration, with shorter intervening periods of rest.

- **Retraction** occurs at the end of each contraction leading to permanent shortening of the muscle leading to descent of the presenting part with upward pulling of the LUS.
- During labor the **UUS** contracts and retracts, while **LUS** stretches and dilates resulting in progressive cervical effacement and dilatation.
- The physiologic **retraction ring**: is the demarcation between the active UUS and the passive LUS. It rises upwards towards the umbilicus with progress of labor.
- Uterine contractions are **painful** due to; compression of nerve ganglia, hypoxia to uterine muscle, stretch of peritoneum over LUS, and the stretch of cervical dilatation.
- Pain is felt at **intrauterine pressure** of 20-25 mm Hg. It reaches 50-60 mm Hg during active 1 ^{s} stage of labor, and reaches 80 mm Hg at the 2nd stage of labor.

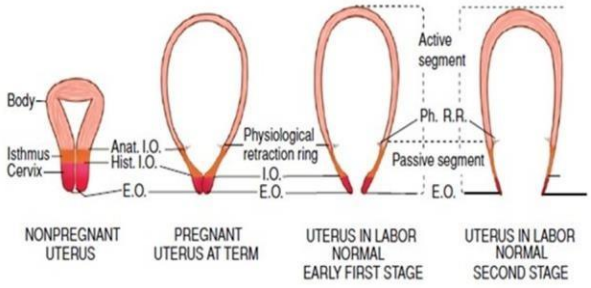


Fig 7-14: Uterine changes prior to and during labor

Comparison between true and false labor pains:

	True labor pains	***False labor pains***
Site of pain	*Abdominal, referred to back*	*Abdominal*
Rhythm	*Regular*	*Irregular*
Frequency, and strength	*Increasing*	*Stationary or decreasing*
Bulging membranes	*Present*	*Absent*
Cervical effacement	*Present*	*Absent*
Effect of sedation	*No effect*	*May relieve pain*

THE AUXILLARY FORCES OF LABOR:

1. Maternal bearing down during the 2nd stage of labor.
2. Unfolding of the fetus during passage through the birth canal.

THE STAGES OF LABOR

Labor is classically divided into **3 stages** that include; cervical effacement and dilatation, delivery of the fetus, and delivery of the placenta. The 4th stage is the 1st 2 hours after delivery.

THE FIRST STAGE OF LABOR

(Stage of Cervical Effacement and Dilatation)

- **Starts** with the onset of labor pains with the cervix formed and closed.
- **Ends** with full cervical effacement (very thin cervix) and full dilatation (almost 10 cm).
- In a primigravida **(PGDA)** effacement precedes dilatation, so its duration is **12- 18 hours**, while in multigravidas **(MGDA),** effacement and dilatation occur simultaneously, so taking half the time of PG (ie: **6- 9 hours**).

PHASES OF FIRST STAGE LABOR:

A) THE LATENT PHASE: the initial slow phase of effacement and dilatation:

- **Starts** with onset of true labor pains with the cervix usually **formed** and **closed** and proceeds slowly **till** a cervical dilatation of around **4 cm** and effacement of almost **50%**.
- It takes a **very variable** period of time.
- **No** cardinal movements of labor occur (descent, engagement, rotation, etc…).

B) THE ACTIVE PHASE: rapid phase of dilatation:

- **Starts** with cervical dilatation (4cm), and **ends** with complete cervical dilatation (10 cm).
- **Rate** of dilatation: 1.2 cm/hour in PGDA and 1.5 cm/hour in a MGDA.
- Cardinal movements of delivery occur in latter half (descent, engagement, flexion….).
- **Abnormalities** include prolonged first stage or failure of progress.

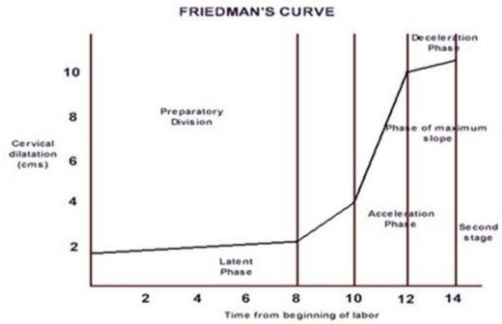


Fig. 7-15: First stage of labor

I. THE SECOND STAGE OF LABOR

(Stage of Delivery of the Fetus)

- **Starts** by full cervical dilatation (10 cm) and **ends** by complete expulsion of the fetus.
- Maternal **pushing down** efforts facilitate descent and delivery of the fetus through birth canal.

- Cardinal movement of delivery ends in this stage (internal rotation) and delivery of the head occurs by a mechanism of extension.
- **Duration** in a **PGDA: maximum** 2hours and in **MGDA: maximum** 1hour.
- **Abnormalities** include: prolonged 2nd stage and arrest of descent (obstructed labor).

II. THE THIRD STAGE OF LABOR (Stage of Delivery of the Placenta)

- ***Starts*** with delivery of the fetus and ***ends*** with expulsion of the placenta.
- ***Mechanism*** *of placental separation:* myometrial **contractions** and **retraction** shear off the anchoring villi from the decidua basalis separating and expelling the placenta.
- ***Signs of placental separation*** *and descent* include: gush of blood, elongation of the cord, and suprapubic bulge (the placenta descends to fill the LUS).
- *Average* ***duration*** is 10-30 minutes both in PGDA and MGDA.
- ***Abnormalities*** include; prolonged 3rd stage (see retained and adherent placenta).



Fig. 7-16: Separation & delivery of placenta

MECHANISM OF NORMAL DELIVERY (Cephalic occipito anterior)

Definition: Occipito Anterior (OA) position is a **longitudinal** lie, cephalic presentation, with the fetal

- **back** directed **anterior**.

Incidence: It is the most commonly encountered position being identified in about **70%** of

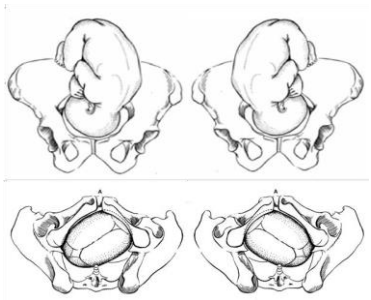
- cases in the 1st stage of labor.

Denominator: the occiput.

- **Positions:** Left Occipito Anterior (LOA) (more common), and Right Occipito anterior (ROA).
- **PV Diagnosis:** palpation of the posterior fontanelle and lambdoid suture to the right or left in
- relation to the symphysis pubis (SP)

The course of labor: in **OA** positions vaginal delivery is usually easily achieved in a reasonable

- period of time in presence of good uterine contractions, normal size gynecoid pelvis and in absence of marked cephalo-pelvic disproportion (CPD). Maternal and fetal complications are usually minimal and thus OA positions are considered the **model for normal labor** and delivery.



LOA (60%) *ROA (20%)*

Fig. 7-17: Left and right Occipito-anterior position

MECHANISM OF DELIVERY IN OCCIPITO ANTERIOR POSITION

A) DELIVERY OF THE HEAD

1. **Descent:** a continuous through the birth canal in late pregnancy, 1st stage and 2nd stage of labor.
2. **Engagement:**
   - Passage of the BPD (9.5 cm) through the plane of pelvic inlet.
   - Clinically the vertex will be felt vaginally at the level of ischial spines by PV exam.
   - Engagement may start at 36weeks of pregnancy in PG.

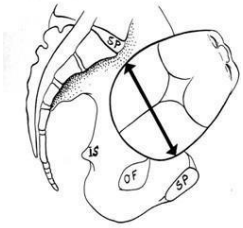


Fig 7-18: Non engaged head

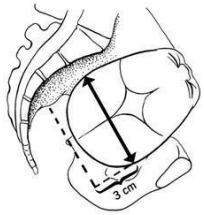


Fig 7-19: Engaged head

3.Increased Flexion:

- Flexion approximates fetal **chin** to the **chest** allowing the head to enter the mid pelvis with the least possible transverse and longitudinal diameters (BPD & SOB **9.5** cm), facilitating rotation.
- Flexion movement is facilitated by the strong downwards forces of uterine contractions against the strong tone of the levator ani muscles downwards.

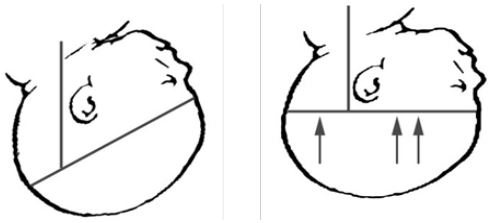


Fig 7-20: Flexion of the fetal head

4.Internal Rotation:

- The occiput rotates anteriorly 1/8 of a circle from either LOA or ROA positions to a direct Occipito anterior position (DOA) in preparation for extension.
- Rotation occurs in the mid pelvis when occiput touches pelvic floor muscles.
- Internal rotation is explained by:
  - The direction of levator ani muscles; that slope downwards forwards.
  - The rifling action of the pelvis; where the largest diameter of the inlet is oblique while that of outlet is longitudinal forcing rotation in the mid-cavity.

5. Extension:

- The sub-occipital region hinges under the symphysis pubis (SP).
- The head is acted upon by two forces finally resulting in extension;
  - downwards forwards force of uterine contractions and maternal pushing,
  - The upwards forwards force of pelvic floor resistance.

6.Restitution:

- The head rotates back 1/8 of a circle in opposite direction to that of internal rotation to undo the twist of the fetal neck caused by internal rotation.

7.External Rotation:

- It is due to internal rotation of the anterior shoulder and occurs for another 1/8 of a circle in the same direction of restitution.

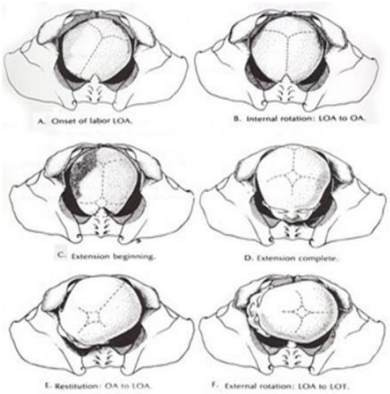


Fig 7– 21: Movements of head during normal vaginal delivery

DELIVERY OF THE SHOULDERS AND BODY

- The shoulders enter the pelvis in an oblique diameter opposite to that previously occupied by the head, descend through the pelvic cavity till

anterior shoulder reaches the pelvic floor muscles then rotate anteriorly at the same time of external rotation of the head bringing the anterior shoulder directly behind the SP.

- The anterior shoulder hinges below the SP.
- The posterior shoulder is delivered first by lateral flexion of the spine (more space).
- The anterior shoulder is then delivered followed by rest of body that slips outwards.

MANAGEMENT OF NORMAL LABOR

Once the patient has been admitted in the 1st stage of labor, the following should be done;

- **Obtaining** comprehensive **data** on the **admission** sheet, to triage patients into normal and high-risk cases and select the appropriate management teams and strategies.
- **Patient preparation;** through reassurance, education, advices for nutrition, cleansing and antiseptic measures, together with pain relief options.
- **Monitoring the progress of labor;** through a **Partogram** to allow for early detection of abnormal and or prolonged labor (see later).
- **Fetal surveillance;** throughout labor through repeated FHR auscultation or continuous electronic FHR monitoring. This will allow picking up cases with fetal distress and offer suitable management options (**see intrapartum fetal surveillance**).

DATA ON AN ADMISSION SHEET:

- *Personal history:* Name, age, LMP, EDD, and Gestational age:
- *Obstetric history:* Numerical 4-digit Code and patterns of previous deliveries.
- *Medical disorders* recent or old complicating pregnancy to document high risk pregnancies.
- *Abdominal examination:* for fundal height, fetal lie and presentation, and FHR auscultation
- *PV examination:* for cervical effacement (0-100 %), dilatation (0-10 cm), head station (-3 to +3), position of the head (LOA, ROA, ROP, LOP).
- *Membranes:* intact or ROM and duration of amniotic fluid leakage.
- *Molding of the skull bones and presence of caput succedaneum*, for possible CPD.
- *Evaluation of the bony pelvis,* to exclude *contracted pelvis* and *CPD*.

DATA ON ULTRASOUND EXAMINATION:

- *Fetal biometry & fetal weight* estimation (BPD, HC, AC, FL, etc.) to exclude SGA & LGA.

- *Confirming presentation and position:* cephalic OA/OP, breech, and transverse lie.
- *Amniotic fluid volume* and *placental localization:* to exclude PL Previa and accreta.
- *Exclude major fetal anomalies:* anencephaly, NTDs, hydrops Fetalis, etc.
- *Presence of associated pathology* as leiomyomas or ovarian masses.
- *Evaluation of fetal wellbeing:* umbilical artery Doppler.

MANAGEMENT OF THE FIRST STAGE OF LABOR

The **1st stage** of labor is the stage of **onset** of **true labor pains** and **cervical dilatation.** Its duration ranges from **12-18** hours in a primigravida and **6-9 hours** in a multipara.

A) Preparation of the patient:

- Ask the patient to evacuate the bladder and the rectum.

B) Pain relief: This may be achieved either by

- *Intermittent IM injections* of Pethidine 1mg/kg/4 hours, **or**
- *Epidural Analgesia* during the active phase of labor and throughout the process of delivery.

C) Nutrition: Light diet, oral fluids, juices or IV fluids; to avoid hypoglycemia.

D) Instructions and Education: reassurance and avoid bearing down in the 1st stage.

E) Monitoring Progress of labor using the PARTOGRAM:

- The Partogram is a graphic record for the progress of labor that allows an instant visual assessment of the rate of cervical dilatation and effacement plotted against descent of fetal head in comparison to an expected norm.
- A Partogram will allow early diagnosis of prolonged and abnormal labor, and whether this will occur in the latent or active phase.
- Other data that can be included in a Partogram include; timing of ROM, medications, frequency of uterine contractions, and basic observations as the Pulse, temp, and BP.

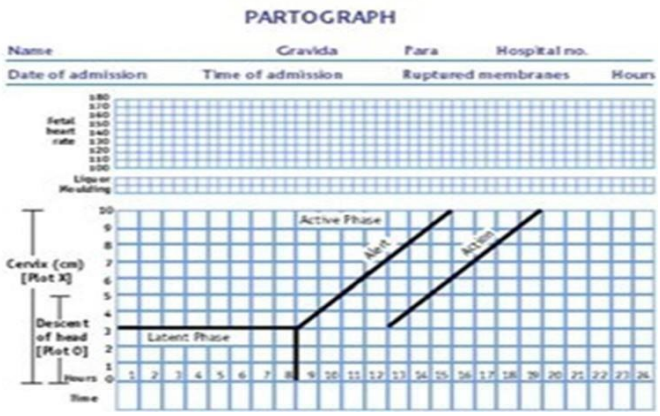


Fig 7 -22: The Partogram

INTRAPARTUM FETAL SURVEILLANCE (IFS):

Fetal surveillance during labor can be achieved via either;

1. **Intermittent** FHR recording by **Sonicaid** in normal low risk cases.
2. **Continuous Electronic FHR Monitoring** by Cardiotography **(CTG)** in high-risk cases (see intrapartum fetal surveillance).

MANAGEMENT OF THE SECOND STAGE OF LABOR

The **2nd stage** of labor is the stage of **delivery of the fetus**. It starts by full cervical dilatation and ends by expulsion of the fetus through the natural birth canal. It has an average duration of 2 hours in a primigravida and 1 hour in multiparas.

- **Place of delivery:** patient is transferred to a specially equipped delivery room.
- **Maternal position:** Lithotomy position is the commonest and most preferable.
- **Ensure** evacuation of bladder by voiding or a urinary catheter.
- **Sterilization:** the vulva and perineum are cleansed with antiseptic solution (Betadine).
- **Sterilized dressings** and **gowns:** The patient is completely covered by sterilized gown from the umbilicus to the legs and feet leaving only the region of vulva and perineum exposed.
- **Maternal bearing down*:*** which was prohibited in the 1st stage, is now encouraged during the peak of uterine contractions with the head moving from station +1 to +2 and +3.
- **Delivery of the fetal head:** adequate *perineal support* (Ritgen's maneuver) is maintained till **after crowning,** then head is allowed for gradual extension, with one hand of the attendant, to prevent sudden expulsion of the head with consequent perineal tears. The diameter distending the vulva is the suboccipito-frontal diameter (SOF) measuring 10 cm.

NB: Crowning: the fetal head remains visible at the vulval ring without slipping back in.

N.B.: allowing head extension **before crowning** will bring the occipito-frontal diameter (OFD) measuring 11.5 cm to distend the vulva with more liability to perineal tears

- **Episiotomy:** is performed -under local infiltration anesthesia (if no other anesthesia was taken)-, **when needed** just before crowning of the head to avoid extensive perineal tears and facilitate head delivery (see episiotomy).

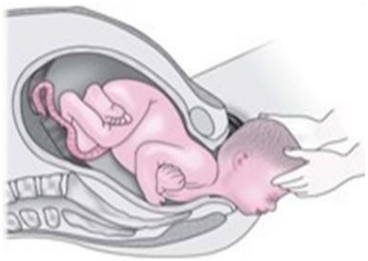


Fig 7-23: delivery of the head



Fig 7-24: Ritgen's maneuver

- **DELIVERY OF THE SHOULDERS AND BODY:**
  - Gentle downwards traction on the head till anterior shoulder appears under SP.
  - The head is then lifted upwards to deliver the posterior shoulder first.
  - The head is again pushed downwards to deliver the anterior shoulder.
  - The rest of the body slips down easily after delivery of the shoulders.

- **Clamping the Umbilical Cord:** after delivery immediate care of the new born starts by clamping the umbilical cord and cleaning the air passages by nasal suction.



Fig. 7-25: Delivery of the anterior and posterior shoulders

MANAGEMENT OF THE THIRD STAGE OF LABOR

The **3rd stage** of labor is the stage of delivery of the stage of **delivery of the placenta.**

- **Duration** of this stage ranges from 10-30 minutes with the **Active management**

mechanism (that entails the use of ecbolics as methergine IM or oxytocin IV drip, to fasten the uterine contractions and so placental separation while reducing the blood loss).

- Wait for signs of **placental separation** and descent: gush of blood, elongation of the cord and suprapubic bulge.
- Gentle **traction** on the cord through **Brandt-Andrews's** method: It entails controlled cord traction during uterine contraction with the right hand, while the left hand is pushing the uterus upwards suprapubically.
- **Inspection** of the **placenta**: is mandatory to detect any missing cotyledon that may cause PPH.
- **Inspection** for **laceration:** for vaginal, cervical, or perineal tears which will be dealt with by immediate surgical layered repair followed by episiotomy repair, if present



Fig 7-26: active methods for delivery of the placenta

MANAGEMENT OF FORTH STAGE OF LABOR

- The **4th stage** of labor is described by some being the **first 1-2 hours** after delivery of placenta. It is the period in which the most serious **complications** of the 3rd stage as postpartum hemorrhage PPH will arise and be detected and managed.
- It is managed by careful observation for the vital signs, amount of vaginal bleeding, and degree of uterine contractility.
- Repeated gentle uterine massage for the first hour together with prophylactic ecbolics are advisable, as they improve the outcome and minimize the risks of PPH.

08 FETAL SURVEILLANCE

Normal fetal development and growth depend on both **genetic** factors and **placental** functions. The placenta offers gas exchanges, nutrient transfer, and elimination of fetal wastes, hence placental **insufficiency** during **pregnancy (chronic fetal distress)** or **labor** (**acute fetal distress**) will end up in a state of fetal deprivation and **hypoxia** that will be reflected on fetal outcome and wellbeing **(fetal distress)**.

Placental insufficiency is the term used to describe in-utero failure of adequate oxygenation and nutrition supply to the fetus. It could be either acute or chronic;

1. **Acute Insufficiency (lead to acute fetal distress):** Suddenly occurring, usually during labor, associated with an otherwise normal fetus, as in; *Cord accidents* as cord prolapse, cord compression, true knots, coils around the neck, and ruptured vasa previa. Sudden *extensive placental separation* (accidental Hemorrhage), *Placental compression* (uterine hypertonicity, obstructed labor).
2. **Chronic Insufficiency (lead to chronic fetal distress):** Occurring over **weeks** and is usually with intrauterine growth restriction **(IUGR)** and a small for gestational age **(SGA)** fetus.

Fetal Response to Chronic Placental Insufficiency:

1. Decreased fetal movement, due to decreased oxygen supply
2. Oligohydramnios due to reduced renal perfusion.
3. Asymmetric IUGR due to redistribution of blood flow preferentially to the brain and fetal heart known as vascular shift phenomenon (see fetal growth disorders)
4. Disturbance of fetal heart rate due to defective autonomic regulation.
5. Abnormal FHR patterns during labor due to severe hypoxia, and acidemia.

Assessment of **fetal wellbeing** can be achieved through a group of objective tests which is known as fetal **surveillance** and can be achieved both **antepartum** (during pregnancy), and **intrapartum** (during labor).

ANTEPARTUM FETAL SURVEILLANCE (APFS)

Antepartum Fetal Surveillance **(APFS)** will allow **identification** of pregnancies at high risk for **placental insufficiency**, and **evaluation** of fetal **wellbeing** to avoid fetal loss. Fetal surveillance is performed during routine ANC in all pregnancies but is done more frequently in pregnancies at a higher risk for placental

insufficiency.

CASES AT HIGH RISK FOR PLACENTAL INSUFFICIENCY

a) Postdate pregnancies.
b) Oligohydramnios.
c) Intrauterine growth restriction (IUGR).
d) Medical disorders as preeclampsia, hypertensive disorders, and gestational diabetes mellitus.
e) Recurrent APH associated with placental abruption or placenta previa.

1. DAILY FETAL MOVEMENT COUNT (DFMC)

Recording the number of fetal kicks over a period of 1 hour after **24** weeks gestation may help in selecting cases at risk for placental insufficiency, for further evaluation.

- **Interpretation of results:**
  - **More** than **10-12** gross movements within **12** hours indicate **good** fetal wellbeing.
  - **Decreased DFMC** calls for further tests for evaluation of wellbeing (see below).

2. THE NON-STRESS TEST (NST):

Electronic monitoring of **FHR changes** in relation to **fetal movements**, known as the NST.

NST is performed for cases with decreased DFMC using the same **Cardiotogography** (CTG) instrument used for continuous FHR monitoring during labor.

A duplex US attached to the mothers' abdomen will record **baseline FHR** together with **FHR changes** in relation to **fetal movements** which are reported by the mother by pressing on a special button with each fetal movement.

- **Interpretation of the results:**
  -Normal NST (reactive fetus)*:* when FHR **accelerations** of **> 15** bpm, lasing **> 15** seconds occur with each fetal **movement** during **15-20** minutes test duration.
  - ***Non-Reassuring NST:*** when movements are minimal or accelerations are not consistent with each movement, these needs to extend to NST for another 20 minutes.
  - ***Abnormal NST (Non-reactive fetus)****:* when FHR accelerations are absent in relation to movements, this needs to do BPP or Doppler tests.

3. THE BIOPHYSICAL PROFILE (BPP) SCORE:

The BPP is a simple test based on **ultrasound** evaluation of: fetal **tone** (flexion / extension), fetal **movements** (limbs and body), fetal **breathing**

activities (expansion lungs movements), **amniotic fluid** volume (using AFI). Each tested parameter is given a **score** (either 0 or 2), with a complete score of **8/8**

- **Interpretation of the test:**
- Scores of **8/8** denote a normal fetal wellbeing, with normal fetal PH.
- Scores **< 8/8** suggests severe hypoxia and need adding Doppler US evaluation.

4. COLOR DOPPLER STUDIES OF FETAL BLOOD FLOW

Measurement of the fetal blood flow indices of the fetal **umbilical** artery**, middle cerebral** artery, **yield** excellent information about the state of fetal organ perfusion.

Color Doppler Indices Suggestive of Placental Insufficiency:

a) **High** resistance for umbilical artery blood flow: indicates poor placental perfusion and the need of close follow up.

b) **Absent** umbilical artery end diastolic flow: indicates very poor or nearly no placental perfusion.

c) **Reversed** flow: is a very dangerous sign and needs immediate termination of pregnancy.

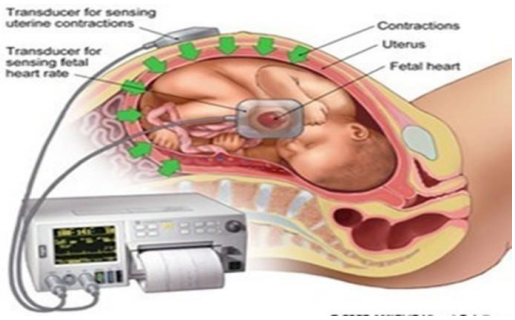


Fig. 8-1: NST using Cardiotocography CTG with electronic FHR Monitoring

INTRAPARTUM FETAL SURVEILLANCE (IPFS)

Childbirth is a normal **physiologic** process that will usually end **safely** both for the mother and her fetus, however perinatal morbidity and mortality may occur in high-risk pregnancies High-risk cases will requires a close **FHR monitoring** during labor to ensure a good fetal outcome other than traditional FHR auscultation by a Sonicaid.

METHODS OF MONITORING FHR DURING LABOR:

A) PASSAGE OF MECONIUM AFTER ROM:

- *In a cephalic presentation,* passage of meconium denotes fetal hypoxia and acidosis that leads to relaxation of the anal sphincter and so passage of meconium.
- *In breech presentations;* meconium passage is not reliable since it may

occur normally with compression of the abdomen prior to delivery.

B) AUSCULTATION OF THE FHR:

Intermittent auscultation of the FHR by the **Sonicaid** during labor.

- Before ROM, it should be done every **30** minutes in the 1st stage and every **15** minutes in the 2nd stage of labor.
- After ROM, it should be done every **15** minutes in the 1st stage and every **5** minutes in the 2nd stage of labor.

Interpretation:

- Normal FHR: 120- 160 bpm
- Fetal bradycardia: FHR <100bpm
- Fetal tachycardia: FHR ≥160 bpm

C) CONTINUOUS ELECTRONIC FHR MONITORING:

By the use of CTG machine, it is indicated in all high-risk pregnancies.

An external **tocodynamometer** is placed on the abdomen to record the frequency and intensity of uterine contractions, and a duplex **US probe** fixed on the abdomen to record FHS in relation to the uterine contractions.

Interpretation:

- **Normally,** FHS shows beat to beat variability (fluctuations of FHS of 5-10 bpm periodically) keeping the baseline between 120- 160bpm. There are also early fetal heart decelerations **(**slowing of FHR with **onset** of contractions with recovery by its end). **Early decelerations are normal fetal response to head compression)** and doesn't denote fetal distress.
- **Variable decelerations:** slowing of FHR which is not related to uterine contractions. This is mostly related to transient cord compression and necessitates changing the maternal position to relieve the cord compression.
- **Late decelerations (acute uteroplacental insufficiency):** slowing of FHR with the uterine contraction, and no recovery by its end, but persistence of the deceleration after the end of the uterine contractions, indicate fetal distress and hypoxia.
- **Decreased or absent beat-to-beat variability:** a prolonged flat baseline with absent beat to beat variability indicates fetal acidosis.

D) FETAL BLOOD SAMPLING DURING LABOR (FBS):

The blood sample is taken via a small needle probe applied to fetal scalp after ROM. Few drops of fetal blood which is known as fetal scalp pH.

- Normal scalp pH ranges from 7.25 to7.35.
- Intrauterine fetal hypoxia leads to acidosis with fetal pH below 7.25 when mild and below 7.2 With severe. This occurs due to accumulation of CO_2 and lactic acid.

MANAGEMENT OF FETAL DISTRESS DURING LABOR:

First Aid Measures:

a) Stop oxytocin,
b) Change position of the mother to left lateral position (allows increased blood supply).
c) O2 mask to the mother (to avoid maternal hypoxia).
d) I.V. Fluids (to avoid maternal dehydration).

Results of the above first aids measures:

- If FHR patterns return to **normal**, labor is allowed to proceed under strict observation.
- If **abnormal** FHR patterns are persistent or recur, immediate delivery is advisable through;
  - **Emergency CS: this** is the safest option whenever cervix is not fully dilated.
  - **Vaginal delivery:** when the cervix is fully dilated and the presenting part engaged, delivery can be aided by a **low forceps** procedure.
  - **Vaginal breech delivery** can be performed via **breech extraction** in cases of complete or frank breech presentations, with a fully dilated cervix and an adequate pelvis.

09 MALPOSITIONS AND MALPRESENTATIONS

OCCIPITO POSTERIOR MALPOSITIONS

Occipito posterior (OP) position is a longitudinal lie**, cephalic** presentation **with the fetal back** directed **posterior**. It is considered a **malposition rather than a malpresentation.**

- **Incidence: OP** positions are encountered in around **20 %** of cases at the onset of labor.
- **Denominator**: the occiput.
- **Positions:** Right and Left Occipito posterior **(ROP – LOP).**
- **ROP** are more common than **LOP** as the descending colon encroaches on the left oblique diameter of the inlet making it slightly smaller than the right oblique diameter.
- **Etiology:**
  - *Idiopathic:* in many cases no underlying etiology can be identified.
  - *Android or anthropoid pelvis;* where the heart shaped inlet allows the larger **BPD** to occupy the wider hind-pelvis, and the smaller bitemporal diameter **(BTD)** occupies the narrower fore-pelvis. This also predisposes to deflexion of the head with the BTD lower than the BPD
  - *More common in PG with strong abdominal wall muscles.*
- **Course of labor:** usually prolonged labor especially in the 2nd stage (more time is needed for long internal rotation).

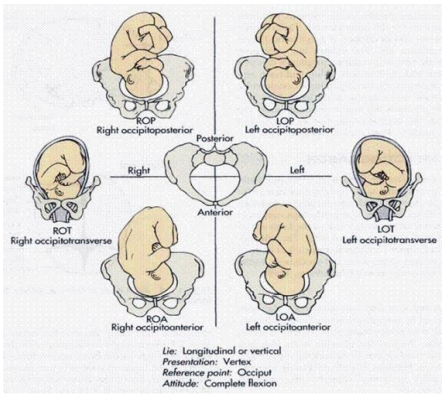


Fig 9-1: Right and left Occipito posterior positions

MECHANISMS OF DELIVERY IN OP POSITIONS

Descent engagement as in OA position with BPD (9.5) cm traversing the pelvic inlet, this is followed by correction of **deflexion**, then long internal **rotation** with the following probabilities;

A. LONG ANTERIOR ROTATION 3/8 OF A CIRCLE TO DOA POSITION (90%):

In around **90% of cases**, there is **mild degree of deflexion**, so the occiput reaches the pelvic floor first, and in presence of good uterine contractions and absence of CPD, correction of deflexion will occur, the occiput rotates anteriorly 3/8 of a circle. Once OA position is achieved, labor will then proceed normally to DOA position, where the occiput hinges below SP and the head will be delivered with a movement of **extension**.

B. POSTERIOR ROTATION 1/8 OF A CIRCLE to a DOP Position (6%):

In almost **6% of cases, severe degree of deflexion** is present, so the sinciput reaches the pelvic floor first and the occiput rotates posteriorly 1/8 of a circle in a DOP position. The head is delivered in **flexion** as **face to pubis**.



Right occiput-posterior (ROP) *Direct occiput-posterior (DOP)*

Fig 9-2: Posterior rotation of the occiput to DOP (face to pubis)

C. FAILURE OF ROTATION AND DEEP ARREST of the head (4%):

In **4% of cases**, there is **moderate degree of deflexion**, so both occiput and sinciput reaches the pelvic floor together and no rotation (persistent OP) or just 1/8 rotation to transverse position (deep transverse arrest). These positions have no mechanism for delivery and **labor will be obstructed.**

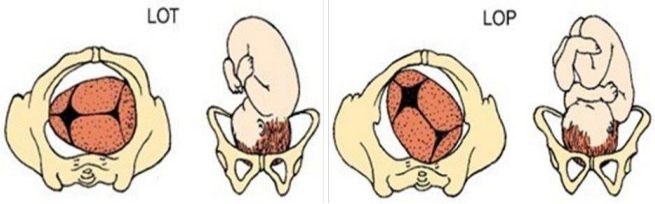


Fig 9-3: Deep arrest of the head in transverse or oblique position

Factors favoring internal rotation	***Factors interfering with internal rotation***
– *Correction of deflexion* – *Strong uterine contractions* – *Adequate pelvis no CPD* – *Good pelvic floor musculature* – *Adequate amount of liquor*	– *Persistent deflexion* – *Weak uterine contractions* – *Contracted pelvis or CPD* – *Poor pelvic floor musculature* – *Early PROM*

MANAGEMENT DURING LABOR:

A) Watchful Expectancy:

- In more than **90%** of OP, correction of deflexion occurs and head **rotates 3/8 of circle anteriorly** and **vaginal** delivery will be achieved same as in LOA in a mechanism of **extension,** (but with more prolonged time needed for the long rotation).
- **In 6% of cases**, the occiput rotates posteriorly, and the head is delivered in a mechanism of **flexion** (face to pubis) with the **need of episiotomy** as the vulva is distended by the occipito-frontal diameter (11.5cm)

B) Instrumental Delivery: (provided that no CPD):

In 4% of cases, with persistent OP or deep transverse arrest, Instrumental delivery may be used for rotation of the fetal head and extraction, this can be achieved by:

- ***Kielland Forceps:*** long forceps with sliding lock to allow rotation and extraction of the fetal head.
- ***The Vacuum Extractor (ventouse):*** can also be used to help rotation and descent of the head with less injury to the vaginal mucosa and pelvic floor muscles.
- ***However, CS is considered safer specially in PG cases with possibility of CPD.***

- ***N.B.: Prerequisites for instrumental delivery include:***
  - Exclusion of contracted pelvis and CPD.
  - Engagement of the head with ROM and full cervical dilatation.
  - Adequate analgesia preferably regional (epidural) analgesia.

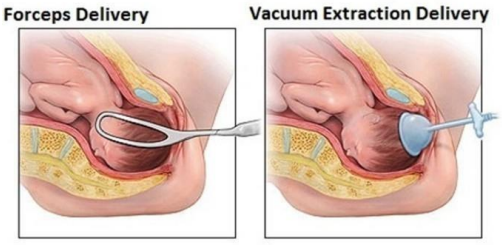


Fig 9-4 Kielland Forceps and Ventouse delivery in OP position

FACE PRESENTATION

Face presentation **is a longitudinal** lie**, cephalic** presentation**, with the head fully extended.**

- **Incidence:** It is an uncommon **malpresentation** occurring in around **1/500** deliveries.
- **Positions:** Right and left mento anterior and mento posterior (RMA - LMA - RMP - LMP).
- **PV Diagnosis:** landmarks (denominator) is the chin (mentum).
- **MA** occur > **MP** positions as they usually develop 2ry to full **extension** of

OP positions.

- **LMA** is the commonest position as it occurs 2ry to full extension of a head in the **ROP** position.
- **Etiology:**
  1. *Idiopathic:* in many cases no underlying factor for face presentation can be identified.
  2. *Anencephaly:* the baby presents by the face instead of the absent vault.
- **Course of labor:** usually prolonged both 1st and 2nd stage with higher CS rates due to; delayed descent and engagement, difficult rotation in android pelvis especially in MP positions, absent molding of maxillary bones, PROM, and delivery of the head in a longer vertical diameter.

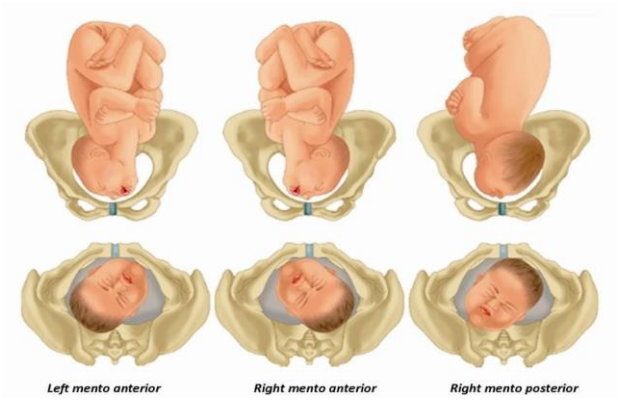


Fig 9-5: Face presentation MA and MP positions

MECHANISM OF DELIVERY IN FACE PRESENTATIONS:

A) MENTO ANTERIOR POSITIONS:

- Delayed **engagement** of the head with **BPD** (9.5 cm) passing through the pelvic inlet.
- **Rotation** of the Mentum **1/8** of a circle anterior from RMA & LMA to **DMA** position.
- **Delivery** of the head in **flexion** with the **SMV** diameter (11.5 cm) distending the vulva.
- Flexion in **DMA** can occur as the longer neck is facing the shorter symphysis pubis (SP).
- Successful delivery will thus be achieved in face MA positions **in absence of CPD.**

B) MENTO POSTERIOR POSITIONS:

- In **2/3** of cases successful long anterior **rotation 3/8** of a circle to a **DMA** position and delivery as above in a mechanism of **flexion.**
- In **1/3** of cases **failure** of anterior **rotation** will lead to **obstructed labor** to either;
  - ***Posterior rotation*** of the Mentum to direct MP **(DMP).** In such cases

the head cannot flex facing the long sacrum, and cannot extend being already fully extended.

- **Arrest** of face in either **oblique** or **transverse** diameters.



Fig 9-6: Face presentation in DMA and DMP positions

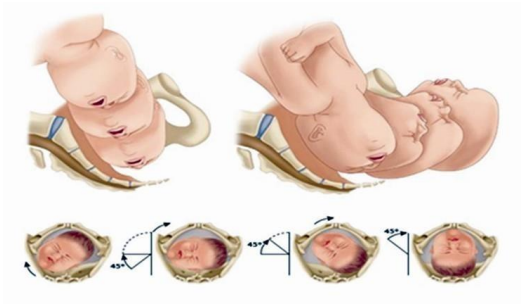


Fig 9-7: Delivery of face MP successful rotation to DMA and delivery in flexion

MANAGEMENT DURING LABOR:

- **Watchful expectancy** as ALL **MA** positions and **2/3** of **MP** positions will deliver vaginally in presence of good uterine contractions, proper sedation and absent CPD.
- **Adequate episiotomy** is mandatory as the diameters distending the vulva are larger than that for occipito-anterior position (SMV= 11.5cm).
- **Forceps delivery:** are sometimes indicated to shorten the 2nd stage of labor in cases where fetal distress or maternal exhaustion are present or strongly anticipated
- However, **vacuum delivery** is totally **contraindicated** on fetal face.
- **Indications of CS in Face Presentation:**
  - Persistent Oblique, Transverse or Direct MP positions predisposing to an obstructed labor.
  - CPD, fetal distress, or marked maternal exhaustion.
  - Other obstetric indications for a CS as a previous CS, myomectomy scar, PL PRV, MFP, etc…

BROW PRESENTATION

Brow presentation is a **longitudinal** lie**, cephalic** presentation, with the head midway between flexion and extension. It is a rare **malpresentation** occurring in **< 1/2000** delivery.

- **Brow:** is that part lying between the root of the nose and the anterior fontanelle.

- **Engaging diameter** in brow is the mento vertical diameter **MVD** (13.5 cm) which is larger than any diameter in the pelvic inlet making engagement impossible. Labor will thus be obstructed.
- **Transient Brow:** usually occurs during conversion of OP position into face presentation.
- **Persistent Brow:** head stops midway between flexion and extension and fails to correct neither to face (complete extension), or to OP (convert to incomplete flexion).

MANAGEMENT DURING LABOR:

CS is indicated in all cases of **Persistent** brow due to failure of engagement and obstructed

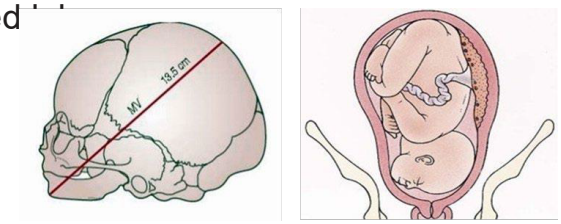


Fig 9-8: Brow presentation and MV diameter

BREECH PRESENTATION

Breech presentation is a **longitudinal lie** in which the **buttocks,** and/or the **lower extremities,** present in to the maternal pelvis.

- It is the commonest **malpresentation** being encountered in **3%** of **term** pregnancies. 25% in **preterm** labor **(≤**28 weeks gestation).
- The **Denominator** in breech is the **sacrum**, and **positions** include;
  - Right and left Sacro-anterior position (RSA & LSA).
  - Right and Left Sacro-posterior positions (RSP & LSP).
  - Sacro-anterior positions are more common than Sacro-posterior.

TYPES OF BREECH:

1. **Complete breech:** hips and knees are both **flexed**.
2. **Incomplete breech: extension** is present at hip &/or knee
   - ***Frank breech:*** hips **flexed** and knees **extended** (buttocks only form the presenting part).
   - ***Footling breech: one or both*** hips and knees **extended.**

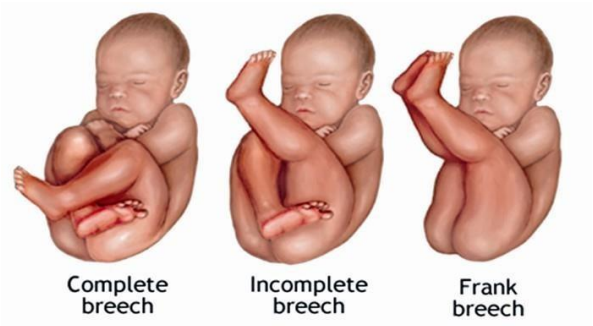


Fig 9-9: Common types of breech presentation

ETIOLOGY:

Any factor that interferes with adaptation of the head and buttocks to the pyriform shape of the uterine cavity may predispose to breech presentation although most are idiopathic:

1. **Idiopathic:** in at least 1/3 of cases no underlying etiology can be detected.
2. **Prematurity:** This is the major factor. This is due to the unstable fetal lie associated with the small size trunk and relatively larger head (that occupies the larger fundus) at preterm.
3. **Hydrocephalus** with large head occupying the large fundus.
4. **Other causes common to all malpresentations as;** twins, uterine abnormalities, polyhydramnios, contracted pelvis, placenta previa, etc…

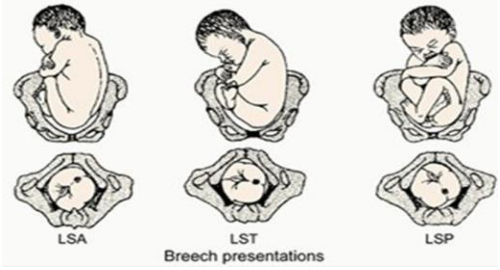


Fig 9-10: SA and SP positions

RISKS ENCOUNTERED WITH BREECH PRESENTATION:

1. Inability to asses **feto-pelvic disproportion** with the fetal head in the fundus.
2. Increased incidence of head entrapment (**retained aftercoming head**), if delivery of the body occurs before full cervical dilatation, predisposing to fetal asphyxia.
3. **Vigorous traction** on the fetal trunk, to accelerate head delivery, may result in serious fetal birth injuries, together with maternal cervical and vaginal lacerations.
   - **N.B.:** Due to various uncalculated maternal and fetal hazards, almost **80%** of breech presentations in PG will be delivered by **Cesarean section** in many centers all over the world.

MECHANISM OF DELIVERY IN VAGINAL BREECH DELIVERY (VBD):

1. **Descent:** during the 1st stage with good uterine contractions.
2. **Engagement:** when bitrochanteric diameter **(9.5cm)** traversing the plane of the inlet.
3. **Internal rotation:** the anterior buttock reaches the pelvic floor first, rotates **1/8** circle anteriorly, hinges below the SP with the posterior buttock facing the sacrum.

4. **Delivery of the buttocks:** the posterior buttock and hip are delivered through the distended vulva by lateral flexion of the spine.
5. **Delivery of the shoulders:** the shoulders enter the pelvis in the same oblique diameter as the buttocks, the anterior shoulder reaches the pelvic floor first, rotates **1/8** circle anterior, hinges below the SP, the posterior shoulder is delivered followed by the anterior.
6. **Delivery of the after-coming head:** the head enters the pelvis in an oblique diameter opposite to that of the buttocks and shoulders. The chin touches the pelvic floor muscles; the head rotates bringing the occiput to hinge below the SP. The head is then delivered in flexion.

N.B.: in Sacro posterior positions the occiput will enter the pelvis in a posterior position and has to rotate 3/8 of circle anteriorly to DOA position to be delivered.

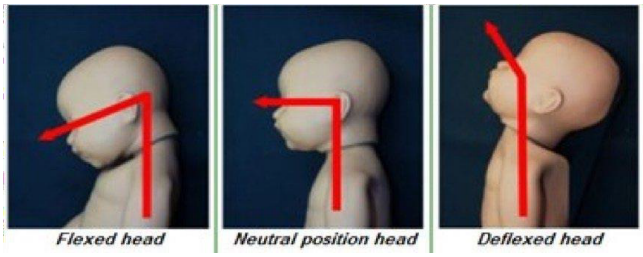


Fig. 9-11: Flexed and deflexed head in breech delivery

DIAGNOSIS OF BREECH PRESENTATION:

A. **Abdominal Examination** (Leopold maneuvers)**:**

- Fundal grip shows a head (hard, rounded, regular, and ballotable).
- Pelvic grip shows a buttock (soft, globular, and irregular).

B. **US Examination:** Longitudinal lie with head at fundus and buttocks at pelvis.

C. **PV during labor:** with the cervix partially dilated and ROM, shows the soft buttocks, with its characteristic bony landmarks (the tip of sacrum and the 2 ischial tuberosities, all at the same level).

MANAGEMENT OF BREECH PRESENTTAIONS DURING PREGNANCY

EXTERNAL CEPHALIC VERSION (ECV)

External cephalic version (ECV) describes conversion of breech into cephalic presentation via rotating the fetal body through manipulations on the maternal abdomen. The aim is to correct the malpresentation, avoid hazards of vaginal breech delivery, and minimize the rate of CS.

- **Timing:** best at 36 weeks. As early ECV <35 weeks may predispose to recurrence of the malpresentation, while late ECV > 37 weeks is more difficult due to larger fetal size and less amniotic fluid.
- **Success rate: 60% of cases** will turn into cephalic presentation avoiding

the maternal and fetal complications of breech presentations.

- **The Procedure:** see figure for description

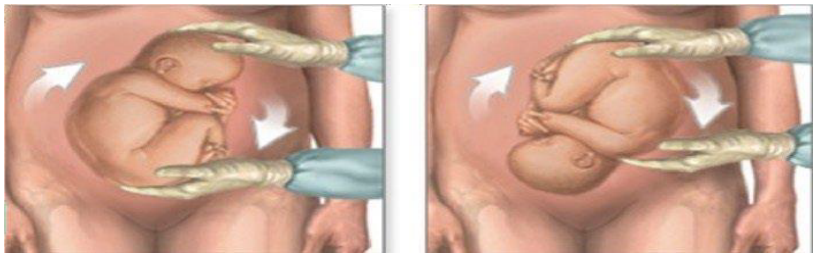


Fig 9:12 external cephalic version ECV

- **Risks and complications of ECV:**
  - Placental abruption and accidental hemorrhage.
  - ROM and initiation of PTL.
  - Cord accidents with loop around fetal neck.
- **Contraindications of ECV**
  - MFP, oligohydramnios, uterine anomalies and leiomyomas.
  - When CS is already indicated: as in previous uterine scat (CS or myomectomy), PI PRV, preeclampsia, fetal macrosomia, contracted pelvis, etc…

MANAGEMENT OF BREECH PRESENTATIONS DURING LABOR:

A) CESAREAN SECTION IN BREECH PRESENTATION

Indications of CS in Breech Presentation

1. Fetal weight > 3.5 kg (LGA fetus) and < 2.0 kg (extremely preterm or SGA fetus).
2. Footling breech presentation: probable ROM, and delivery before full cervical dilatation.
3. Extension of fetal neck by US: to avoid arrest of descent and failure of head rotation.
4. Twins and MFP when first is breech (to avoid locked twins).
5. Associated Other Obstetric indications for a CS as: contracted pelvis, PL PRV, Previous uterine scar, macrosomia, etc…

B) ASSISTED VAGINAL BREECH DELIVERY:

Vaginal breech delivery (VBD) is allowed whenever initial evaluation reveals an average size, full term baby, with normal progress of 1st stage of labor, in absence of CPD. The patient is prepared as in normal labor (see before).

- **Avoid maternal bearing down** before the buttocks reach the vulvar ring.
- **Ensure full cervical dilatation** before attempting delivery.
- **EPISIOTOMY** should be performed in **all cases** when the buttocks distend

the vulva. It should be adequate to allow for lateral flexion of the spine and easy delivery.

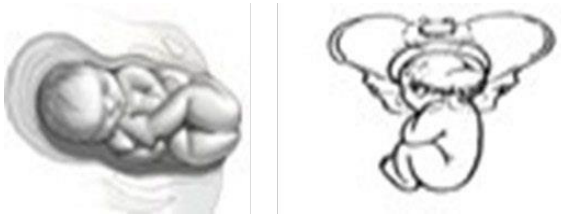
- **Delivery of the Buttocks and Legs:** after episiotomy the buttocks and legs are delivered slowly with controlled perineal support keeping the **back always anterior.**
- **Avoid fetal aspiration of amniotic fluid:** by placing a warm towel on the fetal back to avoid premature stimulation of fetal breathing (due to cold stimulus).
- **Delivery of the shoulders:** *When the anterior scapula hinges below the SP* the posterior arm is delivered by hooking a finger near the elbow and sweeping the arm downwards (Pinard method). This is followed by rotation of the fetal body 180 degrees to bring the anterior arm posteriorly to be delivered as described for the posterior arm.
- **Delivery of the after-coming head:** With the baby hanging down by its weight the head descends until the occiput hinges behind the SP, followed by either;

A. **BURNS- MARSHAL'S METHOD:** when the occiput appears anteriorly hinging below the SP, the infant's legs and feet are held and lifted upwards in a sweeping movement towards maternal abdomen. The head will flex gently and deliver vaginally.

B. **JAW FLEXION SHOULDER TRACTION METHOD** *(Mauriceau-Smellie-Veit maneuver)***:** with the back anterior the fetus is held with the ***left hand*** of obstetrician, the index and middle fingers pressing on the maxilla to promote flexion of the head. The index and middle fingers of the ***right hand*** start gentle traction at the fetal neck downwards (promoting more flexion). When the Occiput appears below the SP, the baby is lifted upwards towards the mother's abdomen, to deliver the head **in flexion**.

C. **FORCEPS DELIVERY:** A specially designed long forceps (**Piper's Forceps**) has been formerly used to promote head flexion and allows traction from the head preventing overstretch of the fetal neck. It is rarely used in modern obstetrics as its use has been associated with maternal birth canal injuries.

C) **BREECH EXTRACTION**: Done ONLY in fetal distress in second stage of labor with NO CPD, or in second twin delivery



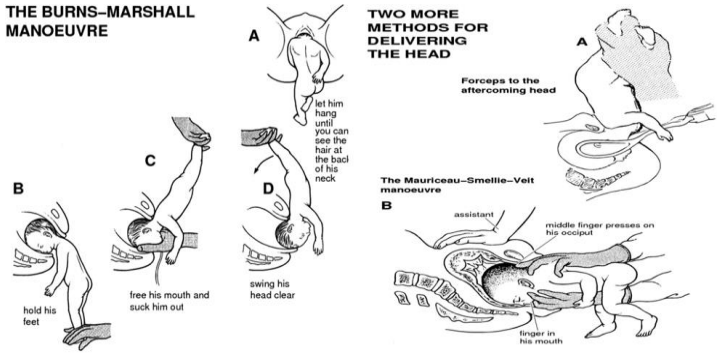


Fig. 9-13: The 3 Methods for Assisted Vaginal Breech Delivery

DIFFICULTIES AND COMPLICATIONS DURING VBD:

A. **Extension of the Arms before delivery of the Shoulders:** managed by rotating the fetal trunk 180 degrees to bring the posterior shoulder anteriorly below the SP and sweeping the anterior arm down, then repeating the same procedure with the posterior shoulder *(Lovset's Maneuver)*.
B. **Posterior Rotation of the Head:** with the occiput directed posterior. Delivery can be accomplished by flexing the whole body towards maternal abdomen, and performing downwards traction on the shoulders from behind by the other hand to deliver the head in flexion *(Prague's Maneuver)*.
C. **Fetal Head Entrapment (Retained aftercoming head):** due to either delivery before full cervical dilatation or an undiagnosed cephalopelvic disproportion (CPD). The consequences are serious and may end in severe fetal birth injuries primarily due to excessive traction on the neck and sudden head compression and decompression. Complications include; intrapartum hypoxia, intracranial hemorrhage, brachial plexus injury, fracture of the clavicle, rupture of the liver or spleen, and hip dislocation.
D. **Maternal complications**: include maternal exhaustion from a prolonged labor, birth canal injuries, postpartum hemorrhage, and puerperal sepsis from extensive maneuvers and prolonged ROM.

SHOULDER PRESENTATION

Shoulder presentation is a **transverse lie** with the **longitudinal axis** of the fetus **perpendicular** to that of the mother. It is a rarely encountered

malpresentation occurring in < **1:200** term deliveries.

There is **no mechanism for delivery** in shoulder presentation and labor will be eventually obstructed. The scapula is the **denominator**, and **positions** include**:** right and left Scapulo (Dorso) anterior and posterior (LDA – RDA – RDP – LDP). Scapulo-anterior are more common than scapulo-posterior positions.

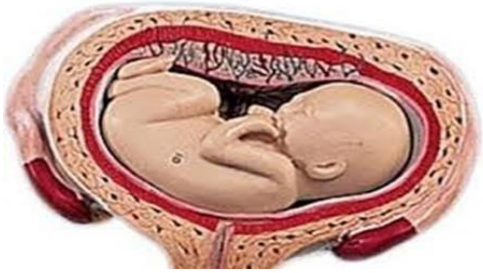


Fig 9-14: Transverse lie (shoulder presentation)

ETIOLOGY:

1. Grand multipara with pendulous abdomen and lax abdominal wall muscles.
2. Prematurity, polyhydramnios, MFP, PL PRV, and contracted pelvic inlet.
3. Septate and bicornuate uterus, and pelvic masses as leiomyomas and large ovarian swellings.

DIAGNOSIS:

A. **Abdominal Examination** (Leopold maneuvers)**:** Fundal level is lower than expected, both fundal and pelvic grip are empty. With the umbilical grip, the head is felt on one side and buttocks on the other side of the midline. FHS is heard at the level of umbilicus, towards the fetal shoulder (head).

B. **PV Examination:** with cervix partially dilated during labor shows a high non engaged presenting part with the ribs felt above the pelvic inlet.

C. **Ultrasound Diagnosis:** gives a rapid easy and definitive diagnosis and helps identifying the etiology, and exclusion of fetal and uterine anomalies and PLPRV.

MANAGEMENT OF SHOULDER PRESENTATION:

1.CESAREAN SECTION:

- CS is **MORE SAFE AND MOST COMMON MODE OF DELIVERY** in shoulder presentation.

2.INTERNAL PODALIC VERSION (IPV) and Breech Extraction During Labor:

- **ONLY DONE** for delivery of the second twin presenting as a transverse lie.

NB: EXTERNAL CEPHALIC VERSION (ECV): ECV entails correcting transverse lie into a cephalic presentation through rotation of the body 90 degrees on its axis. ECV is best performed during late pregnancy at 36weeks gestation, but may be practiced even at onset of labor (not recommended).

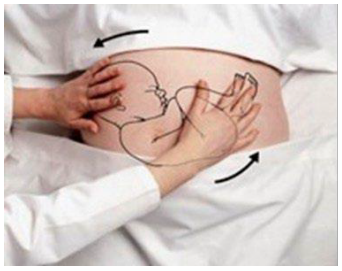


Fig. 9-15: ECV

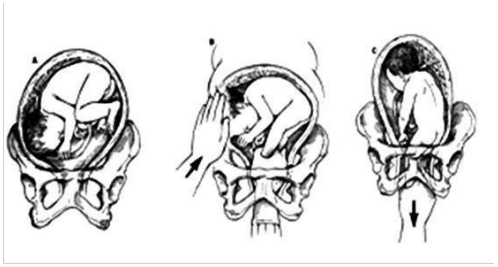


Fig. 9-16: IPV and breech extraction

NEGLECTED SHOULDER: in second stage of labor, after ROM for a longtime, one arm is prolapsed through a fully dilated cervix, into the vagina. Usually the fetus is dead, and signs of obstructed labor are present. CS (usually USCS) is the only solution (even in cases of IUFD) to avoid rupture of the uterus, usually it is an upper segment CS.

CORD PRESENTATION AND CORD PROLAPSE

CORD PRESENTATION:

In cord presentation the cord lies below the presenting part with **intact membranes**. It is a rare condition (< 1/300 deliveries), which once discovered early in labor, the patient should be offered an immediate **CS.** It occurs more commonly in presence of malpresentations as breech and transverse lie in which there is delayed or absent engagement of the presenting part.



Fig 9-17: Cord presentation



Fig 9-18: Cord prolapse

CORD PROLAPSE:

The cord lies below the presenting part **after ROM**. It is considered one of the most acute obstetric emergencies as it is life threatening to the fetus if still living at the time of diagnosis.

Diagnosis:

- PV examination reveals loops of cord through the partially dilated cervix.
- Cord is palpated for presence of fetal pulsations to exclude IUFD.
- If cord is non pulsating IUFD should be confirmed by US.

Management:

a) Cases with confirmed IUFD are left for vaginal delivery in absence of severe CPD.

b) Cases in which the fetus is living are managed according to degree of cervical dilatation:
   - First aid measures as reposition of the cord, with patient in Trendelenburg position or knee chest position to avoid more cord compression by the fetal presenting part.
   - If cervix fully *dilated* immediate vaginal delivery is encouraged (by instrumental delivery in cephalic presentation or breech extraction in breech presentation).
   - If cervix is not fully dilated: immediate CS is the only option

COMPLEX PRESENTATION:

Rarely an arm may present beside the head or buttock during 2nd stage of labor after ROM.

Management: reposition of the arm and allowing labor to continue safely.

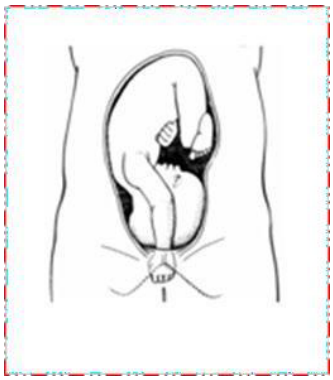


Fig 9-19: Complex presentation

UNSTABLE LIE: when the fetus continuously changes its axis from longitudinal to transverse presentation after 36 weeks. Stabilized induction of labor is done when the fetus is in a cephalic presentation.

MULTIFETAL GESTATION

Multifetal pregnancy **(MFP)** describes pregnancies in which **two** or **more fetuses** are present in the uterus (twins, triplets, quadruplets, quintuplets, etc…).

It may occur either **spontaneously**, with a strong **genetic** and **familial** background, which is the less commonly encountered form, or it may be **iatrogenic induced**, as with the use of multiple ovulation protocols in the management of **infertility**.

INCIDENCE:

Nearly **1-2%** of all pregnancies are MFP. The incidence of iatrogenic twinning has been showing a steady rise in the last decade due to the use of superovulation protocols in ART, especially IVF- ISCI.

- **Twins**: is the commonest form occurring in nearly **97%** of MFP.
- **Triplets**: is rare occurring in only **1%** of MFP.

MECHANISMS AND CLASSIFICATION OF TWINNING:

A) Fraternal or **Dizygotic twins** (DZT); this is the commoner form of twinning that occurs due to fertilization of two or more eggs in a single cycle (> 1 zygote). They are genetically non identical, of different or same sex, and are always dichorionic / diamniotic **(DC/DA)**, i.e. 2 embryos, each having its own amniotic sac and its own placenta.

B) Identical or **Monozygotic** twins (MZT); this is the less commonly encountered form in which twinning occurs due to splitting of a single embryo (one zygote in which cleavage will occur). They are genetically identical having the same sex and same features. The type of twins will depend upon the time duration between fertilization and cleavage as follows;

- ***Day 1-3:*** **Dichorionic / Diamniotic (DC/DA)** 2 placentae / 2 sacs (30%).
- ***Day 4-8:*** **Monochorionic / Diamniotic (MC/DA)** 1 placentae / 2 sacs (65%).
- ***Day 9-12:*** **Monochorionic / Monoamniotic (MC/MA)** 1 placenta / 1 sac (5%)
- ***After 12 days:*** **Conjoined twins** result due to incomplete cleavage division (MC/MA).

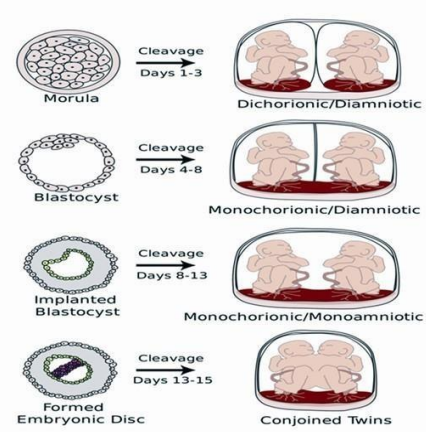


Fig. 9-20: Cleavage effect on type of twin pregnancy

OBSTERICAL RISKS IN MFP:

MFP is considered as a high-risk pregnancy, being associated with several maternal and fetal risks that increase with the increase in the number of fetuses in utero, including;

1. **Preterm labor (PTL):** this is a major risk in MFP. It usually occurs at a mean gestational age of **36** weeks in twins, and even earlier in triplets and quadruplets.

2. **Medical disorders complicating pregnancy;** as hyperemesis gravidarum **(HEG)**, iron deficiency anemia, gestational diabetes mellitus **(GDM)** and preeclampsia **(PE).**
3. **Increased CS rates:** due to polyhydramnios, malpresentations, abnormal labor, and placenta previa.
4. **Other risks** as; spontaneous abortion, low birth weight **(LBW)**, atonic postpartum hemorrhage **(PPH)**, fetal congenital anomalies **(FCA)**, and conjoint twins.

SPECIAL COMPLICATIONS RELATED TO MFP:

1. **Discordant twins; difference in weight between 2 twins is more than 25%:** twin to twin transfusion syndrome **(TTTS)** may occur specially in MZT sharing the same placenta. The **donor twin** gets less blood supply resulting in IUGR, oligohydramnios and anemia. The **recipient twin** gets more blood supply resulting in macrosomia, polyhydramnios, and polycythemia.
2. **Conjoined twins:** twins may be adherent at the chest (thoracopagus), head (craniopagus), or sacrum (ischiopagus). Such cases are usually lethal, however in some cases surgical separation and correction may be successful but, in many instances, sacrificing one baby.

3. **DEATH OF ONE FETUS DURING PREGNANCY:**

During 1st trimester: pregnancy usually continues with minimal risks to the mother and to the other fetus. May present as disappearance of one or more GS in previously diagnosed MFP at early gestation US. Pregnancy usually continues safely with the remaining fetus (**Vanishing twin).**

- **During 2nd trimester:** pregnancy usually continues without significant hazards except for the Increased risk of hypo-fibrinogenemia, DIC. Close observation and early detection of Coagulopathy should be encouraged to determine proper timing for delivery.

4. **THE LOCKED TWINS:** locking of twins during vaginal delivery rarely occurs when the 1st twin is presenting as breech and the second as cephalic. Both heads will be entrapped in the pelvic brim and labor will be obstructed. Emergency CS may salvage the 2ndtwin while the first one will be mostly sacrificed.

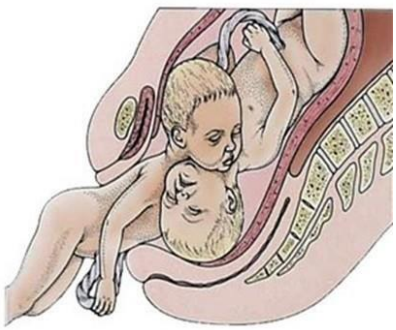
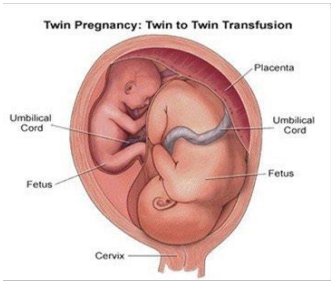


Fig.9-21: Locked twins



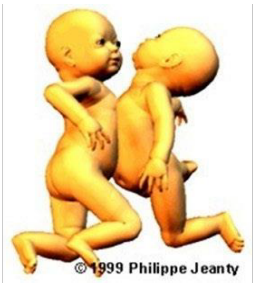




Fig. 9-22 Discordant twins

Fig. 9-23: Conjoined twins thoracopagus and ischiopagus

DIAGNOSIS OF TWINS AND MFP:

- Exaggerated **symptoms** of early pregnancy as nausea, vomiting, and HEG.
- **Inspection and Palpation:** Fundal level is larger than the period of amenorrhea with multiple fetal poles and limbs felt.
- **Auscultation of FHS:** several points of maximum intensity and different FHR (with galloping sign).
- **TAS:** US helps in diagnosis of MFP. benefits include:
  a. *Estimation of number* of fetuses and the true *gestational age* (GA).
  b. *Determination of zygosity* (MZT or DZT), and *chorionicity* (MC or DC).
  c. *Diagnosis of congenital anomalies (FCA)*, *conjoined twins, PL PRV, and polyhydramnios.*
  d. *Evaluation of fetal wellbeing* for each fetus (biophysical profile and Doppler).
  e. *Diagnosis of 2nd twin presentations:* vertex / breech / transverse.





Fig. 9-24: 2D US of DZT (MC/MA)

Fig 9-25 3D US of MZT

MANAGEMENT DURING PREGNANCY:

MFP is managed as a high-risk pregnancy, with more frequent ante-natal care visits, improved nutrition, more vitamin, calcium, and iron supplementation, together with;

- *Early and repeated US scans;* to determine zygosity, and to exclude FCA, polyhydramnios, discordant twins, and placenta previa.
- *Early diagnosis and management of medical disorders;* as GDM and PE,

etc…

- *Avoid factors that increase the risk of PTL;* more rest, progesterone support, corticosteroids for lung maturity at 28 weeks (see preterm labor).
- *Select patients in which delivery by CS will be indicated* and choose the appropriate time for pregnancy termination.

INDICATIONS OF CS IN TWINS AND MFP

1. Other indications of CS as: PL PRV, previous CS or myomectomy, associated severe PE or GDM, etc…
2. Triplets & more MFP: delivery of the 2nd & 3rd twins will be unsafe with malpresentations.
3. Monoamniotic twins: as more incidence of polyhydramnios, malpresentations, and cord entanglement. ROM is an additional risk for the 2nd twin if transverse lie.
4. Conjoined twins: to avoid obstructed labor, rupture uterus and fetal birth injuries
5. Non cephalic presentation of the 1st twin: to avoid locked twins (see later)

MANAGEMENT DURING LABOR:

Vaginal delivery is allowed in twin pregnancy whenever the first baby is presenting as cephalic presentation irrespective of the second twin, which may be either cephalic, breech or transverse lie.

A) **Delivery of the 1st twin:** cephalic presentation (see before).

B) **Delivery of the 2nd twin:** according to fetal presentation:

- *Cephalic presentation:* do artificial ROM and proceed as vaginal delivery.
- *Breech presentation:* do artificial ROM and breech extraction.

-*Transverse lie:* do Internal podalic version (IPD), ROM, and proceed to breech extraction.

C) **Immediate postpartum care:**

- Ecbolics (Oxytocin, Ergometrin, or Misoprostol) for prevention of atonic PP hemorrhage due to uterine overdistension

10 ABNORMAL LABOR

ABNORMAL LABOR PATTERNS

Abnormalities often interact singly or in combination to produce **dysfunctional labor (dystocia).**

Common clinical findings in women with ineffective labor include:

A. **Abnormal cervical dilatation or fetal descent:**
   - Precipitate labor.
   - Protracted or prolonged labor-slow progress.
   - Arrested labor-no progress.
   - Inadequate expulsive effort-ineffective pushing in maternal exhaustion.

B. **Fetopelvic disproportion:**
   - Excessive fetal size.
   - Inadequate pelvic capacity.
   - Malpresentation or malposition of the fetus.
   - Abnormal fetal anatomy.

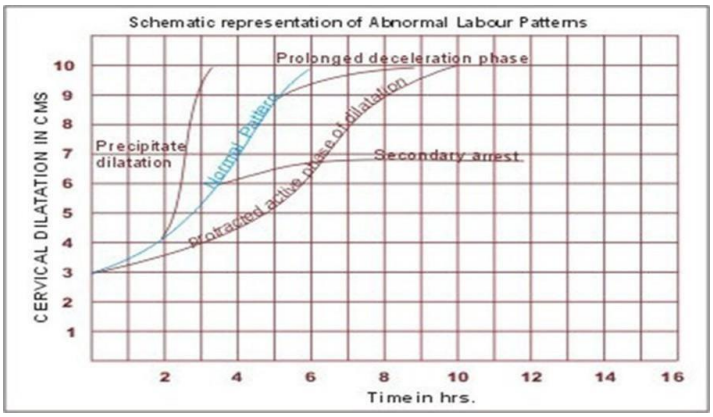


Fig 10-1: Abnormal labor patterns

PRECIPITATE LABOR

- **Precipitate labor** describes a very rapidly occurring labor of duration **< 4 hours** from onset of labor pains till expulsion of the fetus. It is uncommon, occurring relatively more in multiparas, with roomy pelvis, and very strong uterine contractions.

- **Clinical presentation:** Patients usually gives **history** of previous precipitate labor. Commonly present in **late 1st stage** of labor, or in 2nd**stage**, with a short course labor, in which the patient feels only the last

contractions associated with involuntary expulsive forces of the fetus.

MATERNAL AND FETAL RISKS:

A) MATERNAL:

1. **Cervical lacerations:** due to rapid dilatation and forced delivery before full cervical dilatation.
2. **Vaginal and perineal tears:** due to rapid distension of the birth canal without enough time for full stretch of muscles and ligaments together with uncontrolled head delivery
3. **Postpartum hemorrhage (PPH):**
   - *Atonic PPH:* exhaustion of the uterus due to successive rapid contraction.
   - *Traumatic PPH:* due to cervical, vaginal, and perineal lacerations.

B) FETAL:

1. **Intracranial hemorrhage (ICH);** due to rapid head compression and decompression, especially in SGA, IUGR, and PTL.
2. **Fetal distress** due to;
   - Increased frequency of uterine contractions.
   - Increased incidence of placental abruption due to rapid cord traction.

MINIMIZING RISKS AND COMPLICATIONS OF PRECIPITATE LABOR:

1. Patients with past history of a PPT labor should be advised to early hospital admission once they feel early labor pains whatever weak.
2. During the 1st stage of labor: course of labor can be slowed down by use of epidural analgesia.
3. During the 2nd stage of labor: avoid maternal bearing down, and allow for controlled head delivery with adequate perineal support.
4. After delivery; exploration of the cervix and vagina for extensive lacerations with prompt repair.
5. Guard against atonic PPH by use of ecbolics during the 3rd stage of labor.
6. Care of the new born; for early diagnosis of fetal asphyxia and birth injuries.

PROLONGED LABOR

- **Clinical presentation:** according to Partogram during labor.

a) ***Prolonged 1st stage:*** delayed cervical dilatation and head descent in active phase.

- Delayed cervical dilatation in the **active** phase
  - Less than 1.2 cm/hour in PGDA
  - Less than 1.5 cm/hour in MGVDA

- Delayed or arrested descent of the fetal head station.

b) ***Prolonged 2nd stage:*** delayed fetal head descent after full cervical dilatation

- **More than 2** hours in a PGVDA
- **More than 1** hour in MGVDA

ETIOLOGY:

1. Faults in the **powers**:
   - Inadequate uterine contractions (hypotonic inertia).
   - Uncoordinated uterine contractions (hypertonic inertia and dysfunctional labor).
2. Faults in the **passages**: contracted pelvis and cephalopelvic disproportion (CPD).
3. Faults in the **passenger**: fetal macrosomia, malpresentations (face, brow, breech & shoulder).
4. Cervical dystocia (see later)

MATERNAL AND FETAL RISKS:

1. Maternal **exhaustion**; due to prolonged duration of labor pains.
2. Fetal **distress**; due to prolonged duration of uterine contractions and prolonged use of oxytocin.
3. Postpartum **hemorrhage**; due to uterine atony and exhaustion or traumatic difficult delivery.
4. Puerperal **sepsis**; due to prolonged rupture of the membranes (ROM), & excessive manipulation.
5. Risks of prolonged use of **ecbolics.**

MANAGEMENT:

a. ***In cases of hypotonic inertia:*** proved by CTG measurements of the frequency and intensity of uterine contractions, labor pains can be **augmented** by the use of **Oxytocin** given as IV infusion drip in escalating doses till adequate contractions are obtained on CTG, and progress of labor is reassured by repeat PV examination.

b. ***In cases of hypertonic inertia:*** with frequent strong uncoordinated uterine contractions, the use of strong sedation (as Morphia), or regional (epidural) analgesia may help normalizing uterine action and improving cervical dilatation and effacement.

c. ***In cases of marked cephalo-pelvic disproportion (CPD):*** CS is the proper management

d. In cases of prolonged 2nd stage in absence of CPD:

- If head is engaged; instrumental delivery by the **forceps** or **ventouse** may be used to shorten the 2nd stage and avoid maternal and fetal distress.
- If head is not engaged in presence of adequate uterine contractions **CS** will

be the best and safest option for delivery to be accomplished without maternal or fetal hazards.

SHOULDER DYSTOCIA

Shoulder dystocia occurs when delivery of the fetal shoulder is **delayed** after delivery of the head. It is a **rare** condition complicating **< 1%** of vaginal deliveries.

Shoulders are usually **entrapped** in the anteroposterior diameter of the cavity with the anterior shoulder impacted behind the symphysis pubis **(SP)**. The head recedes backwards shortly after its delivery (**turtle sign**).

Fetal macrosomia is the most important risk factor. It is mostly encountered in association with uncontrolled gestational diabetes mellitus **(GDM)** and fetal **hydrops** (immune or nonimmune).

Although shoulder dystocia is usually attributed to a large for gestational age **(LGA)** fetus, yet
50% occur with fetal birth weight **< 4.0** kg.

RISKS AND COMPLICATIONS:

1. **Fetal asphyxia and IUFD:** due to brain hypoxia associated with delayed delivery
2. **Fetal birth injuries:** Attempts at delivery of the impacted shoulder may be associated with fracture clavicle, fracture humerus and or brachial plexus injury (Erb's palsy).
3. **Maternal birth injuries:** due to rough intrauterine manipulations including rupture uterus, cervical, and vaginal lacerations.

PREVENTION AND PREDICTION:

In many cases it is difficult to predict occurrence of shoulder dystocia to avoid its complications, however the following measures may be of help;

1. Proper assessment of fetal weight before delivery by clinical and US evaluation.
2. Proper control of maternal GDM especially in 3rd trimester.
3. CS should be offered to LGA and cases with established macrosomia.

MANAGEMENT DURING LABOR:

Shoulder Dystocia (Shoulder Dystocia Drill): to better organize emergency management:

1. **Call for help:** Senior obstetrician, anesthesia and neonatology personnels. Initially, a gentle attempt at traction is made. Evacuate the bladder if it is distended.
2. **A generous episiotomy** to afford room posteriorly.
3. The **McRoberts maneuver:** hyperflexion and abduction of the maternal thighs

to increase the A/P diameter of the pelvis and allow dis-impaction of the anterior shoulder.

4. **Suprapubic pressure**, while normal downward traction is applied to the fetal head.

These maneuvers will resolve most cases of shoulder dystocia. If the above listed steps fail, the following steps may be attempted:

5. **Delivery of the posterior arm**, however, this is usually difficult to accomplish.
6. **Woods screw maneuver** is applied: anterior shoulder is pushed to baby's chest and posterior shoulder to the baby's back.
7. **Rubin maneuver** is attempted: rotation of the head 180, trying to disimpact the shoulder.

Other methods that may be tried if all the above methods fail,

- Posterior axilla sling traction, but with very high possibility of Erb's palsy.
- Deliberate fracture of the anterior clavicle using the thumb (inward outwards) to press it against the pubic ramus.
- Zavanelli maneuver: try to repush the head back in the uterus and do CS.
- Symphysiotomy, to widen the symphysis pubis,
- Cleidotomy consists of cutting the clavicle with scissors or other sharp instruments and is usually done for a dead fetus.

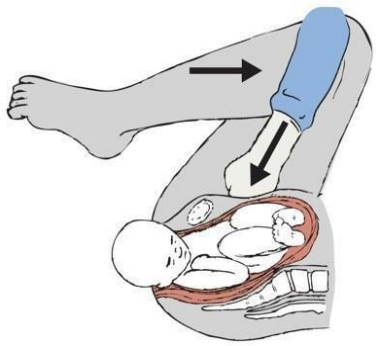


Fig.10-2: Mc Roberts maneuver



Fig.10-3: suprapubic pressure

OBSTRUCTED LABOR

Obstructed labor describes failure of delivery of the fetus due to **mechanical** obstruction.

ETIOLOGY:

1. *Contracted pelvis* and marked degrees of *Cephalo-pelvic Disproportion (CPD)*.
2. *Fetal macrosomia* and extremely *LGA fetus* > 4.5 kg.
3. *Non corrected malpresentations and malpositions* as:

- Persistent oblique and transverse Occipito Posterior positions (OP).
- Direct face Mento Posterior (MP).
- Brow and shoulder presentation.
- Breech impacted in the pelvic inlet.

CLINICAL PICTURE AND DIAGNOSIS:

A) **HISTORY:** of prolonged labor with long ROM in presence of adequate contractions.

B) **GENERAL EXAMINATION:** Signs of maternal exhaustion including; dehydration, fever, tachypnea, tachycardia, oliguria, and dry tongue.

C) **ABDOMINAL EXAMINATION:**

- ***Palpation:*** tender hard uterus with persistent tonic contractions (except if ruptured).
- ***Pathological Retraction ring (Bandl's ring);*** it is a transverse groove that can be seen and felt abdominally at the demarcation between the upper and lower uterine segments. It shows a rising level upwards towards the umbilicus in cases of impending rupture of the uterus.
- ***Auscultation:*** FHS usually show abnormal patterns denoting fetal distress or Inaudible FHS in cases of intrauterine fetal death (IUFD).

D) **VAGINAL EXAMINATION** may reveal one or more of the following signs:

- *Non engagement of the presenting part* (vertex, face, brow, breech, or shoulder).
- *Edematous vagina, which may be also dry and hot (fever).*
- *The cause of obstructed labor* may be elicited (malpresentation, malposition, contracted pelvis, CPD)
- *Caput succedaneum* in vertex cephalic presentation due to scalp edema

- **N.B.** *caput succedaneum may give a false impression of head engagement where the scalp is felt low in the vagina while the skull bone is still at a higher non engaged level (-2, or -3).*
- **N.B.** *Bandl's ring should be differentiated from a full bladder, a uterine leiomyoma, contraction ring.*

Differentiation between Retraction ring and Contraction ring

Pathological Retraction Ring (Bandl's ring)	Contraction Ring

– *Occurs with arrest of 2nd stage of labor (obstructed labor).* – *Lies always between UUS and LUS.* – *Rises upwards towards umbilicus.* – *Felt and seen abdominally.* – *Uterus is tense and tender.* – *Maternal and fetal distress are common.* – *Relieved only by delivery.*	– *Occurs during any stage of labor.* – *Occurs at any level (due to spasm of circular muscle fibers)* – *Does not change its position.* – *Not seen, and only felt vaginally.* – *Uterus is not tonically retracted.* – *Distress may not be present.* – *Relieved by deep anesthesia or vertical LSCS (if persistent).*

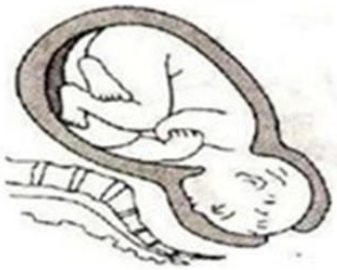


Fig. 10-4: Pathological retraction ring

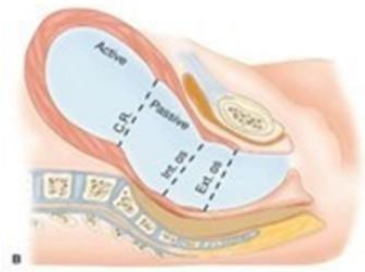


Fig. 10-5: Contraction ring

COMPLICATIONS OF OBSTRUCTED LABOR:

1. Complications of prolonged labor; maternal exhaustion, fetal distress, ROM, chorioamnionitis, neonatal asphyxia, and IUFD.
2. Rupture of the uterus which may necessitate laparotomy for repair, or hysterectomy.
3. Cervical, vaginal, and perineal lacerations: from traumatic manipulation.
4. Necrotic vesico-vaginal fistula (VVF): due to prolonged head compression on the bladder.
5. Puerperal sepsis: due to prolonged ROM and traumatic manipulations.

MANAGEMENT

1. **Emergency CS** is the best and **safest** option even in cases with IUFD to avoid rupture of the uterus
2. Instrumental delivery (obstetric forceps or ventouse) for cephalic presentations is **contraindicated** as any attempt will be complicated by rupture of the uterus, with high maternal morbidity and mortality from bleeding and sepsis.

11 CONTRACTED PELVIS ANDCEPHALOPELVIC DISPROPORTION

CONTRACTED PELVIS

From the **clinical** point of view a **contracted pelvis** is a pelvis in which one or more of its **main diameters** are **reduced** to the extent that **interferes** with **normal** delivery.

ETIOLOGY OF CONTRACTED PELVIS:

1. **Idiopathic** small pelvis: gynecoid, android, anthropoid, etc…
2. **Metabolic disease**: as Rickets and Osteomalacia.
3. **Trauma** or **diseases affecting pelvic bones:** as fractures, tumors, and TB joints.
4. Diseases of the **spine:** severe dorso-lumbar kyphoscoliosis, and spondylolisthesis.
5. Diseases of **lower limbs (LL)**: hip dislocation, fracture femur, poliomyelitis, etc…

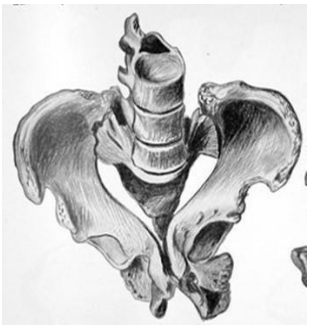


Fig. 11-1: Osteomalecic pelvis



Fig. 11-2: Scoliosis

DIAGNOSIS OF CONTRACTED PELVIS:

A) HISTORY:

- History of pelvic trauma or disease of the spine or LL.
- History of previous prolonged difficult or obstructed labor.

B) GENERAL EXAMINATION:

- Short stature < 150 cm, limping and abnormal gait, spine and LL deformities.
- Malpresentations as face, brow, transverse lie.
- Non engagement of the head during the 1st stage of labor.

C) CLINICAL PELVIMETRY (ON performing PV examination):

1. ***Sacral promontory is felt.***
2. ***Jutting ischial spines.***
3. ***The Sub-pubic angle:*** is acute and cannot accommodate 2 fingers.

4. ***The Bituberous diameter (11 cm):*** (transverse diameter of the outlet) roughly cannot admit the 4 knuckles of the closed fist of the hand.
5. ***The Diagonal conjugate DC (12.5 cm):*** is measured between the lower border of the SP and the promontory of sacrum.

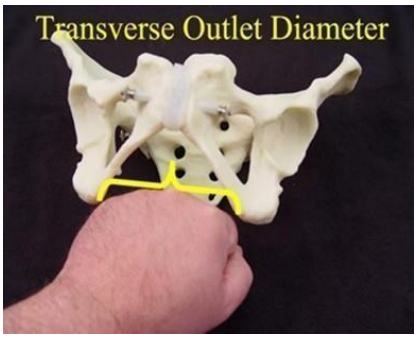


Fig 11-3: Bituberous diameter

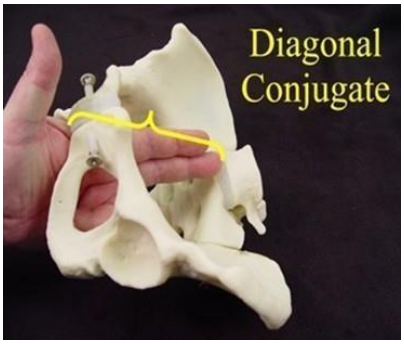


Fig. 11-4: Diagonal conjugate

D) CEPHALOPELVIC DISPROPORTIONTESTS:

Cephalopelvic disproportion **(CPD)** tests are **clinical** tests to assess possibilities of the fetal head to be delivered vaginally in **primigravida**, based on the concept that **the head is the best Pelvimeter.**

1. PINARD'S METHOD:

- Patient in semi-sitting position with bladder empty.
- The operator with his **left hand** grasps the **fetal head** and tries to push itdownwards and backwards in the pelvis along the pelvic axis.
- The **fingers** of the operator's **right hand** placed over the **SP** can determine the degree of disproportion and whether head overrides SP or passes smoothly through the inlet.

2. MULLER-KERR METHOD:

- Patient in dorsal position with bladder empty.
- The operator performs a digital **PV** examination with the fingers of the **right hand** performing the steps of internal **clinical pelvimetry.**
- The **thumb** of the **right hand** is placed over the **SP** to determine the degree of disproportion if present.
- The operator with his **left hand** grasps the **fetal head** and tries to push it down through the pelvic inlet with the thumb of the right hand in over the SP trying to evaluate the presence and degree of CPD.



Fig. 11-5: Muller- Kerr test for CPD

INTERPRETATION OF THE RESULTS OF CPD TESTS:

A) **No CPD:** the head can be pushed into the maternal pelvis passing behind the SP.

B) **Moderate CPD**:

- The head **stops** at the same level of the **SP**.
- Vaginal delivery is possible after **trial labor** (testing undetermined factors of labor: molding of the head, yielding of the pelvis and the strong uterine contractions pushing the head in the pelvis)

C) **Marked CPD:**

- The head **overrides** the anterior surface of the SP.
- Vaginal delivery cannot occur and trial of labor should not be allowed and **CS** performed.

12 OBSTETRIC TRAUMA

RUPTURE OF THE UTERUS

Rupture of the uterus is one of the most **life-threatening** obstetric emergencies that may lead to serious maternal morbidity, or even mortality due to massive **obstetric hemorrhage**.

In many cases rupture of the uterus can be avoided or prevented by adequate antenatal care, proper selection of high-risk cases, and adequate intrapartum care during labor and delivery.

TYPES OF RUPTURE:

A) **Complete Rupture:** describes complete tear of **all layers** of the uterine musculature with the fetus being **totally** or **partially extruded** within the peritoneal cavity.

B) **Scar Dehiscence:** describes separation of the uterine musculature at the site of a **previous LSCS** scar that includes **all** layers **except** the visceral peritoneum. The fetus will thus be kept in utero with no fetal parts extruded into the general peritoneal cavity.

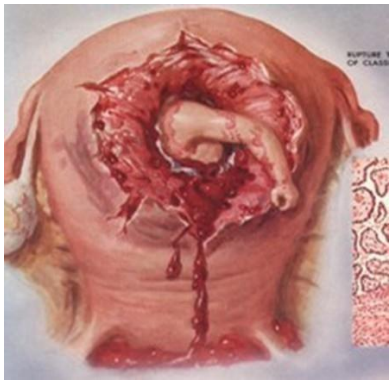


Fig 12-1: Rupture UUS CS scar

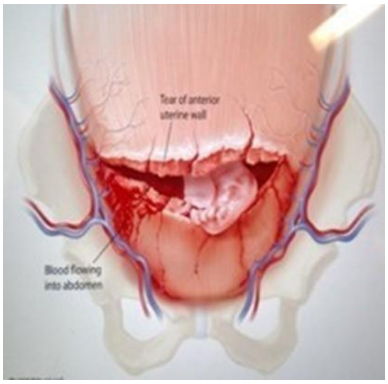


Fig 12-2: Rupture LUS CS scar

RISK FACTORS:

1. **UTERINE SCAR:**
   - *Especially Upper uterine segment:* as classical CS scar, myomectomy, and hysterotomy scar it has (2- 9%) 10 times possibility of rupture over LSCS scar (0.2- 0.9%).
2. **GRAND MULTIPARA:** (> 5 deliveries): (see before)
3. **TRAUMATIC RUPTURE:**
   - *External Trauma:* as car accidents, or

- *Rough Obstetric manipulations:* as in instrumental deliveries.

4. **OBSTRUCTED LABOR: due to** as in pathological retraction ring of Bandl's…

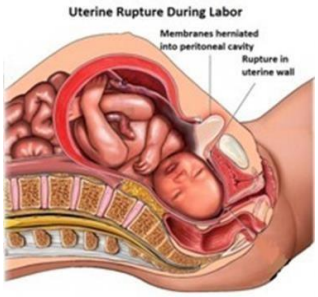


Fig 12-3: Rupture LUS during labor

5. **Improper us of oxytocin and or PGL** for labor induction or augmentation especially in presence of undiagnosed CPD or unknown prior uterine scar.

DIAGNOSIS:

A) SYMPTOMS:

- Sudden severe abdominal **pain** followed by cessation of uterine contractions.
- Sudden onset of **bleeding** that may be internal hemorrhage or external vaginal bleeding.
- Sudden cessation of **fetal movements** (usually associated with IUFD).

B) SIGNS:

- **General Examination:** may reveal signs of **hypovolemic shock** due to rapid blood loss.
- **Abdominal Examination:** may reveal easily palpable fetal parts just below abdominal wall, abnormal fetal lie, severe fetal bradycardia or absent FHS.
- **Vaginal Examination:** recession of the presenting part to a higher pelvic station or even to an abdominal position. Vaginal bleeding may be present.

MANAGEMENT OF RUPTURE OF THE UTERUS:

1. RESUSCITATION AND ANTISHOCK MEASURES:

This is the ***first step*** prior to any surgical intervention in cases with massive blood loss.

2. EXPLORATORY LAPAROTOMY AND SURGICAL REPAIR:

Aiming at arresting uterine bleeding, repairing the torn myometrium, and reforming the uterus back to its normal configuration. Successful pregnancy usually occurs after adequate surgical repair however, delivery by elective ***CS will be mandatory*** for fear of rupture uterine scar.

3. ABDOMINAL HYSTERECTOMY with preservation of the ovaries:

Hysterectomy is indicated whenever rupture of the uterus is extensive beyond proper repair, and bleeding is severe risking the life of the patient, especially in elderly grand multiparas with no desire for future fertility.

CERVICAL LACERATIONS

ETIOLOGY& RISK FACTORS:

1. **Precipitate labor:** due to rapid cervical dilatation and rapid delivery of the head.
2. **Delivery through incompletely dilated cervix** as in Preterm delivery and aftercoming head of breech.
3. **Difficult Instrumental delivery:** forcible forceps or ventouse application and traction especially before full cervical dilatation (wrong application).
4. **Excessive use of oxytocin or PGL:** leading to rapid dilatation & excessive expulsive forceps.
5. **Previous scarring of the cervix:** as in previous cerclage procedures, cervical amputation, cervical conization, or previously repaired cervical lacerations during previous deliveries.

DIAGNOSIS: (See traumatic post-partum hemorrhage)

1. Inspection: excessive vaginal bleeding immediately after delivery of the fetus or placenta.
2. Abdominal examination: fundus of the uterus is well contracted.
3. Vaginal examination: cervical tear is sometimes palpable on PV examination.
4. Exploration of the cervical lips under good light using vaginal speculum and ring forceps to visualize the cervical laceration and start immediate repair.

MANAGEMENT:

Surgical repair to stop bleeding and restore normal anatomy.

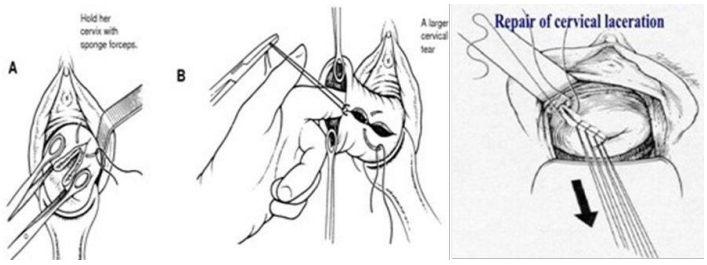


Fig 12-4: Repair of cervical laceration

VAGINAL AND PERINEAL LACERATIONS

Most vaginal deliveries will be associated with minor vaginal tears that do not necessitate surgical repair without causing late side effects or complications. However severe lacerations must be promptly dealt with as they may cause serious morbidity and late complications.

ETIOLOGY:

1. **Precipitate labor:** due to the rapid passage of head through the birth canal, without adequate perineal support (Ritgen's maneuver).

2. Delivery of an **LGA or** a **macrosomic fetus** without an adequate episiotomy.
3. **Instrumental delivery** especially in absence of an adequate episiotomy.
4. **Rigid perineum** with tight vaginal and strong perineal muscles.

CLASSIFICATION OF TYPES OF LACERATIONS:

A. **First degree tears:** involve only **vaginal mucosa.**
B. **Second degree tears:** involve the vagina and the muscles of the **perineal body.**
C. **Third degree tears:** involve the vagina, perineal muscles, and the **anal sphincter.**
D. **Fourth degree tears:** involve the vagina, perineal muscles, anal sphincter, and **anal mucosa.**

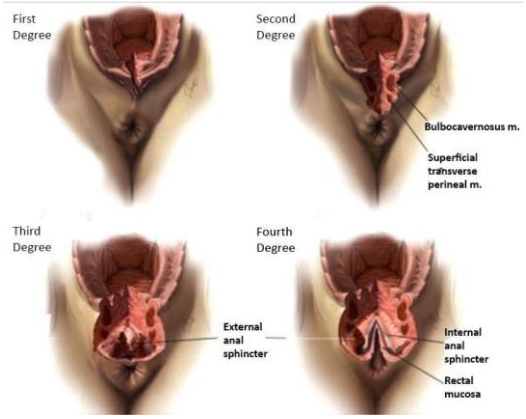


Fig.12-5: Degrees of Perineal Tears

PREVENTION OF PERINEAL LACERATIONS:

1. Proper management of 2nd stage of labor, by maintaining flexion of the fetal head until crowning occurs, then allowing for slow delivery of the head in between uterine contractions.
2. Episiotomy, when the perineum threatens to tear and routinely with instrumental delivery.

MANAGEMENT OF PERINEAL LACERATIONS:

Primary suture – in layers- is done within the first 24 hours. If the case is seen later than that, it is considered as an infected wound, and left to heal by granulation, Repair in such cases is postponed until all signs of infection have disappeared, usually 3-6 month later.

The post-operative care after repair aims at keeping the wound DRY AND CLEAN to encourage healing by primary intention.

- The vulva is regularly washed with antiseptic solutions then dried (at least 2times daily).
- The low residue diet is continued, as well as the intestinal antiseptic.

- Antibiotics against wound infection (systemic and possibly local creams or gel).
- The vaginal pack is removed after 24 hours.
- On the fifth night the patient is given oral purgative solutions (as 50 ml. castor oil) in order to lubricate the stools. After that the patient is given daily oral laxatives and stool softeners to avoid constipation.

N.B.; *in the event of subsequent pregnancy, a postero-lateral episiotomy should be done before delivery of the head to avoid recurrence of the laceration*

COMPLICATIONS OF PERINEAL TEARS:

1. Postpartum hemorrhage, due to bleeding from lacerations.
2. Infection may occur in the laceration site (puerperal sepsis).
3. Patulous vaginal introitus with unsatisfactory sexual function.
4. Incomplete tears; may predispose to genital prolapse (due to loss of pelvic floor support).
5. If unrepaired, it will lead to old complete perineal tear (see below).
6. Residual rectovaginal fistula.
7. Dyspareunia from a tender scar in the vagina.

13 OBSTETRIC COMPLICATIONS

The **3rd stage** of labor is the stage of **delivery** of the **placenta**. Its duration is variable ranging from **10-30** minutes. Adequate **myometrial contractility** is the key to successful placental separation and expulsion. The most common **obstetric complications** encountered in the **3rd stage** of labor include postpartum hemorrhage **(PPH), retained placenta** and rarely **uterine inversion.** These complications may be associated with hemorrhagic shock, DIC, and hypofibrinogenemia.

POST PARTUM HEMORRHAGE

PPH is defined as excessive blood loss after delivery that affects the patients' general condition. It is estimated to be around **> 500 ml** following vaginal delivery and **> 1000** ml following CS.

The **general condition** of the patient is affected by severable factors including:

- The **amount** of blood loss.
- The **Rate** of blood loss and the **duration** of bleeding.
- The presence of a **prior medical condition** as; maternal anemia, cardiac, renal, and hepatic disease, hypertension, preeclampsia, and bleeding disorders.

CLASSIFICATION OF PPH:

A) **PRIMARY PPH:** occurring **immediately** after delivery, and throughout the first **24**hours postpartum. It is mostly related to uterine atony, genital tract trauma, retained placenta, and coagulation disorders.

B) **SECONDARY PPH:** This is excessive bleeding occurring any time during the puerperium starting > 24 hours after delivery. It is less commonly encountered and is mostly associated with retained placental fragments, and puerperal sepsis.

CAUSES OF 1ry PPH:

1. Uterine **Atony (most common).**
2. Genital tract **Trauma (second most common).**
3. **Retained placenta or its fragments (mainly causes 2ry PPH).**
4. **Bleeding disorders (very rare).**

N.B. many cases PPH are mixed, that may start traumatic and after more bleeding, the uterus fails to contract and it becomes atonic.

RISK FACTORS OF 1ry PPH:

A) **RISK FACTORS FOR UTERINE ATONY:**

Uterine **atony** is the **most common** cause for **1ry PPH** encountered in **> 75 %** of cases. Risk Factors for ATONIC PPH:

- History of *previous atonic PPH* (possibly genetic or familial).
- *Grand multipara* with 5 or more deliveries (laxity in uterine musculature and larger fetuses).
- *Overdistension* of the uterus as in twins & MFP, polyhydramnios and fetal macrosomia.
- *Myometrial Exhaustion* in prolonged & precipitate labor, and improper use of oxytocin.
- *Placenta Previa (PL PRV)* because of the relatively deficient muscular content of the lower uterine segment (LUS) compared to that of the upper segment.
- *Placental abruption* (concealed & mixed types) where retroplacental hematoma formation dissects in-between myometrial layers interfering with contractility.
- *Large uterine leiomyomas* (interfere with efficient myometrial contractility).
- *Chorioamnionitis* and intrauterine infection (weak uterine muscle).
- *Severe maternal anemia* and *vitamin D deficiency* (poor contractility).

B) RISK FACTORS FOR GENITAL TRACT TRAUMA:

Trauma during labor and delivery is the **2nd most common** cause for **1ry PPH,** mostly in association with traumatic cervical, vaginal, and perineal **lacerations**, or **rupture** of the uterus; Risk Factors for TARUMATIC PPH:

- *Previous uterine scar;* CS, myomectomy, or hysterotomy.
- *Traumatic delivery of an oversized baby* as in macrosomia and shoulder dystocia.
- *Insufficient episiotomy;* especially with a large fetal head and rigid perineum.
- *Delivery before full cervical dilatation* as in precipitate labor and preterm breech.
- *Difficult instrumental delivery* as mid-forceps procedures.

C) RETAINED PLACENTA or PLACENTAL FRAGMENTS:

This is the **3rd common** cause for **1ry** PPH and the **leading** cause for **2ry PPH,** where retention of the whole placenta or undelivered placental fragments impairs myometrial contractility.

Risk factors for RETAINED PLACENTA:

- *Uterine atony;* leading to incomplete separation and failure of expulsion.
- *Adherent placenta;* as focal or partial placenta accreta.
- *Rupture uterus;* where placenta may be expelled in peritoneal cavity.
- *Contraction ring;* preventing placental expulsion (very rare).

D) COAGULATION DISORDERS: these are **rare** but may be **fatal** causes of PPH (see obstetric shock):

- *Idiopathic thrombocytopenic purpura (ITP)* associated with abnormal platelets.
- *Von Willebrand disease* (factor VIII deficiency).
- *Hypofibrinogenemia and DIC* due to consumption coagulopathy associated with placental abruption or prolonged IUFD.
- *Amniotic fluid embolism:* (see later).

CLINCAL PICTURE OF 1RY PPH:

A)ATONIC PPH:

- Bimanual examination reveals: uterus is **soft** and **hypotonic** with a rising fundal level due accumulating hematometra and dark retained blood and blood clots coming out on pressing on the uterus.

B) TRAUMATIC PPH:

Bimanual examination reveals: uterus is **firm contracted** and **fresh** bright **red** bleeding coming out of the traumatic part, So vaginal exploration is essential to determine the traumatic area (perineal, vaginal, cervical, uterine) and so deal with it.

C) RETAINED PLACENTAL OR PLACENTAL FRAGMENTS:

- *Sub-involuted uterus* with a high fundal level and inefficient contractility.
- *Ultrasonography* will show retained placental tissue in utero confirming diagnosis.

DIAGNOSIS OF A CASE OF 1ry PPH:

- **General** examination and **vital signs** will diagnose hypovolemic **shock** if present and its degree.
- Abdominal **palpation** of the uterine fundus will diagnose uterine **atony** and subinvolution.
- **Speculum** vaginal examination will reveal genital tract **lacerations** by naked eye inspection.
- **Digital vaginal** examination allows for exploration of the uterine cavity for **retained** fragments, rupture of the uterus, or dehiscence of a CS scar.
- **Ultrasonography** for diagnosis of retained placental cotyledons. It can also detect intraperitoneal hemorrhage and broad ligament hematoma in uterine rupture.
- **Laboratory tests** are important for evaluation of the effect of the amount of bleeding on the blood picture, the severity of hypovolemic shock, and the diagnosis of **coagulopathies**.

MANAGEMENT OF A CASE OF 1ry PPH:

A) RESUSCITATION AND CORRECTION OF HYPOVOLEMIC SHOCK if present:

- Insertion of a wide bore IV cannula and urinary catheter.
- Continuous monitoring of Vital signs (pulse, temp, BP, & RR) and urine

output.

- IV infusion with saline/dextrose or lactated Ringer solution.
- Blood or blood components is mostly needed.

B) ECBOLICS: **immediately started to induce myometrial contractility, eg of ecbolics:**

- ***Oxytocin IV infusion***.
- ***Ergometrine IM***.
- ***Prostaglandin (PG) E1;*** Misoprostol transrectal.
- ***Prostaglandin (PG) F2 a***; Carboprost IM.

Combined ecbolics are usually used and maintained all through the following methods of bleeding control.

C) UTERINE MASSAGE: starts together with ecbolic medications as follows:

- ***Bimanual uterine compression*** by placing the closed fist of one hand vaginally and the other hand abdominally squeezing the uterus between both hands. This is maintained for at least 30minutes (while the resuscitative measures are going on), so long the patient is responding, to stimulate regain of uterine contractility.

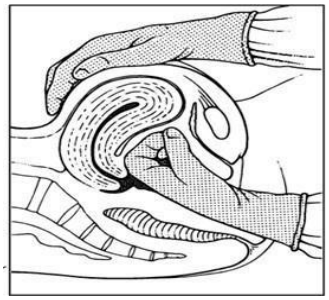


Fig 13-1: Bimanual uterine compression

D) EXPLORATION OF THE GENITAL TARCT; **(*start by this after the resuscitative measures in cases with well contracted uterus and suspecting obstetric trauma***)

- ***Vaginal exploration: Inspection*** of the ***perineum, vagina***, and ***cervix*** for extensively bleeding lacerations.
- ***Exploration of the uterine cavity;*** digitally by 2 fingers introduced through the dilated cervix to detect retained placental fragments, ***rupture*** of the uterus or ***dehiscence*** of a prior CS scar.

E) INTERVENTIONAL METHODS: If the above methods fail to control atony, one of the following interventions may be of help in the following sequence without delay;

- ***Intrauterine Balloon Catheter*** (Bakry's balloon) inflated at an adequate pressure to induce *internal tamponade* to stop bleeding from atonic uterus.
- ***Bilateral uterine arteries ligation*** to minimize uterine blood flow.

- **B-lynch sutures** embracing the uterus by absorbable sutures may control bleeding by acting as an *external tamponade*.
- ***Bilateral internal iliac artery ligation*** to decrease pelvic blood flow minimizing the amount and rate of blood loss.
- ***Subtotal hysterectomy*** with ovarian preservation may sometimes be a last step performed as a lifesaving procedure especially in multiparous women.

NB: in cases of traumatic Rupture of the uterus is managed primarily by surgical repair and reconstruction. If failed Supracervical hysterectomy is performed as an emergency procedure.



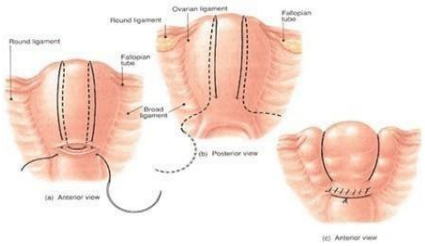


Fig. 13-2: Bakry's balloon (internal tamponade) *Fig. 13-3: B-Lynch sutures (external tamponade)*

COMPLICATIONS OF 1ry PPH:

1. Complications of severe hypovolemic shock (acute renal failure, DIC,)
2. Complications of massive blood transfusion (DIC, DVT, pulmonary embolism,)
3. Complications of acute blood loss (Sheehan's syndrome)
4. Maternal morbidity (hysterectomy), and mortality in case of delayed management, ***Obstetric hemorrhage is the leading cause of maternal mortality in Egypt.***

PREVENTION OF PRIMARY PPH:

a. Proper antenatal care and selection of cases with high risk factors (as prevention and treatment of anemia and all risk factors….)
b. Proper management of 1st stage of labor (as proper use of oxytocin, prevent precipitate and prolonged labor…..)
c. Proper management of 2ndstage of labor (as proper use of oxytocin, make episiotomy whenever needed and do Ritgen maneuver to prevent perineal lacerations, proper use of instrumental delivery)
d. Proper management of 3rdstage of labor (do active management and explore the vagina for any laceration to be repaired immediately, proper inspection of placenta after delivery to detect a missing cotyledon….)
e. Proper management of 4th stage of labor and observation to guard against uterine atony.

SECONDARY POST PARTUM HEMORRHAGE

Secondary **PPH** describes **excessive** vaginal bleeding commencing **> 24 hours** after delivery, occurring or recuring any time throughout the **puerperium** (6 weeks after delivery).

ETIOLOGY:

1. **Retained placental tissue** and fragments is by far the most common cause (> 50%).
2. **2ry Hemorrhage** from cervical, vaginal, or perineal lacerations.
3. Sloughing of an infected **submucous fibroid polyp (SMFP).**
4. Undiagnosed **chronic inversion** of the uterus.
5. **Choriocarcinoma** (rare).

DIAGNOSIS:

- **History:** severe bleeding within the period of puerperium.
- **Inspection:** may reveal infected lacerations, SMF Polyp, or chronic inversion of the uterus.
- **Palpation:** the uterus may be sub-involuted due to retained blood and tissue fragments.
- **Ultrasound:** may diagnose retained placental tissue, placental polyp, or SMF Polyp.

MANAGEMENT treat the cause;

a. **Antibiotics** in cases of infection.
b. **Ecbolics** may help to extrude and expel some retained placental tissue.
c. **Surgical evacuation** under anesthesia using ring or ovum forceps for retained missed placental fragments or cotyledons.
d. **Secondary sutures** in cases with gaping vaginal or perineal lacerations
e. **Chronic inversion of the uterus** is treated surgically.

RETAINED PLACENTA

Retained placenta describes a condition in which the placenta **fails to be delivered** within **30 minutes** after delivery of the fetus. It may complicate **0.5%** of vaginal deliveries.

ETIOLOGY:

A) **FAILURE OF EXPULSION** after placental separation due to**:**

- Uterine atony.
- Contraction ring.
- Rupture of the uterus.

B) **FAILURE OF EXPULSION** due to a **MORBIDLY ADHERENT PLACENTA**

(MAP) OR PLACENTA ACCRETA SEPCTRUM (PAS): This is a condition in which the decidua basalis is totally or partially defective leading to direct adherence and/or invasion of the chorionic villi into the myometrium. Absence of a plane of cleavage which interfere with placental separation. There are 3 types of PAS:

a. **Placenta Accreta:** chorionic villi become **adherent** to the myometrium (most common type).
b. **Placenta Increta:** chorionic villi **invade** into the myometrium (less common type).
c. **Placenta Percreta:** chorionic villi **penetrate** through the myometrium reaching serosa and may invade the bladder wall (rarest and most serious type, we can't separate the placenta from the myometrium and even the bladder may be injured by the placental invasion and hematuria is present).

CLINICAL PICTURE:

Failure of placental expulsion within **30 minutes** from delivery of the fetus, inspite of attempts at gentle cord **traction** and suprapubic pressure.

The clinical picture will vary according to underlying etiology whether associated with uterine **atony** or **rupture** of the uterus (see atonic and traumatic PPH). **Rarely** placental retention is due to;

- **Contraction Ring;** vaginal examination with 2 fingers of one hand inside the uterus through the dilated cervix will reveal the condition. Severe PPH is usually present.
- Morbidly **Adherent Placenta (MAP)**; bleeding is usually **not severe** because no placental separation has occurred. Digital exploration through the open cervix will reveal absent plane of cleavage between the placenta & myometrium. Attempts at manual removal of an adherent placenta may provoke severe PPH.

MANAGEMENT OF RETAINED PLACENTA:

1. **Manual separation of the placenta** under anesthesia by introducing one hand inside the uterus creating a plane of cleavage separating the placenta and delivering it.
2. **Dealing with retained placental fragments***;*
   - *Digital evacuation* by 2 fingers exploring the uterine cavity through the cervix, **or**
   - *Surgical evacuation* under anesthesia using a ring or ovum forceps.
3. If due to **Contraction ring**; use deep general Halothane anesthesia to relax the uterus followed by steps of manual separation and cord traction.



Fig. 13-4: Manual removal of the placenta 1, 2, &3

4.**Placenta PAS (Increta, and Percreta):** are best managed by an immediate total abdominal hysterectomy, to avoid intractable bleeding during useless attempts of separation. This major surgical procedure usually requires an experienced team supported by vascular surgery and urology team. Most cases of PAS occur in association with a placenta previa implanted on a scar of prior repeated CS. Diagnosis should be made antepartum, and cases managed electively by **CS hysterectomy (classical CS + TAH)** with adequate blood transfusion facilities.

NB: Conservative management: in cases of PAS in primigravidae, and leaving the placenta in situ to undergo autolysis, under cover of broad-spectrumantibiotics and methotrexate, is associated with very high incidence of septicemia and septic shock and is not recommended due to its high maternal mortality rates.

INVERSION OF THE UTERUS

Uterine inversion is a condition in which the uterus is turned inside out. It may occur in the 3rd stage of labor. It is very **rarely encountered** occurring in <1:20.000 deliveries.

RISK FACTORS FOR UTERINE INVERSION:

- *Vigorous Pressure* on the soft uterine fundus from *above.*
- *Excessive Traction* on the cord of a partially adherent placenta from *below.*
- *Fundal submucous leiomyoma,* of moderate size, facilitates dipping of the uterine fundus in to the uterine cavity by pressure of its weight and uterine contractions.

DIC IN OBSTETRICS

(Disseminated Intravascular Coagulation (DIC) or Consumptive Coagulopathy).

PHYSIOLOGICAL BACKGROUND:

In late pregnancy, there is increased concentration of coagulation factors I (fibrinogen: level is 350-650mg/dL), VII, VIII, IX and X. Other plasma factors and platelet count do not change remarkably.

DEFINITION OF DIC:

A wide spread hematological condition characterized by *accelerated fibrin*

formation and lyses.

- There is consumption of coagulation factors in variable quantities.
- Signs of hypofibrinogenemia develop when fibrinogen level goes below 100 mg/dL.

ETIOLOGY: Pregnancy related ***Common causes***

1. Massive blood loss with inadequate replacement and massive rapid blood transfusion too.
2. Concealed accidental hemorrhage.
3. Severe pre-eclampsia/eclampsia or HELLP syndrome
4. Retained IUFD for more than 3-4 weeks.
5. Amniotic fluid embolism.
6. Sepsis.
7. Acute fatty liver of pregnancy.
8. Adult RDS, acute hemolytic transfusion reactions, autoimmune disease, hematological malignancies and solid tumors.

CLINICAL PICTURE

1. Postpartum or antepartum hemorrhage.
2. Persistent bleeding from venipuncture sites or after catheter insertion.
3. Spontaneous bleeding from gums and nose.
4. Generalized oozing in surgical fields.
5. Purpuric areas at pressure sites (thrombocytopenia and incoagulable blood).

INVESTIGATIONS: aiming to detect fibrinolysis.

1. FDPs and fibrin D-dimer (normally absent).
2. Prolonged PT and PTT (PTT may be normal).
3. Low fibrinogen level.

TREATMENT

1. It must be directed to the underlying cause to reverse defibrination. Two wide-bore IV cannulas are inserted.
2. Transfuse fresh frozen plasma (FFP).
3. If fibrinogen level is < 100 mg/dL, transfuse cryoprecipitate or give Fibrinogen 4-10 g IV.
4. Platelets should be transfused if the count is < 20 000/cmm or if clinically significant bleeding occurs with a platelet count between 20 000 and 50 000/cmm.

   N.B Heparin infusion to stop coagulation is contraindicated as it increases the bleeding.

AMNIOTIC FLUID EMBOLISM

The classic triad of abrupt **hemodynamic** and **respiratory** compromise along with **DIC** points its diagnosis.

Incidence is rare, 1 in 50.000, yet the case-fatality may reach 80%.

Predisposing conditions: rapid labor, meconium-stained liquor, labor induction or augmentation, placental abruption and tears into uterine and other large pelvic veins that permit an exchange of fluids between the maternal and fetal compartment

Diagnostic Criteria for Amnionic Fluid Embolism:

- Abrupt onset in late stage of labor or early postpartum of cardiorespiratory arrest, or both hypotension and respiratory compromise.
- Documentation of overt DIC. Coagulopathy must be detected prior to loss of sufficient blood to cause dilutional or shock-related consumptive coagulopathy.
- No fever ≥ 38°C.

Management:

- Immediate high-quality **CPR** (cardiopulmonary resuscitation).
- **Intubation** is usually necessary.
- **Inotropic** agents and **vasopressors** to improve cardiac output, and treat systemic hypotension.
- Excess fluid administration is discouraged due to risks of worsening dilatation.
- Coagulopathy needs early aggressive treatment, including **fresh frozen plasma**.

PREMATURE RUPTURE OF MEMBRANE (PROM)

PROM describes spontaneous rupture of the fetal membranes before the onset of labor. It may complicate up to **10%** of all pregnancies.

If PROM occurs **before 36** weeks' gestation (**preterm PROM**), it may be associated with significant maternal and/or fetal morbidity resulting in one of the most challenging obstetrical complications.

ETIOLOGY

1. **Idiopathic:** in many cases PROM will occur spontaneously without an identifiable etiology.
2. **Infection:** ascending infection from the lower genital tract (LGT) is the **most common** risk factor causing PROM mostly due to decreasing the tensile strength of fetal membranes. Common organisms include Group B streptococci (GBS) and Trichomonas Vaginalis (TV).
3. **Cervical Incompetence:** bulging membranes through the dilated cervix are exposed to trauma and infection that may cause erosion and PROM.
4. **Polyhydramnios and Multifetal pregnancy (MFP):** are commonly associated with PROM mostly due to chronically increased intra-amniotic

pressure.

5. **Local membrane defects** and **cigarette smoking** have been proposed as risk factors.

HAZARDS AND COMPLICATIONS

A) PRETERM LABOR (PTL) (most common sequelae):

In most cases labor will start within **few hours** after ROM. If this occurs < 36 weeks gestation, it will lead to PTL with all serious fetal hazards related to prematurity (see preterm labor).

B) INFECTION (CHORIOAMNIONITIS) (most serious complication):

Prolonged ROM may be complicated by ascending **LGT** infection. The most serious infection is caused by **GBS** that may lead to *Maternal hazards:* chorioamnionitis, septicemia, septic shock and puerperal sepsis. And untreated infection may be transmitted to the fetus.

C) PLACENTAL ABRUPTION:

ROM in cases of polyhydramnios and MFP may result in a sudden decrease in intra-amniotic pressure leading to placental separation and accidental hemorrhage (see placental abruption).

D) OLIGOHYDRAMNIOS:

- *Hazards of prolonged ROM with oligohydramnios:* if pregnancy continues for few weeks or months after preterm PROM the fetus will be at risk for all complications of oligohydramnios (see amniotic fluid disorders).

DIAGNOSIS:

- ***Symptoms***: Sudden gush of a significant amount of colorless fluid through the vagina
- ***Signs:*** *PV examination:* should be minimized and done under complete aseptic conditions to decrease the risk of infection:
  - *Sterile speculum examination:* clear, colorless, odorless fluid flowing out of the vagina.
  - ***Nitrazine paper test or Fern test:*** clean watery amniotic fluid collected in the posterior vaginal fornix, will change the color of *nitrazine paper* to blue or purple due to alkaline pH of amniotic fluid, and may show *arborization* when put on glass slide due to NaCl content of amniotic fluid.
  - ***AmniSure test:*** highly accurate test that measures placental alpha macroglobulin, it should be done in case of the pooling of fluid is not observed.
  - ***Ultrasound:*** may reveal decreased amount of liquor amnioii around the fetus.

DIAGNOSIS OF CHORIOAMNIONITIS:

A) **Clinical Diagnosis:** maternal **fever** and uterine **tenderness** in confirmed case of **PROM** in absence of urinary tract infection (UTI) or respiratory tract infection (RTI).

B) Laboratory Diagnosis: increased **TLC**, abnormal **DLC** with increase staff/ segmented WBC ratio (**left shift),** with **toxic granules**, and increased **CRP.**

MANAGEMENT OF PROM:

Management of PROM will vary according to gestational age, presence of uterine contractions or evidence of infection.

1. In presence of ***uterine contractions:*** labor is allowed to proceed, **regardless** gestational age, as tocolytic agents are *contraindicated* for fear of masking evidence of maternal infection.
2. Evidence of starting ***chorioamnionitis:*** IV therapeutic broad spectrum antibiotics started, and pregnancy is promptly **terminated** (induction of labor or CS).
3. If **NO** labor pains and **NO** chorioamnionitis, management will be according to gestational age;
   - ***Term PROM > 36 weeks:*** pregnancy should be promptly terminated.
   - ***Preterm PROM:*** Conservative management including hospitalization, IM Betamethasone for lung maturity, and Ampicillin / Erythromycin prophylactic therapy and follow up with lab tests for early detection of any subclinical signs of chorioamnionitis.

PRETERM LABOR

Preterm labor **(PTL)** is defined as onset of **true** labor pains any time **after** fetal **viability** but **before 36**weeks gestation. PTL is the commonest underlying factor for **prematurity** which is the most common cause for neonatal **morbidity** and **mortality**. PTL may complicate **5-10%** of pregnancies:

- ***Early PTL:*** is PTL occurring **< 34** weeks gestation. It is the more serious type of PTL.
- ***Late PTL:*** is PTL occurring **> 34** but **< 36** weeks. It carries less fetal risks than early PTL.

ETIOLOGY AND RISK FACTORS:

In most cases there is **NO** identifiable etiology making it mostly unpredictable, however **RISK FACTORS** for PTL include;

A) OBSTETRIC RISK FACTORS:

- Previous PTL.
- Cervical incompetence and recurrent pregnancy loss (RPL).
- Short cervical canal < 2.5 cm by TVS > 24 weeks gestation.
- Uterine anatomic abnormalities; as septate, bicornuate, unicornuate uterus and large leiomyomas
- Uterine over-distension as: twins & MFP, polyhydramnios, and fetal macrosomia.

B) MEDICAL FACTORS AND CONDITIONS:

- Genital tract and urinary tract infections (GTI & UTI) during pregnancy.

- Uncontrolled medical condition as: hypertension, preeclampsia, and gestational DM necessitating early pregnancy termination.
- Severe anemia and malnutrition.

c) ENVIRONMENTAL AND BEHAVIORAL FACTORS

- Successive short interval deliveries < 18 months.
- Excessive smoking and substance abuse.
- Excessive physical activity, competitive sports, heavy exercise, and psychosocial stress.

COMPLICATIONS OF PREMATURITY:

1. **Respiratory Distress Syndrome (RDS):** decreased lung surfactant (mainly Lecithin), will predispose to lung collapse with loss of alveolar wall elasticity in terminal air spaces.
2. **Birth Trauma:** fragile cerebral vessels may easily lead to intracranial hemorrhage (ICH) and intraventricular hemorrhage (IVH) 2ry to head compression and hypoxia.
3. **Neonatal Sepsis:** due to immature and relatively incompetent immune system.
4. **Necrotizing Enterocolitis (NEC):** bacterial infection in small intestine leading to necrosis.
5. **Retinopathy of prematurity.**
6. **Cerebral Palsy (CP):** more liable to occur 2ry to birth trauma, hypoxia, ICH, IVH, infection, and perinatal infections.

PREDICTION OF PTL (Difficult):

However, in most cases PTL will be **unpredictable**. Irregular repeated uterine **cramps** and excessive vaginal **discharge** may raise suspicions; however, the following may help prediction;

a. **TVS** showing short **cervical canal** length < 2.5 cm after 24 weeks gestation.

b. **Fetal Fibronectin (FFN)** in cervical discharge: FFN is a protein matrix produced by fetal cells that acts as a glue binding trophoblasts to maternal decidua. It leaks into the vagina if PTL is likely to occur. It is measured by repeated vaginal swabs from vaginal pool at posterior fornix. The test It is valid between 24-32 weeks;

- *Negative test (absence of FFN):* < 1% chance for a PTL within 2 weeks.
- *Positive test (presence of FFN):* 50% chance for PTL within 2 weeks.

MANAGEMENT OF PTL:

A) CONFIRM DIAGNOSIS OF PTL: True labor pains with progressive cervical dilatation and effacement is confirmed by;

- ***Cardiotocography (CTG):*** recording at least 4 contractions in 20 minutes.
- ***PV examination:*** showing cervical effacement 50% or more and dilatation

at least 2 cm.

B) HOSPITALIZATION: bed rest, sedation, and adequate hydration with IV Dextrose 5%, together with monitoring the frequency and intensity of uterine contractions with **CTG.**

C) MANAGEMENT OF A CORRECTABLE CAUSE as **GTI** and **UTI** via proper antibiotic therapy according to culture and sensitivity obtained by vaginal swabs and urine samples.

D) ULTRASOUND EXAMINATION: for confirming gestational age, fetal weight, fetal presentation, placental localization, amniotic fluid volume, and evaluation of fetal wellbeing by biophysical profile and umbilical artery color Doppler velocimetry studies. Ultrasound can also diagnose many risk factors as presence of polyhydramnios, fetal macrosomia, MFP, and leiomyomas.

E) INITIATING CORTICOSTEROID THERAPY: for minimizing hazards of prematurity through:

- ***Enhancing lung maturity:*** IM corticosteroid act via stimulation of fetal type II pneumocyte surfactant production < 34 weeks gestation thus minimizing the risk of neonatal RDS.
- ***Minimizing incidence of NE, ICH, and IVH.***
- ***Dose Regimens:*** either
  - *Betamethasone (24 mg/24 hours):* 2 IM doses **12** mg each **24** hours apart, **OR**
  - *Dexamethasone (24 mg/48 hours):* 4 IM doses **6.0** mg each **12** hours apart

F) TOCOLYTIC THERAPY IN PTL:

Aim of Tocolysis: drugs are used in attempt to stop uterine contractions and prolong pregnancy for at least **48-72** hours to get **benefit** from administration of IM corticosteroids, and transfer the patient to a tertiary center d with advanced neonatal ICU facilities.

Contraindications to Tocolysis:

- *Obstetric conditions as:* placental abruption, ROM, and chorioamnionitis.
- *Fetal conditions as:* fetal distress, and lethal anomalies.
- *Maternal conditions as:* severe preeclampsia (PE), eclampsia, and advanced cervical dilatation and effacement.

TOCOLYTIC AGENTS

1. B2- adrenergic agonist (Ritodrine):

- It is a selective B2-agonist causing uterine relaxation; however, it may have some B1 effect on the heart causing maternal and fetal tachycardia.
- Given by IV infusion, and may continue orally.

2. Progesterone:

- Natural progesterone relaxes the uterus
- Can be given IM, suppository (vaginal or rectal), or orally.

3. Anti-PGs (indomethacin):

- Not to be given after 32 weeks gestation as it causes premature closure of the ductus arteriosus

4. Calcium Channel Blockers (Nifedipine):

- Given orally, may cause headache, flushing, hypotension and tachycardia (vasodilatation effect).

5. Anti-Oxytocin (Atosiban):

- Act by blocking intra-myometrial oxytocin receptors, given as IV infusion with titrated doses that inhibit uterine contractions.

6. $MgSO_4$ Magnesium Sulphate:

- *Mode of action:* $MgSO_4$ is a competitive inhibitor of calcium.
- *Route of administration:* Loading dose IV followed by IV infusion.
- *$MgSO_4$ Toxicity* includes; muscle weakness and respiratory depression.
- *Dosage monitoring:* monitor deep tendon reflexes (kneejerk), urine output, and respiratory rate
- *Antidote for toxicity:* IV calcium gluconate
- $MgSO_4$ reduces the risk and severity of cerebral palsy **(CP)** in surviving extremely preterm neonates < 28weeks (neuro-Protection).

MANAGEMENT DURING LABOR AND DELIVERY

- *Vertex presentations:* vaginal delivery is allowed with generous episiotomy (the smaller the fetus, the wider the episiotomy to avoid sudden compression and decompression of the fetal head and so ICH).
- *Breech presentations:* CS gives better neonatal outcome with fetal weight < 1500 gm.

POST TERM PREGNANCY
(Prolonged Pregnancy)

Prolonged or **Post term** pregnancy **(PTP)** is a pregnancy that continues beyond **42** weeks gestation, estimated by the **LMP.** Nearly 10% of pregnancies may extend post term.

Accurate pregnancy **dating** supported by 1st trimester **US** are the golden standard for diagnosis of a PTP, however both will be lacking in the majority of cases making sure diagnosis quite difficult.

ETIOLOGY:

Most cases with PTP are **idiopathic** with **NO** identifiable **etiology**, while in at least 20-30% of cases pregnant women proper history taking reveals an **unreliable** LMP.

- **N.B.:** lack of a proposed **labor initiating factor** produced by fetal adrenals has been implicated as an etiological factor. This has been related to cases with **anencephaly** due to absent fetal hypothalamic-pituitary regulation.

RISKS OF POSTMATURITY

A) **Large for Gestational Age fetus (LGA):** In almost 80% of PTP normal placental function is maintained leading to advanced fetal growth and a LGA fetus with a higher risk for a difficult vaginal delivery, and both maternal and fetal traumatic birth injuries.

B) **Placental Insufficiency:** In almost 20% of cases placental function declines by ageing, scarring, and fibrin deposition, leading to fetal **dysmaturity** syndrome and intrauterine growth restriction **(IUGR),** with hazards of fetal acidosis and hypoxia during labor compromising fetal outcome.

C) **Higher CS delivery rates:** LGA, and IUGR are associated with higher CS rates to improve fetal and maternal outcome by avoiding birth trauma, oligohydramnios, meconium aspiration, and neonatal hypoxia.

- **N.B.:** PTP has a **2-3 fold** risk of increased **perinatal mortality**.

MANAGEMENT OF PTP

1. ***Confirming diagnosis:*** via proper dating of pregnancy through meticulous history taking, reaching a reliable LMP or an accurate 1st trimesteric US examination.
2. ***US:*** using reliable parameters as BPD, AC, HC, FL, together with long bones ossification centers and placental grading.
3. ***Evaluation of fetal wellbeing:*** to facilitate decisions for pregnancy prolongation or termination
4. **Induction of labor** is weighed against **CS** according to favorability of the cervix, estimated fetal weight and presence or absence of evidence of placental insufficiency.

- **NB: Bishop scoring** is the method to evaluate favorability of the cervix for induction (see induction of labor).

THE AMNIOTIC FLUID

SOURCES OF AMNIOTIC FLUID

- **Fetal (Main source):** urine production, amniotic membrane, and transudation from fetal circulation.
- **Maternal:** placental chorionic membrane and transudation from maternal circulation.

AMNIOTIC FLUID VOLUME AND COMPOSITION

The amniotic fluid (AFL) is continuously produced and absorbed from both fetal and maternal surfaces with a **more pronounced fetal contribution**.

Amniotic fluid volume (AFV) shows gradual increase with gestational age reaching a maximum volume of 1-1.5 L at 34-36 weeks. This is followed by a slight decline the last 4 weeks, then a sharper and more rapid decline occurs after 40 weeks gestation.

The AFL is composed of **99% water**, and is characterized by having a **clear** aspect, **pale** color, and slightly **alkaline PH**. It contains **carbohydrates, lipids, proteins, enzymes**, and **mediators** as cytokines. It also contains **fetal excretory products** as; uric acid, urea, creatinine, and sodium chloride, together with vernix caseosa and lanugo hair.

FUNCTIONS OF THE AMNIOTIC FLUID

1. Protection of the fetus from external trauma.
2. Maintains constant surrounding temperature suitable for fetal growth.
3. Provides a suitable medium for fetal movement for muscular development and neuromuscular coordination.
4. Provides a source for fetal excretion and a source of fetal nutrients.
5. Allows for fetal lung distension and development.
6. Prevents occurrence of intra-amniotic bands, adhesions, and fetal compression deformities.

EVALUATION OF AMNIOTIC FLUID VOLUME

Clinical evaluation of AFL is only subjective especially in presence of maternal obesity, twins and MFP, PI PRV, and uterine leiomyomas. US is the gold standard in evaluation of AFV and diagnosis of volume disorders as oligohydramnios and polyhydramnios

A. AFL pockets measurement longitudinally in cm length.

B. AFL Index (AFI) in which the vertical diameter of largest pocket is measured in each of the four abdominal quadrants. The sum of 4 quadrants gives an estimate of AFI.

- **Normal liquor (**AFI: 10-15), **oligohydramnios** (AFI < 5.0), and **polyhydramnios** (AFI > 25).

OLIGOHYDRAMNIOS

Oligohydramnios is defined as **diminished AFV**. By US it is diagnosed when measurement of the deepest **AFL pocket is < 2 cm**, or an **AFI < 5.**

INCIDENCE:

- Oligohydramnios affects almost 3-4 % of all pregnancies.
- Incidence is significantly increased in association with IUGR and placental insufficiency.

ETIOLOGY:

1. Undiagnosed **ROM.**
2. **Placental insufficiency:** as in IUGR, Preeclampsia (PE), and post term pregnancy (PTP).
3. **Fetal causes:** Fetal congenital anomalies (FCA) as renal agenesis and renal hypoplasia
4. **Drugs;** Indomethacin Is known to decrease AFV due to decreased renal

perfusion

SYMPTOMS AND SIGNS:

1. History of leaking amniotic fluid (ROM).
2. Inappropriate growth of maternal abdominal girth especially in the 3rd trimester
3. Inspection: reveals smaller than expected abdominal contour.
4. Abdominal examination: lower than expected fundal level for gestational age.

RISKS AND COMPLICATIONS:

A) During pregnancy:

- Complications related to IUGR and placental insufficiency (see later)
- Pulmonary hypoplasia, compression limb deformities, and amniotic adhesions.

B) During labor

- Cord compression with fetal hypoxia and fetal distress.
- Meconium passage and fetal meconium aspiration.

MANAGEMENT

Termination of pregnancy: in cases that reached near term especially if associated with placental insufficiency or lethal anomalies or if the fetus is compromised as per the assessment of fetal well-being measures.

POLYHYDRAMNIOS

Polyhydramnios is defined as **excessive AFV, measured** By US when the deepest vertical **AFL pocket is > 8 cm**, or an **AFI > 25.**

INCIDENCE: it may be associated with **0.5-1.5 %** of pregnancies.

ETIOLOGY

1. **Idiopathic:** most cases are possibly due to an imbalance between AFL production and absorption of an unknown etiology.
2. **Fetal causes:**
   - Twins and MFP: due to large surfaces producing AFL (placenta and membranes).
   - Placental tumors: as chorioangioma.
   - FCA: as anencephaly, open NTDs, esophageal atresia, and duodenal atresia.
   - Hydrops fetalis: due to large placenta & excessive transudation from fetal circulation.
3. **Maternal causes:**
   - Gestational diabetes mellitus (GDM): most common cause.

SYMPTOMS AND SIGNS

1. Progressive rapid enlargement of abdominal girth especially in 3rd trimester.
2. Respiratory embarrassment and pressure symptoms only in severe cases.

3. Inspection: shows larger abdominal contour than expected for gestational age.
4. Fundal level: is higher than expected for gestational age.
5. Fetal parts are not easily palpated due to excessive surrounding AFL.

RISKS AND COMPLICATIONS

1. PTL due to uterine overdistension, with fetal prematurity complications.
2. Dysfunctional labor due to malpresentations, macrosomia, PROM, and associated MFP.
3. Placental abruption due to suddenly decreased intraamniotic pressure after PROM.
4. Risks of associated fetal anomalies or maternal GDM.

MANAGEMENT

1. **Mild and Moderate Polyhydramnios:**
   Mild and moderate cases are best managed conservatively till spontaneous labor pains start. During this period the patient will need, reassurance, follow up, and exclusion of MFP, FCA, and GDM by US.
2. **Severe Polyhydramnios:**
   - ***Term pregnancy > 36 weeks:*** termination of pregnancy by either induction of labor or CS according to fetal presentation, pelvic adequacy, and bishop score
   - ***Preterm pregnancy < 36 weeks:*** Repeated amniocentesis removing 1-1.5 liters every time to prolong pregnancy until lung maturity is established



Fig. 13-5: Polyhydramnios

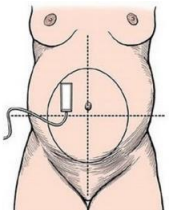


Fig. 13-6: 4 quadrant assessment of AFI by US

14 THE FETUS AND NEONATE

FETAL GROWTH DISORDERS

Fetal birth weight a function of both gestational age and rate of fetal growth.

Estimation of gestational age is based on a *reliable LMP* dating and/or an *early US* scan, while fetal growth rates are affected by both *placental function* and *genetic* or constitutional factors.

- ***Appropriate for gestational age (AGA):*** fetuses between 10 th- 90 th percentile for their GA.
- ***Small for gestational age (SGA)*** fetuses are <10thpercentile weight for gestational age
- ***Large for gestational age (LGA)*** fetuses are >90th percentile weight for gestational age.
- ***Low birth weight (LBW):*** Infants with birth weight <2500 gm regardless their gestational age.

ESTIMATION OF FETAL WEIGHT

A) ***Clinical Estimation:*** via clinical assessment of fundal level and abdominal girth is an unreliable subjective method with low accuracy.

B) ***US Estimation:*** via combined measurements of BPD, HC, AC & FL, is an objective method giving rather accurate fetal weight estimates with less than 10% error range.

I- FETAL GROWTH RESTRICTION (FGR)

FGR refers to growth patterns in which the fetus fails to reach its full growth potential.

These fetuses are generally **SGA** with weights <10th percentile according to fetal growth curves. SGA fetuses may be either constitutionally small babies or growth restricted ones.

FGR may be **symmetrical** or **asymmetrical.** These two types are likely to be the consequence of different causes, time of onset, and duration of events.

A) SYMMETRICAL (TYPE I) 20%:

Symmetric FGR usually results from fetal **injury** very **early** in development that is **intrinsic** to the fetus as with intrauterine infections and chromosomal abnormalities.

B) ASYMMETRICAL (TYPE II) 80%:

Asymmetric FGR usually results from fetal **injury later** in pregnancy,

commonly due to maternal disease that is **extrinsic** to the fetus. The fetal size is affected via some degree of **placental insufficiency** secondary to reduced placental blood flow.

ETIOLOGY OF FGR:

A) SYMMETRICAL FGR:

1. **Intrauterine Infections:** *(Congenital infections) as* Rubella, Cytomegalovirus, Hepatitis, Varicella, Influenza, Listeriosis, T.B. and syphilis. In Syphilis, the placenta almost always increases in size and weight due to edema and perivascular inflammation.
2. **Chromosomal Abnormalities:** as trisomies 13, 18 and 21.
3. **Congenital Malformation:** as cardiovascular malformation, renal hypoplasia, and skeletal anomalies (osteogenesis imperfecta) and many other inherited disorders.

B) ASYMMETRICAL FGR:

1. **Maternal vascular disease:** Chronic hypertension, Preeclampsia, and Diabetic vasculopathy.
2. **Antiphospholipid syndrome:** antibodies cause vascular lesions in uterine and placental bed leading to diminished blood supply and impaired placental infusion.
3. **Chronic renal disease:** usually associated with type II IUGR.
4. **Chronic Hypoxia:** heavy smoking, poor nutrition, drug abuse and cyanotic heart disease

- **N.B:** In cases with chronic placental insufficiency, the fetus reacts to restricted placental perfusion by redirecting its blood flow to be maintained to the brain and decreased to most visceral organs, a condition known as **"*Brain sparing phenomenon*".** This will result in:
  a. Preserved HC while decreased AC measurements by US ***(increased HC/AC ratio)*.**
  b. Reduced renal perfusion and decreased urine output, leading to ***oligohydramnios*.**

DIAGNOSIS OF FGR

1. **Clinical Examination:** a fundal level lower than expected (2 finger breadths lower than GA) and longitudinal symphysial fundal height (2 cm less than GA) can only **suspect** SGA fetus, given a reliable GA dating by an accurate LMP and/or early 1st trimesteric US scan.
2. **Ultrasound Assessment:**
   a) ***Fetal Biometry:*** measurements of the BPD, HC, AC & FL in the 2nd & 3rd trimester.
      - Type I FGR: Symmetrically SGA fetus with reduction in all biometry measurements (BPD, HC, AC & FL) giving an estimated fetal weight < 10th percentile for GA.

- Type II FGR: biometry measurements reveal discrepancy between HC and AC measurements with increased HC/AC ration due to brain sparing phenomenon.

b) ***Oligohydramnios*** with decreased AFI measurements is common with type II FGR.

c) ***Accelerated placental aging*** (with calcifications).

d) ***Abnormal umbilical and cerebral artery Doppler flow indices*** (decreased umbilical and preserved or increased cerebral vascular flow in asymmetrical FGR).

COMPLICATIONS OF FGR

1. **During pregnancy and labor:**
   - Intrapartum Hypoxia leading to Intrauterine fetal death (IUFD).
   - FHR abnormalities during labor, asphyxia and intrapartum fetal death (IPFD).
2. **During Neonatal period:**
   - *Immediate:* meconium aspiration, polycythemia and pulmonary hemorrhage.
   - *Late:* Cerebral dysfunction (mild cerebral palsy).

MANAGEMENT OF FGR

A) Prevention:

- Identification of risk factors during the prenatal and antenatal period and correcting those factors, if possible, for example improving nutrition, avoiding short interval pregnancies and smoking cessation.
- Starting low dose aspirin 75 mg before 16 weeks of age for high risk group, for example history of preeclampsia, history of SGA, those with chronic kidney disease or hypertension.

B) Fetal Surveillance

- Fetal growth monitoring every 2weeks using BPD, HC, AC and FL and amniotic fluid index.
- Doppler studies:
  - Biweekly Doppler studies as long as the UA doppler is normal.
  - Weekly Doppler if there is increased UA Doppler resistance.

C) Pregnancy Termination

Before 36 weeks:

1. Abnormal Doppler studies i.e reversed end diastolic flow, abnormal MCA.
2. Associated Severe preeclampsia: Administration of betamethasone for fetal lung maturation.

Administration of magnesium sulphate for neuroprotection for fetuses <28 weeks.

FETAL MACROSOMIA

The term fetal **macrosomia** is used to describe a newborn who is significantly larger than average (more than 90th centile) with a birth weight

exceeding 4500 gm regardless its gestational age.

RISK FACTORS

1. Maternal **diabetes** is the most common and most important risk factor for fetal macrosomia. (GDM) and uncontrolled pre GDM are of more risk than well controlled diabetic patients.
2. **Other risk factors** include; post-term pregnancy (PTP), maternal obesity (pre-pregnancy weight
   > 90 kg), rapid weights gain during pregnancy, grand multipara, and a previous LGA baby.

COMPLICATIONS

B) *Fetal Complications:*

- IUFD (in diabetic macrosomia or serious congenital malformations).
- Birth trauma (shoulder dystocia and brachial plexus palsy).

C) *Maternal:*

- Higher incidence of C.S. deliveries (due to prolonged or obstructed labor).
- Traumatic injuries of birth canal: (rupture uterus, cervical and vaginal lacerations).
- Postpartum hemorrhage (PPH), whether atonic (due to uterine overdistension), or traumatic (due to lacerations).

DIAGNOSIS

1. **Clinical** estimates of fetal size and weight based on fundal height measurements are often unreliable and are markedly affected by clinical experience and maternal obesity.
2. **Ultrasound** estimates of fetal weight; however, US estimates of fetal weight are more difficult in obese women.

PREVENTION

1. Proper evaluation of risk factors during pre-conception visits weight loss in obese women.
2. Early diagnosis of GDM and meticulous control of established cases.
3. Control of weight gain during pregnancy at regular antenatal care visits.

MANAGEMENT

2. **During pregnancy:** regular antenatal care, control excessive weight gain, repeated investigations to discover GDM, serial US estimates for growth curves, and anomaly scans.
3. Elective cesarean section **(C.S.)**: If USEFW ≥ 4500 especially in diabetic pregnancies.
4. **Vaginal delivery:** If attempted, anesthesia staff and neonatal resuscitation team must be available. Assisted instrumental vaginal delivery MUST BE AVOIDED.

NB: **Induction of labor >36 weeks:** to minimize the need for a CS delivery, is very controversial, and should be reserved only to highly selected cases with uncontrolled medical conditions.

INTRAUTERINE FETAL DEATH (IUFD)

IUFD describes fetal death in utero **> 24 weeks** gestation but before onset of labor. It may complicate **<1%** of all pregnancies. Prolonged IUFD carries maternal hazards that should be dealt with to avoid severe maternal morbidity.

ETIOLOGY:

1. **Idiopathic:** In at least **50%** of cases no etiology can be identified.
2. **IUGR,** Preeclampsia **(PE),** and Gestational Diabetes Mellitus **(GDM).**
3. Placental **abruption, umbilical cord** accidents (as true knots, and tight loops around the neck).
4. Erythroblastosis Fetalis **(EBF)**, and some serious fetal congenital anomalies **(FCA).**
5. Fetal and maternal **Infections.**
6. Antiphospholipid syndrome antibodies **(APS),** and **Thrombophilias.**

DIAGNOSIS:

- IUFD is **suspected** when the mother notices cessation of previously felt fetal movements.
- Examination reveals **absent FHS** by Sonicaid, with a fundal level less than gestational age.
- **US** is the **gold standard** for confirming diagnosis of IUFD;
  - *Recent IUFD:* absent FHR is the only finding.
  - *Old IUFD;* SGA fetus with collapsed skull bones and oligohydramnios.
  - *US may suggest* a probable cause for IUFD (FCA, EBF, IUGR, Fetal macrosomia, etc…).

MATERNAL RISKS ASSOCIATED WITH IUFD:

a. ***Risk of the etiological factors;*** as PE, GDM, Placental abruption, infections, etc…
b. ***Risk of coagulopathy:*** prolonged IUFD is associated with thromboplastin release from the placenta and dead tissues, leading to DIC and consumption coagulopathy.

MANAGEMENT

A. **Watchful Expectancy:**
Labor pains usually start spontaneously 2-3 weeks from IUFD. However, the risk of coagulopathy increases if labor is delayed beyond 48hrs. Women choosing expectant management should be offered testing for DIC twice weekly.

B. **Active Management:** termination of pregnancy by **induction** of labor or **CS** is indicated in the following conditions;
1. Difficulties in controlling maternal disease: as GDM, HTN, and PE.
2. Clinical or laboratory evidence of an impending coagulopathy or DIC
3. Women who cannot psychologically tolerate watchful expectancy period.

- **N.B.:** Non sensitized RH –negative women should receive an IM injection of 300 ug Anti-RhD immunoglobulin till 72 hours after delivery.

FETAL ASPHYXIA

INTRODUCTION:

Intrauterine fetal asphyxia (IUFA) is a serious situation, which might put the fetus at risk for loss of life or permanent CNS disability. Therefore, properly trained personnel in well-equipped delivery rooms are essential to the anticipation, recognition and prompt management of these serious conditions.

DEFINITION:

IUFA is a state of inadequate oxygenation and inadequate elimination of CO_2, which if allowed to be continued, will result in metabolic acidemia (umbilical arterial blood pH < 7.2).

CAUSES OF IUFA

I. Maternal causes: (conditions leading to imperfect oxygenation of maternal blood) i n severe anemia, Hemorrhage & shock, Respiratory failure, and heart failure.

- Eclamptic convulsions, advanced pulmonary T.B., pneumonia, and pulmonary edema.

II. Placental causes:

- Placental compression: interfering with its circulation as in tonically contracted uterus, prolonged labor after rupture of the membranes.
- Placental separation as in accidental hemorrhage.
- Placental insufficiency: (extensive degeneration, multiple infarcts & abnormally small placenta).

III. Causes in the umbilical cord: Obstruction of the circulation, which may be due to:

1. Tight coils around the neck.
2. True knots of the cord.
3. Prolapsed cord.
4. Compression of the vessels by hematoma of the cord or by blades of the forceps.
5. Rupture of vasa previa.

IV. Prolonged fetal head compression:

Prolonged compression may be due to:

1. Contracted pelvis (C/P disproportion).
2. Rigid perineum.
3. Intracranial hemorrhage.
4. Forceps application for a long time.
5. Depressed fracture.

CLINICAL FEATURES OF IUFA (FETAL DISTRESS FD):

1. **Abnormal Fetal heart rate (FHR)**
   - *Bradycardia* (FHR < 100 bpm); most dangerous sign
   - *Tachycardia* (F.H.R. >160 bpm) - *Irregular FHR* patterns.

2. **If continuous electronic FHR monitoring** and **CTG** would suggest fetal distress:
   a. *Late deceleration* (see assessment of fetal well-being).
   b. *Persistent Variable deceleration (even after changing the maternal position*.
   c. Loss of beat-to-beat variation in FHR.
3. **Passage of meconium in cephalic presentations**, due to relaxation of the anal sphincter due to hypoxia/anoxia and increased intestinal peristalsis.
4. **Fetal acidosis:** Detected by taking blood samples from the scalp of the fetus during labor; pH below 7.25 indicates fetal asphyxia (N. 7.25- 7.35).

MANAGEMENT OF INTRAUTERINE ASPHYXIA (FETAL DISTRESS):

1. *Elimination of the cause of asphyxia* if possible:
   a. Oxytocin infusion, if started, should be discontinued to decrease uterine activity and improve placental perfusion.
   b. The patient should be turned onto her side: this may relieve either umbilical cord compression, or alleviate poor return of blood to maternal heart caused by occlusion of the maternal aorta or IVC by the gravid uterus.
   c. Intravenous hydration or vasopressor treatment in cases of severe hypotension.
2. *Oxygen (100%)* should be administered to the mother by facemask.
3. *Atropine* given to the mother may be beneficial in some cases of fetal bradycardia.
4. ***If FD is relieved:*** careful observation by EFHR monitoring till delivery is accomplished
5. ***If fetal distress is not relieved***, immediate delivery is indicated;
   - If the cervix is not fully dilated: C.S. is immediately performed regardless presentation.
   - If the cervix is fully dilated; delivered by:
     - Forceps: in cephalic presentation, engaged head, CPD excluded.
     - Breech extraction: in breech presentation.

FETAL BIRTH INJURIES

Fetal birth injuries represent an important commonly avoidable cause of neonatal morbidity and mortality. They vary from minor skin abrasions to severe intracranial hemorrhage. Prevention of serious birth trauma depends mainly upon the art and the experience of the obstetrician, and is considered a reflection of the improvements in antenatal and perinatal care.

TYPES OF FETAL BIRTH INJURIES:

A) <u>Bone injuries</u>:

1.Skull fracture:

- *Etiology:*
  - Difficult forceps delivery.
  - Delivery through a contracted pelvis.
- *Types:*
  - Fracture vault: linear or depressed (associated with intra-cranial hemorrhage so needs surgical intervention).
  - Fracture base: usually associated with intra-cranial hemorrhage.
  - Fracture mandible.

2.Others bone injuries:

- Spine injuries, Fracture humerus, Fracture femur.
- Fracture clavicle.
- Dislocation of hip.
- Dislocation of shoulders.

B) <u>Soft tissue birth injuries</u>:

1. Intra-cranial hemorrhage (over compression of cranial bones).
2. Cephalhematoma (instrumental trauma especially ventouse).
3. Nerve Injuries (undue traction on neck, shoulders, and arms).
4. Visceral and Muscle Injuries.
5. Eye Injury.
6. Injury of hymen, or anal sphincter especially in breech presentation, during examination.
7. Skin and scalp Injuries by the scalpel on opening the uterus in a C.S.

1. CEPHALHEMATOMA (Sub-periosteal hematoma)

		Caput succedaneum	**Cephalhematoma**
Cause		*Obstructed labor*	*Forceps or ventouse*
		Ventouse	*Difficult delivery through contracted pelvis*
Appearance		*At birth*	*Few hours after birth*
Character	*Edges*	*ILL defined*	*Well defined*
	Skin	*May be ecchymotic*	*Normal*
	Sutures	*Overlap sutures and cover more than 1 bone*	*No overlap & limited to one bone*
Complications		*No*	*Calcification, infection and hyperbilirubinemia*
Treatment		*No treatment (disappears after 1 -2 days)*	*Expectant treatment (disappears within few weeks)*

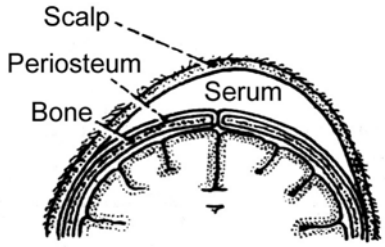


Fig. 14-2: Caput succedaneum

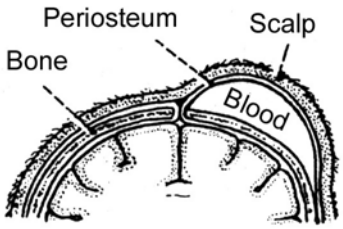


Fig. 14-3: Cephalhaematoma

2. NERVE INJURIES

I. Facial Nerve Palsy:

Cause: Compression of the nerve by blade of forceps results in edema and hematoma around the nerve.

Clinical picture: *unilateral and temporary*

- Absent nasolabial fold.
- Angle of the mouth is deviated to the healthy side.
- Absent blinking on the affected side.

Treatment:

Conservative management. May need corticosteroids.

II. Brachial Plexus Injury:

Cause: Forcible lateral flexion of the head during delivery causes damage of the roots of brachial plexus (edema and hematoma around the nerves) especially in cases of shoulder dystocia.

Clinical picture:

A) Upper injury (Erb's palsy):

- Injury to C5 and C6.
- Characters: Policeman tip position:
  1. The affected limb is adducted to the body and internally rotated.
  2. Elbow is extended.
  3. Wrist is flexed.

B) Lower injury (Klumpke's palsy):

- Injury to C8 and T1.
- Characters:
  1. Wrist drop.
  2. Absent grasp reflex.
  3. Paralysis of small muscles of hands (atrophy).


Erb' palsy

Fig. 14-4:

Treatment:

- *Upper injury (Erb's palsy):* Fixation of the affected limb in pharaoh's position.
- *Lower injury (Klumpke's palsy):*
  - Physiotherapy.
  - Rarely, it may need plastic correction.

15 THE PUERPERIUM NORMAL AND ABNORMAL

NORMAL PUERPERIUM

Puerperium is the period of time that **starts** in the early **post-partum** period after delivery and extends for around **4-6 weeks later**.

During the puerperium involution of the genital organs occur hand to hand with several maternal physiologic, metabolic, endocrine, and mental changes.

INVOLUTION OF GENITAL ORGANS

A. THE UTERUS:

- ***Size:*** gradual shrinkage in size together with myometrial autolysis. The fundus reaches the umbilicus immediately after delivery, and is felt at the symphysis pubis (SP) 2 weeks later then finally regains its pre-pregnancy size **4 weeks'** post-partum.
- ***Endometrium:*** the decidua gradually casts off and is replaced by regenerating endometrium in 4-6 weeks.
- ***Cervix:*** gradually reforms and closes within 3-4 weeks. Stretched cervical ligaments involute and become thickened preventing uterine prolapse.
- ***Lochia:*** lochia is the postpartum genital **discharge** that consists of blood and necrotic decidual cast.
  - Lochia rubra: profuse and red postpartum discharge in the first 2 weeks.
  - lochia serosa: small amount yellowish discharge in 3rd week.
  - Lochia alba: scanty whitish discharge that continues till end of puerperium.

B. THE VAGINA:

- Vaginal rugae reappear in 3 weeks.
- Pelvic floor muscles regain their usual tone and strength in 4-6 weeks.

C. THE VULVA:

- The gapping that occurs during delivery remains for 2-3 weeks then gradually disappears but not completely to pre-pregnant size and shape.
- Episiotomy scar is completely healed in 4 weeks if had been performed.

D. THE BREASTS:

- Prolactin (PRL) and nipple suckling initiate milk secretion from the breast glands previously primed by both estrogen (E) and progesterone (P).
- For the first 2 days the breast secretes colostrum (the first milk produced by the breasts) which is rich in protein and immunoglobulin, while low in carbohydrates and fat.
- Proper milk secretion commences at the 3rd day and may be associated with breast engorgement in the form of pain and congestion.

- Lactation suppresses ovulation and fertility and may be associated with amenorrhea in up to 60% of lactating women. However, this does not preclude the need for the use of an appropriate method for contraception.

PHYSIOLOGIC CHANGES

A. **Blood Pressure:** cardiac output diminishes rapidly after labor with normalization of BP.
B. **Temperature:** transient rise < 38°C may occur in 1st 24 hours in prolonged labor, and with breast engorgement in the 3rd day with starting lactation.
C. **Pulse:** Transient rise occurs during labor with rapid decline after delivery.
D. **Urinary bladder:** transient postpartum diuresis occurs in 1st few days after delivery sometimes associated with mild stress urinary incontinence (SUI). Retention may occur after use of regional anesthesia till bladder regains its normal tone.
E. **Bowel:** constipation is common due to lax abdominal muscles and weak pelvic floor. Anal incontinence may occasionally occur due to overstretching of pelvic floor muscles and ligaments but is usually spontaneously corrected by the end of puerperium.

MENTAL CHANGES

1. Emotional liability (postpartum blues) commonly occur and is managed by family support.
2. Sometimes mild degree of depression occurs and may need psychiatric consultation and medications.
3. Severe postpartum psychosis rarely occurs but might bear serious consequences to the mother and her newborn and may need hospitalization.

POSTPARTUM CARE (PPC) and MANAGEMENT DURING PUERPERIUM:

Postpartum acre **(PPC)** for women during the puerperium is an extension of both antenatal care **(ANC)** provided during pregnancy and intrapartum care **(IPC)** during labor and delivery.

1. **General health care:** as physical and mental rest and reassurance.
2. **Observation:** for amount of lochia, breast feeding, and occurrence of bleeding or fever.
3. **Balanced diet:** providing at least 2500 kcal/day, rich in fibers to minimize constipation.
4. **Breast feeding:** encourage suckling by the neonate every 2-3 hours together with additional amounts of water, fluids, and juices.
5. **Breast care:** using cleansing and moisturizing agents to avoid infection and nipple cracking.
6. **Abdominal wall exercise:** active exercise starts few days after delivery and throughout puerperium. Abdominal binder is not necessary except in multiparous women with pendulous abdomen.
7. **Perineal care:** keeping a dry pad on the region of perineum is advisable

with regular cleansing with antiseptic solutions (Betadine) minimized infection in genital wounds.

8. **Care for episiotomy:** regular cleansing with antiseptic solutions every 4-6 hours together with use of oral antibiotics twice daily for the first 4-6 days.
9. **Bladder care:** encourage the mother for frequent emptying of the bladder till it regains its normal tone and capacity.
10. **Bowel care:** avoid constipation by taking extra fluids and high fiber diet. Some laxatives are allowed without affecting the new born especially locally acting glycerin suppositories.
11. **Immunization;** Anti-D immunoglobulin is given to RH negative mothers that deliver an RH positive baby preferably within the first 24-48 hours after delivery (RhoGAM 300 µg/IM).
12. **Discharge from Hospital:** usually within 24 hours from delivery in absence of maternal complications. CS may need 48 hours before discharge until bowel movements return.
13. **Postpartum Exercise:** The aim of puerperal exercise is to restore venous circulation, help genital involution, prevention of genital prolapse, and restoration of normal posture.
14. **Post partum visits:** the patient is best seen 3-4 weeks after delivery in order to reassure general health, follow up genital involution, promote breast feeding, and plan for contraception.
15. **Contraception counselling:** once marital sexual life is resumed, there is a need for a method of contraception to allow proper spacing. Barrier contraception should be started immediately till a choice for hormonal contraception or IUD is taken according to each cases individually.

PUERPERAL PYREXIA

DEFINITION:

- A **temperature of 38°C** or higher which lasts for **>24 hours** and or **recur within 24 hours** during the **period of puerperium** excluding of the first 24hours.

ETIOLOGY OF PUERPERAL PYREXIA:

1. Puerperal sepsis (**most serious**): ***it still remains the leading cause of maternal mortality in certain developing countries, and the third leading cause of maternal mortality in Egypt.***
2. Breast engorgement or mastitis (**most common**).
3. Urinary tract infection (UTI).
4. Respiratory system infection.
5. Thrombophlebitis and DVT.
6. General causes of fever as typhoid or malaria.

Any case of puerperal pyrexia is considered puerperal sepsis until proved otherwise.

PUERPERAL SEPSIS

DEFINITION

It is a type of infection of the genital tract that during the postpartum period.

ETIOLOGY

A. Predisposing factors:

- ***General factors*** (leading to decreased immunity):
  - Anemia.
  - Antepartum or postpartum hemorrhage.
  - Diabetes mellitus.
  - Septic focus.
- ***Local factors*** "in the genital tract":
  - Lack of antiseptic measures.
  - Premature rupture of membranes.
  - Prolonged labor with excessive vaginal examination.
    Retained parts of placenta or membranes. Intrauterine manipulations (e.g. manual separation of placenta).
  - Instrumental delivery with possible genital tract lacerations.
  - Cerclage sutures.

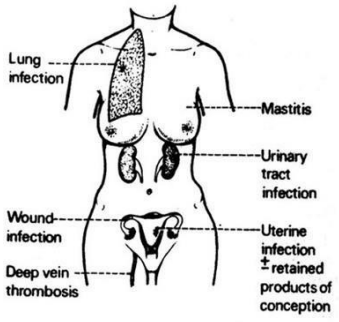


Fig 15-1: Common causes of puerperal pyrexia

B. Sources of infection:

- ***Exogenous:*** from attendants by droplet infection or from unsterilized instruments.
- ***Endogenous:*** organisms are already present in the genital tract before delivery, e.g. vaginitis or cervicitis.
- ***Autogenous:*** organisms reach the genital tract from remote site via blood stream e.g.
  tonsillitis or respiratory tract infection, carious teeth……etc.

C. Causative organisms:

- ***Anaerobic streptococci:*** It is nonpathogenic and in the presence of dead tissue, it becomes pathogenic and produces mild infection.
- ***Group A hemolytic streptococci:*** Causes severe infection.
- ***Others:*** As Staphylococci, E. coli, non-hemolytic Streptococci, Pseudomonas and Clostridia Welchii.

CLINICAL PICTURE:

1. **History and symptoms:**
   - Pre-existing infection in the genital tract or in the body in general.
   - Diseases that lower the patient's resistance as diabetes or anemia.
   - Mode of delivery, premature rupture of membranes and the place of delivery.
   - Fever, lower abdominal pain, foul smelling vaginal discharge.
2. **Signs (Examination):**
   a. ***General:*** aiming to
      1. Assess the extent of the disease.
      2. Exclude other causes of puerperal pyrexia.
      3. Discover possible source of infection outside the genital tract. So, look for:
         - Vital signs (pulse, temperature, blood pressure and respiratory rate).
         - Lower limbs for deep venous thrombosis.
         - Cyanosis and purpura (in septicemia).
         - Breast, chest and throat (for infection).

   b. ***Abdominal:***
      - Tenderness and rigidity.
      - Uterine size: the uterus may be sub-involuted.
      - Renal angles for pyelonephritis.

   c. ***Local:***
      - Lochia: amount and nature (excessive, foul odor).
      - Infected lacerations.
      - Uterine size and tenderness.
      - Pelvic swellings.

DIAGNOSIS

1. Urine culture (Catheter specimen).
2. Complete blood picture.
3. Blood culture at the peak of the fever.
4. Doppler ultrasound for venous thrombosis (for all pelvic vessels and IVC).
5. Chest X-ray for chest infection and blood film for malaria.
6. Swabs from vaginal fornix and cervix for aerobic, anaerobic cultures and antibiotic sensitivity (in resistant cases).

PREVENTION OF PUERPERAL SEPSIS:

1. **During pregnancy and Antenatal care:**
   - Treat any infection especially in the genital tract.
   - Improve the general condition of the patient by treating anemia and diabetes.
   - Treat any septic focus.
2. **During labor:**
   - Proper aseptic and antiseptic measures.
   - Avoidance of unnecessary repeated vaginal examination.

- Antibiotics in case of premature rupture of membranes.
- Avoid prolonged labor.
- Proper use of instrumental delivery if indicated.
- Proper management of genital lacerations.
- Avoidance of retained placental or membrane fragments in the uterus.

3. **During puerperium:**
   - Maintenance of aseptic & antiseptic measures.
   - Prevent infected personnel or visitors from visiting the puerperal mothers.
   - Sitting in bed to promote lochia drainage.
   - Isolation of suspected cases.
   - Respiratory and pelvic floor muscle exercises.

TREATMENT:

a. ***General measures:***
   - Isolation.
   - Light nutritive diet, vitamins, iron and plenty of fluids.
   - Frequent measurements of vital signs.
   - Analgesics and antipyretics to assure good sleep.

b. ***Antibiotics:***

A combination of broad-spectrum antibiotics is given:
- Cephalosporins for gram +ve organisms.
- Gentamycin for gram -ve organisms.
- Clindamycin or Metronidazole for anaerobic organisms.

c. ***Promotion of Drainage:***
- Fowler's position (Semi-sitting with flexed knees).
- Ergometrin, which increases uterine ability to expel retained placental parts or membranes.
- Removal of retained parts in the uterine cavity, if present (under anesthesia & antibiotic cover).
- Removal of sutures in infected wounds to allow drainage.
- If pelvic abscess: surgical drainage if antibiotics fail to cure.

d. ***Treatment of Complications:***

1. **Septic thrombophlebitis and DVT:**
   - Antibiotics.
   - Anticoagulants. Limb immobilization.

Generalized peritonitis:
- No oral feeding.
- IV fluids.
- Fix a Ryle tube for gastrointestinal drainage.
- Massive IV antibiotics.

2. **Septic shock:** See shock in obstetrics.

16 OBSTETRIC PROCEDURES AND OPERATIONS

ANALGESIA AND ANESTHESIA IN LABOR AND DELIVERY

CAUSES OF PAIN DURING LABOR:

1. Stretch of the cervix during dilatation.
2. Distension and ischemia of the muscle wall of the uterus with buildup of lactate.
3. Stretch of the vagina and perineum is the second stage.

IDEAL PAIN RELIEF should provide *good analgesia*, be *safe* for the mother and fetus, and in the same time *reversible* if necessary.

PHARMACOLOGICAL METHODS FOR PAIN RELIEF:

I. SYSTEMIC ANALGESIA AND SEDATION:

(a) Narcotic Drugs:

1. Meperidine hydrochloride (pethidine):

- Given 50mg IM or intermittent I.V. doses.
- May cause neonatal respiratory depression if given 2-4 hours from delivery.
- Antidote is Naloxone

2. Morphine Sulphate:

- Rarely used as it is associated with higher incidence of respiratory depression although 10 times more potent than Meperidine.

(b) Non Narcotic Drugs:

Usually given in combination with narcotic drugs to produce good analgesia and sedation.

1. Benzodiazepines: Diazepam (Valium) and Midazolam (Dormicum). Both may cause neonatal hypotonicity, hypoactivity and impaired temperature regulation if given in repeated doses.

(c) Inhalation Drugs:

1. **Nitrous oxide gas:** can be inhaled periodically with uterine contractions in a mixture with oxygen (50:50).

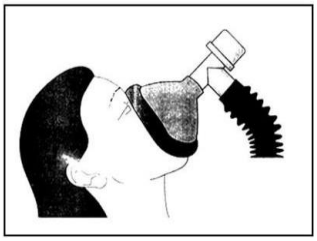


Fig 16-1: Inhalation analgesia

II. REGIONAL ANALGESIA:

This is pain relief (nerve block) without loss of consciousness.

(a) ***Epidural Analgesia:*** It provides the most effective pain relief

- A plastic catheter is introduced into the epidural space (outside the dura) through a needle with a curved tip.
- Intermittent doses of a local anesthetic are injected through the catheter.

Disadvantages:

- Motor block with weakness of the lower limbs. Decreasing the dose of the local anesthetic and adding an opiate (e.g. fentanyl) spare the motor function (mobile epidural).
- Loss of the urge for straining increasing the duration of the first and second stages of labor and increase the need for instrumental delivery.
- Accidental puncture of the dura.

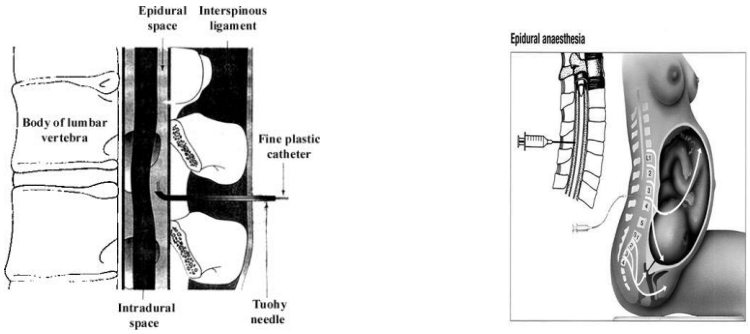


Fig 16-2: Regional Epidural Analgesia

(b) ***Local infiltration anesthesia:***

A local anesthetic (e.g. lidocaine) is injected at the site of episiotomy or perineal lacerations.

(c) ***Pudendal nerve block:***

A local anesthetic is injected through the vagina, behind the ischial spine where the pudendal nerve crosses.

(d) ***Paracervical block:***

A local anesthetic is injected on each side of the cervix through the lateral fornices.

It may cause fetal bradycardia, so rarely used nowadays.

B) NON-PHARMACOLOGICAL METHODS FOR PAIN RELIEF:

- **Prepared childbirth:**
  Educating the woman in antenatal classes about the birth process, how to relax, exercising and specific breathing patterns to be used during labor pains.

- **Trans Cutaneous Electrical Nerve Stimulation (TENS):**
  Application of a mild electric current to the skin can result in the reduction of pain.
  Stimulation of cutaneous nerve fibers closes a gate in the spinal cord preventing pain impulses transmission.

- **Acupuncture:** Acts through the gate theory.

- **Warm water baths** (underwater birth): heat is analgesic and the buoyancy of water is relaxing.

ANESTHESIA FOR CESAREAN SECTION

A. **Spinal (subarachnoid) anesthesia:**
- The local anesthetic is injected into the subarachnoid space
- It is considered nowadays as the **first choice for C.S.** anesthesia for its strong and rapid action although in some cases it may cause hypotension and headache, so has to preceded by I.V. infusion to increase blood volume (preload) and avoid hypotension.

B. **General anesthesia:**

General anesthesia using I.V. anesthetics (as Thiopental sodium or Ketalar) in addition to inhalation gas drugs (as Nitrous oxide and Oxygen). However, it crosses the placenta and may affect the fetus.

NB: General anesthesia may be also used for: repair of episiotomy, instrumental (forceps) delivery, and manual removal of the placenta.

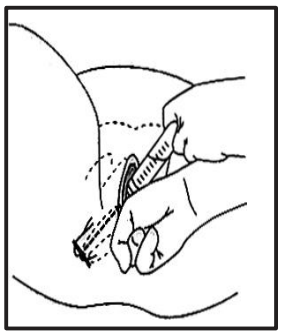


Fig 16-3: Pudendal nerve block

C. **Epidural analgesia:**

It is a safe method both for the mother and fetus both during vaginal delivery and C.S. It has a slow onset of action and has to preceded by I.V. infusion to increase blood volume (preload) to avoid commonly occurring hypotensive episodes. It may be used as analgesia and anesthesia.

INDUCTION OF LABOR

DEFINITION: Termination of a pregnancy after fetal viability.

INDICATIONS: Whenever continuing pregnancy will harm the mother or the fetus: e.g.;
- Hypertensive disorders (PE / eclampsia).
- Prolonged pregnancy (Post date).
- Compromised fetus (e.g. IUGR).
- Maternal diabetes mellitus.
- Rhesus iso-immunization.

PREREQUISITES FOR SUCCESSFUL INDUCTION OF LABOR:

- Ripening of the cervix before the onset of labor makes it favorable for easy dilatation and effacement. An Unfavorable cervix increases the chances of a difficult prolonged labor, failed induction, fetal hypoxia and operative delivery.
- The Bishop score takes account of the cervical length, cervical dilatation, cervical consistency, cervical position and station of the fetal head. Scores of < 5 indicates and unfavorable cervix that need PGs use for ripening. While a Bishop score more than 5, indicates a favorable cervix, so the use of oxytocin is sufficient to induce labor (no need for cervical ripening).

The Bishop score

Parameter/score	0	1	2	3
Position	Posterior	Central	Anterior	
Consistency	Firm	Intermediate	Soft	
Effacement	Formed 0-<30%	30- <50%	50-80%	> 80%
Dilatation	Closed	1-2 cm	3-4 cm	> 5 cm
Head station	-3	-2	-1, 0	+1, +2

METHODS OF INDUCTION OF LABOR:

I. Cervical ripening by Prostaglandins (PGLs):

Prostaglandins are used in inducing cervical ripening (softening of cervical collagen fibers) which effectively results in success of labor induction. PG F2alpha, PGE2 vaginal pessaries or *PGL E1 (Misoprostol 25 µg);* vaginal tablet. Repeated doses every 6 hours for maximum 24 hours may be needed.

N.B.; *Prostaglandins should never be used to augment labor. It might be used to induce labor or to accomplish cervical ripening in term pregnancies.*

II. Amniotomy (Artificial rupture of membranes):

- This is done to induce or augment labor.
- The cervix must be at least 2.0 cm dilated to allow performing amniotomy.
- Amniotomy release an endogenous PGLs.
- It is done by passing a special hook along the fingers or by direct vision using a speculum to rupture the membranes overlying the presenting part, but care must be taken not to damage cervical or fetal tissues.
- If Labor is not established after amniotomy, IV infusion of oxytocin may be needed to stimulate the uterus to contract.

- ***Complications***:
  a. Placental separation (abruption): may be caused by sudden reduction in intrauterine pressure especially in polyhydramnios.

b. Vaginal bleeding: soft tissue and fetal injury should be excluded.
c. Prolapse of the cord: Especially in ill-fitting presenting pare. FHR monitor is a must after amniotomy.
d. Infection: Intra-amniotic infection may occur if the interval from amniotomy to delivery is excessive. Careful antiseptic techniques and prophylactic antibiotics are recommended whenever amniotomy is performed.

III.I.V. Oxytocin drip:

- Synthetic oxytocin by IV drip is commonly used to stimulate uterine contractions.
- Doses should be carefully titrated and monitored as the drug is powerful and uterine response to stimulation is generally unpredictable.
- A solution of 2 units of syntocinon in 500 ml of lactated Ringer is used beginning at a dose of 1 mIU/min. (10-15 drops/min). This is increased by 1 mIU/min. every 15 minutes until satisfactory contractions are established.
- ***Complications of Oxytocin infusion***:
  a. Failure of induction: ripening of the cervix with PGLs should be used first.
  b. Hyperactive uterine action: Resulting in dysfunctional labor.
  c. Abnormal FHR patterns: due to fetal hypoxia by over-stimulation of the uterus.
  d. Rupture of the uterus: It is rare in a primigravida, but is more to be expected in grand multiparas, and patients with previous uterine scar (as in cases of VBAC).

EPISIOTOMY

DEFINITION:

It is an incision of the perineum during labor.

TYPES:

- ***Median (Midline) Episiotomy:*** Midline incision of the perineum directly backwards towards the anus.
- ***Mediolateral Episiotomy:*** Directed posterolateral 45˚towards the buttocks away from the anus.

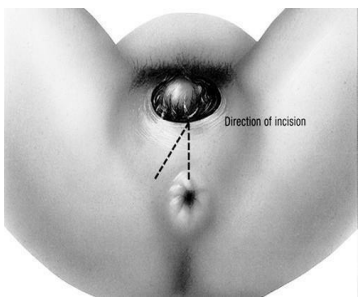


Fig 16-4: Episiotomy incisions

INDICATIONS:

A) Maternal:

- Short rigid perineum as in elderly primigravida.
- Perineal scarring such as in previous perineal or pelvic floor repair.
- Mild degree of pelvic outlet contractions.

B) Fetal:

- Face to pubis delivery.
- Vaginal breech delivery.
- Shoulder dystocia.
- Oversized fetus.
- Forceps or ventouse delivery.
- Preterm labor (the smaller the fetus, the wider the episiotomy).

TIMING OF EPISIOTOMY:

- Best time to perform episiotomy is when the head is visible during a contraction just before crowning.
- Before traction by forceps or vacuum extractor.
- Too early episiotomy causes bleeding
- Too late episiotomy is useless.

TECHNIQUES OF EPISIOTOMY:

- ***Median type:*** A vertical incision is made in the perineal body. The incision starts at the fourchette till approximately half the length of the perineal body. The incision should extend into the vagina approximately 2 to 3 cm.
- ***Mediolateral type:*** Incision starts at the fourchette at a 45-degree angle to the midline of the perineum. The incision should extend into the vagina approximately 2 to 3 cm. When the incision reaches the ischiorectal fossa (with its fat) it is termed ***generous episiotomy***.

Median Episiotomy	Mediolateral Episiotomy
Extension to the anal sphincter and rectum is more common	*Extension to the anal sphincter and rectum is less common*
Easier to repair	*More difficult to repair*
Rare faulty healing	*Faulty healing is more common*
Less pain in the perineum	*More pain in the perineum*
Dyspareunia is rare	*Dyspareunia is more common*
Less blood loss	*More blood loss*



Fig 16-5: Crowning of head and episiotomy

Episiotomy Repair:

1. Vagina is closed continuously starting 1 cm beyond the visible apex till the hymeneal ring.
2. Interrupted sutures are used for the perineal muscles.
3. Closure of the skin by interrupted simple or inverted sutures or subcuticular continuous stitches.

Advantages of episiotomy:

1. Clean cut incision which is easy to repair compared to irregular vaginal lacerations.
2. Shorter second stage of labor, thus less fetal and maternal distress.
3. Reduce intracranial hemorrhage in PTL by decreasing compression decompression effect.
4. Reduce damage to maternal pelvic floor predisposing to vaginal prolapse, stress incontinence and ano-rectal dysfunction.

Complications of episiotomy:

1. Increased blood loss.
2. Extension to anal sphincter (median type).
3. Hematoma formation.
4. Infection.
5. Perineal pain, dyspareunia and ugly scar.



Fig. 16-6: Episiotomy

Contraindications of episiotomy:

1. Adequate perineal size. Episiotomy is not routine even in primigravidae.
2. High suspicion of Cesarean delivery.
3. Extensive perineal lesions such as condyloma accuminata or severe varicosities.

OPERATIVE (INSTRUMENTAL) VAGINAL DELIVERY (OVD)

Operative vaginal delivery (OVD) *refers to the use of specially designed instruments, namely the obstetric forceps and the vacuum extractor, to facilitate delivery of the fetal head.*

THE SHORT OBSTETRIC FORCEPS

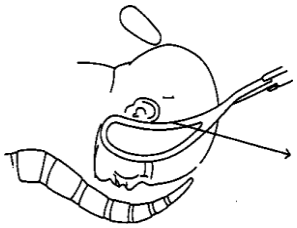


Fig 16-7: short forceps DOA

INDICATIONS FOR THE USE OF FORCEPS:

1. Prolonged second stage of labor, when maternal exhaustion is eminent.
2. To shorten second stage of labor, in maternal cardiac or hypertensive disorders.
3. Inadequate maternal expulsive forces, as with the use of epidural analgesia.
4. Fetal distress if the cervix is fully dilated with engaged head.
5. Prolapsed pulsating cord when the cervix is fully dilated and engaged head.
6. Some malpositions & malpresentations:
   - Occipito posterior position after failure of spontaneous rotation.
   - After coming head in breech, to promote flexion and avoid skull compression.

PREREQUISITES FOR THE USE FORCEPS:

1. The cervix should be fully dilated.
2. The head should be engaged and rotated.
3. No cephalopelvic disproportion.
4. The membranes (forewater) should be ruptured.
5. Presence of adequate uterine contractions.
6. The bladder & rectum should be evacuated.
7. Adequate anesthesia and asepsis.

COMPLICATIONS OF FORCEPS PROCEDURES:

A) Maternal complications:

1. ***Maternal birth injuries:***
   - Vaginal, perineal, and cervical tears.
   - Complete perineal tear (Injury of rectum and anal sphincter)
   - Rupture of the uterus.
2. ***Postpartum hemorrhage (PPH):***
   - Traumatic PPH, due to maternal birth injuries.
   - Atonic PPH, if applied in cases of poor uterine contractions.

B) Fetal complications:

1. ***Intracranial hemorrhage:***

   Liable to occur with wrong forceps application, leading to marked fetal head compression. This leads to tear of the vein of Gallen at the junction of the falx cerebri and tentorium cerebelli.

2. ***Head & Skull injuries: e.g.***
   - Skull fractures.
   - Cephalhematoma.
   - Facial nerve injury.
   - Facial skin bruises and lacerations.

THE VACUUM EXTRACTOR

VACUUM EXTRACTOR: is used for rotation and extraction of non-rotated occipito anterior (or occipito-posterior) position.

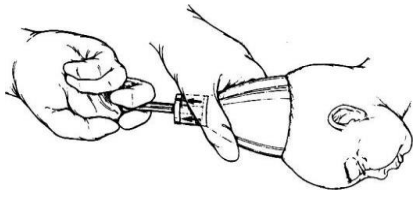
PREREQUISITES FOR THE USE OF VACUUM EXTRACTOR (in addition to forceps pre-requisites)

1. Suitable presentation: In contrast to forceps, vacuum is not suitable for face or after coming head in breech presentation.
2. Living fetus: Vacuum is not suitable for dead fetus.
3. Term fetuses: vacuum is not suitable for premature fetus.
4. No fetal distress: Vacuum is time consuming.
5. In contrast to forceps: vacuum can be used on unrotated head, as it causes rotation and extraction.

Complications of the vacuum extractor:

1) ***Fetal birth injuries including:***
   - Cephalhematoma (bleeding into the scalp).
   - Scalp lacerations (excessive force and repeated slipping of the cup).
   - Cerebral hemorrhage due to excessive negative pressure especially in preterm and asphyxiated fetus.

2) ***Maternal birth injuries lead to PPH including:***
   - Vaginal and perineal lacerations.
   - Cervical lacerations (inclusion of cervical tissue within the cup).
   - Rarely rupture of the uterus.



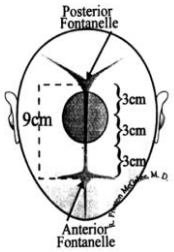


Fig 16-8: application of the ventouse cup on the fetal head and direction of traction

CESAREAN SECTION (CS)

DEFINITIONS

- ***Cesarean section:*** is the operation done for delivery of a viable fetus (whether living or dead), through an abdominal and uterine incisions.
- ***Hysterotomy:*** is the operation done for delivery of the fetus before viability < 24 weeks, through an abdominal and uterine incision.
- ***Repeat CS:*** CS in patients with one or more previous C.S.
- ***Elective CS:*** CS in a patient not in labor
- ***Emergency CS:*** CS in a patient undergoing trial vaginal delivery or an acute obstetrics complication as APH, eclampsia, etc

TYPES OF CS

1. ***Lower Segment (LSCS):*** using a transverse incision in the LUS, most common type.
2. ***Upper segment (USCS):*** using a longitudinal incision in the upper uterine segment (UUS), rarely indicated.



Fig 16-9: LSCS

NB: INDICATIONS OF USCS:

Placenta accreta spectrum, usually followed by hysterectomy.

Neglected shoulder, even if the fetus is dead (for fear of rupture uterus).

- Previous successful repair of high VVF (for fear of recurrence).
- Large fibroids or excessive varices on the LUS.

INDICATIONS of C.S.:

I. Previous CS is nowadays the most common indication.

II. **Faults in the Passages *(maternal indications):***

1. Contracted pelvis or cephalopelvic disproportion.
2. Cervical dystocia.
3. Placenta Previa (PL PRV): total or partial central PL PRV, or marginal posterior PL PRV.
4. Perineal and vulval lesions as HPV infection forming large condylomata.
5. Pelvic tumors such as low corporeal fibroid or ovarian tumors obstructing labor.
6. Maternal medical condition that contraindicates vaginal delivery.

III.*Faults in the Passenger *(fetal indications):

1. Macrosomia: oversized fetus (the most common cause in multipara).
2. Malpresentations as in:
   - Cephalic OP positions with failed long rotation.
   - Face DMP or with failed long rotation.
   - Persistent Brow presentation.
   - Breech presentation when vaginal delivery is contraindicated.
   - Transverse lie.
3. Fetal distress.

COMPLICATIONS OF CS:

I. Intra-operative:

1. Anesthetic complications such as; cyanosis, cardiopulmonary complications.
2. Primary hemorrhage: due to injury of inferior epigastric vessels, uterine vessels, uterine atony or DIC in concealed accidental hemorrhage.
3. Injury to the bladder, colon or ureters.

I. Early post-operative (within 24h):

1. Reactionary hemorrhage: due to slipping of a ligature.
2. Bladder or ureteric injuries.

II. Late post-operative:

1. Wound complications: infection, burst abdomen or incisional hernia.
2. Infections: generalized peritonitis, parametritis.
3. GIT: Late diagnosis of intestinal injuries, paralytic ileus, acute gastric dilatation, and intestinal obstruction.
4. Thrombo-embolic complications.
5. Rupture scar in subsequent pregnancy.
6. Abdominal adhesions leading to infertility or intestinal obstruction.

ADVANTAGES OF LSCS OVER USCS:

1. Stronger scar: due to:

- Better healing as the lower segment is less active in puerperium.
- Better coaptation of the edges (lower segment is thin).
- Less hematoma in the suture line (less vascularity).
- In subsequent pregnancies, erosion of the incision site by the chorionic villi is rare.

2. Less Hemorrhage: because

- The placental site is away from the operation area.
- Lower segment is thin and less vascular.

3. Less abdominal distension and ileus: because the intestines are away and not manipulated during the operation.

4. Less infection due to: better peritonization, and better healing.

5. ***Less adhesions and intestinal obstruction.*** Wound is low and is covered by peritoneum.
6. ***Less mortality rate.***

PREREQUISITES FOR TRIAL VAGINAL BIRTH AFTER CS (VBAC):

- Non-persistent cause of previous CS as severe degrees of contracted pelvis and CPD.
- No more than one previous LSCS.
- Previous normal puerperium.
- No tenderness over the CS scar.
- Vertex presentation, with head engaged.
- No associated other obstetric or medical complications.

N.B.: a previous normal vaginal delivery followed by C.S. improves the chances of a safe and successful VBAC

CESAREAN HYSTERECTOMY:

- Perform CS then remove the uterus.
- Indications:
  - Uterine atony associated with uncontrollable severe postpartum hemorrhage.
  - Couvelaire uterus (severe concealed accidental hemorrhage).
  - Tumors associated with pregnancy as operable cancer cervix.
  - Placenta accreta spectrum (Increta and Percreta)

17 OBSTETRIC HISTORY AND Examination

OBSTETRIC HISTORY

The most important evidence is always provided by history given by the patient. In many cases a provisional diagnosis can be reached before starting detailed physical examination.

1. PERSONAL HISTORY:

Including; Patient's name, age, address, occupation, marital status, gravidity and parity.

2. COMPLAINT: *in the patient's own words*

- The patient may have ***no complaint*** and is coming for ***routine antenatal care*** (ANC).
- The patient may complain of ***symptoms related to her present pregnancy*** (in the patient own words), as nausea, vomiting, vaginal bleeding, pelvic pain, etc.…).
- The patient may complain from ***symptoms related to a previously known medical disorder*** dating before the present pregnancy (as DM, chronic hypertension, cardiac disease, bronchial asthma, autoimmune disease, etc.….)

3. PRESENT HISTORY:

- ***Calculate the Expected Delivery Date (EDD):***
  - Naegele's rule (LMP + 7 days + 9 months).
- ***Calculate the duration of pregnancy*** using the following steps:
  - Subtract the today's date from the date of LMP (months and days).
  - Convert the result into weeks.
- ***Analysis of the present complaint;***
  - Onset, Course, and duration of each complaint starting with the most prominent one
  - Symptoms relevant to the present complaint or disease.
- Ask about ***important warning symptoms*** during pregnancy;
  - **In 1st trimester:** vaginal bleeding, nausea, and frequency.
  - **In 2nd trimester:** quickening, relieve of symptoms of first trimester.
  - **In 3rd trimester*:*** headache, vomiting, blurring of vision, epigastric pain, Rt hypochondrial pain, perception of Fetal movements, vaginal bleeding, vaginal gush of fluid, LL edema.
- Ask about **symptoms of other systems** involvement.

- Ask about ***special investigations and treatment in pregnancy*** specially Hb, Rh, blood glucose, urine analysis and ultrasound.

4. PAST, FAMILY, MENSTRUAL, and OBSTETRIC HISTORY:

See gynecological history and diagnosis.

OBSTETRIC EXAMINATION

1. GENERAL EXAMINATION: (see gynecological history and examination)

2. ABDOMINAL EXAMINATION: as gynecology plus:

A. FUNDAL LEVEL:

- ***Manual method;*** using the ulnar border of the left hand, the fundal level correlated with duration of pregnancy in relation to symphysis pubis (SP), umbilicus, and xiphisternum (XS):
  - At 12 weeks; fundal level is at the upper border of the symphysis pubis.
  - At 20-24 weeks: fundal level is at the umbilicus.
  - At 36 weeks: fundal level is at the XS.

  N.B.: after engagement, the fundal level descends below the XS.
- ***Metric method:*** the distance between symphysis pubis and fundal level, in cm, correlates with duration of gestation.



Fig 17-1: Fundal level manual and metric assessment

B. LEOPOLD MANEUVERS: (see figure)

- ***Fundal grip:*** to determine the **fetal pole** occupying the uterine **fundus**. Normally the fundus contains the fetal breech near term.
- ***Umbilical grip:*** to determine the direction of the **fetal back** and spine;
  - *Right or left:* in relation to maternal midline
  - *Anterior or posterior:* in relation to the back and spine of the mother
  - *Presence of a fetal head* at either side denotes transverse lie
- ***1st pelvic grip:*** gripping LUS above the SP (done if presenting part is not engaged);
  - *Cephalic presentations:* head is felt small, globular, hard, ballotable and tender.
  - *Breech presentation:* breech is felt large, soft, not ballotable, and not tender.

- *Shoulder presentation:* the LUS feels empty.

➢ ***2nd pelvic grip:*** Done if presenting part is engaged to confirm if it is head or buttocks.



Fundal grip Umbilical grip 1st pelvic grip 2nd pelvic grip

Fig 17-2: Leopold maneuvers for abdominal obstetric examination

3. AUSCULTATION OF FETAL HEART SOUNDS:

➢ In the 2nd trimester FHS can be audible by sonicaid which will also detect FHR.

➢ This is the simplest accurate method to ensure fetal life throughout pregnancy.

➢ Point of maximum intensity will depend upon gestational age, fetal lie, and presentation.

➢ FHS are most commonly heard below the umbilicus near to the middle line or on either side.



Fig 17-3: auscultation of FHS by sonicaid

4. LOCAL EXAMINATION:

➢ Local examination is ***not routinely*** performed during pregnancy except when needed

➢ It may be performed to **ensure** diagnosis in certain obstetric conditions as; vaginal bleeding during early or late pregnancy, PROM, diagnosis of onset of labor, evaluation of cephalopelvic disproportion and contracted pelvis, and associated conditions as uterine leiomyomas.

➢ ***During 2nd trimester:*** to obtain vaginal swabs for culture in certain cases

➢ ***During 3rd trimester:*** at 36-38 weeks: to confirm fetal presentation, determine head engagement, pelvic capacity, and cervical status (ripening, effacement, and dilatation).

➢ ***During labor:*** to confirm onset of labor, cervical status, and station of fetal head.

Part II

Gynecology

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01 ANATOMY AND EMBRYOLOGY

ANATOMY OF THE FEMALE GENITAL TRACT

I. ANATOMY OF THE VULVA

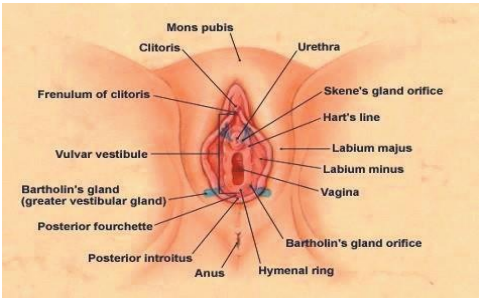


Fig 1-1: The vulva

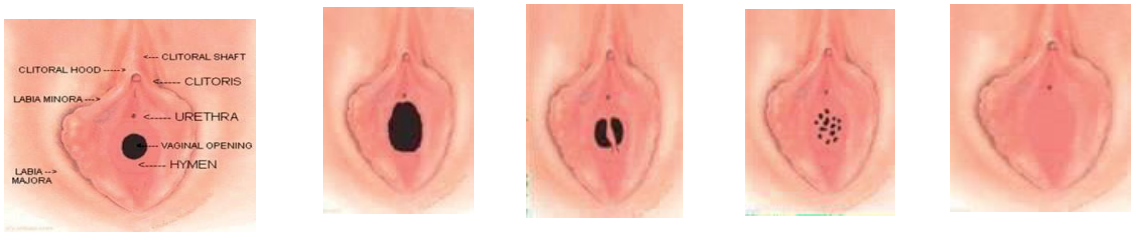
Female external genital structures are called the vulva and consist of:

1. **Mons Veneris**: Pad of fat overlying symphysis pubis and covered by hairy skin.
2. **Clitoris:**
   - An erectile cavernous structure. It is homologous to the penis in the male.
3. **Labia Majora:**
   - They are the outer two skin folds, raised by underlying fat, and passing back from the mons veneris to the perineum. The outer skin is covered by hairs while the inner medial surface is smooth, hairless and contains sebaceous and sweat glands.
   - They are homologous to the scrotum in the male, and contain fat and Bartholin's gland.
4. **Labia Minora:**
   - These are two thin folds of modified skin situated medial to the labia majora.
   - They vary in size and contain loose connective tissue devoid of fat.
   - The labia minora are very vascular and become turgid during coitus.
5. **Vestibule:**
   - The vestibule is the area between the inner aspects of the labia minora and the fourchette.
   - It is continuous posteriorly with the vaginal opening (vaginal orifice or introitus).
   - Structures that open in the vestibule: 1. The urethra. 2. The vagina 3. Bartholin's glands duct.

6. The Hymen:

- The hymen is a membrane, situated about 2 cm deep to the vestibular orifice it demarcates the external from the internal genital organs, and partially closes the vaginal orifice.
- It is formed of connective tissue covered by stratified squamous epithelium on both sides.
- The hymen is usually torn by the first intercourse unless the opening is unusually large or the hymen itself is elastic. Further lacerations occur on repeated intercourse.
- Remnants of the hymen in parous women are called “carunculae myrtiformis”.

- **Types of hymen:**
  - It is usually perforated and according to the shape of its opening; it may be; crescent, annular, fringed, cribriform, or bi-perforate.
  - Rarely the hymen may be imperforate resulting in accumulation of the menstrual blood in the vagina after puberty resulting in cryptomenorrhea with hematocolpos (see amenorrhea).



Virgin *Deflorated* *Bi-perforate.* *Cribriform* **Imperforate**

Fig 1-2: Types of hymen

7. Bartholin's Glands: (Greater Vestibular Glands):

- Bartholin's glands are bilateral compound racemes glands situated deep in the labia majora, at the junction of its posterior and middle thirds.
- They secrete mucus during sexual excitement to act as a lubricant.
- Its duct is 2 cm long and opens between the hymen and the labium minus.
- If the duct is obstructed it may form a vulval swelling (Bartholin's cyst) (see swellings of the vulva).
- The gland is rarely a site for a Bartholin's gland tumor.

8. The urethra and external urethral meatus:

- In the female the urethra is a tubular structure, 4 cm in length, extending from the bladder neck to the external urethral meatus that opens in the vestibule anteriorly below the clitoris.

BLOOD SUPPLY AND LYMPHATIC DRAINAGE OF THE VULVA

Arterial Supply: **Mainly** Internal pudendal artery: One of the two terminal branches of the anterior division of the internal iliac artery that ends as the dorsal artery of the clitoris.

Lymphatic Drainage of the vulva

- From the skin and appendages, to the superficial inguinal lymph nodes, to the deep inguinal and femoral lymph nodes of which the lymph node of Cloquet drains the clitoris directly.
- From the former superficial group, lymphatic channels pass to the deep pelvic nodes including; the external iliac, common iliac, then para-aortic lymph nodes.

II. ANATOMY OF THE VAGINA

- **The vagina** is a fibro-muscular tube that extends upwards and backwards from the vulva to the uterus forming an angle of 60° with the horizontal plane. Its mucous (skin) is elevated into folds or rugae allowing the vagina to be distended during labor.
- **Vaginal Length:** The anterior wall is 8-9 cm and the posterior wall is 10 - 11 cm long.
- **Vaginal fornices:** The cervix projects in the upper blind end of the vagina that forms a pouch (vaginal pouch) around the cervix and is divided into four fornices: two lateral, anterior and posterior (deeper) fornices.

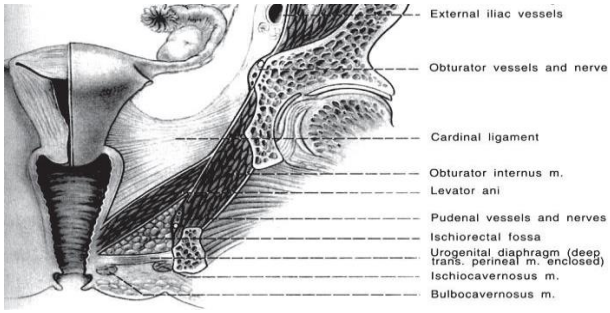


Fig 1-3: Anatomy of the vagina

BLOOD SUPPLY AND LYMPHATIC DRAINAGE OF THE VAGINA

I. **Arterial supply:** **Mainly** The vaginal artery (from internal iliac artery) is the main blood supply to the upper vagina

II. **Lymphatic drainage of the vagina**

- The upper 1/3 follows lymphatic drainage of the cervix (see later)
- The lower 1/3 drains to the inguinal lymph nodes.
- The middle 1/3 drains in both upper and lower directions.

III. ANATOMY OF THE UTERUS

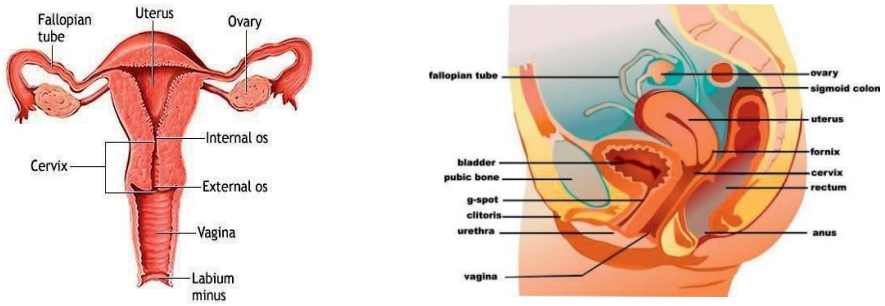


Fig1-4: Anatomy of the uterus

The adult uterus is a pear shaped hollow muscular organ measuring around 7.5 x 5.0 x 2.5 cm in the longitudinal, transverse, and antero-posterior diameters. It is slightly larger in the multipara than in the nullipara.

Histology of the uterus:

1. **Endometrium: The lining of the uterus.**
2. **Myometrium** (musculosa); formed of 3 layers: Outer longitudinal, Middle layer of interlacing criss-cross muscle fibers surrounding the blood vessels (its contraction controls blood loss during menstruation and after labor) and Inner circular muscle layer which is well developed over tubal openings and isthmus uteri.
3. **The perimetrium** (the peritoneal covering) **which is firmly adherent to the uterus.**

Histology of the cervix:

A) **Endocervix:** Lined by simple columnar epithelium with compound racemose glands or crypts that is liable to chronic infection. It secretes alkaline cervical mucus.

B) **Muscle layer:** Outer longitudinal and inner circular muscles. (2 layers only)

C) **Ectocervix:** Formed of stratified squamous epithelium covering the outer portion of the cervix. The junction between squamous and columnar epithelium at the external os forms a transitional zone 1-3 mm known as the transformation zone.

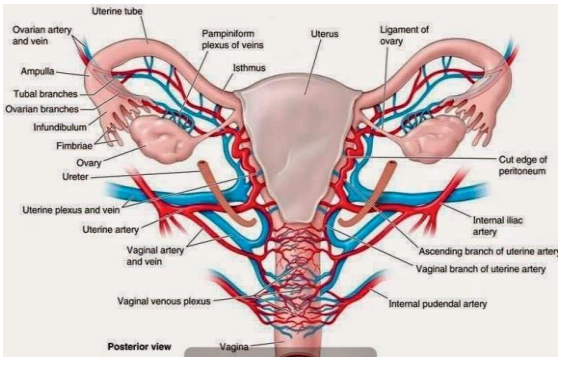


Fig 1-5: Blood supply of the uterus

Blood Supply and lymphatic drainage

I. Arterial supply:

The uterine arteries form the main supply of the uterus. They arise from the anterior division of internal iliac artery and run medially towards the isthmus, in the base of the broad ligament, crossing above the ureter 1/2 an inch lateral to the supravaginal cervix.

N.B.; the tortuous course of the uterine artery allows for increase in the size of the uterus without overstretch of the artery (no ischemia).

II. Lymphatic drainage:

- Fundus: To the para-aortic lymph nodes via ovarian vessels.
- Cornu: To the superficial inguinal lymph nodes via lymphatics of the round ligament.
- Body: To the internal then external iliac lymph nodes via the uterine vessels.
- Isthmus: As that of the cervix.
- Cervix: Two groups of lymphatics:
  - Primary groups: Paracervical, parametrial, obturator, internal and external iliac nodes.
  - Secondary groups: Common iliac, para-aortic, and lateral sacral lymph nodes.

IV- ANATOMY OF THE FALLOPIAN TUBE

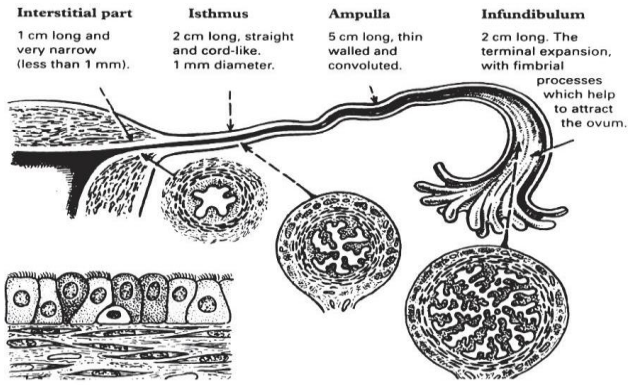


Fig 1-6: The Fallopian tube

The Fallopian tubes are two tortuous tubes that lie in the free upper part of the broad ligament. They blend medially with the cornu of the uterus, while laterally their free outer end curves backwards towards the ovary.

Tubal functions:

The major functions of the Fallopian tubes include ***ovum pick up***, at the time of ovulation, by their free fimbrial end, ***transport of the ova*** through the tubal lumen, by their peristaltic and ciliary movements, and ***production of secretions*** necessary for capacitation of the sperm and nutrition of the ova during their journey, by their lining cells.

Blood supply and lymphatic drainage:

I. **Arterial supply:** Each Fallopian tube is supplied by branches from both the *uterine artery*, and the *ovarian artery*. That is why gangrene is rare to occur in the tube unless completely twisted.

II. **Lymphatic drainage:** to the para-aortic lymph nodes directly via ovarian lymphatics.

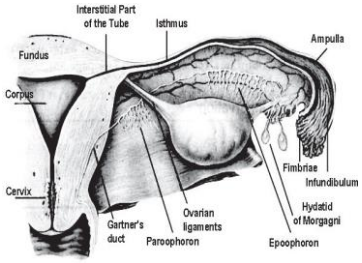
V- ANATOMY OF THE OVARY

The ovary is an almond shaped organ lying in the fossa ovarica on the lateral pelvic wall, measuring 3 x 2 x 1 cm. It is the only organ in the abdomen that is ***not covered by peritoneum***. Its surface is pearly white and corrugated by the effect of the monthly ovulatory activity that leaves surface gyrii indentations.

Ovarian attachments:

The ovary is fixed in the pelvis by three attachments:

Fig 1-7: Attachments of the ovary



1. The *mesovarium:* A peritoneal fold that suspends the ovary to the back of the broad ligament.
2. The *infundibulopelvic ligament:* suspends the upper pole of the ovary to the lateral pelvic wall and carries the ovarian vessels, nerves and lymphatics.
3. The *ovarian ligament:* attaches the lower pole to the cornu of the uterus.

Blood supply and lymphatic drainage

I. **Arterial supply:**

a. *Ovarian artery:* Arises from the *aorta*.
b. *Ovarian branch* from the *uterine artery;* which anastomose with the ovarian vessels at the broad ligament.

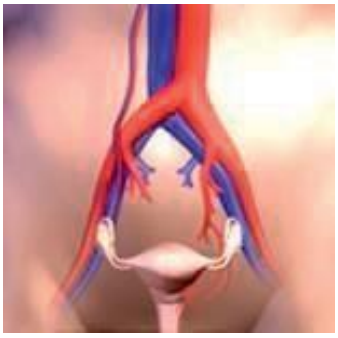


Fig 1-8: Blood supply of the ovary

Lymphatic drainage: Lymphatic drainage of the ovaries is directly to the para-aortic lymph nodes via the ovarian vessels.

VI. ANATOMY OF THE PELVIC PART OF THE URETER

The ureter is a narrow muscular tube, about 25 cm in length. It runs retroperitoneal from the kidney to the urinary bladder. Its wall undergoes peristaltic movement to propel urine downwards.

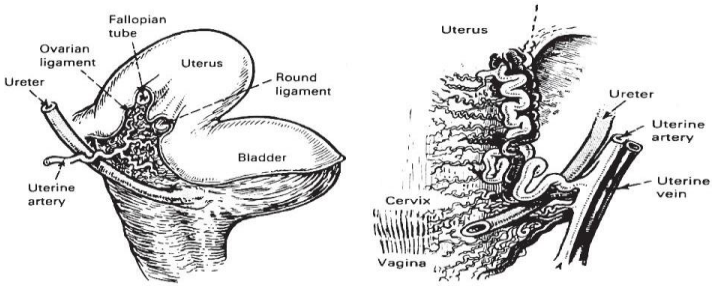


Fig 1-9: pelvic part of the ureter

- **At the pelvic inlet:** The ureter enters the pelvis above the bifurcation of the common iliac artery anterior to the sacroiliac joint.
- **In the pelvis:** It runs downwards lying in front of the internal iliac artery.

At the base of the broad ligament, it runs medially and forwards through the parametrium till it reaches about 1 cm lateral to the supravaginal cervix where it passes below and at right angle to the uterine artery (i.e. the artery crosses over the ureter = water under the bridge). The ureter then passes forwards to enter the trigone of the urinary bladder.

THE UTERINE AND CERVICAL LIGAMENTS

Ligaments attached to the uterine fundus and body on each side include; the broad ligament, the round ligament, and the ovarian ligament. These ligaments being soft, lax, and easily stretchable, play almost no role in the pelvic support of the uterus in contrast to the strong, tough cervical ligaments.

A) THE BROAD LIGAMENT:

The broad ligament is a double sheet of peritoneum that extends from the lateral wall of the uterus to the lateral pelvic wall.

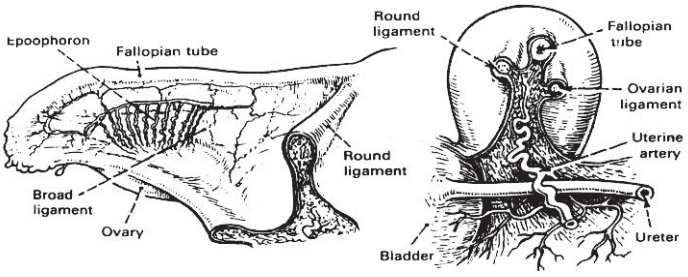


Fig 1-10: The broad ligament

Contents of the broad ligament:

- *The uterine vessels, the ureter, parametrial lymphatics and lymph nodes, sympathetic and parasympathetic nerves, parametrial pelvic cellular tissue and fascia.*

B) THE ROUND LIGAMENT:

This is a fibromuscular ligament attached to the uterine cornu. It runs

downwards and forwards to enter the inguinal canal to be inserted in the upper part of the labium majus. It pulls the uterus forwards and help keeping it in an anteverted position.

C) THE OVARIAN LIGAMENT:

It is a fibromuscular ligament that attaches the inner lower pole of the ovary to the cornu of the uterus. It plays no role in pelvic support of the uterus.
It contains the ovarian vessels. Its lateral extension to the lateral pelvic wall is called the ***infundibulopelvic ligament*** (that contains also the ovarian vessels).

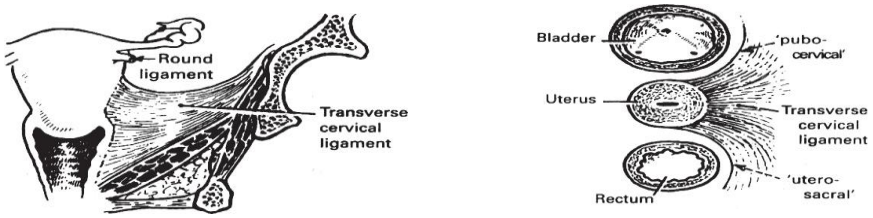
THE CERVICAL LIGAMENTS

The cervical ligaments are ***condensed thickening of the pelvic cellular tissue***,

Three pairs of ligaments can be distinguished, which *extend from the supravaginal part of the cervix and upper part of the vagina* to the pelvic walls:

A. **Mackenrodt's ligaments (lateral cervical or cardinal ligaments of the cervix):** spread out on either side from the lateral surface of the cervix and vagina, in a fan-shaped manner, and are inserted in the lateral pelvic wall. These are ***the strongest*** cervical ligaments.
B. **Utero-sacral ligaments:** From the posterior aspect of the cervix and vagina, backwards surrounding the rectum, below the utero-sacral folds of peritoneum, to become inserted in the third piece of the sacrum.
C. **Pubo-cervical ligaments:** extend from the anterior surface of the cervix and vagina, forwards beneath the bladder and surrounding the urethra, to the posterior surface of the pubis.

Fig 1-11: Attachments of the cervical ligaments



VII. THE PELVIC FLOOR MUSCLE (LEVATOR ANI)

The levator ani muscles together with the fascia which covers their upper and lower surfaces are collectively called the pelvic diaphragm. In the midline, the pelvic floor is pierced by the urethra, the vagina and the rectum.

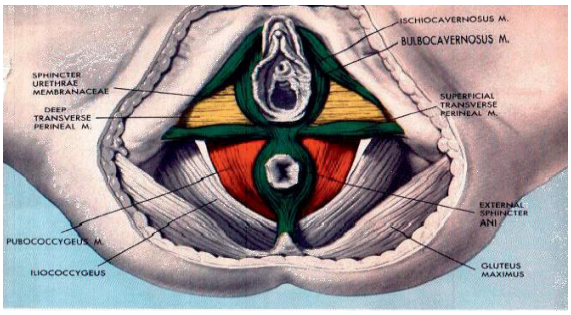


Fig 1-12: Levator ani muscles

THE LEVATOR ANI MUSCLE:

A. **The ischiococcygeus** part;
B. **The iliococcygeus**
C. **The pubococcygeus** muscle is the ***most important*** of the three muscles:
   - The pubococcygeus sends fibers to the urethra (*pubourethralis)*, to the vagina (*pubovaginalis)*, and rectum (*puborectalis)* to blend with their intrinsic muscles, giving them muscular support.

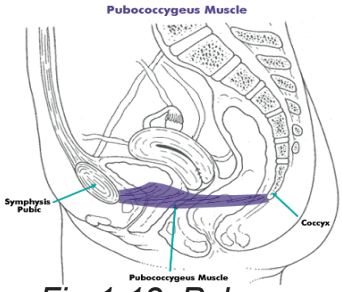


Fig 1-13: Pubococcygeus muscle

THE PERINEUM

The perineum is the area between the vaginal orifice and the anus. It covers the perineal body and characterized by less hairy skin with less subcutaneous tissue. Normally it is 5cm, if less, it is called deficient perineum.

DEVELOPMENT OF FEMALE GENITAL ORGANS
DIFFERENTIATION OF THE EXTERNAL GENITALIA

External genitalia start to develop from the **urogenital sinus.** The basic parts of the undifferentiated external genitalia are:

1. A medial pair of swellings called genital folds.
2. A lateral pair of swellings called genital swellings.
3. A central swelling called genital tubercle.

In the male: (in presence of androgens)

- The genital tubercle enlarges markedly forming the penis
- The genital folds fuse to become the penile part of the male urethra.
- The genital swellings enlarge, fuse and form the scrotum.

In the female (in absence of androgens);

- The genital tubercle enlarges slightly and becomes the clitoris.
- The genital folds become the labia minora.
- The genital swellings enlarge to form the labia majora.

Factors needed for Male sexual differentiation:

1. Presence of Y chromosome (for **TDF)** and so normal testicular function (producing androgens).
2. Responsive male end organs for testicular testosterone (responsive androgen receptors).

Factors needed for Female sexual differentiation: just absence of Y chromosome

DIFFERENTIATION OF THE INTERNAL GENITALIA

A) DEVELOPMENT OF THE OVARIES

The ovaries develop from the **genital ridges.**

B) DEVELOPMENT OF THE FALLOPIAN TUBES

The fallopian tubes develop from the upper parts of the **Mullerian** ducts after their canalization.

C)DEVELOPMENT OF THE UTERUS

The uterus develops from the middle part of the **Mullerian** ducts after their fusion and canalization. Uterine anomalies may be due to failure of Mullerian duct development, failure of fusion of both Mullerian ducts, or failure of their canalization.

I) IMPROPER MULLERIAN DUCT DEVELOPMENT

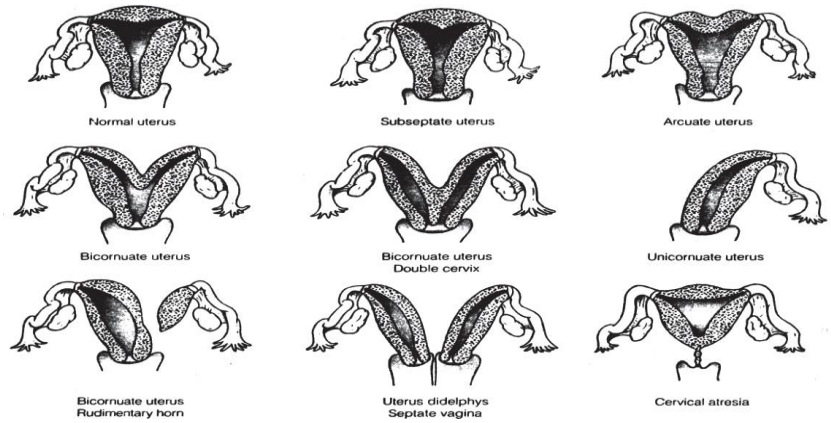
1. **Uterine aplasia:** complete absence of the uterus (rare anomaly). Cases will present after puberty with primary amenorrhea and well developed secondary sexual characters.
2. **Uterine hypoplasia:** small solid uterus with no cavity, or small infantile uterus with a body to cervix ratio of 1: 2. Cases will present after puberty with primary amenorrhea, hypomenorrhea, infertility, or rarely recurrent pregnancy loss (RPL).

II) IMPROPER MULLERIAN DUCT FUSION AND CANALIZATION

1. **Arcuate uterus**: broad fundus with a depression from outside. It is of no clinical significance
2. **Subseptate uterus:** the uterine cavity is divided by a longitudinal incomplete median septum starting at the fundus downwards not reaching the isthmus.
3. **Septate uterus:** the uterine cavity is divided by a longitudinal complete median septum extending from the fundus to the isthmus, not reaching the cervix.
4. **Bicornuate uterus:** the uterine body is divided into two horns, both

attached to one cervix, joining one vagina.

Fig 1-14: Anomalies of the uterus



5. **Bicornuate uterus with double cervix** (pseudo-didelphys); two uterine bodies (2 horns) each attached to a cervix (2 cervices) attached to each other and both open together in one vagina.
6. **Uterus didelphys:** there are two separate uteri (two bodies & two cervices) with two vaginae (separated by a longitudinal septum).
7. **Rudimentary horn:** one horn of the uterus is well developed, but the other is rudimentary.
8. **Unicornuate uterus:** One Mullerian duct is well developed forming a well canalized Fallopian tube connected to one horn of the uterus opening into the cervix and vagina. The other Mullerian ducts failed to develop with absent other tube, and second uterine horn.
9. **Cervical atresia;** Improper canalization of the cervical canal may lead to cervical atresia with entrapment of menstrual blood after puberty leading to hematometra and hematosalpinx (see amenorrhea).

COMMON CLINICAL PRESENTATIONS OF UTERINE ANOMALIES:

a. Primary amenorrhea: in cases with uterine aplasia, hypoplasia, and cervical atresia.
b. Severe dysmenorrhea: in cervical atresia, and non-communicating rudimentary horn
c. Primary infertility: in cases with uterine aplasia, hypoplasia, cervical atresia, and less commonly with septate uterus.
d. Recurrent pregnancy loss (RPL) in first trimester; may be associated with incompetent isthmus, infantile uterus or septate uterus.
e. Preterm labor, malpresentations, and dysfunctional labor: may occur with unicornuate and bicornuate uteri.
f. Cervical atresia may present with; primary amenorrhea, spasmodic dysmenorrhea, hematometra, hematosalpinx, endometriosis and infertility.
g. Pregnancy in a rudimentary horn is a type of ectopic pregnancy.

C) DEVELOPMENT OF THE VAGINA

The upper 3/4^{ths} of the vagina develops from the lower parts of the **Mullerian**

ducts after their fusion and canalization, while the lower 1/4th develops from the urogenital sinus.

DEVELOPMENTAL ANOMALIES OF THE VAGINA:

1. **Aplasia of the vagina:**
   - It usually affects the upper 3/4^{ths} of the vagina while the lower 1/4th develop giving rise to a small vaginal pouch.
   - After puberty, they will present by 1ry amenorrhea as the uterus is usually absent too. It may also present with dyspareunia and 1ry infertility after marriage.
   - Operative correction of the vagina aims at allowing sexual intercourse. Mc Indoe's operation (creation of a new vagina) can be done using a skin graft taken from the thigh over a mold, thus fashioning a new vagina.
2. **Transverse vaginal septum:**
   - Results from improper fusion between the urogenital sinus and the Mullerian ducts (see amenorrhea)
3. **Imperforate hymen:**
   - Occurs due to improper canalization of the urogenital sinus (see amenorrhea).
4. **Longitudinal vaginal septum:**

   Occurs due to improper fusion of the two Mullerian ducts.

   Clinically it may lead to dyspareunia and is treated by surgical excision of the septum.

02 REPRODUCTIVE ENDOCRINOLOGY

Physiology Of the Menstrual Cycle

OVERVIEW

The average menstrual cycle recurs every 21-35 days (mean **28** days), the average **blood flow** lasts for 3-7 days (mean **5** days), with an average blood loss of about **30-50** ml/cycle. The first few cycles following menarche and those in the few years prior to menopause are usually irregular mostly due to **anovulation**. Along their lifetime from puberty to menopause women will have around **400** menstrual cycles. Each cycle represents a complex interaction between the hypothalamus, pituitary gland, ovaries and the endometrium.

THE HYPOTHALAMUS

Hypothalamic Releasing Hormones

Releasing hormones are small peptides characterized by having short half-lives of few minutes due to their rapid degradation (rendering them undetectable in serum). They are released in minute quantities in a pulsatile manner, and reach the anterior pituitary via the hypothalamic portal circulation.

Hypothalamic releasing hormones include:

1. **Gonadotrophin-releasing hormone (GnRH):** It is responsible for synthesis and release of **pituitary gonadotropins (FSH and LH).**
2. **Thyrotropin-releasing hormone (TRH);** stimulates the release of **TSH**
3. **Corticotrophin-releasing hormone (CRH);** stimulates the release of **ACTH**.
4. **Growth hormone-releasing hormone (GHRH);** stimulates the release of **GH**
5. **Prolactin inhibiting factor (PIF or dopamine);** it inhibits production of prolactin (PRL)

THE PITUITARY GLAND

The pituitary gland lies at the base of the brain, below the hypothalamus within a bony cavity known as the sella turcica. It is divided into two major portions:

A. The Anterior Pituitary Gland (adenohypophysis)

The anterior pituitary serves primarily in the production of six trophic hormones produced through five hormone producing cell types:

1. **Follicle stimulating hormone (FSH)** and **Luteinizing hormone (LH)**

(gonadotrophins) produced by cells known as Gonadotrophs. (FSH) and (LH) are glycoproteins responsible for the process of growth, maturation and rupture of Graafian follicles (ovulation), and maintenance of corpus luteum function.

2. **TSH** (Thyroid stimulating hormone).
3. **ACTH** (adreno-cortico-trophic hormone).
4. **GH** (growth hormone).
5. **PRL** produced by cells known as lactotrophs.

Pituitary hormones are all **stimulated** and controlled by hypothalamic releasing hormones (GnRH, GHRH, TRH, and CRH) with the exception of PRL which is under chronic **inhibition** by prolactin inhibiting factor **(PIF)**. Consequently, damage to the pituitary stalk will result in hypopituitarism for FSH, LH, GH, ACTH, and TSH, but with an associated increase in PRL secretion.

B. The posterior pituitary gland (neurohypophysis):

The posterior pituitary serves primarily in the transport of **oxytocin** and **vasopressin** (antidiuretic hormone: ADH) from the hypothalamus to their release into the circulation.

THE OVARIAN CYCLE

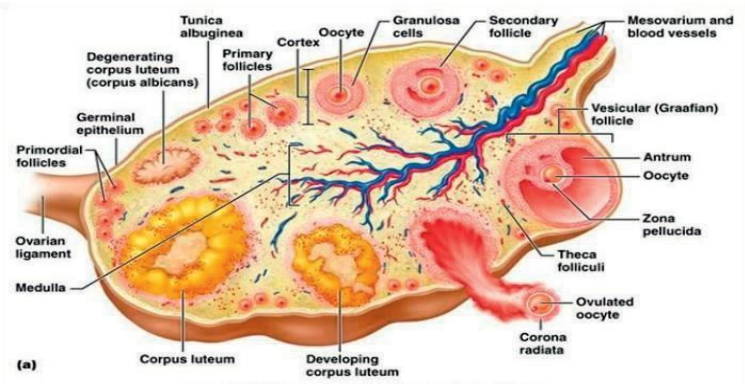


Fig 2-1: histology of the ovary at different phases of ovarian cycle

The ovarian cycle is divided into **follicular phase**, and **luteal phase**, separated by **ovulation**.

A.FOLLICULAR PHASE:

This phase lasts from the **1st day** of menses till **ovulation**. It is of **variable** length (mean **14** days). Graafian follicles grow and mature under the influence of follicle stimulating hormone **FSH**. They produce increasing amounts of **E2** till becoming a mature dominant Graafian follicle.

THE TWO-CELL, TWO-GONADOTROPHIN hypothesis of estrogen production

A) **Within theca cells;** LH stimulates conversion of cholesterol to

androstenedione, which are then transported from theca cells to granulosa cells

B) **Within granulosa cells;** FSH stimulates aromatization of androgens into estrogens (estradiol E2) by the enzyme aromatase.

B. OVULATION

- **Late in the follicular** phase, high **E2** levels exert a positive feedback effect on **LH** produce an **LH surge.**
- **Ovulation (oocyte release)** occurs approximately 36 hours after the onset of the **LH surge** and about 12 hours after the **LH peak.**

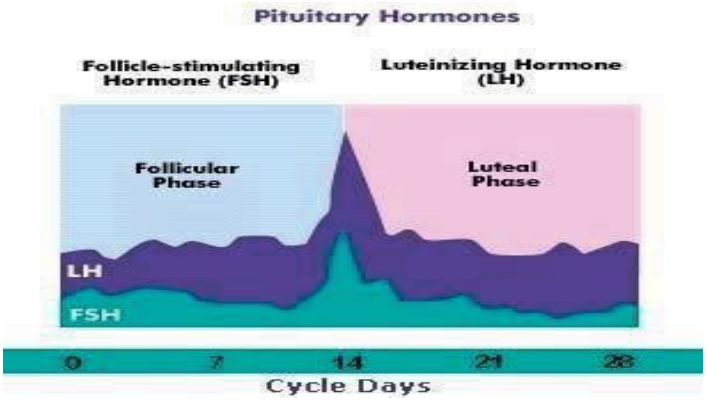


Fig 2-2: Diagram showing LH surge

C. THE LUTEAL PHASE

- The luteal phase extends from the time of ovulation till the onset of menstruation. It is of fixed duration lasting for **14 days**.
- **Corpus luteum (CL) formation:** After ovulation, and under the influence of **LH**, the granulosa cell of the ruptured follicle undergoes **luteinization.** These luteinized granulosa cells, form the **CL**.
  - **The CL** will produce copious amounts of Progesterone **(P).**
  - **Progesterone** produced by the CL will lead to secretory changes in the endometrium, which is necessary to prepare the endometrium for implantation of the embryo.
- **Fate of the CL:** if pregnancy doesn't occur, CL undergoes **luteolysis** and becomes replaced by the avascular corpus albicans.
- **As the CL dies**, the pituitary gland becomes released from the negative feedback effect of E2 and P.

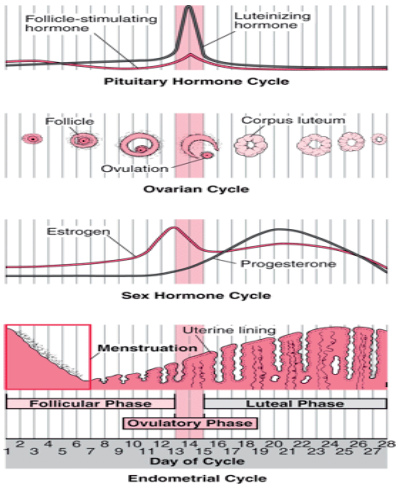


Fig 2-3: hormonal patterns in the menstrual cycle

THE ENDOMETRIAL CYCLE

Changes in the endometrium in response to ovarian hormones throughout the menstrual cycle are divided into three phases; proliferative phase, secretory phase, and menstruation.

1. THE PROLIFERATIVE PHASE

- The proliferative phase of the endometrium coincides with the follicular phase of the ovarian cycle, and is characterized by both glandular and stromal growth under the effect of E2.
- The endometrial thickness grows from 0.5 mm at the end of menstruation to about 8- 12 mm at the end of the proliferative phase.

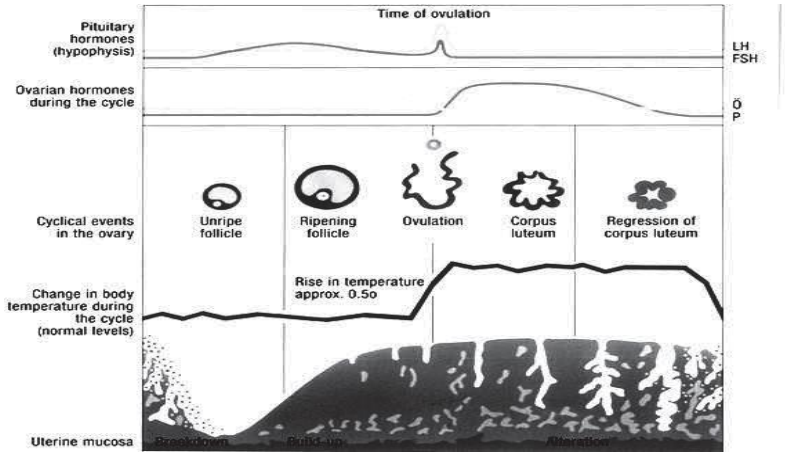


Fig 2-4: the endometrial cycle

2. THE SECRETORY PHASE

The secretory phase of the endometrium coincides with the luteal phase of ovarian cycle, and is characterized by endometrial glandular secretory activity under the effect of **P** produced by the **CL**, till menstruation starts.

3. MENSTRUATION

- Premenstrual sharp and rapid decline in P & E2 levels result in coiling and vasoconstriction of the endometrial spiral arteries, with ischemia to the functional portion of the endometrium (superficial and intermediate layers) and endometrial shedding.
- Within 2 days from the onset of menstruation the surface epithelium begins to regenerate under the influence of follicular phase estrogen.

CERVICAL MUCUS CHANGES DURING THE MENSTRUAL CYCLE

The physical and chemical properties of the cervical mucus vary in relation to the changing estrogen and progesterone levels in the follicular and luteal phases of the menstrual cycle.

Cervical mucus changes in the menstrual cycle:

Follicular phase (estrogen effect)	Luteal phase (progesterone effect)
• Profuse	• Scanty
• Decreased viscosity (thin-watery)	• Increased viscosity (thick-viscid)
• Decreased leucocytes	• Increased leucocytes
• Positive Ferning test (arborization test)	• Negative Ferning test (arborization test)
• Positive Spinbarkeit test (Stretchability test) (threads 7-10 cm)	• Negative Spinbarkeit test (Stretchability test)

- **Ferning test**: Microscopic examination of a drop of cervical mucus left to dry for 10 minutes
  on a glass slide in the **follicular phase** will reveal an **arborizing palm leaf pattern**, due to its high sodium chloride and potassium content in response to a high estrogenic level (+ve test). In the luteal phase the arborizing pattern is lost giving a negative test.
- **Spinbarkeit test** is positive when the cervical mucus can be drawn between two slides into threads stretching up to 10 cm due to high mucus content in response to high estrogen levels

Vaginal Epithelial Cellular changes during the menstrual cycle

The vaginal epithelium is composed of basal, intermediate, and superficial (mature) cells.

- **In the follicular phase** under estrogenic effect superficial cells predominate
- **In the luteal phase** progesterone causes shedding of more intermediate cells

N.B.; Maturation index calculates the ratio of basal, intermediate, to superficial cells on vaginal cytology of the vaginal epithelium in different phases of the menstrual cycle

Follicular phase vaginal cytology	Luteal phase vaginal cytology
• Few lymphocytes	• Many lymphocytes
• Maturation index 0-30-70	• Maturation index 0-70-30

DYSMENORRHEA

DEFINITION

Dysmenorrhea is defined as **pain** and **cramping** during menstruation that **interferes** with normal activities, and requires the use of medications to control the symptoms. **Pain** may range from **mild** discomfort, to **severe** pain that causes some patients to be bedridden for 1-3 days each month.

INCIDENCE

Dysmenorrhea affects **45-90%** of women in their reproductive age with variable degrees of severity. Almost **50%** of women will suffer significant pain, with **10%** of these become incapacitated during the first few days of the cycle.

CLASSIFICATION

- **Primary** dysmenorrhea is idiopathic menstrual pain without identifiable pathology
- **Secondary** dysmenorrhea is painful menses due to underlying pathology (endometriosis, fibroids, adenomyosis, PID, cervical stenosis).

PRIMARY DYSMENORRHEA

Primary Dysmenorrhea is almost always associated with ovulatory cycles, with *no obvious organic cause.*

Etiology

- Mostly results from increased levels of endometrial prostaglandins **(PGLs)** released as a result of secretory endometrium separation.

Diagnosis

- *Age;* usually occurs in younger women <20 years of age
- *Pain;* occurs on the 1st or 2nd days of ovulatory cycle
- *Associated symptoms;* nausea, vomiting, and headaches
- *Physical examination;* reveals no obvious abnormalities or pelvic pathology
- *Pelvic US;* reveals normal uterus, ovaries, and adnexa.

Treatment

1. ***Non-steroidal anti-inflammatory drugs (NSAIDs):*** this is the first and most important line of treatment (Ibuprofen, ketoprofen, and naproxen). They are taken at onset of menses, continued for 1-3 days then taken as needed.
2. ***Combined oral contraceptive pills (COCs):*** OCPs are the second line of treatment in cases which do not get adequate pain relief with the use of NSAIDs alone, or cannot tolerate their side effects.

SECONDARY DYSMENORRHEA

Secondary dysmenorrhea occur in both ovulatory and anovulatory cycles.

Etiology

Symptoms are 2ry to an **identifiable cause**, such as endometriosis, uterine leiomyomas, cervical stenosis, pelvic adhesions, and PID.

Diagnosis

- *Age;* women are commonly middle aged (30-35 years).
- *Parity;* it occurs in both parous and nulliparous women.
- *Pain;* starts few days before menses and gradually decreases by onset of menses. It is dull aching or colicky suprapubic pain commonly associated with low back ache that usually occurs after several years of relatively painless menstruation.
- *Physical examination;* may detect sizable uterine or adnexal masses.
- *Pelvic US;* may diagnose smaller uterine myomas, ovarian endometriomas, and tuboovarian masses
- *Laparoscopy;* for diagnosis of endometriosis, pelvic adhesions, and PID.

Treatment: *Treatment of the cause*

THE PREMENSTRUAL SYNDROME

The premenstrual syndrome **(PMS)** describes a group of **physical** and/or **emotional** changes that constantly **occur** and **recur** in the **luteal phase** of successive cycles, severe enough to interfere with the patient's regular life style.

Symptoms include; headache, weight gain, bloating, breast tenderness, mood fluctuation, restlessness, irritability, anxiety, depression, and fatigue.

Symptoms should occur in the **2 weeks prior to menstruation** (luteal phase), with at least a *7-day symptom free interval* in the first half of the menstrual cycle.

The **highest incidence of PMS** occurs in women in their late 20's to early 30's.

Up to **80%** of women will report uncomfortable symptoms in the premenstrual period.

However, in only **5-10%** symptoms will be **severe** enough to interfere with daily activities.

Etiology

The causes of PMS are **not completely understood**.

- PMS may be due to the interaction between the **neurotransmitter serotonin** and cyclic changes in **ovarian steroids**.
- PMS is more pronounced in **genetically** and **psychologically** susceptible women.
- Although women with PMS have normal levels of E2 & PRG, they may have an **abnormal response** to normal hormonal changes.

TREATMENT

Treatment of PMS is based on alleviating symptoms with special consideration to the psychological aspects of PMS. They include;

1. **Selective serotonin reuptake inhibitors (SSRI)** have demonstrated reasonable efficacy in treating both physical and mood symptoms in women with severe PMS, taken as daily oral tablets throughout the menstrual cycle.
2. **Changes in the life style:** as regular exercise, balanced diet, avoid premenstrual stress, eliminate caffeine, cigarette smoking, and alcohol ...etc
3. **Calcium**, **Vitamin D**, **Vitamin B6** & **Magnesium** all may slightly improve PMS symptoms in many women.
4. **OCP:** may improve some cases as it minimizes fluctuations in ovarian steroids.

PUBERTY

DEFINITION

Puberty is the period of life that marks the normal physiologic transition from childhood to sexual and reproductive maturity (physical, mental, sexual growth).

Normally, pubertal changes start by the age of 8-9 years, and are completed by the age of 12-14 years.

CLINICAL MANIFESTATIONS OF PUBERTY

1. **Growth spurt:** is the first sign of puberty to occur with a peak growth velocity at age of 11 years, followed by a slower growth rate until cessation usually by age of 15 years, where closure of bone epiphysis occurs in most girls as an effect of increased estrogen levels.
2. **Thelarche: (Breast Development):** The breast grows in phases (5- Tanner stages) under the effect of estrogen. This starts by growth of the body of the breast (budding), followed by pronounced areolar development and ends by full breast development, where the breast tissue grows to become confluent with the areola.

3. **Pubarche:** Growth of pubic hair occurs under effect of ACTH and androgens. Pubic hair growth is classified into (5- Tanner stages) according to degree of distribution.
4. **Menarche:** This is the onset of first menstruation. Its average age in Egypt is 12.5 years.
5. **Axillarche:** growth of axillary hair, appears later also under effect of androgens.

DISORDERS OF PUBERTY

I. PRECOCIOUS PUBERTY

Precocious puberty describes **appearance of secondary sexual characters before 8 years** of age, with or without onset of menstruation. Besides the psychological impact of the syndrome, early estrogen production will carry **the risk of permanent short stature** from early epiphyseal bone closure.

A) ISOSEXUAL PRECOCIOUS PUBERTY (IPCP):

The secondary sexual characters are in agreement with genetic and phenotypic sex.

1. **True IPCP:** (GnRH dependent, associated with ovulation)

- ***Mostly Constitutional:*** due to idiopathic premature activation of the GnRH cell, leading to early FSH production, and early onset of ovarian E2 secretion.

2. **Pseudo IPCP:** (GnRH independent, not associated with ovulation)

a. Estrogen secreting ovarian tumors (granulosa cell tumors)
b. Primary hypothyroidism; elevated TSH may induce FSH production
c. Exogenous estrogen intake

B. HETEROSEXUAL PRECOCIOUS PUBERTY (HPCP):

The secondary sexual characters are in disagreement with genetic and phenotypic sex.

a. Androgen secreting tumors (virilizing ovarian or adrenal neoplasia)
b. Congenital adrenal hyperplasia (CAH)
c. Exogenous androgen intake

DIAGNOSIS OF PRECOCIOUS PUBERTY

- **History and Physical examination:** Age, height, growth spurt, breast development, and hair distribution, according to Tanner stages of pubertal development.
- **Hormonal assays:**
  - Pituitary gonadotropins (serum FSH & LH)
  - Serum E2, and DHEAS, to detect ovarian and adrenal contribution.
  - Thyroid function tests (TSH, T3, & T4) to exclude 1ry hypothyroidism.

- **Radiological investigations:**
  - X-ray to the wrist of the non-dominant hand to determine bone age.
  - CT & MRI for brain tumors and intracranial masses.
  - Pelvic ultrasound for exclusion of ovarian tumors or cysts.

MANAGEMENT OF PRECOCIOUS PUBERTY

The chief goal of therapy aims at slowing down accelerated growth and stopping the development of secondary sex characters through inhibiting pituitary and ovarian hormones release, thus avoiding premature epiphyseal closure which predisposes to **short stature.**

a. Idiopathic PCP: long-acting GnRH agonists will suppress pituitary FSH & LH, and ovarian E2, halting the progression through puberty. Treatment is continued until an appropriate age has been reached for resumption of pubertal development.

b. Functioning ovarian tumors are surgically removed.

PERIMENOPAUSE

- **Climacteric** is the phase of the aging process during which a woman passes from the reproductive to the non-reproductive stage *(FIGO definition)*.
- **The Perimenopause**, also known as the **menopausal transition**, refers to the part of the climacteric before the menopause (*2-8 years prior to the FMP),* in which there is transition from normal ovulatory cycles to permanent amenorrhea of menopause.
  - This period is characterized by irregular menstrual cycles together with some of the symptoms associated with the menopause as; hot flushes, night sweats, and mood swings.
  - During this period, **inhibin-B** secretion from granulose cells falls due to diminished follicular number, and as result **FSH** levels rise.
  - Ovarian **E2** secretion is preserved until late perimenopause.

MENOPAUSE

Menopause is the final menstruation that occurs at the end of climacteric. It is defined by *12 months of amenorrhea after the final menstrual period* ***(FMP)***, in the absence of any other pathological or physiological cause. It occurs due to depletion of ovarian primordial follicles due to their consumption since menarche.

- **Natural Menopause:** occurs due to intrinsic ovarian failure that usually occurs between 50-54 years with a median age of **52** years. It is characterized by complete ovarian follicular depletion, with subsequent cessation of ovarian estrogen (E2) secretion.

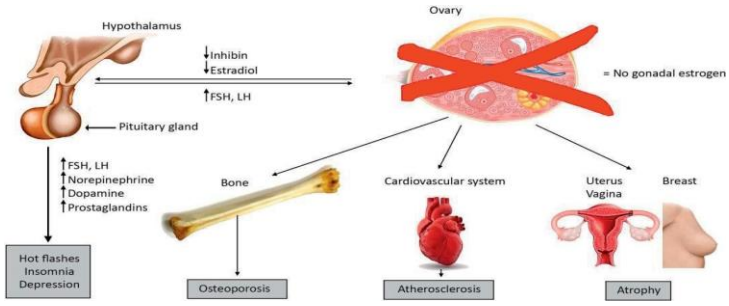
- *Late or delayed menopause:* menopause occurring > 55 years of age
- *Early or premature menopause:* menopause occurring before the age of 45 years of age

- **Induced Menopause:** menopause can be artificially induced;
  - *Surgically;* as *after bilateral oophorectomy*
  - *Ablation of ovarian function;* as pelvic irradiation or systemic chemotherapy
  - *Medically:* during the use of long-acting GnRH agonists as in the management of endometriosis.

ENDOCRINE CHANGES CHARACTERISTIC OF MENOPAUSE:

- *Decreased* serum **inhibin-B** levels that started at the perimenopause.
- Marked *increase* in serum **FSH** & **LH** levels.
- Marked and persistent *decrease* in Ovarian **E2** levels
- *Decrease* in sex hormone binding globulin **(SHBG)**
- *Increased* free testosterone **(FT)** levels

Fig 2-5: Changes in both the ovary and the hypothalamus contributing to changes of the menopause



CHANGES IN BODY SYSTEMS ASSOCIATED WITH MENOPAUSE

They are almost all related to **estrogen deficiency**, and are therefore gradually progressive and time related.

- ***The vagina*** becomes smaller, thinner, with gradual loss of its rugae, decreased vascularity, and increased vaginal PH. Vaginal smears become atrophic.
- ***The pelvic ligaments*** become weaker predisposing to pelvic organ prolapse (POP).
- ***The uterus*** becomes smaller in size. If myomas are present they undergo atrophy.
- ***The endometrium*** becomes thin and atrophic (< 5 mm thickness)
- ***The cervix*** becomes gradually flushed with vaginal fornices, and the squamo-columnar junction migrates higher in the endocervical canal.
- ***The urethra*** and ***bladder mucosa*** show loss of elasticity, bladder dysfunction, and stress incontinence (due to relaxation in the weak pelvic ligaments).

- ***The Breasts*** become gradually smaller and flabby with progressive fatty replacement of breast tissue and atrophy of active glandular elements.
- ***The skin*** shows gradual decrease in thickness and collagen content.
- *Increased* ***facial hair*** and androgenic alopecia in response to increased androgens.
- Gradual changes in ***cognitive function*** and ***mood swings***.
- *Decreased* ***mineral bone density*** leading to osteoporosis (see later)
- *Nervous* and *Psychological changes.*

CLINICAL FEATURES OF THE MENOPAUSE (Symptoms & Signs)

Menopausal manifestations are secondary to low serum E2 levels.

1. **Vasomotor Symptoms (VMS); Hot flushes & Cold sweating**
   - *Hot flushes:* are recurrent waves of heat over the chest, neck, and face, followed by *cold sweating*. A flush may last for 1-5 minutes, and may be associated by palpitation, dizziness or headaches.
   - *Flushes affect at least 50%* of menopausal women but with variable grades of severity.
   - *Flushes start in the perimenopause* and become more aggressive in the menopause.
   - *Flushes result from* inappropriate stimulation of the thermoregulatory centers in the hypothalamus with vasodilatation of the skin over the head, neck, and chest, causing a skin temperature rise although core body temperature does not change.
2. **Nervous and psychological symptoms:** Anxiety, irritability, mood changes, insomnia, and lack of concentration, are common although with variable degrees.
3. **G.I.T. symptoms:** Constipation, abdominal distension, and tendency to weight gain.
4. **Dyspareunia:** due to vaginal atrophy, dryness, and senile vaginitis.
5. **Urinary symptoms:** as frequency, dysuria, and stress urinary incontinence, may occur alone or in association with pelvic relaxation and genital prolapse.
6. **Tendency towards pelvic organ prolapse (POP)**: uterine and vaginal prolapse occur more frequently due to weakness and atrophy of pelvic and cervical ligaments.
7. **Androgenic manifestations:** increased facial hair and baldness affect women variably.
8. **Symptoms related to osteopenia and osteoporosis** (see later)
9. **Cardiovascular risks due to decreased HDL secondary to decreased E2.**

DIAGNOSIS OF MENOPAUSE

- Amenorrhea for 1 year after the FMP in premenopausal women.
- Elevated serum FSH levels (> 20 mIU/ml).
- Low E2 levels (<20mIU/ml).
- Menopausal symptoms; as flushes, cold sweats, irritability, insomnia, and weight gain.

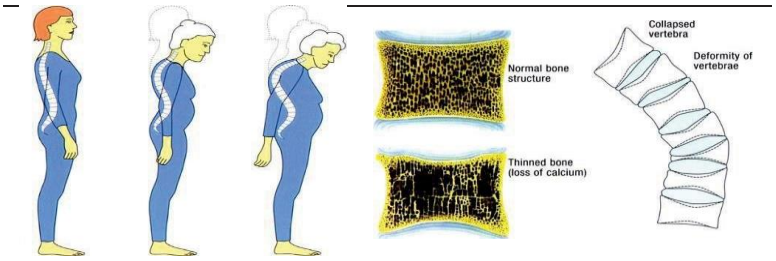


Fig 2-6: Osteoporotic changes in menopause

MANAGEMENT OF THE MENOPAUSE:

- *Reassurance* about the physiological nature of the menopausal symptoms.
- *Education for healthy life style;* Calcium rich diet, suitable exercise, and avoid smoking.
- *Symptomatic treatment for some menopausal symptoms;* Sedatives, tranquilizers, or antidepressants on individual basis whenever needed.
- *Periodic tests* for early detection of premalignant and malignant lesions including; Mammography, TVS, Pap smear cytology.
- *Diagnosis, prevention, and treatment of osteoporosis* (see later).
- *Hormone therapy (HT)* in selected cases with severe menopausal symptoms (see later).

REMOTE HEALTH HAZARDS RELATED TO MENOPAUSE

A. CARDIOVASCULAR SYSTEM CHANGES (CVS)

- ***Pathogenesis:*** Estrogen deficiency may lead to hypercholesterolemia with increased LDL and decreased HDL (reversed LDL/HDL ratio).
- ***Clinical manifestations:*** reversed LDL/HDL ratio may predispose to ischemic coronary heart disease (ICHD), myocardial infarction (MI), atherosclerosis, hypertension (HTN), and cerebrovascular strokes.

B. OSTEOPOROSIS

- ***Definition:*** Osteoporosis is a disorder characterized by decreased bone mineral density leading to compromised bone strength with increased risk of bone fractures.
- ***Pathogenesis:*** Estrogen deficiency results in accelerated bone mineral calcium loss, and increased activity of osteoclasts (bone destroying cells), affecting mainly the lumbar vertebrae, femoral neck and distal radium.
- ***Clinical manifestations:*** bone demineralization is usually a silent disease that

manifests years after menopause with; decreased height, increased curvature of the spine, silent fractures of the vertebrae, or fractures of the hip and long bones on exposure to mild trauma

- ***Risk factors:*** premature menopause, heavy smoking, lack of exercise, low body weight, together with hereditary and genetic factors (white race).
- ***Diagnosis:*** by X-ray bone densitometry (DEXA).
- ***Prevention of osteoporosis:***
  - Balanced diet with rich calcium intake of 1500 mg daily.
  - Vitamin D 600-800 IU/day.
  - Healthy life style; including weight bearing exercises, stop smoking, and avoid long term corticosteroid therapy.
- ***Treatment of Osteoporosis:***

A) Drugs that slow bone breakdown during bone remodeling

- *Biphosphonates:* orally once per week, decrease non vertebral fractures.
- *Calcitonin:* nasal spray increases the vertebral bone mass and reduce fractures risk.

B) Drugs that have anabolic effect that stimulate bone remodeling

- *Teriparatide:* IM injections over a period up to 18-24 months. It decreases vertebral and non-vertebral fractures.

C) Hormone Therapy (HT)

- *Hormone Therapy (HT):* although effective in prevention and management of osteoporosis, is rarely used as a primary therapy, except in cases which need HT for control of menopausal flushes, or in cases of premature menopause.
- *Selective estrogen Receptor Modulators (SERM)* e.g.; Raloxifene; has a combined estrogen effect on bone and anti-estrogen effect on the breasts and uterus. It is approved for prevention of osteoporosis but may induce hot flashes.
- *Phyto-estrogens:* plant substitutes that have a weak estrogen action

HORMONE THERAPY IN MENOPAUSE

Types of Hormones used

A) ***Estrogen + Progestin therapy (Combined HRT)***: where a progestin is added to avoid estrogen induced endometrial hyperplasia (EH) and possible endometrial carcinoma.

B) ***Estrogen only therapy (ERT):*** suitable for women who have undergone hysterectomy (no risk for estrogen induced endometrial hyperplasia or carcinoma).

Benefits of estrogen in the Menopause

1. It reduces menopausal symptoms as; hot flashes, sleepiness, and mood disorders.
2. Treats vaginal dryness and atrophy which may cause dyspareunia and senile vaginitis.
3. Prevents or reduces the risk of osteoporosis during the period of therapy.

Risks of Hormone Therapy

- Small but significant increase for CVS disease, stroke, and venous thromboembolism.
- Small but significant increase in the risk of breast cancer.
- Endometrial cancer risk: with prolonged estrogen only therapy (> 5 years).

AMENORRHEA

DEFINITION: The term **amenorrhea** describes absence or cessation of menstruation; it may be either;

A. **Primary Amenorrhea:** Menstruation has never occurred before:

- Menses delayed **>14** years of age in absence of developed 2ry sexual characters.
- Menses delayed **>16** years whether 2ry sexual characters has started development or not.

B. **Secondary Amenorrhea:** Cessation of menstruation for 3 successive cycles, after a period of regular cyclic bleeding.

CAUSES OF AMENORRHEA

A. PHYSIOLOGIC AMENORRHEA

1. ***Before puberty:*** due to absence of pituitary FSH / LH stimulation.
2. ***After menopause:*** due to exhaustion and depletion of oocytes in the ovaries.
3. ***During pregnancy:*** due to continuous production of estrogen and progesterone.
4. ***During lactation:*** high prolactin levels interfere with LH surge and cause anovulation.

B. PATHOLOGIC AMENORRHEA

1. ***Outflow Tract Disorders:*** imperforate hymen, transverse vaginal septum, and cervical atresia.
2. ***Uterine Disorders:*** as Mullerian agenesis, androgen insensitivity, and Asherman's syndrome.
3. ***Ovarian Disorders:*** as Turner's syndrome, premature ovarian insufficiency (POI), and PCOS.
4. ***Pituitary Gland Disorders:*** pituitary adenomas, empty sella syndrome, pituitary insufficiency.

5. ***Hypothalamic and CNS Disorders:*** as psychiatric disorders, rapid weight loss, excessive exercise, drug induces, hypothalamic tumors and infiltrative disease.
6. ***Endocrine Disorders:*** as hypothyroidism, hyperthyroidism and Cushing's syndrome.

I. OUTFLOW TRACT DISORDERS

These disorders describe cases with obstruction of the menstrual flow at the level of: hymen, vagina, or cervix. Such cases present with **1ry Amenorrhea** usually at a young age, with normal development of their 2ry sexual characters.

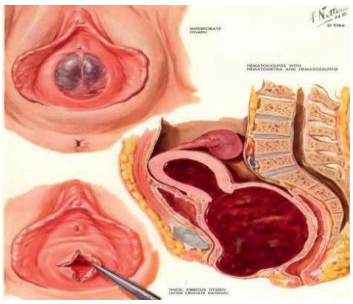
Because menstruation occurs but is not revealed this type of 1ry amenorrhea is commonly known as false amenorrhea or **cryptomenorrhea**.

A) IMPERFORATE HYMEN:

Imperforate hymen is a congenital anomaly in which the hymeneal orifice is absent, it is the commonest cause for crypto- menorrhea.

- ***Pathogenesis:*** at puberty, menstruation will start, but blood will accumulate behind the imperforate hymen leading to;
  - *Hematocolpos;* blood entrapped and accumulating inside the vagina.
  - *Hematometra;* in long standing cases blood will extend to the uterine cavity.
  - *Hematosalpinx;* blood collecting in the fallopian tubes retrograde from the uterus.
- ***Symptoms:*** 1ry amenorrhea and well developed 2ry sexual characters, lower abdominal pain with the timing of menstrual period. Progressive abdominal enlargement due to more blood accumulated each month. Urine retention may occur due to compression of the urethra from the huge hematocolpos.
- ***Signs***:
  - *Inspection and palpation of the lower abdomen;* may reveal a suprapubic bulge only if the hematocolpos is large enough to extend upwards to the suprapubic region.
  - *Inspection of the vulva & vagina:* hymeneal orifices are absent, and if hematocolpos is large the hymen will bulge at the hymeneal ring with a slight bluish color.
  - *PR Examination:* a sizable hematocolpos can be easily palpated by PR.
- ***Special Investigation:*** pelvic ultrasound is the gold standard in diagnosis.
- ***Treatment:*** once diagnosed, surgical cruciate incision is performed dividing the imperforate hymen, allowing drainage of the dark altered accumulated

blood.



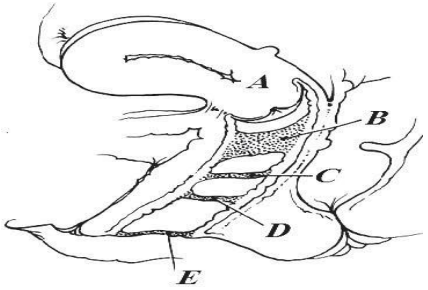


Fig 2-7: Imperforate hymen with hematocolpos and hematometra *Fig 2-8: Transverse vaginal septum site from above downward*

B. TRANSVERES VAGINAL SEPTUM

A transverse vaginal septum may be **congenitally present** at any level in the vagina (usually between the upper part developed from Mullerian duct and the lower part developed from the urogenital sinus). The condition is rare, but is the 2nd most common cause for **cryptomenorrhea.**

- ***Clinical Presentation:*** cryptomenorrhea as seen with imperforate hymen (see above)
- ***Diagnosis:*** hematocolpos diagnosed by US, while hymen is apparently normal.
- ***Treatment:*** surgical excision of the transverse septum.

C. CERVICAL ATTRESIA

Congenital cervical atresia is a very rare condition, in which there is failure of canalization of the cervical canal as a part of Mullerian dysgenesis.

- ***Clinical Presentation:*** picture of cryptomenorrhea as seen with imperforate hymen (see above)
- ***Diagnosis:*** Pelvic US reveals marked hematometra, absent cervix, and no hematocolpos.
- ***Treatment*:** is difficult and controversial. Cervical reconstruction and canalization are attempted, but with poor results, and if fails hysterectomy may be the only option.

II. UTERINE ABNORMALITIES AND DISORDERS

Uterine causes of amenorrhea are not uncommon, they include

- Complete or Partial Mullerian Agenesis (congenital)
- Complete Androgen Insensitivity Syndrome CAIS (congenital)
- Asherman Syndrome (acquired)

A) MULLERIAN AGENESIS (COMPLETE or PARTIAL)

(Mayer Rokitansky Kuster Hauser Syndrome: MRKH Syndrome):

These are **females** (**46 XX** karyotype) with congenital failure of

development of the Mullerian duct (para-mesonephric duct) structures (the uterus, cervix, and upper vagina).

- *Incidence:* accounts for almost 20% of cases with 1ry amenorrhea (2nd most common after Turner syndrome)
- *Clinical Presentation:* 1ry amenorrhea with normal 2ry sexual characteristics.

- *PV Examination (if not virgin):* reveals a blind ended short vagina, with absent cervix and uterus.
  - *Pelvic US:* reveals absent uterus, and upper vagina, with normal ovaries.
  - *Management:* a short vagina can be managed surgically by lengthening the vaginal canal through vaginoplasty as with Mc Indoe procedure.
    - **N.B.:** Associated urinary tract (mesonephric duct anomalies) is a common association.

B. COMPLETE ANDROGEN INSENSITIVITY SYNDROME (CAIS) (Testicular feminization Syndrome):

These are **male** subjects (**46 XY** karyotype) with androgen resistance due to a defect in peripheral androgen receptors.

These male subjects fail to develop testosterone dependent male sexual characteristics, inspite of normal male testosterone levels. They therefore develop phenotypically as females. *They present by* 1ry amenorrhea, female external genitalia, and a blind ended vagina.

- *Incidence:* 3rd common cause of 1ry amenorrhea after Mullerian agenesis.
- *Pathogenesis:* male gonadal tissue may be present in the labia or inguinal canal, but incapable of spermatogenesis. The presence of a Y chromosome, and Mullerian inhibiting factor (MIF) leads to failure of Mullerian system development (uterus, cervix and upper 1/3 vagina).
- *Management:*
  - Gonadectomy must be performed in such patients as they carry a risk of gonadoblastoma, however treatment is usually deferred until puberty is complete to allow for normal bone growth.
  - Creation of a neo-vagina surgically via McIndoe procedure if the patient requires, as such cases are reared as females.

- **N.B.:** Hormone therapy in the form of estrogen preparation will help keeping the external female appearance, and prevent accelerated bone loss (osteoporosis) after gonadectomy.

C. ASHERMAN'S SYNDROME (Intrauterine Synechiae)

Asherman's syndrome describes acquired intrauterine adhesions which prevent endometrial proliferation, leading to 2ry amenorrhea.

- *Etiology:* Iatrogenic via vigorous curettage during D&C procedures

especially with surgical evacuation or *TB* Endometritis (as TB affects the basal layer of endometrium).

- *Clinical presentation:* 2ry amenorrhea following curettage or endometritis.
- *Diagnosis:* both HSG and hysteroscopy are excellent diagnostic tools.
- *Treatment:* Lysis of adhesions under vision via hysteroscopy. This is followed by cyclic combined EST/PRG therapy for at least 3 cycles in an attempt to restore endometrial regeneration.

III. OVARIAN DISORDERS

A) TURNER SYNDROME: (Gonadal Dysgenesis)

In Turner syndrome, an X chromosome is missing (45-XO) and the ovaries will be replaced by fibrous tissue (streak gonads).

- ***Incidence:*** It is the **commonest cause of 1ry amenorrhea**.
- ***Clinical presentation:*** 1ry amenorrhea with low hair line, short stature, webbing of the neck, increased carrying angle at the elbow, Cardiac anomalies as; coarctation of the aorta.

Pathogenesis: Absence of Y chromosome will allow development of Mullerian elements (the vagina, uterus, and fallopian tubes), however, at puberty the streak gonads fail to produce enough E2 inspite of elevated FSH and LH levels, leading to 1ry amenorrhea.

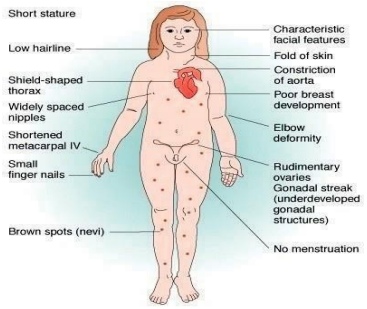


Fig 2-9: Features of Turner syndrome,

Treatment: HRT, in the form of cyclic combines EST/PRG will induce regular cycles, and preserve the 2ry sexual characters, and act as prophylaxis from bone osteoporosis. But pregnancy can't happen due to congenitally absent oocytes.

- **N.B.:** In some cases, with a mosaic karyotype (45X0/46XX), spontaneous menstruation and in some cases, pregnancy might occur, but most cases will develop premature ovarian failure (POF) and premature menopause.

B) PREMATURE OVARIAN INSUFFICIENCY (POI):

Premature ovarian insufficiency occurs due to early exhaustion of ovarian primordial follicles before the age of 45 years leading to premature menopause.

- *Etiology:*
  - Idiopathic; associated with autoimmune ovarian destruction.

- Karyotype abnormalities; as loss of a small portion of the X chromosome
- Viral infections; as mumps
- Induced; by radiation therapy, chemotherapy, or surgical removal

– *Clinical Presentation:* Premature menopause (2ry amenorrhea with high FSH & LH levels)

– *Treatment:* cases of POI should be offered HRT (combined EST/PRG) to avoid hazards and complications of premature menopause (see menopause)

- **Resistant Ovary Syndrome:** 2ry amenorrhea with high FSH and LH levels may occur when viable ovarian follicles fail to respond to pituitary gonadotropins due to a defect in their FSH/LH receptors. In few cases the condition is temporary and some may ovulate and conceive. They are differentiated from cases of POI, by the presence of normal AMH levels.

C) POLYCYCTIC OVARY SYNDROME (PCOS)

It is the **commonest cause** for **2ry amenorrhea** due to pathologic anovulation.

Clinical Presentation, Diagnosis, and treatment (see PCO)

IV. ANTERIOR PITUITARY GLAND DISORDERS

A. PITUITARY ADENOMAS (PROLACTINOMAS):

Prolactinomas are the commonest cause for hyperprolactinemia, and the most common pituitary cause for 2ry amenorrhea.

Classifications, Diagnosis and Treatment of Prolactinomas: (see hyperprolactinemia)

B. EMPTY SELLA SYNDROME:

The condition refers to enlarged sella turcica that is not entirely filled with pituitary tissue.

– *Pathogenesis:* It occurs mostly due to a defect in the diaphragma sella that allows cerebrospinal fluid (CSF) pressure to enlarge the sella.

– *Clinical Presentation:* 2ry amenorrhea with galactorrhea.

– *Diagnosis:* MRI or CT scan is the mainstay in its diagnosis.

C. PITUITARY INSUFFICIENCY

Pituitary insufficiency will result in diminished FSH and LH secretion, leading to a state of chronic anovulation with hypogonadotrophic amenorrhea. Causes are rare but include;

- Sheehan's Syndrome: (anterior pituitary necrosis following severe postpartum hemorrhage)

- Simmond's Disease.
- Radiation necrosis or pituitary infarctions.
- *Management:* according to the cause, but generally HRT with combined EST/PRG is suitable. Induction of ovulation can be attempted using FSH/hCG preparations (if pregnancy is desired), but clomiphene citrate is contraindicated due to inability of the pituitary to produce FSH and LH.

V. HYPOTHALAMIC DISORDERS

A. KALLMANN'S SYNDROME (CONGENITAL GnRH DEFICIENCY):

This is a rare condition that presents with 1ry amenorrhea in association with infantile sexual development (hypogonadotropic hypogonadism).

- Kallmann's syndrome describes congenital GnRH deficiency in association with anosmia due to congenital defect in olfactory placode.
- *Treatment:* HRT with combined EST/PRG therapy is essential to help preservation of 2ry sexual characters, uterine size, and attempt to induce uterine withdrawal bleeding. If pregnancy is desired, induction of ovulation may be attempted using FSH/LH/hCG protocols.

B. PSYCHIATRIC DISORDERS:

1. ***Emotionally stressful events:*** as family problems, work, study, travel, death or severe illness may cause 2ry amenorrhea in some women.
2. ***Pseudocyesis:*** (false pregnancy): is a rare condition that may occur in emotional women extremely desirous of pregnancy. Prolactin levels may be elevated enough to cause galactorrhea.
3. ***Anorexia nervosa:*** and Bulimia (induced vomiting) may be associated with 2ry amenorrhea.

C. RAPID WEIGHT LOSS AND EXCESSIVE EXERCISE:

1. ***Rapid and Excessive weight loss:*** a minimum of 20% of body fat is required for initiation of menarche and maintenance of menses. Rapid weight loss (below 20% of ideal body weight) may cause 2ry amenorrhea.
2. ***Excessive stressful Exercise:*** is usually associated with increased circulating endorphins and may cause 2ry amenorrhea in female athletes as marathon runners, ballet dancers, etc..., especially those with markedly decreased body fat.

D. DRUG-INDUCED AMENORRHEA:

1. ***GnRH Agonists; the*** use of long-acting GnRH agonists will induce initial stimulation of GnRH for a short period, followed by a longer suppression of FSH & LH due to a down regulation receptor effect.
2. ***Progestins;*** continuous synthetic progesterone therapy will prevent endometrial shedding, and inhibit GnRH pulses, leading to delay in the cycle and 2ry amenorrhea. Normal menses is resumed after stopping therapy.

3. ***Combined EST/PRG therapy:*** continuous therapy will also prevent endometrial shedding and inhibit GnRH pulses as long as therapy continues. **Post pill amenorrhea** may occur in 1% of women after long use of combined OCPs, due to chronically decreased gonadotropin levels. The condition is self-limiting, and normal menses is usually spontaneously resumed in 2-6 months.
4. **Androgenic drugs** as Danazol (formerly used in treatment of endometriosis) induces atrophic endometrial changes by its androgenic and progestational effects.
5. ***Antipsychotics and tricyclic antidepressants:*** stimulate prolactin secretion.

HYPERPROLACTINEMIA AND GALACTORRHEA

Prolactin is a polypeptide hormone secreted by the lactotrophic cells of the anterior pituitary gland. Its secretion is controlled by hypothalamic Prolactin inhibiting factor **(PIF)** known as **dopa-mine**. It is the hormone responsible for initiation and maintenance of lactation in females.

Hyperprolactinemia; elevated serum Prolactin levels (N: 2.9-29 ng/ml.)

Galactorrhea; refers to the continuous extrusion of milk from the nipples in the absence of recent pregnancy or lactation. It is mostly secondary to hyperprolactinemia.

ETIOLOGY

1. ***Physiologic causes:*** During pregnancy and lactation.
2. ***Drug induced:*** Phenothiazine derivatives, Reserpine, psychotropic drugs, Metoclopramide, and estrogens. These drugs act by reduction in hypothalamic secretion of dopamine (PIF).
3. ***Primary hypothyroidism:*** due to persistently elevated TRH levels.
4. ***Prolactin secreting pituitary adenomas*** (Prolactinomas);
   - Microadenomas (tumors < 1 cm) are a common cause for hyperprolactinemia.
   - Macroadenomas (tumors > 1 cm) are rare; may be associated with symptoms and signs of increased intracranial tension (headache, vomiting, and diminution in the field of vision).
5. ***Hypothalamic disorders:***
   - Severe stress, psychological conditions or tumors.

CLINICAL PICTURE

Hyperprolactinemia may clinically present by one or more of the following;

a. **Mastalgia** (breast pain and tenderness), with or without galactorrhea.
b. **Menstrual disorders** (irregular cycles, 2ry amenorrhea); due to chronic anovulation.
c. **Infertility*:*** due to anovulatory dysfunction (PRL interferes with GnRH pulses and with ovarian sensitivity to pituitary gonadotrophins).

DIAGNOSIS

1. **History:** of breast pain (mastalgia), and breast milky secretions (galactorrhea), with or with-out menstrual disturbance and/or infertility.
2. **Clinical Examination:** gentle pressure on the nipples results in extrusion of milky secretion.
3. **Diagnostic Investigations:**
   - *Laboratory:* elevated serum PRL levels (N: 2.9-29 ng/ml).
     - N.B.: markedly elevated levels of PRL > 100 ng/ml suggest PRL secreting adenomas.
   - C.T scan and MRI: in cases of persistently high serum PRL levels, or in cases with signs of increased intracranial tension suggestive of brain tumors.

TREATMENT

a. **Stop medications** that may cause hyperprolactinemia.
b. **Treat primary hypothyroidism** by thyroid hormone replacement therapy (e.g.; Eltroxin)
c. **Drugs used for treatment of hyperprolactinemia** (dopamine agonists)
   - *Bromocryptine:* 2mg (1-2 tablets daily, until normal PRL levels restored).
   - *Cabergoline:* 0.5 mg (1/2 tablet twice weekly for 4 weeks, until normal PRL levels restored).

d. Treatment of pituitary adenoma:
   - *Medical treatment:* using dopamine agonists; is the primary treatment for most cases.
   - *Surgical treatment:* are reserved to cases with failure of response to medical treatment, or Macroadenomas with CNS pressure symptoms.

OVULATORY DISORDERS AND POLYCYSTIC OVARIAN SYNDROME

Definition of ovulation: release of oocytes from mature Graffian follicle.

Etiology or FIGO ovulatory disorders classification: four Types

❖ ***Type I: Hypothalamic:***

- **Genetic**
- **Autoimmune**
- **Iatrogenic :**
  - OCPs; inhibit Gn-RH, FSH and LH production.
  - Progestin therapy and androgenic drugs as Danazol in endometriosis treatment.
  - Gn-RH agonists; as in treatment of endometriosis.
  - Drugs that induce hyperprolactinemia; e.g.; psychotropic agents, etc.
  - Antihypertensive drugs; (reserpine and alpha methyl dopa), opiates and barbiturates.
  - Chemotherapeutic agents; are sometimes followed by a transient period of anovulation
- **Neoplastic**

❖ ***Type II: Pituitary:***

- **Functional**
- **infection – inflammatory**
- **Traumatic - vascular pituitary affection**

❖ ***Type III: Ovarian:***

- **Physiologic**
- **Idiopathic**
- **Endocrine: Thyroid disorders; as hypothyroidism, Adrenal disorders; as Cushing's disease (adrenal hyperplasia)**

❖ ***Type IV: PCOS***

Clinical presentation of anovulation:

❖ Menstrual disturbances that range from amenorrhea to infrequent or irregular menstruation

❖ Primary or secondary infertility

❖ Hirsutism or other features or findings associated with PCOS.

DIAGNOSIS OF ANOVULATION

A. **History:** suggestive of irregular cycles, with periods of 2ry amenorrhea or oligomenorrhea is strongly suggestive of anovulatory dysfunction, but not diagnostic**.**

B. **Symptoms:** Absence of symptoms suggestive of ovulation, as primary dysmenorrhea, midcycle pain, or spotting, is only assumptive but not reliable in diagnosis of anovulation.

C. Test for detection of ovulation.

TESTS FOR DETECTION OF OVULATION

1. Basal body temperature charts (BBT):

A daily record of the patient's oral temperature is taken in the morning just before rising from bed in two or three successive cycles. A temperature rise is expected to occur in the luteal phase of ovulatory cycles due to **progesterone induced thermogenic effect**. These charts although cheap and easily applicable, yet have the disadvantage of being easily biased by other factors that may cause any rise in BBT throughout the month.

- ***Ovulatory cycles*** show a **biphasic BBT** chart with a rise in temp (0.2–0.3 C) in the luteal phase of the cycle.
- ***Anovulatory cycles*** will show a flat **monophasic BBT** chart due to absence of thermogenic progesterone thermal effect.

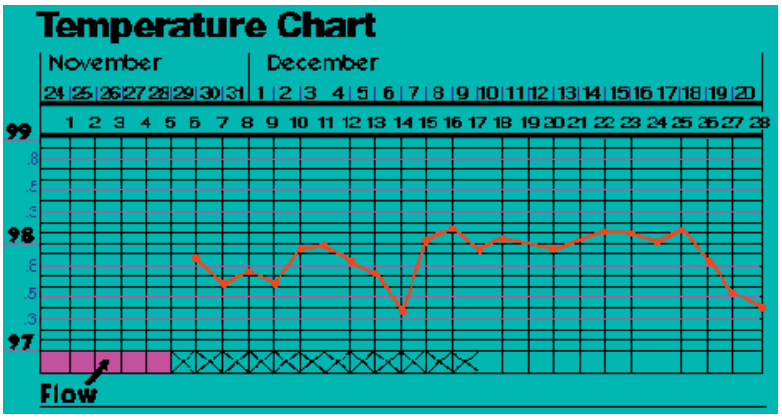


Fig. 2-10: Biphasic BBT

2. Ultrasound monitoring of ovulation (Folliculometry):

Serial TVS scans, in the follicular phase, allow monitoring growth, maturation, and ovulation (rupture of dominant follicles at a mean diameter of 18-24 mm), and tracing early CL formation.



Fig. 2-11: Preovulatory follicle by TVS

3. Mid luteal serum progesterone assay (MLSP):

MLSP is performed at day 21 of the cycle (maximum CL activity).

- Levels <5 ng/ml suggest anovulation.
- Values >10 ng/ml suggest ovulation with adequate CL function.
- Levels 5-9 ng/ml suggest ovulation but with inadequate CL function (LPD).

4. Urinary LH kits: detect timing of LH surge that immediately precedes ovulation.

5. Premenstrual endometrial biopsy (PEB):

PEB can be performed without cervical dilatation as an office procedure, using a Pipelle, or under general anesthesia if cervical dilatation is mandatory (see D&C operation). PEB done 2-3 days prior to menstruation will reveal either;

- *Secretory endometrium,* in ovulatory cycles due to progestational effect ofCL.
- *Proliferative endometrium*, in anovulatory cycles.
- *Poor secretory changes,* in cases of LPD (luteal phase defect).

TREATMENT OF ANOVULATION

Induction of ovulation can be achieved using oral drugs as Clomiphene Citrate or IM injection of human Pituitary Gonadotropins & Human Chorionic Gonadotropins. It can also be achieved surgically via laparoscopic ovarian drilling (LOD) in special cases.

A) MEDICAL INDUCTION OF OVULATION

1. CLOMIPHENE CITRATE: (Oral tablets)

Clomiphene Citrate (CC); is a non-steroidal compound with estrogenic and non-estrogenic effect.

- ***Mode of action:*** It competes with endogenous E2 for its hypothalamic receptors leading to increasing pituitary FSH production, that directly stimulates follicular growth and maturation.
- ***Dosage:*** 50 mg oral tablets, twice daily for 5 days starting day 2- 5 of menstruation.
- ***Side effects:***
  - Vasomotor flushes, nausea, headache, and visual disturbances.
  - Increased risk of twin pregnancy (10%), and of multifetal pregnancy (1%).
  - Ovarian hyperstimulation (OHSS), grade I-II, with mild pelvic pain and discomfort.
  - Anti-estrogenic effects on the endometrium may cause LPD, and on cervical mucous may cause hostile cervical mucous.

2. PITUITARY GONADOTROPINS (FSH & LH DERIVATIVES): (repeated I.M. injections)

A. **Human menopausal Gonadotrophins (HMG):** derived from urine of menopausal women. It contains **75** IU FSH + **75** IU LH.

B. **Purified Urinary FSH:** derived from urine of menopausal women and contains only one IU of LH (**75** IU FSH + **one** IU LH). They are more suitable for cases with high endogenous LH as those of PCOD, to minimize possibility of severe OHSS, and in IVF/ICSI protocols.

C. **Synthetic FSH:** containing **no** LH is prepared by **recombinant DNA technology**, used mostly in IVF/ICSI protocols in same indications as purified urinary FSH.

- ***Mode of action of gonadotrophins****:* direct stimulation of growth and maturation of ovarian primordial follicles.
- ***Indications:***
  - CC resistant cases
  - Cases with hypogonadotrophic anovulation (type I & II) (low FSH / LH levels)
  - In ICSI/IVF/ET protocols to stimulate growth of multiple follicles.
- ***Dosage:*** repeated I.M. Injections given from early-follicular phase of the cycle until complete follicular maturation. Doses are repeated with TVS monitoring of follicular growth to avoid the complications of severe OHSS (see later).

- ***Side effects:*** **Same as those with CC however;**
  - **OHSS is more frequent especially** severe forms (grade III-IV OHSS).
  - Marked increased risk of twin and multifetal pregnancy (10-30%)
  - No adverse effects on cervical mucous or endometrium.

3.HUMAN CHORIONIC GONADOTROPINS (hCG): (single I.M. injection)

- ***Preparation and action:*** hCG is prepared from urine of pregnant women. It has a strong LH action that induces an artificial LH surge leading to ovulation & oocyte release.
- ***Indication:*** to assist ovulation especially in CC or HMG induced cycles.
- ***Dosage and timing:*** 2 ampoules 5000 m/IU each, given by I.M. injection as one shot after full oocyte maturation (size of dominant follicle > 18 mm by TVS).
- ***Side effects:*** OHSS, especially if given after HMG (see later).

4.GONADOTROPIN RELEASING HORMONE (GN-RH) ANALOGUES:

- ***Preparation:*** **GnRH agonists** are synthetic decapeptide forms of GnRH, given by repeated S.C. injection.
- ***Dosage & Mode of action:*** They induce initial FSH stimulation, followed by down regulation of hypothalamic receptors, with consequent suppression of FSH & LH production.
- ***Indications:*** They are used in IVF & ICSI protocols.

5.COMBINED THERAPY

- **CC/HMG/hCG:** Adding CC to HMG/hCG has the benefit of minimizing the dose of HMG thus reducing both the cost and chanced for OHSS.
- **GnRH/HMG/hCG** is used in practice in various doses and different protocols of stimulation especially in IVF/ICSI stimulations protocols.

6. ADJUVANT DRUGS USED TO ASSIST IN INDUCTION OF OVULATION:

A. **Bromocryptine:** or **Cabergoline:** in cases of hyperprolactinemia.
B. **Metformin:** in cases with insulin resistance as PCOS.
C. **Thyroid extract:** Eltroxin in cases of hypothyroidism.
D. **Corticosteroids:** in cases of Addison's disease, adrenogenital syndrome, and some cases of PCOS.

B) SURGICAL PRODECURES USED FOR INDUCTION OF OVULATION

- *Laparoscopic Ovarian drilling (LOD):* LOD is a procedure in which a diathermy needle is used to make multiple small punctures in the surface of the ovary via laparoscopy.

- *Effect of drilling:* LOD may decrease ovarian androgen environment resulting in spontaneous ovulation.
- *Indications:* LOD is reserved to selected PCOS cases that are resistant to ovulation induction.
- *Disadvantages:* The use of LOD is hindered by its time limited effect on ovulation (3-6 months), its potential for creating peritubal adhesions, and decreasing the ovarian follicular reserve.

POLYCYSTIC OVARY SYNDROME (PCOS)

- In 1935 Stein and Leventhal described a group of patients showing **cystic changes in their ovaries** in association with **amenorrhea, hirsutism, obesity,** and **infertility.**
- **PCOS** is the more recent description of the syndrome which is characterized, according to ***Rotterdam criteria***, by **2** or more of:
  - **Evidence of oligo-anovulation:** presenting clinically by 2ry amenorrhea or oligomenorrhoea
  - **Clinical or biochemical Hyperandrogenism**; (hirsutism, elevated serum LH, and increased free Testosterone levels)
  - **Characteristic ultrasound morphology**; increased ovarian size and volume, with peripherally arranged small follicles in a dense stroma (necklace appearance), all are called Adam's criteria.



Fig. 2-12: PCOS: The peripherally arranged multiple small follicles (Necklace appearance) Enlarged ovaries with pearly white smooth surface as seen during laparoscopy

PREVALENCE AND ETIOLOGY: PCOS may affect up to **5-10%** of women in the reproductive age.

- It is **the commonest** ovarian cause for 2ry amenorrhea, chronic anovulation, and infertility.
- The exact **etiology** of PCOS remains unclear however a familial background is suggested with a *special gene* related to its development.

PATHOGENESIS OF PCOS:

PCOS is characterized by **a triad of**; high LH, high serum androgen, and high plasma insulin:

A.High LH:

- Etiology: chronic anovulation (so absent PRG, ie unopposed E2), so no

negative feed-back on LH, resulting in an abnormally high LH/FSH ration (> 2:1).

B. Hyperandrogenemia:

- Etiology: it is due to stimulation of theca cells (by both high LH and high insulin levels), together with inhibition of aromatase enzyme.

C. Hyperinsulinemia:

- Etiology: PCOS may be associated with peripheral insulin resistance disorder leading to hyperinsulinemia.

DIAGNOSIS OF PCOS:

A. Clinical presentation:

- 2ry amenorrhea, oligomenorrhea, and or infertility (chronic anovulation).
- Hyperandrogenism (Hirsutism affecting face, chest and inner thighs)
- Obesity with tendency to abnormal glucose tolerance.
- DUB may occur in cases with chronic anovulation associated with endometrial hyperplasia.

B. Laboratory investigations:

- Elevated LH levels with normal FSH lead to an abnormal LH/FSH ration of >2.
- Elevated levels of plasma E1, androstenedione, and free Testosterone.
- Hyperinsulinemia due to associated increased insulin resistance.

C. Ultrasound diagnosis: *(Adam's criteria)*

- The ovaries are increased in size and volume.
- The ovaries show central dense stroma surrounded by small follicles (2-10 mm in diameter) peripherally arranged giving the characteristic *necklace appearance*.
- No dominant or mature follicles are present due to chronic anovulation.
- US picture of PCOS may be encountered in up to 25% otherwise normal females.



Fig. 2-14: PCOS by TVS



Fig 2.13: PCO by US



Fig. 2-15: PCOS by Laparoscopy

D. Laparoscopic appearance: Enlarged ovaries with thick capsule, and absent gyrii due to absence of ovulation stigma on ovarian surface (*oyster shell*

ovary).

LONG TERM RISKS OF PCOS:

1. Increased risk for diabetes mellitus, obesity, hyperlipidemia, and cardiovascular disease.
2. Increased risk of endometrial hyperplasia (EH); due to increased estrogen effect on the endometrium unopposed by progesterone.
3. Increased risk of endometrial carcinoma if atypical EH develops.

MANAGEMENT OF PCOS

1. **Weight reduction;** In obese females a reduction of 5-10% of body weight, reduces insulin and androgen and improves response to therapy, and may by itself re-establish ovulation.
2. **Hormone therapy:** In cases with menstrual disorders as amenorrhea or DUB;
   a. Cyclic gestagen therapy for 10 days every cycle (day 16-25), to induce a regular cycle.
   b. Combined OCP (day 5-25), to establish regular cycles in cases not requesting pregnancy.
3. **Induction of ovulation for infertility** (see ovulatory disorders).
4. **Insulin sensitizing drugs** (metformin 500 mg/day orally) improve insulin sensitivity.
5. **Corticosteroid therapy** to improve folliculogenesis in resistant PCO.
6. **Surgical laparoscopic ovarian drilling (LOD)**; aims at decreasing ovarian androgen production.
7. **Treatment of Hirsutism: see later**

LUTEAL PHASE DEFECT

Luteal phase defect (LPD); is a condition in which the endometrium shows poor secretory changes of ovulatory cycles. These cycles are characterized by either low progesterone levels in the luteal phase (5-10ng/ dl at D21), or short luteal phase duration (<11 days) or both.

Sporadic LPD may occur in isolated cycles in normally ovulating women while recurrent LPD may be a cause of infertility or recurrent first trimesteric abortion.

Etiology

a. Inadequate follicular phase FSH production or inadequate FSH/LH ratio at time of ovulation.
b. Hyperprolactinemia (decrease FSH and blunted LH surge)
c. Prolonged use of CC for induction of ovulation (decreased endogenous FSH release)

Diagnosis

- Biphasic BBT chart showing a short luteal phase.
- Low midluteal serum P levels (5-10 ng/ml).

- Dated PEB showing secretory changes with a lag > 2 days behind that for a normal cycle.

Treatment

- Progesterone support throughout luteal phase.
- Bromocryptine or Cabergoline for cases with elevated serum PRL

HIRSUTISM

DEFINITIONS

- **Hirsutism;** Excessive growth of *androgen dependent sexual hair* (present on the upper lip, chin, inner thighs, limbs, chest, abdomen and pubic triangle).
- **Hypertrichosis;** Excessive growth of *androgen independent hair* (as in the forearm and legs)
- **Virilization;** is hirsutism associated with other signs of **hyperandrogenism** such as; increased muscle mass, clitoromegaly, temporal baldness, and voice deepening.

ETIOLOGY

1. **Idiopathic:** increased hair follicles sensitivity to normal female androgen levels
2. **Adrenal gland causes:**
   - Congenital adrenal hyperplasia.
   - Adrenal tumors: secrete DHA, DHAS, and rarely Testosterone.
3. **Ovarian causes:**
   - PCOS.
   - Androgenic ovarian tumors as Sertoli Leydig cell tumor or hilar cell tumor.
4. **Mixed ovarian & adrenal hyperandrogenism: (30-40% of cases)**
5. **Pituitary gland:**
   - Cushing's syndrome due to increase production of ACTH.
   - Acromegaly due to increased production of GH.
6. **Androgenic drugs:** Rarely, long term use of drugs with androgenic side effects (as Danazol in endometriosis) may cause hirsutism and virilizing effects.
7. **Hypothyroidism and hyperprolactinemia.**

INVESTIGATIONS OF A CASE OF HIRSUTISM

a. **Hormonal assays**
   1. Plasma androgen hormones: total and free testosterone.
   2. Pituitary hormones: FSH, LH to detect PCOS
   3. DHEA and DHEAS, their elevation suggests adrenal tumor.
   4. Thyroid function tests and prolactin level.

b. **Radiological investigations:**
   - CT or MRI on the pituitary gland.
   - IVP and abdominal US for adrenal tumor.
   - Pelvic US, for PCOS, and virilizing ovarian tumors.

TREATMENT OF HIRSUTISM

- A combination of hormonal suppression of hair growth and mechanical hair removal offer the most complete and effective treatment for patients with hirsutism.
- Response to treatment starts after 6-8 months of continuous treatment due to hair growth cycle.
- Permanent hair removal is best postponed until suppression of excess serum androgens.

1.Elimination of specific causes

- Removal of androgen secreting ovarian or adrenal tumors.
- Elimination of drugs suspected to contribute to the abnormal hair growth.
- Treatment of Cushing syndrome, thyroid disease, or hyperprolactinemia.

2.Hair removal techniques

- Shaving, tweezing, waxing, and use of depilatories, performed at repeated intervals.
- Permanent destruction of hair follicles by electrolysis or by laser is reasonably effective.

3.Suppression of androgen synthesis

- Oral contraceptive pills (OCPs), containing combined low dose E and P, decrease ovarian androgen production, increases SHBG, and decrease free T levels.
- Spironolactone is an aldosterone antagonist used frequently as a diuretic that also inhibits 5 alpha reductase and variably suppresses ovarian and adrenal synthesis of androgen.
- Cyproterone acetate is a potent progestin and antiandrogen that inhibits LH and decreases androgen levels. **Diane 35** is an OCP that uses cyproterone acetate as a progestin and is widely used in treatment of hirsutism in females that request contraception.

4.Androgen receptor blockers

- Androgen receptor blockers inhibit binding of DHT to androgen receptors thus directly inhibiting hair growth as Cimetidine or Finasteride.

3 INFERTILITY AND ASSISTED REPRODUCTION

OVERVIEW

Infertility (or subfertility) is the term that describes **failure** of a couple to achieve **conception**
after one year of regular sexual intercourse, without using any method of contraception.

Almost **15%** of couples may suffer infertility after the first year of marriage. Problems may be found either in the **female** or in the **male** partner. In about **25%** of cases problems will be found in **both partners** at the same time.

The **cumulative pregnancy rates** increase by time to reach up to **50%** in 6 months, and up to
85% after one year of regular marital life.

- **Infertility** may be either primary or secondary;
  - ***Primary infertility:*** pregnancy has never been achieved before.
  - ***Secondary infertility:*** history of previous one or more pregnancies, but failure to conceive again.

Investigations for infertility should be carried out on both male and female partners, and should cover all the possible factors that may be responsible for delaying pregnancy in each of them.

FACTORS RESPONSIBLE FOR INFERTILITY

- Male factor (40%)
- Female factor (50%)
- Coital dysfunction
- Unexplained Infertility (10%)

MALE INFERTILITY

The male is responsible alone for up to 40% of infertility cases, and should be evaluated separately and thoroughly in every couple presenting with infertility, before going into invasive female investigations.

ETIOLOGY:

A) Abnormal Spermatogenesis

1. *Increased Scrotal temperature:* (scrotal temperature should be 1□C less than body temperature): as in Undescended testes, varicocele, mumps orchitis.
2. *Genetic Causes:* as in micro-deletion of the Y chromosome.
3. *Drug Induced:* as psychotropic or chemotherapeutic agents.

B) Failure of Sperm Transport through the Vas deferens**:**

1. *Bilateral Epididymal Obstruction:* Gonorrheal inflammation is the commonest cause.
2. *Bilateral Vas Ligation:*
   - Intended: as in male sterilization.
   - Iatrogenic: rarely faulty ligation during bilateral inguinal hernioplasty operations.
3. *Immotile Cilia Syndrome:* All sperms are immotile.

C) Failure of Semen Deposition: (due to ejaculatory dysfunction):

- Erectile dysfunction.
- Premature ejaculation.
- Retrograde ejaculation, and anejaculation.

DIAGNOSIS OF MALE FACTOR:

A) History:

- Related operative procedures: repair of inguinal hernia or varicocele.
- Infection by Mumps, STDs; as gonorrhea, chlamydia, etc...
- Coital problems; as erection and ejaculation disorders.
- Heavy smoking; due to decreased oxygen tissue perfusion.
- Chronic disease as Diabetes Mellitus.
- Chronic drug intake; as psychotropic drugs, chemotherapeutics etc...

B) Clinical Examination:

- To exclude: undescended testicles, testicular masses, and small testes.
- To detect: the presence of hydrocele, and/or varicocele, and their extent.

C) Special Investigations:

1. *Semen Analysis:* should be obtained from all males during infertility investigations. Samples are collected by masturbation after 3-4 days of abstinence from intercourse.
2. *Hormonal assay:* FSH, LH, Prolactin, Testosterone.
3. *Doppler US:* on the testicles to detect varicocele.
4. *Testicular biopsy:* in cases with azoospermia, to differentiate between defective spermatogenesis and obstructive disorders.

CRITERIA FOR NORMAL SEMEN ANALYSIS (WHO 2010)

Parameter	Normal Criteria
Volume	Equal or more than 2ml
Count	Equal or more than 15 million/ ml
PH	Alkaline: 7.2 – 7.8
Motility	Equal or more than 50% with active forward progressive motility
Morphology	Equal or more than 30% of normal morphology

Term	Definition
Aspermia	Absence of semen
Azoospermia	Zero sperm count
Oligospermia	Count < 15 million / ml
Asthenospermia	< 50% with forward progressive motility
Teratospermia	> 70% abnormal forms

MANAGEMENT OF MALE INFERTILITY:

Improvement in poor semen parameters may take from 3-6 months of appropriate therapy.

A. **Stop** smoking (to improve oxygenation).

B. Use of **Multivitamins** and **Antioxidants,** are thought to improve sperm count and motility.

C. **Treat erection disorders;** using medications (as Viagra), or synthetic prosthesis.

D. **Shift** to other types of psychotropic drugs if they were a cause.

E. **Hormonal treatment:** CC & HMG are useful in cases of defective spermatogenesis.

F. Surgical treatment:

- Ligation of varicocele: may improve sperm count and motility in some cases.
- Artificial prosthesis: in some erection disorders

G.*Assisted Reproductive Techniques (ART): *see later for details

FEMALE INFERTILITY

Females are **responsible alone** for delayed conception in nearly **50%** of cases. **Anovulation, tubal damage**, and **peritoneal adhesions** are the most frequently encountered causes. **Uterine** and **endometrial** factors are less common, while **cervical factor** is rare.

FACTORS CAUSING FEMALE INFERTILITY

I. OVARIAN FACTOR

. **Ovulatory disturbance** is the most commonly encountered factor for female infertility being the primary cause in almost 40% of cases: (See ovulatory disorders)

- **Luteal phase defect:** (see LPD)
- **Age of the female** is the single most important factor in anovulatory infertility, where fertility declines rapidly after the age of 35 years and more sharply after the age of 40 years.

II. TUBAL AND PERITONEAL FACTORS

Tubal and Peritoneal factors may be responsible for almost **30** % of female infertility.

- ***The following mechanisms may be responsible:***
  - *Impaired ovum pick-up;* due to fimbrial destruction and / or peritubal adhesions.
  - *Impaired oocyte or embryo transport;* due to tubal obstruction.
  - *Tubal epithelial damage;* with absent nourishment to the oocyte during transport

1. **Salpingitis, Salpingo-oophoritis, and Pelvic Peritonitis:** causing tubal damage & pelvic adhesions
2. **Surgical Trauma (Iatrogenic):** leading to direct tubal damage or peritubal adhesions
   - Salpingectomy (ectopic pregnancy), tubal ligation (for sterilization), or tuboplasty.
   - Gynecologic surgery; after myomectomy, and ovarian cystectomy.
   - Pelvic surgery; as during surgery for intestinal obstruction and appendectomy.
3. **Pelvic Endometriosis:** extensive disease may cause pelvic and peritubal adhesions.

III. UTERINE AND ENDOMETRIAL FACTORS

Uterine and endometrial factors may be responsible for **10-15%** of female causes of infertility, mostly via the following mechanisms;

- Impairment of blastocyst implantation within the endometrial cavity.
- Bilateral cornual obstruction of tubal ostia.
- Interference with transport of sperms through the endometrial cavity.

1. **Congenital Uterine Anomalies:**
   - Septate uterus (may cause infertility and or RPL)
   - Uterine hypoplasia and or aplasia (may cause amenorrhea and or infertility)

2. **Uterine Leiomyomas:** especially SMF (submucous fibroid) causing failure of implantation
3. **Uterine Polyps:** SMF polyps or endometrial polyps.
4. **Intrauterine Synechiae** predisposed to by;
   - Over curettage of basal endometrial layer (Ashermann's syndrome)
   - Chronic specific infections (e.g.; T.B. endometritis)

IV. CERVICAL FACTOR

Abnormalities of cervical mucus that affect its amount, quality, may impair sperm transport through the cervical canal making it incapable of fertilization, via the following mechanisms;

- Changes in physical and chemical properties, e.g.; scanty amount or increased viscosity.
- Presence of anti-sperm antibodies in cervical mucus.

INVESTIGATIONS FOR FEMALE INFERTILITY

In the **apparently normal couple**, investigations for infertility do not usually start before **1 year** of regular marital life without contraception in 1ry infertility, and 2 years in 2ry infertility. **Earlier investigations** may be started whenever;

- *Age* of the female partner is *above 35 years.*
- History reveals marked *menstrual disorders.*
- History strongly suspicious of *uterine, tubal, or peritoneal pathology.*
- Suspicion of a severe *male factor.*

Female evaluation usually starts by a detailed **history**, general and local **examination**, pelvic **US**, **HSG,** and a **hormonal assay**. If no apparent cause was revealed, and the male **semen analysis** was normal, then more invasive tests as **Laparoscopy,** and **hysteroscopy,** and will be required to evaluate **uterine**, **tubal** and **peritoneal factors**.

In 10-15% of cases investigating both partners by standard tests may not reach a cause, a condition known as unexplained infertility (see later).

I. ASSESSEMENT OF OVARIAN FACRTOR: (see ovulatory disorders)

II. ASSESSMENT OF TUBAL AND PERITONEAL FACTORS

1. **HYSTEROSALPINGOGRAPHY (HSG):** HSG is considered **an essential primary investigation** in evaluation of tubal factor of infertility, it also gives an idea of uterine factors.

   This consists of the injection of radio-opaque substance into the uterine cavity & fallopian tubes using a special cannula introduced into the cervical canal & an X-ray film is taken after the injection & another film could be taken after 24 hrs to show spill of the dye from the tubes to study tubal & peritoneal factors.

The dye used is either oil soluble e.g. Lipiodol (40% organic iodine in poppy seed oil) or water soluble e.g. Urographin or renographin (in which situation the second film is taken just 20 minutes after removal of the cannula as they are rapidly absorbed)

Timing: immediately postmenstrual **Contraindications:**

1. Pelvic infection 2. Suspicion of pregnancy
3. Allergy to iodine 4. During menstruation or uterine bleeding

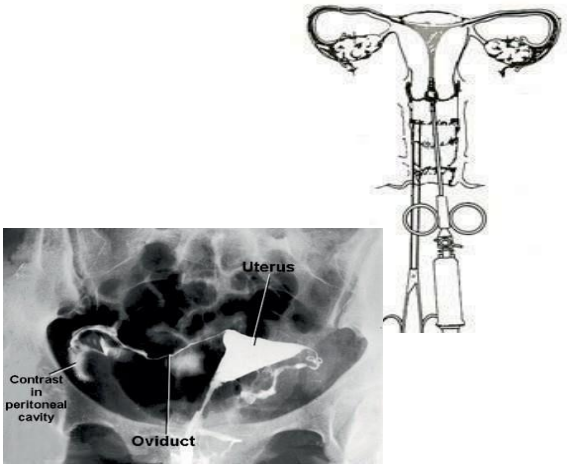


Fig 3-1: HSG

Complications of H.S.G.:

1. Shock.
2. Oil embolism (with lipiodol).
3. Disturbance of an undiagnosed pregnancy
4. Flaring up or introduction of infection. 5. Iodine allergic reactions.
6. Intravasation (the oil passes into the uterine vessels during injection which is mainly lymphatic & rarely venous)

HSG allows diagnosis of;

- Mullerian anomalies: as septate, bicornuate, and unicornuate uterus.
- Endometrial pathology: as polyps, SMM, and intrauterine adhesions.
- Diagnosis of tubal obstruction: with delineation of its site, nature, and bilaterality
- Diagnosis of peritubal and peritoneal adhesions, in the post dye injection control film.

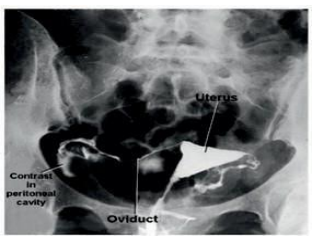


Fig 3-2: HSG showing normal uterine cavity with no filling defects, and bilateral tubal patency

2. **LAPAROSCOPY AND DYE INJECTION: (**see endoscopy in gynecology)

Laparoscopy is **the gold standard** for evaluating tubal factor of infertility.

It allows direct visualization of the pelvic peritoneum, uterus, and external surface of the fallopian tubes. Injection of methylene blue dye through the cervix, allows for observing its spillage through the fimbrial ends of fallopian tubes to ensure tubal patency, and evaluate its ciliary function. The procedure is performed under general anesthesia.

III. ASSESSMENT OF UTERINE FACTOR:

1. **PELVIC ULTRASOUND (US):**

TAS and **TVS** are considered the *gold standard* in evaluation of uterine factor. Introduction of

3D US and Saline Infusion Sonography **(SIS)**, has improved the accuracy of US diagnosis:

- Congenital uterine anomalies; as septate and bicornuate uterus (TVS & 3D US)
- Uterine leiomyomas, endometrial polypi: (TAS, TVS, & 3D US)
- Intra uterine adhesions (SIS)
- Associated adnexal masses (TAS & TVS)

2. **HYSTEROSALPINGOGRAPHY: see before**

3. **HYSTEROSCOPY:** see endoscopy in gynecology

Hysteroscopy allows visualization of the inside of the uterine cavity, the presence of any polypi, in addition to visualization of the tubal ostia.

ASSESSMENT OF CERVICAL FACTOR

The Post Coital Test (PCT): rarely performed nowadays

- Examination of the cervical mucus directly after intercourse, at time of ovulation, to assess the number of living and dead sperms and presence of leucocytes, affecting sperm motility within the cervix.
- An abnormal PCT may point to hostile cervical mucus, or presence of antisperm antibodies, yet its value has been recently questioned being of limited prognostic value.

HORMONAL ASSAYS COMMONLY REQUIRED FOR INFERTILITY

- **Serum FSH & LH** (day 1-3 of cycle): abnormal LH/FSH ratio in PCOS.
- **Mid luteal Serum PRG:** usually day 21 of the cycle (> 10ng/ml denote ovulation)
- **Serum PRL: at any time** (N: 2.8-29 ng/ml); high levels denote hyperprolactinemia
- **Thyroid function tests: at any time** (TSH, T3, T4) to detect major thyroid disorders

COITAL DUSFUNCTION AFFECTING FERTILITY

Coital dysfunction may prevent penetration and deposition of male sperms into the vagina. It may also affect the frequency and efficiency of sexual intercourse.

a. ***Superficial Dyspareunia:*** painful or difficult coitus at the level of the vulva and lower vagina.

b. ***Vaginismus:*** violent reflex spasm of levator ani, gluteus muscle, and adductors of the thigh on any attempt at intercourse.

c. ***Male Erectile dysfunctions:*** and premature ejaculation, etc…

N.B.: Effluvium seminis (excessive escape of the seminal fluid from vagina after intercourse) is a common complaint that does not really affect or delay fertility.

MANAGEMENT OF FEMALE INFERTILITY

After the cause of female infertility has been apparently revealed, treatment will be initiated according to the cause discovered.

MANAGEMENT OF OVARIAN FACTOR:

Induction of ovulation (See ovulatory disorders)

MANAGEMENT OF TUBAL AND PERITONEAL FACTORS

Tubal occlusion can usually be anatomically corrected by various surgical techniques however restoration of tubal function 2ry to tubal damage could not be easily achieved.

a. Laparoscopic Procedures:

- **Lysis of Peritoneal Adhesions (Adhesiolysis)**
- **Excision & Ablation of Pelvic Endometriosis**

B. Laparotomy for Tuboplasty:

The role of laparotomy for performing tuboplasty procedures, as fimbrioplasty and neosalpingostomy, has largely diminished in the last decade in favor of much better success rates for **IVF- ICSI procedures** after tubal disconnection (TD) or salpingectomy for hydrosalpinx.

MANAGEMENT OF UTERINE FACTOR

1) ***Operative hysteroscopy*** can deal successfully with many uterine abnormalities in infertility;

- *Hysteroscopic Septoplasty:* resection of uterine septum (septate uterus)
- *Hysteroscopic Polypectomy:* excision of endometrial polyps
- *Hysteroscopic Myomectomy:* excision of certain types of SMM and SMF polyp
- *Hysteroscopic Adhesiolysis:* division of thin intrauterine synechiae.

2) ***Myomectomy via Laparotomy or laparoscopy:*** indicated in large

leiomyomas causing infertility.

MANAGEMENT OF CERVICAL FACTOR

- ***Treatment of the cause***; as chronic cervicitis, hostile cervical mucus.
- ***Use of estrogen and mucolytic drugs*** to improve cervical mucus physical properties.
- ***Intrauterine Insemination IUI:*** injection of processed semen directly within the endometrial cavity bypassing the cervix.

UNEXPLAINED INFERTILITY

Unexplained infertility is a diagnosis of exclusion. It may be present in almost 15% of infertile couples when all standard male and female infertility investigations fail to detect a possible cause. Almost 60% of these couples will conceive within 3 years without treatment. This diagnosis is reached after having normal primary infertility workup, in addition to a normal diagnostic laparoscopy/ hysteroscopy.

Possible Contributing Factors

- Immunological and psychological factors.
- Defective sperm fertilization capacity.
- Decreased ovarian reserve: decreased number and quality of oocytes (day 3 FSH level > 10 mIU/ml, and low AMH < 1ng/ml).
- Occult cervical infection.

MANAGEMENT OF UNEXPLAINED INFERTILITY

- Reevaluation of both partners and obtaining more recent investigations.
- Further tests, not previously performed, as diagnostic laparoscopy and hysteroscopy.
- Serum Chlamydia antibodies, and cervical mucus assessment.
- Repeated IUI, for at least 3 cycles, may improve the chances of conception.
- Empirical Ovulation induction protocols.
- IVF & ICSI procedures are the last resort, but with best prognosis.

ASSISSTED REPRODUCTIVE TECHNIQUES

Assisted reproductive techniques (ART), will be indicated in cases with severe male factor, severe tubal and peritoneal factors, and in some cases of unexplained infertility. ART includes;

- ***IUI:*** Intra Uterine Insemination
- ***IVF-ET:*** In Vitro Fertilization & Embryo Transfer, in which Intra Cytoplasmic Sperm Injection **(ICSI)** is the most commonly applied technique.

OVARIAN HYPER-STIMULATION SYNDROME (OHSS)

OHSS is an **iatrogenic disorder** that describes the occurrence of ovarian

multicystic enlargement secondary to the use of **drugs for induction of ovulation**, namely HMG/hCG. It is classified into four grades according to severity.

- **Mild forms (Grades I-II);**
  - These cases may be associated with variable degrees of abdominal pain and distension, edema, mild ascites, hypotension, and oliguria.
  - Cases are managed by rest, reassurance, and symptomatic treatment
- **Severe forms (Grades III-IV);**
  - These cases may be associated with fluid shift from intravascular to extravascular compartment leading to severe ascites, pleural effusion, hypoalbuminemia, electrolyte imbalance, hemoconcentration and changes in blood coagulability with tendency to DIC and increased risk of pulmonary embolism.

    These life-threatening cases are managed by **hospitalization**, restoration of fluid and electrolyte imbalance, and sometimes tapping for aspiration of ascites. Diuretics are avoided as they will increase hemoconcentration with risk of thrombosis.
  - **Prevention of severe OHSS** could be achieved by both careful TVS monitoring of HMG doses during stimulation, and withholding hCG injections in cases with Gr II-III hyperstimulation.



Fig 3-4: OHSS

4 FEMALE SEXUAL DYSFUNCTION AND FEMALE GENITAL MUTILATION

OVERVIEW OF SEXUAL HEALTH

Sexual health is an essential component of overall health and well-being. According to WHO, sexual health is defined as physical, psychologic, and sociocultural wellbeing related to sexuality.

Unfortunately, women's sexual concerns often go unassessed. Patients may hesitate to initiate conversations regarding this topic. And for clinicians, insufficient medical training can lead to a lack of confidence in their knowledge or ability to respond to a patient's sexual concerns.

FEMALE SEXUAL RESPONSE

Female sexual functioning encompasses multiple aspects of the sexual response cycle including desire, arousal, lubrication, and orgasm. FSD is characterized by a distressing impairment in a specific domain of sexual response.

FEMALE SEXUAL DYSFUNCTION (FSD)

Etiology and classification of FSD:

There are **six categories** of FSD:

1. *Hypoactive sexual desire disorder (HSDD)*: persistent lack of desire.
2. *Sexual aversion (avoidance) disorder:* persistent aversion to sexual activities.
3. *Female sexual arousal disorder (FSAD)*: persistent lack of lubrication in response to sexual excitement.
4. *Female orgasmic disorder:* persistent lack of orgasm following sexual excitement.
5. *Dyspareunia:* persistent genital pain with intercourse.
6. *Vaginismus:* persistent involuntary muscle spasm in the outer third of the vagina with intercourse.

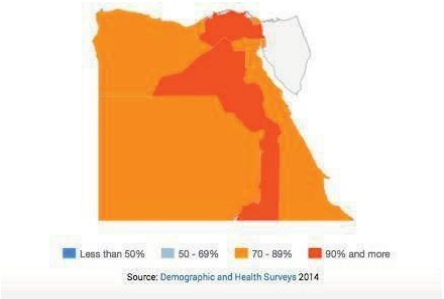
FEMALE GENITAL MUTILATION (FGM) IN EGYPT

Female genital mutilation (FGM) comprises all procedures that involve partial or total removal of the external female genitalia, or other injury to the female genital organs for **non-medical reasons.**

FGM is recognized internationally as a violation of the human rights of girls and women.

FGM is still widespread. The reasons why FGM is performed is a mix of sociocultural factors within families and communities.

In Egypt, 92% of women and girls aged 15-49 have undergone some form of FGM, though increasingly condemned and criminalized.



Types of FGM:

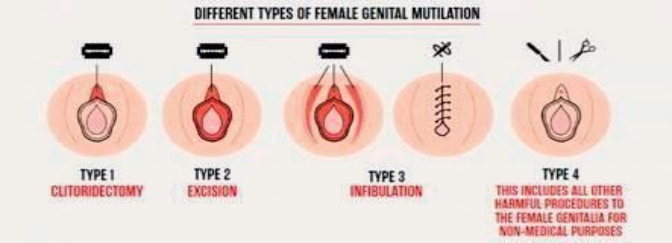
FGM is classified into 4 major types:

Type 1: Partial or total removal of the clitoral glans and/or the prepuce/ clitoral hood.

Type 2: Partial or total removal of the clitoral glans and the labia minora.

Type 3: Also known as infibulation, this is the partial or total removal of the clitoris, labia minora, labia majora with closure of the raw area formed and so narrowing of the vulval opening.

Type 4: This includes all other harmful procedures to the female genitalia e.g. pricking, piercing, incising, scraping and cauterizing the genital area.



Effects of FGM:

No health benefits, only harm

FGM has no health benefits, and it harms girls and women in many ways. It involves removing and damaging healthy and normal female genital tissue, and interferes with the natural functions of girls' and women's bodies. Although all forms of FGM are associated with increased risk of health complications, the risk is greater with more severe forms of FGM.

***Short- term complications of FGM* can include:**

- Severe pain.
- Excessive bleeding (hemorrhage).
- Fever, Infections e.g., tetanus.
- Urinary problems as retention.
- Wound healing problems.
- Injury to surrounding genital tissue.
- Shock, Death.

***Long-term complications* can include:**

- Urinary problems (painful urination, recurrent urinary tract infections)
- Vaginal problems (discharge, itching, bacterial vaginosis and other infections)
- Menstrual problems (painful menstruations, difficulty in passing menstrual blood, etc.)

- Scar tissue and keloid
- Sexual problems (pain during intercourse, decreased satisfaction, etc.)
- Increased risk of obstetric complications (difficult delivery, excessive bleeding, caesarean section, need to resuscitate the baby, etc.) and newborn deaths
- Need for later surgeries: for example, the sealing or narrowing of the vaginal opening (Type 3) may lead to the practice of cutting open the sealed vagina later to allow for sexual intercourse and childbirth (deinfibulation).
- Psychological problems (depression, anxiety, post-traumatic stress disorder, low self-esteem, etc.)

Legal actions against FGM in Egypt:

1- In Egypt, Al-Azhar Supreme Council of Islamic Research has declared that FGM/C has no basis in Islamic core or any of its partial provisions and that it is harmful and strictly prohibited.

2- The Egyptian parliament passing a law banning FGM in 2016, with a minimum penalty of five years imprisonment and more if there is resultant infirmity or death.

5 CONTACEPTION AND FAMILY PLANNING

OVERVIEW

Family Planning is practiced by many couples aiming at spacing pregnancies and choosing the proper time for conception and childbirth.

Delaying conception is usually a **temporary** need for most couples, however **permanent contraception** may be required due to the presence of severe chronic medical disorder that may complicate or be complicated by pregnancy.

An ideal method for contraception should be cheap, readily available, easy to use, with minimal side effects and complications, and with maximum efficiency in protection from unwanted pregnancies.

Occasionally, an urgent contraceptive method may be required whenever a coital relationship has occurred in absence of a suitable protection, in an attempt to minimize the risk of unwanted pregnancy.

Contraceptive methods are grouped according to their effectiveness:

- **Top-tier methods** are those that are most effective and are characterized by their ease of use. The first-tier methods provide the ***longest duration*** of contraception after initiation and require the fewest number of return visits. Top-tier include various methods of male and female sterilization, **intrauterine contraceptive devices, and contraceptive implants**. The last two are considered long-acting reversible contraceptives (LARCs).
- **Second-tier methods** include systemic hormonal contraceptives that are available as oral tablets, intramuscular injections, transdermal patches, or transvaginal rings.
- **Third-tier methods** include condoms for men and women, withdrawal, and fertility awareness methods such as safe periods.
- **Fourth-tier methods** include spermicidal preparations.

PHYSIOLOGIC CONTRACEPTION

Physiologic or natural methods for contraception are easy, cheap, readily available methods but are not reliable when used alone. Combining more than one method will improve the efficacy without additional costs or hazards.

1. COITUS INTERRUPTUS:

Withdrawal and ejaculation outside the vagina during intercourse is one of the oldest most commonly practiced methods for contraception.

Efficacy of this method is not high, as in some cases the pre-ejaculatory fluid may contain sperms capable of fertilization. It is best coupled with the safe period method.

2. THE SAFE PERIOD

Intercourse is totally prevented, or protected (by a condom or coitus interrupted) at the time of expected ovulation (day 10-18 of a 28 days cycle), while allowed for the rest of the month without protection methods. This method is suitable only for intellectual couples, with regular cycles.

Efficacy is good, and is improved when coupled with additional methods that allow for more accurate prediction of ovulation as the **BBT** chart and **urinary LH** tests.

3. LACTATION

During lactation, at least 60 % of females will experience amenorrhea, anovulation, and oligoovulation during the first 6 months of the puerperium, due to elevated prolactin levels.

- Efficacy: is not high (nearly 50% protection if used alone), due to the fact that some women may resume ovulation at variable unpredictable periods of time. Efficacy is improved by adding other methods as barrier, spermicidal, and coitus interruptus.

BARRIER CONTRACEPTION

These methods work by physically or chemically preventing sperm/ egg interaction.

To be effective, barrier methods need to be used correctly and consistently with every act of sexual intercourse. Correct use requires motivation, skills and, communication between partners.

Advantages of Barrier Methods	*Disadvantages of Barrier Methods*
– *Easy to initiate and discontinue* – *No systemic side effects* – *No effects on future fertility* – *No effect on lactation* – *Some may prevent transmission of STDs*	– *Difficult to use consistently and correctly* – *May interrupt sexual activity and pleasure* – *Some require partner's participation* – *Less effective than other modem methods*

1. MALE AND FEMALE LATEX CONDOM:

Latex condoms, used by the male, are the most widely used barrier contraceptive method all over the world. The female latex condoms are not widely available in many countries.

- *Efficacy:* relatively high pregnancy prevention rate which is additionally increased when used with spermicidal creams, coitus interruptus, or the safe period.
- *Advantages:* high efficiency, ready availability, easy use, cheap price.
- *Medical benefits:* prevention of STDs (as HIV & HPV), HCV, and treatment

of premature ejaculation.

– *Disadvantages:* rarely becomes torn, or slips out, thus losing protection.

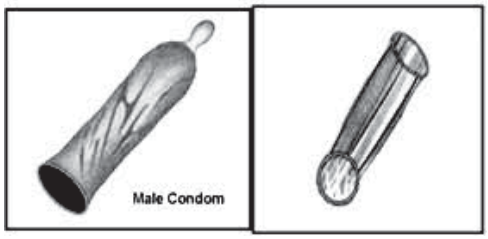


Fig. 5-1: Male condom



Fig 5-2: Spermicide cream

SPERMICIDES (NONOXYNOL-9):

Spermicides inactivate sperms deposited in the vagina during intercourse.

– ***Forms*:** vaginaltablets, creams, gel, or sponge applied 10 minutes before intercourse.

– ***Efficacy:*** failure rates reach up to 30 /HWY if used alone however they are more effective when used with condoms or vaginal diaphragms.

– ***Disadvantages:*** spermicides may cause allergic vaginitis in some women.

2. CERVICAL CAP AND VAGINAL DIAPHRAGM

– Spermicidal agents may be added to them to increase the efficacy.

– They don't protect against STDs.

INTRA-UTERINE CONTRACEPTIVE DEVICE (IUCD or IUD)

The **IUD** is the most commonly used method in national family planning programs, in developing countries, due to its high efficacy and long-term duration of use.

The **IUD** is made of a **polyethylene** material that is Barium Sulphate coated to render it radio- opaque on X-ray.

The **stem** is usually covered with a thin **Copper** (Cu) wire, but may also contain **silver** or **Progesterone impregnated IUD,** as the Levo-Norgestrel Releasing Intrauterine System **(LNG-IUS)** have been introduced in the last decade, with increased **efficiency**, and excellent control of uterine bleeding. They are nowadays commonly used in treatment of **DUB**.

The **limbs** of the IUD emerge from its **upper stem**. They are usually plain, but in some IUDs may be covered by Copper bars. The limbs serve as an anchor holding the device properly at the fundus of the uterine cavity, avoiding its expulsion during menstruation.

Two **nylon threads** are attached to its **lower stem** and protrude out of the cervix. They serve as **markers** for its presence and at the same time facilitate IUD **removal** by simply pulling on the threads out- wards.

IUD is the **preferable** method for **long term contraception** especially for **multiparous** women after completing their family.



Fig. 5-3: IUD

MECHANISM OF ACTION:

A) **Interference with Blastocyst Implantation**, rendering the endometrium unsuitable, via:

- *Local foreign body inflammatory responses.*
- *Inhibition of Carbonic anhydrase and Alkaline phosphatase activity (Copper IUD).*
- *Induction of atrophic endometrial changes (LNG-IUS).*

B) **Interference with Sperm Motility** rendering them incapable of fertilization.

- *Inhibits sperm motility through the endometrial cavity and fallopian tubes.*
- *Induces and antisperm phagocytic activity (Cu IUD).*
- *Produces thick cervical mucus interfering with transcervical sperm penetration (LNG-IUS).*

TYPES OF IUDS:

1. *Copper medicated IUD:* (Cu T 380) is the most commonly used type with a surface area of the copper 380 mm^2. It lasts effectively for 7-10 years then changed by a new device.
2. *Copper+ Silver medicated IUD* (Nova T): Cu T 200 IUD in which silver is added to minimize fragmentation of copper.
3. *Progesterone medicated IUD:* (LNG-IUS Mirena IUD) with sustained release of levonorgestrel 35 ug/day. It acts effectively for 3-5 years then changed by a new device.

IUD INSERTION:

A) **Timing of Insertion:**

- Best inserted by end of menstruation to ensure a slightly dilated cervix, absence of pregnancy, and minimal post-insertion bleeding.
- It can be inserted 4-6 weeks after delivery, or 3-4 weeks after an abortion.

B) **Technique of insertion:**

The contraceptive device is composed of a polythene IUD, an insertion tube and a plunger rod.

- A bimanual examination is performed to ensure the size and direction of

the uterus, and to exclude myomas, polyps, and adnexal masses (TVS may be also done).

- Cusco speculum is introduced into the vagina with the patient in the lithotomy position.
- The anterior lip of the cervix is grasped by a multiple toothed Volsellum.
- A uterine sound is passed through the cervix to reassure uterine length.
- The IUD is launched in the insertion tube which is gently passed through the cervical canal reaching just short of the uterine fundus.
- *Withdrawal Technique:* the plunger rod is used to fix the IUD near the fundus, while the insertion tube is gently rotated and withdrawn back releasing the limbs of the IUD with the T arms spreading towards each uterine cornu (see diagram).
- The long IUD threads are then cut 2 cm from the external os.

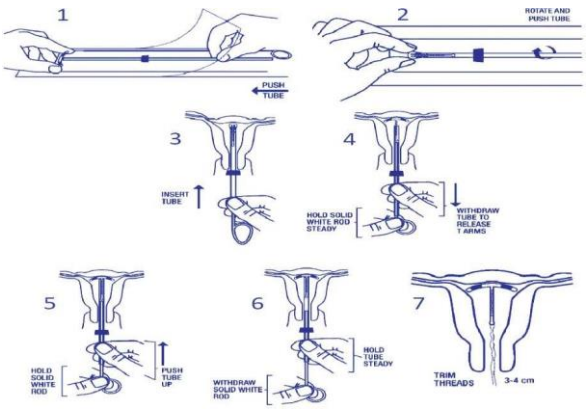


Fig 5-4: IUD insertion

c) Follow up after insertion:

- Periodic yearly follow up visits after IUD insertion are recommended to ensure its proper fitting.
- Threads can be felt on PV examination, or seen on speculum examination.
- TVS or TAS is generally the best method to reassure proper IUD fitting within the uterine cavity.
- Dealing with associated side effects or complications.

ADVANTAGES OF IUD:

- Single choice method, with a long-term protection (7- 10 years).
- It does not affect lactation, in recently delivering women.
- It does not affect or interfere with sexual intercourse
- Very low failure rate: 0.5/HWY (Hundred Women Yearly).

PROBLEMS AND COMPLICATIONS ASSOCIATED WITH IUDs:

1. Abnormal Uterine Bleeding AUB:

a) Post Insertion Bleeding:

Mild bleeding, or spotting, may occur in the first few days following insertion. Bleeding usually stops spontaneously, but if persists use antifibrinolytics and venotonics.

b) Menorrhagia:

Heavy menstrual bleeding **(HMB)** is common with IUD (except LNG - IUS)

- Etiology: mostly due to increase **PGLs** production, or increased **fibrinolytic** activity. Copper IUD increases blood loss by almost 35% in many women.
- Management:
  - *Exclude uterine and pelvic pathology;* by clinical examination and pelvic US
  - *Anti-fibrinolytic agents* (Tranexamic acid 500 mg orally 2-3 times daily) and/or
  - *NSAIDs* (Brufen 600 mg orally 2-3 times daily) till bleeding stops.
  - *Remove IUD;* if displaced, or discovered significant uterine or pelvic pathology.
  - *LNG-IUS* induces endometrial atrophic changes, leading to a **70%** reduction in menstrual blood loss, and hence can be used as an alternative in cases who suffered intractable bleeding with the use of regular Cu IUD.

c) Metrorrhagia:

Irregular intermenstrual bleeding is usually associated with anovulatory dysfunction, pelvic infection, or partial expulsion of an IUD.

- Management: Treatment of the cause of metrorrhagia

2. Pelvic Pain:

- *During insertion:* cramping pain is common due to forcible insertion. Give NSAIDs
- *After insertion:* possibly due to incorrect placement. Check position and place another IUD properly
- *Backache:* due to pelvic congestion, or chronic cervicitis.
- *Acute Abdominal pain:*
  - With IUD in situ; as with abortion, tubal ectopic pregnancy and PID, or
  - With a missed IUD; as in cases with uterine perforation (see later).

3. Pelvic Infection:

- Chronic cervicitis; are more common in presence of an IUD.
  - Antibiotics should be given, then IUD removed if persistent infection.

- PID is more common in presence of IUD s; the threads of an IUD may facilitate ascending infection, especially with STDs, causing endometritis and salpingitis
  - Managements; Antibiotics and removal of IUD.

4. Vaginal Discharge:

- Increased watery cervical discharge, due to pelvic congestion.
- Mucopurulent discharge may occur in chronic cervicitis.
  - Management: treat the cause

5. Expulsion of an IUD:

Spontaneous expulsion may occur usually in the first few cycles after insertion due to;

- Intermittent uterine contractions especially during menstruation, or
- Misplacement at the time of insertion.
  - Management: In both cases IUD should be removed and correctly insert another IUD.

6. Perforation:

Perforation occurs at the time of insertion.

- *Symptoms:* sharp stabbing or colicky pelvic pain, together with persistent vaginal bleeding.
- *Signs: (if passed un-noticed during insertion)* the threads will not be felt by the patient, nor seen by speculum examination.
- *Investigations:* TVS will fail to detect the IUD within the uterine cavity, while a plain X-ray will reveal its presence in the pelvic cavity.
  - *Management* (see later in missed IUD).

PREGNANCY ON IUD:

INTRAUTERINE PREGNANCY:

Most intrauterine pregnancies occur when the IUD is displaced downwards, leaving the fundus of endometrial cavity unprotected.

- Management of pregnancy with IUD:
  - *If threads are visible;* IUD removed, pregnancy continues with an abortion rate 25%
  - *If threads are not visible;* IUD is left in place, pregnancy continues with an abortion rate of around 50%, with higher incidence of septic abortion.

ECTOPIC PREGNANCY

IUD prevents intrauterine but does not prevent extra uterine pregnancy. Incidence of ectopic gestation may be slightly increased especially in the presence of PID.

Management (see ectopic gestation)

MANGEMENT OF MISSING IUD THREADS:

If IUD threads are not felt by PV nor seen at speculum examination, a TVS should be performed to confirm the presence or absence of the IUD within the uterine cavity, and exclude pregnancy

a) **If IUD is intrauterine;** leave it in place if there is no other indication for removal.

b) **If IUD is not detected by TVS;** a plain X-ray to the abdomen and pelvis is performed:

- *Absence of IUD on plain X-ray* denotes expulsion outside the body. A new IUD may be inserted if the patient asks for one.
- *Presence of IUD on plain X-ray* denotes perforation and presence inside the abdominal or pelvic cavity. Such cases are managed by removal of the IUD via laparoscopy or laparotomy, as its presence will provoke an inflammatory reaction leading to peritoneal, omental, and intestinal adhesions, and / or injury.

CONTRAINDICATIONS FOR THE USE OF IUD:

1. *Undiagnosed vaginal bleeding.*
2. *Uterine anomalies* that interfere with proper insertion (as septate or bicornuate uterus).
3. *Uterine pathology* that interferes with proper insertion or cause complications (leiomyomas, polyps).
4. *History of PID;* as IUD may provoke or aggravate PID.
5. *History of ectopic pregnancy;* relative contraindication for fear of higher incidence of repeat ectopic gestation.

HORMONE METHODS FOR CONTRACEPTION

Hormone contraception methods are among the most commonly used methods for contraception around the world, owing to its very high efficiency and its ease for its administration or its discontinuation.

HORMONAL CONTRACEPTION:

ESTROGEN- PROGESTOGEN CONTRACEPTION: OCPs, Vaginal rings, Monthly IM Injectables, and dermal patches.

PROGESTERONE ONLY CONTRACEPTION; POPs, IM injectables, subdermal implants, and LNG- IUS

I. COMBINED HORMONE CONTRACEPTION

MODE OF ACTION:

This type of hormone contraception contains a ***combination*** of an ***estrogen*** and a ***gestagen*** in the form of oral pills, vaginal ring, monthly IM injections, adhesive

dermal patches. They exert their contraceptive effect through the following mechanisms;

1. **Estrogen:** Inhibits ovulation via suppression of GnRH, FSH, LH, and LH surge.
2. **Gestagen:** synthetic progesterone exerts their contraceptive effect through;
   - *Endometrial changes unfavorable for implantation:* due glandular atrophy and stromal oedema.
   - *Cervical mucus changes* resulting in thick cervical mucus hostile to sperm penetration
   - *Altered tubal motility and secretions;* unfavorable for oocyte transport.
   - *Interference with ovulation;* through FSH & LH suppression (but less than estrogen)

N.B.; In addition to its contraceptive effect, gestagens prevent estrogen induced endometrial hyperplasia.

TYPES OF COMBINED HORMONE CONTRACEPTION

A. *Combined Oral Contraceptive Pills COCPs*: taken orally on D3, for 21 days. It contains a combination of;

- **Estrogen** (Ethinyl Estradiol EE), **and**
- **Gestagen** (synthetic progesterone)

B) Vaginal Contraceptive Ring VCR:

The ring is introduced vaginally by the patient by D3, left for3 weeks then removed followed by a 7-day ring free period, to allow for menstruation. The mode of action of VCR is similar to that for the COC pills.

C)Contraceptive Adhesive Skin Patches:

The contraceptive patch is applied to the skin of the abdomen, buttock, or thigh, by D3, changed on the same day every week for three consecutive weeks, with one week free of patches to allow withdrawal bleeding.

D)***Monthly IM Injection:*** taken by D3 and repeated monthly.

ADVANTAGES OF COMBINED HORMONE CONTARCEPTION (mainly COCs):

1. ***Contraceptive benefits of COCs:***
   - Most effective method of contraception 0.1-1/HWY. If pregnancy occurs it is said that it is rather patient failure than method failure.
   - Excellent cycle control, especially in patients with previously irregular cycles.
   - No long-term adverse effects on fertility, with rapid recovery after pill discontinuation.
   - No effects on sexual intercourse compared to coitus interruptus, spermicidals, condoms, and calendar methods for the safe period.

2. ***Non contraceptive benefits of COCs:***
   - Treatment of DUB.
   - Postponing next menstruation (delaying an expected cycle).
   - Treatment of spasmodic dysmenorrhea.
   - Lessrisk for PID, functional cysts of the ovary, endometrialcarcinoma, and epithelial ovarian cancer.

DISEASE RISK ASSOCIATED WITH THE USE OF COCS

Disease	Relative risk
1. Ischemic Coronary Heart Disease (ICHD) and ischemic stroke	**increased** risk
2. Venous thrombo-embolism VTE	**increased risk**
3. Breast cancer	**increase** risk in genetically predisposed patients
4. Ovarian cancer	**decreased** risk due to anovulation
5. Endometrial cancer	**decreased** risk due to gestagens content
6. Colorectal cancer	**decreased** risk

SIDE EFFECTS OF OCPS:

1. **Spotting:** mostly due to inappropriate hormone content of the pill. In such cases shift to other preparation with higher dose of gestagen in the following cycles.
2. **Breakthrough Bleeding:** intermenstrual bleeding during the course of pill taking. If persists more than 3 cycles, then shift to higher hormone content pills.
3. **Hypomenorrhea:** It is common with OCP as gestagens cause glandular atrophy. Pills is stopped if the symptom is unacceptable by the user.
4. **Amenorrhea:** after stopping the pills (post pill amenorrhea), is managed by excluding pregnancy, inducing withdrawal bleeding, and shifting to another non hormonal method.
5. **Thrombo-embolic Disorders:** increased due to increased platelet adhesiveness, increased level of factor II, VII, IX, and X. The risk for DVT and pulmonary embolism is increased 4 to 6 folds if OCP were not stopped at least 4 weeks before surgery.
6. **Hypertension (HTN):** increased incidence in predisposed patients after prolonged use.
7. **Diabetes mellitus (DM);** impaired carbohydrate metabolism may predispose to DM, and in diabetic patients will lead to difficult glycemic control.
8. **Liver:** transient impairment of liver functions is not uncommon, with

increased hepatic cholestasis, higher incidence of gall stones, and rarely benign hepatocellular adenoma.

9. **Effect on Lactation:** decreased milk production & quality protein, only with COCs.

OTHER MINOR SIDE EFFECTS:

a. ***Nausea and vomiting:*** may be encountered in the first few cycles.
b. ***Migraine headache:*** if severe needs discontinuation and shift to non-hormonal method.
c. ***Irritability and depressive mood:*** shift to lower progestogen content pills.
d. ***Weight gain:*** usually minor, mainly due to salt and water retention.
e. ***Breast tenderness, and enlargement:*** shift to lower E/P pills
f. ***Acne:*** may worsen during OCP intake.
g. ***Skin pigmentation:*** Chloasma similar to pregnancy due to increased pigment deposition.
h. ***Change in Libido:*** decreased sexual desire may rarely occur in some women.
i. ***Vaginal Discharge:*** Leucorrhea may occur due to associated cervical congestion.
j. ***Eye Symptoms:*** corneal oedema, contact lens wearers suffer from blurring of vision & corneal irritation.

CONTRAINDICATIONS FOR USE OF OCP

Absolute Contraindications	Relative Contraindications
1. *Thrombophlebitis or thromboembolic disease.* 2. *History of DVT.* 3. *Coronary heart disease.* 4. *Cerebrovascular accidents or strokes.* 5. *Liver disease (impaired function).* 6. *Malignancy of the female genital system.* 7. *Abnormal bleeding from the genital tract.* 8. *Benign or malignant liver disease.* 9. *Suspected or known carcinoma of breast or history of benign neoplasms of the breast.*	1. *Superficial thrombophlebitis.* 2. *Varicose veins.* 3. *Migraine headache.* 4. *Hypertension.* 5. *Diabetes mellitus.* 6. *History of liver disease.* 7. *Gall bladder stones.* 8. *Age > 35 years specially if smokers.* 9. *History of: Pre-eclampsia, Diabetes with pregnancy, Cholestasis with pregnancy.*

II. PROGESTOGEN ONLY CONTRACEPTION

A) PROGESTOGEN ONLY PILLS (POPS)

In POP each pill contains the same amount of gestagen, administered orally, daily, on the same time every day, without a pill free interval irrespective of menstrual cycle.

- **Indications:**

Although POP are suitable for most women, they are most often used by women for whom COC is contraindicated (e.g.; breast feeding women, hypertension, age above 40, women who smoke regularly, and those who commonly get severe migraine headaches especially with aura).

- **Mode of Action:**

POP affects mainly the cervical mucus rendering it thicker preventing sperm penetration. It has an atrophic endometrial effect, together with inhibiting of ovulation in only 60% of cases.

- **Side Effects:** POP have minimal side effects compared to COC, but are generally associated with higher incidence of menstrual irregularities, and ectopic pregnancy.

B) PROGESTOGEN ONLY INJECTABLE CONTRACEPTION

The most commonly used injectable progestogen is composed of depot medroxy-progesterone acetate (DMPA), given I M every 12 weeks.

- **Mode of Action:** as POPs.
- **Side Effects:** Vaginal spotting is common in the first few months, while up to 70% of women will have amenorrhea at the end of the first year.

C) SUBDERMAL PROGESTAGEN IMPLANTS

A subdermal implant (Implanon) is made from a non-biodegradable polymer which contains an active slow release progestogen formulation and is about the size of a matchstick.

- **Application & Mode of Action:**

It is placed subdermally by a special introducer, and will provide contraception for almost 3 years, mainly via inhibition of ovulation.

- **Side Effects:**

About 20% of women will have amenorrhea within three months, while up to 50% will have frequent or prolonged cycles.

- **Advantages:**

Implants are associated with significant weight &mood changes, loss of libido, their side effects cannot be easily dealt with owing to their prolonged slow hormone release

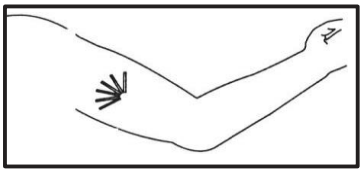


Fig 5-5: Subdermal implants



Fig 5-6: Tubal Ligation

FEMALE STERILIZATION

Female sterilization is a permanent method for contraception that entails surgical occlusion of both fallopian tubes to prevent egg fertilization.

- **Indications:**
  - Its use should be reserved only to those cases with medical or surgical diseases or conditions that contraindicate pregnancy, and necessitate pregnancy termination if it occurs.
  - It should not be advised for cases who have just completed their family and can use another contraceptive method of long duration.
- **Techniques for tubal occlusion:**
  - ***Laparoscopy:*** using clips, rings, or bipolar diathermy (commonest method used).
  - ***Laparotomy***: using delayed absorbable sutures, performed via a mini-laparotomy incision, or more often during repeat CS procedure.
  - ***Vaginal tubal ligation***; performed via a posterior colpotomy in the vaginal fornix
- **Side Effects:**

Some cases may suffer pelvic congestion (post ligation syndrome), leading to pelvic congestion with heavy menstrual bleeding, leucorrhea and pelvic pain.

MALE STERIIZATION

- **Surgical:** via ligation of the vas deferens. The most commonly practiced method
- **Medical:** Gossypol (the male pill), inhibits spermatogenesis. Not commonly available.

CONTRACEPTION FOR LACTATING FEMALES

During lactation around 40-60% of females will experience amenorrhea and anovulation in the 1st six months, mostly due to the anti-gonadotrophic effect exerted by the markedly elevated PRL levels. However, lactation alone is not considered a reliable method of contraception, as it is difficult to predict when ovulation will be restored.

Methods used:

1. **IUD:** is an ideal method as it gives long term contraception without interference with lactation.
2. **Progestogen only injectables:** DMPA: 150 mg I.M./3 months are suitable in most cases.
3. **Progestogen only pills (POP):** daily administered in the same time, without pill free intervals, regardless presence or absence of menstruation, are a good alternative to injections.

4. **Barrier Method:** male and female condom and diaphragm are suitable especially with back-up methods as spermicidals or coitus interruptus.
5. **Spermicidal gel or foam:** used in cases with infrequent intercourse (husband travelling, etc.)
6. **Physiologic Methods:** Coitus interruptus is a popular method **widely practiced. The** safe period is not always suitable due to the high incidence of menstrual irregularity with lactation.

EMERGENCY OR POST COITAL CONTRACEPTION

It is the use of methods to prevent pregnancy after an unprotected intercourse has occurred whenever no method has been previously used, or a method did not function properly, such as a torn condom, or a missed pill.

Options for Emergency Contraception

IUD Insertion within 4-5 days of unprotected intercourse.

OCPs: within 48 hrs of the unprotected intercourse.

- ***POP high dose pills regimen****:* Each oral tablet contains 0.75 mg of levonorgestrel. The first tablet should be taken within 48 hours after unprotected intercourse, and the 2nd dose tablet is taken 12 hours after the first one.
- ***COCs regimen:*** 4 tablets of the standard low-dose COCs taken together, must be taken within 48 hours after unprotected intercourse, with a repeat of same dose 12 hours after first dose.
- **Side Effects:** include, nausea, vomiting, headaches, dizziness, fatigue, breast tenderness, irregular bleeding and spotting.

6 GYNECOLOGIC INFECTIONS

VAGINAL FLORA AND VAGINAL PH

- **Vaginal flora** (ecosystem) of a normal asymptomatic reproductive age woman includes;
  - *Multiple aerobic or facultative species*
  - *Obligate anaerobic species*. Ratio of anaerobes to aerobes is approximately **(10:1).**
- **Nonpathogenic bacteria;** normally inhabit the vagina without causing infection include; **Lactobacillus acidophilus** is the dominant bacterium in a healthy vagina. They play a critical role to maintain the normal vaginal environment through:
  - *Production of lactic acid (by acting on the glycogen content of the vaginal mucosa), this lactic acid is* responsible for acidic vaginal environment **(pH: 3.8-4.5).**
  - *Lactic acid and hydrogen peroxide* produced are toxic to anaerobic bacteria.

N.B.: Decreased glycogen content of vaginal mucosa (as in menopause or childhood) will result in diminished acid production, leading to a *rise in vaginal pH* that becomes alkaline**.**

NATURAL DEFENSE MECHANISMS AGAINST GENITAL TRACT INFECTION (GTI)

A. **Vaginal PH:** The slightly acidic vaginal pH acts as an unfavorable medium for many pathogenic organisms.

B. **Vaginal epithelium:** the thick stratified squamous epithelium lining of the vagina is protective against pathogenic organisms and toxic substances. Normal estrogen level further protects it against atrophic changes.

C. **Cellular and Humoral immunity** play a role in normal vaginal defense mechanisms.

FACTORS THAT PREDISPOSE TO VAGINAL INFECTIONS

1. **Alterations in the normal vaginal pH:**
   - *Menstrual blood flow;* results in a more alkaline vaginal pH.
   - *Vaginal douching;* unnecessary routine vaginal douching alters vaginal pH.
   - *Decreased estrogen;* as in childhood and menopause, lead to decreased glycogen content and so higher vaginal pH.

- *Sexual intercourse;* may lead to higher vaginal pH due to the alkaline pH of the semen.

2. **Imbalance between vaginal bacterial flora and Candida species**
   - *As with recent treatment with broad spectrum antibiotics.*
3. **Decreased resistance of vaginal epithelium**
   - *Atrophic vaginal epithelium:* as in childhood and menopause.
   - *Introduction of new organisms;* as with STDs that alter vaginal microbiology.
   - *Foreign bodies;* as vaginal tampons, IUD threads, and neglected pessaries in cases of POP.
4. **Suppression of immune system:** as with HIV infections, autoimmune disorders, corticosteroid therapy, and diabetes mellitus, may all result in recurrent vaginal infections especially candidiasis.

COMMON TYPES OF VAGINITIS

Vaginitis, with or without vaginal discharge, is one of the commonest complaints encountered in gynecologic practice. It commonly occurs in different periods of women's life:

a. **In the child bearing period:**
- Bacterial vaginosis (BV).
- Candida vaginitis (CV).
- Trichomonas vaginalis (TV).

b. **In childhood:** as Vulvovaginitis of childhood.

c. **In menopausal women:** as Senile or Atrophic vaginitis.

Lower Genital Tract Infections in Childbearing period

1. BACTERIAL VAGINOSIS

Bacterial vaginosis (BV) is the **commonest bacterial vaginal infection** encountered in gynecologic practice (previously known as non-specific vaginitis or Gardnerella vaginitis).

It is also the **commonest cause** for non-irritant, malodorous **vaginal discharge** encountered in women attending routine gynecologic and obstetric clinics.

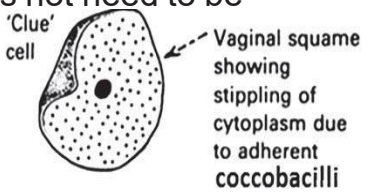
ETIOLOGY AND CAUSATIVE ORGANISMS

Anaerobic bacteria normally found in the vagina including:
- Gardnerella vaginalis (GV)
- Bacteroides species and Mycoplasma hominis.

- **BV** represents a state of **unexplained alteration of the normal vaginal flora**, where; **Anaerobic organisms** markedly **increase** in concentration (thousands of times)

- Hydrogen peroxide producing lactobacilli markedly **decrease** (or even disappear)
  - The **vaginal pH rises**, ranging **4.7-7.0** (alkaline PH).

- Anaerobic bacteria produce enzymes that break peptides to amino acids and amines, resulting in its characteristic fishy odor discharge.
- BV is not sexually acquired hence the sexual partner does not need to be treated.



CLINICAL PICTURE (Symptoms & Signs)

- **Asymptomatic** in 50% of women.
- **Vaginal discharge** is the commonest symptom. It is usually profuse, thin, homogenous, non-irritant, malodorous discharge. It is yellow or greyish in color with milky consistency.
- **Characteristic fishy amine smell** is especially noticeable around the time of menses or following sexual intercourse.

Fig 6-1: Clue cells in BV

DIAGNOSTIC INVESTIGATIONS:

It requires at least 3 of the following four symptoms orsigns:

1. **Characteristic vaginal discharge**: thin homogenous milk like discharge.

2. Microscopic examination of a saline wet preparation of vaginal secretions; reveal characteristic clue cells which are vaginal epithelial cells heavily coated with bacteria obscuring its borders *(fig.6-1)*.

3. **Vaginal pH > 4.5:** results from diminished acid production due to markedly decreased number of lactobacilli.

TREATMENT OF BV (CDC 2021)

- **Oral Preparations:**
  - *Metronidazole tablets:* 500 mg twice a day for 7 days,
- **Intravaginal Preparations:**
  - *Clindamycin cream vaginally* daily 7 days
- **During Pregnancy:**
  - *Vaginal Clindamycin* may be used from first trimester throughout pregnancy.
  - *Oral Metronidazole* may be used after the first trimester.

N.B.:

- Women with BV are at increased risk for: second trimester abortion, preterm labor, endometritis, and pelvic inflammatory disease (PID).
- Screening for BV should be offered for women with a history of idiopathic preterm labor or second trimester abortion. Treatment for positive cases is best initiated before a new pregnancy is allowed.

2. CANDIDA VAGINITIS

Candida vaginitis CV (Candidiasis or Moniliasis) is the **second most common** cause for childbearing period vaginitis. It flourishes in acidic media.

Candida albicans on the vaginal walls

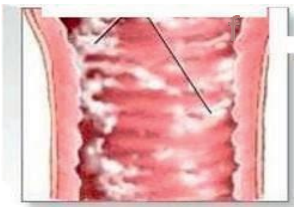


Fig 6-2: Candida albicans on the vaginal walls



Fig 6-3: Candida albicans vaginal discharge

ETIOLOGY AND CAUSATIVE ORGANISMS

- **Candida albicans:** (a **fungus** causing yeast infection) is the causative organism in **> 80%** of cases. It is a normal inhabitant of the bowel and peri-anal region. Almost *30%* of women may have vaginal colonization with *no symptoms of infection*.
- **Non albicans strains:** as Candida tropicalis and Torulopsis glabrata, are **less common** being encountered in **< 20%** of cases. They are usually *resistant to standard antifungal treatment* regimens.

PREDISPOSING FACTORS FOR SYMPTOMATIC INFECTION

- **Alteration of vaginal pH to a more acidic due to**
  - *Repeated use of vaginal douches.*
  - Repeated use of some *vaginal spermicidals* as a method for long term contraception.
- **Altered host immunity:**
  - *Pregnancy*, *diabetes mellitus*, long term use of *corticosteroid* therapy, and *HIV infection*.
- **Altered microbiology:**
  - Long term use of *antibiotics* or *chemotherapy*, thus disturbing the balance between different types of bacterial inhabitants and vaginal flora.
- **Wetness and moisture:** as with tight clothing, nylon underwear, and bathing suits.

CLINICAL PICTURE (Symptoms & Signs)

- **Vulvar pruritis:** Itching and progressive scratching is usually the main symptom.
- ***Vaginal soreness:*** Burning *sensation* that may cause discomfort and dyspareunia.
- ***Vaginal discharge;*** scanty thick odorless white discharge (cottage

cheese discharge).

- **Vulvar erythema:** red and swollen, sometimes showing fissures and excoriations.

DIAGNOSTIC INVESTIGATIONS

- Vaginal pH is slightly acidic (< 4.5).
- **Microscopic examination** with **saline**; reveals hyphae or pseudo-hyphae, with budding yeast.
- Vaginal swabs for culture are not necessary except in resistant or recurrent infection.

TREATMENT OF VAGINAL CANDIDIASIS (CDC 2021)

Non complicated cases

- **Oral preparations:**
  - *Fluconazole;* Single oral dose 150 mg.
- **Intravaginal preparations**
  - *Clotrimazole* cream, **OR** *Tioconazole* ointment **OR** Terconazole vaginal suppository.
- **During pregnancy**; Only intravaginal treatment for **CV** is considered safe throughout pregnancy.

Recurrent vulvovaginal candidiasis: Fluconazole 150 mg orally weekly dose for 6 weeks.

N.B.:

- Routine **treatment of sexual partners** is not recommended.
- **Recurrent infection**: may be due to:
  - Infection with non albicans strains (C. tropicalis & T. glabrata).
  - Immunocompromised hosts; as pregnancy, DM, HIV, or chemotherapy treatment.

3. TRICHOMONAS VAGINALIS VAGINITIS

Trichomoniasis is the **3rd most common** cause for vaginitis, accounting for almost 25% of cases presenting with vulvovaginitis accompanied by vaginal discharge. It is a multifocal infection that involves the vaginal epithelium, Skene's glands, Bartholin gland and urethra.

ETIOLOGY AND CAUSATIVE ORGANISM:

- **Trichomonas vaginalis (TV):** an ovoid, motile, flagellated **protozoon**, with 4 anterior flagellae and an axostyle, which traverses its body to end in a spike. It is slightly larger than a leucocyte (20mm length & 10mm width).
- **Sexual transmission:** In 70% of cases TV can be recovered from male partners, hence it is considered a sexually transmitted disease (STD).

CLINICAL PICTURE (Symptoms and Signs):

- ***Asymptomatic*** in 70%- 85% of women.

- ***Vaginal discharge****:* Copious, yellow or green, frothy, offensive.
- ***Vulval irritation*** causing burning pain and dyspareunia.
- **Speculum examination:** reveals vaginal or cervical punctuate hemorrhages in 25% of cases also known as strawberry spots.

DIAGNOSTIC INVESTIGATIONS:

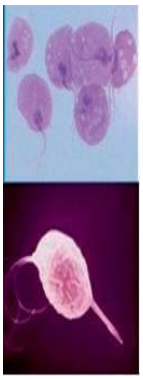


Fig 6- 4: Flagellated organism in TV

- **Saline wet mount preparations:** of the vaginal discharge reveal highly motile flagellated trichomonads.

TREATMENT (CDC 2021)

- *Recommended regimen: Metronidazole:* 2g single oral dose for both partners.
- **During Pregnancy:** Oral Metronidazole can be used after the first trimester.
- **The male sexual partner** should be promptly treated by oral Metronidazole to prevent re infection.
- **Re testing for the female partner** is recommended after 3 months due to the high rates of re infection.

CHILDHOOD VAGINITIS

Childhood vulvovaginitis is **rare**, however may occur mostly due to decreased vaginal wall resistance secondary to low estrogen levels and low vaginal PH.

RISK FACTORS

- *Poor hygiene;* leading to Infection by pathogenic organisms.
- *Foreign body* introduced inside the vagina, usually accidentally.
- *Pinworms* (Enterobius vermicularis), may reach the vagina from anus as they migrate by night, causing intense pruritus and itching by night.

CAUSATIVE ORGANISMS:

- **Bacterial infection**: E. coli, streptococci, staphylococcus aureus, and gonococci.
- CV and TV (rare causes).

CLINICAL PICTURE (symptoms and signs)

- **Vaginal discharge**: which is purulent, foul odor, and sometimes bloody.
- **Pruritus** vulvae: itching, scratching, pain, and sometimes dysuria.

DIAGNOSTIC INVESTIGATIONS

- Culture and sensitivity of discharge to choose the appropriate antibiotic.
- U.S. and /or X-ray to detect foreign body.
- Investigations to detect Enterobius vermicularis and Oxyuris worms.

TREATMENT

1. **Local** cleanliness with diluted antiseptics.
2. **Systemic antibiotics** according to culture and sensitivity of discharge.
3. **Treatment** of Enterobius and oxyuris **worms**.
4. Removal of **foreign body** in the vagina under anesthesia.

ATROPHIC VAGINITIS

Atrophic vaginitis occurs due to loss of the normal protective thickness of the vaginal epithelium associated with decreased estrogen levels. It is often seen in postmenopausal women **(senile vaginitis)**, but can be seen in breast feeding women.

The vaginal epithelium becomes thin, the amount of glycogen content decreases, and PH becomes alkaline. The vagina becomes pale with predominance of parabasal cells.

CLINICAL PICTURE (Symptoms, signs and investigations)

- Leucorrhea, **pruritus**, vaginal **burning**, **tenderness**, and **dyspareunia.**
- **Speculum examination** shows atrophic vagina, inflamed walls, and sometimes discharge that may be offensive if secondary infected.
- Vaginal **PH** is alkaline > 4.5.
- Vaginal infection is usually not identified on wet mount preparation.

TREATMENT

- **Systemic antibiotics;** according to culture and sensitivity, in 2ry infected vaginal discharge.
- **Intravaginal estrogen cream** administered every night for 2 weeks, then once weekly.
- **Systemic hormone therapy (HT):** is given only if otherwise medically indicated for treatment of menopausal symptoms (as hot flashes) or osteoporosis.

ACUTE AND CHRONIC CERVICITIS

1. ACUTE ENDOCERVICITIS

Acute endocervicitis describes acute inflammation of the endocervical glands and underlying tissues, mostly secondary to infection by sexually transmitted pathogens.

ETIOLOGY AND CAUSATIVE ORGANISMS

- **Sexually transmitted pathogens:** *Neisseria gonorrhea, Chlamydia*

Trachomatis, and TV.

- **Other bacterial organisms**: as *streptococci* and *staphylococcus aureus*, in relation to IUD insertion, puerperal, post-abortive, or post-operative infection (D&C procedures).

CLINICAL PICTURE (symptoms and signs)

- ***Vaginal discharge***: Mucopurulent discharge, backache, and dyspareunia.
- ***On speculum examination:*** the cervix is red swollen with mucopurulent discharge.
- ***On bimanual examination:*** tenderness on moving the cervix may be elicited.

DIAGNOSIS

- ***Culture and sensitivity*** of the discharge may be needed in resistant infection.

TREATMENT (CDC 2010)

- *Doxycycline* 100 mg orally twice a day for 7 days **OR**
- *Azithromycin* 1 g orally in a single dose (in case of pregnancy).

COMPLICATIONS

1. **Chronic Cervicitis:** high tendency for chronicity due to the racemose (branching) nature of the cervical glands, and lack of cyclic shedding of cervical mucosa.
2. **Spread of the infection** to the upper genital tract, may lead to **pelvic inflammatory disease (PID).**

2. CHRONIC CERVICITIS

Chronic cervicitis describes chronic inflammation of the endocervical glands. It usually occurs as a common sequel of acute Cervicitis.

CLINICAL PICTURE

- **Symptoms:**
  - ***Vaginal discharge****:* mucopurulent offensive discharge in association with endocervicitis.
  - ***Backache*** due to spread of infection along the uterosacral ligament.
  - ***Dyspareunia*** due to parametrial infection if present.
  - ***Dysmenorrhea*** due to pelvic congestion.
  - ***Contact bleeding*** in some cases in association with cervical erosion.
  - ***Frequency of micturition*** due to cystitis caused by lymphatic spread.
- **Signs on Speculum examination:**
  - **Mucopurulent offensive discharge** coming out from the cervix (Endocervicitis).
  - **Cervical erosion** *(see later)*
  - **Chronic hypertrophic cervicitis:** the cervix becomes swollen and hyperemic.
  - **Mucous polyp:** reddish pedunculated small polyp protruding from the

endocervical canal, due to proliferation of the endocervical epithelium.

- **Nabothian cysts:** small blue or yellowish cysts within the substance of the cervix, projecting on the portio-vaginalis. It represents distended cervical glands with secretions due to blockage of their ducts.

DIAGNOSIS

1. Culture and sensitivity of the discharge in cases of secondary infection.
2. Exclusion of malignancy by vaginal and cervical smear in cases of suspicious cervix.

TREATMENT (CDC 2021)

Oral or vaginal antimicrobials according to the causative organism in cases of acute on top of chronic infection.

CERVICAL EROSION

Cervical erosion (ectopy) is a bright red area around the external os due to replacement of the stratified squamous epithelium of the ectocervix with the endocervical columnar epithelium, which is thin and shows the underlying blood vessels.

ETIOLOGY

1. **Chronic Cervicitis:** The infected cervical discharge from the endocervix produces a devitalized denuded area of the stratified squamous epithelium around the external os.
   - *First healing phase:* columnar epithelium from the endocervix grows to cover the denuded area on the ectocervix (excessive proliferation of endocervical epithelium leads to mucous polyp formation).
   - *Second healing phase:* the stratified squamous epithelium from the ectocervix grows under the columnar epithelium leading to blockage of the ducts of cervical glands and formation of retention (Nabothian) cysts.



Fig 6-5: Cervical erosion

2. **Hormonal erosion:** Excess estrogen (as during pregnancy and with oral contraceptive use) causes growth of columnar epithelium replacing the stratified squamous epithelium.
3. **Congenital erosion:** Persistence of the intra-uterine condition where the columnar epithelium covers an area on the ectocervix (very rarely seen after puberty).

CLINICAL PICTURE

- **SYMPTOMS**
  - *Vaginal discharge:* excessive mucoid in nature.
  - *Contact bleeding:* manifested after douching or intercourse.
  - *Symptoms of chronic cervicitis* if complicated by associated infection (mucopurulent foul discharge, backache, dysmenorrhea, dyspareunia, etc...)
- **SIGNS**
  - *Vaginal examination:* velvety sensation and occasional contact bleeding.
  - *Speculum examination:* reveals red flat area of erosion.

DIAGNOSIS

- Vaginal and cervical smears exclude malignancy.

TREATMENT

- **Antimicrobials** to treat **associated infection** as in cases of acute cervicitis.
- **Resistant or recurrent infections** are mainly treated by repeated antibiotic courses. Destruction of cervical tissue by cauterization, cryotherapy, or laser is not commonly done nowadays.
- **Hormonal erosion:** require **no treatment** except if they persist for more than 3 months.

UPPER GENITAL TRACT INFECTIONS

PELVIC INFLAMMATORY DISEASE (PID)

Pelvic inflammatory disease (PID) is an infection of the **upper reproductive tract organs**. The fallopian tube is the most common and most important organ affected hence PID is commonly used to describe acute salpingitis and/or salpingo-oophoritis. PID is usually caused by bacterial pathogens ascending from the lower to the upper female genital tract.

DEFINITIONS AND TYPES

A. **Acute PID:** acute infection ascending from the cervix, to the tubes and ovaries.

B. **Chronic PID:** chronic pelvic infection that can follow an acute episode of PID, or may occur as a sequel to an inflammatory pelvic lesion or genital T.B.

N.B: PID may be silent, occurring with no or mild symptoms. It is usually diagnosed only when the sequelae of tubal damage is found later, as with formation of fine peritubal adhesions detected at Laparoscopy during investigation for chronic pelvic pain or infertility.

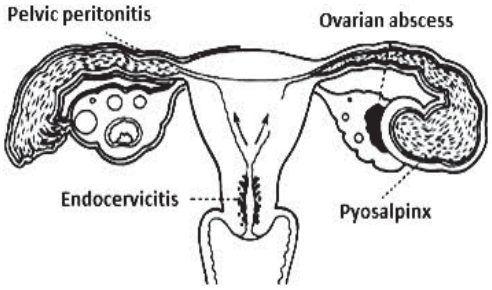


Fig 6- 6: PID

1- ACUTE PELVIC INFLAMMATORY DISEASE (PID)

MICROBIOLOGY OF PID

Acute PID is a polymicrobial infection that may be caused by;

- ***Neisseria gonorrhea:*** a Gram-negative diplococcus. It is the most common cause for acute salpingitis.
- ***Chlamydia trachomatis:*** Chlamydial Trachomatis usually causes silent PID. N. gonorrhea and C. trachomatis may coexist in many cases of PID.
- ***Endogenous bacteria;*** which may be:
  - *Aerobic* as E. coli, Streptococcus species, bacillus proteus, and Klebsiella.
  - *Anaerobic* as Bacteroides fragilis, and Peptostreptococcus.

ROUTES OF INFECTION:

1. Ascending infection. It is **commonest route** of infection.
2. Direct or lymphatic extension from inflammatory GIT conditions.
3. Rarely the tubes may be affected by blood borne extension from pulmonary T.B.

RISK FACTORS

- ***Factors associated with increased risk for PID***
  - Young, sexually active women, mostly in the age group ranging from 25-35 years.
  - Women with multiple sexual partners.
  - Use of **intrauterine contraceptive device (IUCD)**.
- ***Factors associated with decreased risk for PID***
  - Barrier methods for contraception; as the male condom.
  - Oral contraceptive pills (OCPs); probably due to its progestational effects on the cervical mucus rendering it thick and less penetrable by bacterial pathogens.
- ***Factors facilitating ascending infection:***
  - **Menstrual periods:** due to alkaline menstrual blood, degenerating endometrium and retrograde menstruation.
  - **Sexual intercourse:** alkaline semen and bacteria may be pushed into the cervix during intercourse.

- **Iatrogenic procedures:** as D&C operations, IUD insertion, hysterosalpingography (HSG), hysteroscopy, and during dye tubal patency testing in laparoscopy.

REMOTE SEQUELAE AND COMPLICATIONS OF ACUTE PID:

1. **Tubal obstruction and infertility:** such risk correlates with the severity and the number of PID episodes. It reaches up to 11% after one episode, 23% after 2 episodes, and 54% after the third episode.
2. **Ectopic pregnancy:** the rate increases 10-fold in women with PID. Approximately 50% of tubal ectopic pregnancy is thought to result from tubal damage caused by PID.
3. **Chronic pelvic pain:** develops in almost 20% of women with acute PID, sometimes associated with dyspareunia.
4. **Mortality:** is rare, less than 1% only in neglected cases with septic shock and ruptured tubo-ovarian abscess

CLINICAL PICTURE (Symptoms & Signs):

A. **History:** recent IUD insertion, D&C procedure, recent delivery or abortion, or recently per-formed HSG for infertility, in a young sexually active woman.

B. **Symptoms:** Symptoms often start in the postmenstrual period. Women with chlamydial PID usually have clinically milder disease than those with gonococcal PID. Symptoms are generally non-specific including:
   - Acute lower abdominal pain is the most common symptom
   - Fever > 38 C, headache, and malaise
   - Increased vaginal discharge, mucopurulent foul smelling and sometimes vaginal bleeding
   - GIT symptoms; as nausea and vomiting.

C. **Signs:** lower abdominal tenderness, cervical motion and adnexal tenderness, mucopurulent discharge.

DIAGNOSTIC INVESTIGATIONS

1. **Blood tests:**
   - *Increased WBCs (Leucocytosis);* is not accurate for diagnosing acute PID.
   - *Increased ESR, CRP;* are non-specific findings.
2. **Ultrasound:**
   - Findings may be normal, the presence of fluid filled thickened tubes or tubo-ovarian complex is very suggestive of PID.
   - To exclude other causes of acute abdomen as ectopic pregnancy and ovarian accidents.
3. **Examination of Cervico-vaginal discharge:**
   - *Gram stain:* is useful in diagnosis of N. gonorrhea.
   - *Testing for Chlamydial and gonorrheal antibodies.*
4. **Laparoscopy:**

- Whenever in doubt, laparoscopy is done to confirm the diagnosis or in cases with poor response to parenteral antibiotic therapy after 48-72 hours of initiating treatment.
- Laparoscopy will reveal; tubal serosal hyperemia, tubal wall oedema, and purulent exudates from the fallopian tubes pooling in the cul-de-sac, peri-tubal adhesions, peritonitis, pelvic abscess and tubo-ovarian complex.

5. **Testing for HIV and STDs:** should be offered to all women diagnosed as PID.
6. **Pregnancy test** should be performed to exclude ectopic pregnancy.

CDC Criteria for Diagnosis of Acute PID (2021)

- One or more of the following clinical criteria are present on pelvic examination: Cervical motion tenderness **or** uterine tenderness **or** adnexal tenderness.
- One or more of the following additional criteria can be used to support a diagnosis of PID:
  - Oral temperature >38.3˚C
  - Abnormal cervical mucopurulent discharge or cervical friability.
  - Presence of abundant numbers of WBC on saline microscopy of vaginal fluid.
  - Elevated erythrocyte sedimentation rate.
  - Elevated C-reactive protein.
  - Laboratory documentation of cervical infection with *N. gonorrhoeae* or *C. trachomatis*.

DIFFERENTIAL DIAGNOSIS FOR ACUTE PID

1. Ectopic pregnancy.
2. Torsion or ruptured ovarian cyst.
3. Degenerating fibroids.
4. Inflammatory bowel disease; as appendicitis and diverticulitis.

TREATMENT OF ACUTE PID

Empiric treatment is immediately started. The primary goal of therapy is to eradicate bacteria, relieve symptoms, and prevent adverse sequelae especially tubal damage.

- **Rest in bed**
- **Analgesics and antipyretics** are important to alleviate pain and fever.
- **Antibiotic treatment**

CDC antibiotic regimens (2021)

A. **IM/ Oral regimen** is indicated in **mild-moderate PID cases**. The recommended drugs are:
Ceftriaxone 250- 500 mg IM single dose **PLUS**
Doxycycline 100 mg orally twice a day for 14 days **WITH/ WITHOUT**
Metronidazole 500 mg orally twice a day for 14 days

B. **IV regimen** is used in **severe PID cases**, which require hospitalization. IV regimens are continued for at 24-48 hours after clinical improvement then oral

treatment is continued to complete 14 days. The recommended drugs are:
Ceftriaxone 1gm IV every 24 hrs **PLUS** *Metronidazole 500 mg IV every 12 hrs*

- If no improvement occurs within 72 hours of start of standard treatment, the patient should be hospitalized (if not) for further investigations and surgical treatment may be indicated.
- If the woman is using an IUCD, it should be removed if no improvement after 48-72 hrs after starting the antibiotic cover.
- Evaluation and treatment of male partner is essential.

Treatment of Tubo-Ovarian Abscess (TOA):

A. **Medical:**
   - Clindamycin 900mg IV/8 hours + Metronidazole 500 mg IV/8 hours + Doxycycline 100mg orally/12 hours (triple therapy).

B. **Surgical:** Considered in severe cases whenever the patient's condition worsens or does not improve within 72 hours of treatment, or whenever diagnosis is not definitive.
   - **Laparoscopy:**
     - *Diagnostic laparoscopy* confirms diagnosis whenever in doubt.
     - *Operative laparoscopy* may assist drainage or for performing unilateral salpingo-oophorectomy.
   - **Laparotomy:** To perform drainage, or salpingo-oophorectomy.
   - **Vaginal drainage** rarely done in selected cases of localized TOA.



Fig 6-7: Tubo-ovarian complex by US



Fig 6-8: Tubo-ovarian mass

CHRONIC PELVIC INFLAMMATORY DISEASE (PID)

Chronic PID is an ill-defined term including all sequelae of acute PID as incomplete bacterial resolution or chronic inflammatory pathology affecting the upper genital organs and the surrounding structures.

ETIOLOGY

1. **Sequelae of acute PID** (inadequate treatment or neglected cases).
2. **TB** starts as chronic disease (it is considered to be a separate entity).

PATHOLOGY

1. **Hydrosalpinx:** sequelae of acute catarrhal salpingitis. The tube is retort shaped, thin walled, transparent, containing serous fluid, with no or minimal

surrounding adhesions.

2. **Pyosalpinx:** sequelae of acute suppurative salpingitis. The tube is thick walled, containing pus, and surrounded by dense adhesions.
3. **Chronic interstitial salpingitis.**
4. **Tubo-ovarian cyst:** communicating between hydrosalpinx and ovarian cyst.
5. **Tubo-ovarian abscess:** connecting between pyosalpinx and ovarian abscess.

CLINICAL PICTURE (Symptoms & Signs)

A. **Symptoms**

- History of acute PID.
- Dull aching lower abdominal pain.
- Pelvic congestive symptoms; as vaginal discharge, menorrhagia, congestive dysmenorrhea and Chronic pelvic pain
- Infertility (primary or secondary) due to tubo-peritoneal factor whether associated with dense adhesions or tubal occlusion.

B. **Signs:** Adnexal swelling; usually bilateral, tense cystic, tender, fixed, with fixed RVF.

DIAGNOSTIC INVESTIGATIONS

- **Pelvic ultrasonography (TAS / TVS):** to detect adnexal masses.
- **Diagnostic Laparoscopy:** to confirm the nature of adnexal mass.
- **For TB** *(see later).*

TREATMENT OF CHRONIC PID

- **Antibiotics** are used only in acute exacerbations or when bacterial existence is diagnosed.
- **Symptomatic treatment** for pain (analgesic), inflammation (NSAID) and congestion (anticongestives).
- **Infertile patients;** are best advised for ICSI/IVF procedures (after tubal disconnection) since tubal surgery (salpingolysis or salpingostomy) have poor prognosis.
- **Elderly patients and cases resistant to medical treatment** that have completed their family may benefit from surgery with total abdominal hysterectomy and bilateral salpingo-oophorectomy (TAH-BSO).

N.B.: Surgery in cases with chronic PID carry a high risk for intestinal and ureteric injury due to associated dense adhesions.

TUBERCULOSIS OF THE FEMALE GENITAL TRACT (TB)

Tuberculosis (TB) is a chronic granulomatous disease caused by Mycobacterium tuberculosis. Female genital TB is always secondary to a lesion elsewhere in the body, most commonly the lungs. Spread of infection through the genital tract is mainly hematogenous. Fallopian tubes are affected in 100% of cases of genital TB, followed by the uterus in 80%, ovaries in 20% of cases and vulva may show ulcerations.

CLINICAL PICTURE (Symptoms & Signs)

A. Symptoms:

- Early cases are usually asymptomatic.
- More advanced disease:
  1. Infertility: mostly due to; tubal destruction, peritoneal adhesions, or endometrial synechiae.
  2. Chronic pelvic pain associated with low grade fever, is a common presentation.
  3. 2ry amenorrhea or oligohypomenorrhoea may occur in about 10% of cases due to intrauterine synechiae, and/or ovarian destruction.

B. Signs

- Most cases are normal on gynecologic examination.
- A Tender, fixed, adnexal swelling, with low grade fever, especially in an infertile patient with no previous history of pelvic infection, should be strongly suspicious for TB.
- Genital ulcerative lesions: with serpiginous margin and undermined edges, characteristic of TB, may develop on the vulva and cervix, with later scarring or sinus formation.

DIAGNOSTIC INVESTIGATIONS

1. **Chest x-ray** to look for evidence of pulmonary TB
2. **Plain x-ray pelvis:** may detect calcification at tubal sites or pelvic nodes, in chronic cases.
3. **Ultrasonography:** may detect solid adnexal masses with scattered calcification.
4. **HSG:** (not done if active T.B or 2ry infection is suspected to avoid flaring up infection): usually diagnosis of TB is made accidently. HSG characteristic findings for TB are:
   - The endometrial cavity may be normal, or may be small showing linear filling defects due to intrauterine synechiae.
   - The tubes are usually affected showing beading, pipe-stem appearance.
5. **Endometrial Biopsy:**
   - PEB (premenstrual endometrial biopsy) obtained from the uterine fundus and cornu and subjected to histopathologic examination is strongly suggestive.
6. **Laparoscopy:**
   - Usually indicated to confirm abnormal HSG findings suggestive of TB in an infertile patient.
   - Performed also in presence of an adnexal mass suspicious of TB detected on pelvic examination or ultrasonography.

N.B.: Tuberculin test is good negative test for exclusion of the disease rather than for

establishing the diagnosis.

GUIDELINES IN THE TREATMENT OF FEMALE GENITAL TB

- **General treatment;** for anemia and malnutrition and to increase the general resistance.
- **Antituberculous drugs:** The treatment should last for 18-24 months.
- **Surgical treatment:** Indicated only in cases of Adnexal mass not responding to medical treatment.
- Post-operative antituberculous treatment is mandatory.

SEXUALLY TRANSMITTED DISEASES

Sexually transmitted diseases (STDs) are infections acquired principally through sexual contact or sexual intercourse. They are among the most common encountered infectious diseases.

RISK FACTOR FOR STDS

- Young sexually active females especially between 25- 35 years of age.
- Multiple sexual partners.
- Lack of sexual hygiene.
- Previous or present infection of one STD increases the risk for other STDs.

CAUSATIVE ORGANISMS FOR STDS:

- **Bacterial:** N. Gonorrhea, C. Trachomatis, H. ducreyi (Chancroid), Klebsiella (Granuloma ingui-nale).
- **Viral:** Human papilloma virus (HPV), Herpes simplex virus (HSV), Human immune deficiency virus (HIV), and Hepatitis B virus (HBV).
- **Spirochetes:** Syphilis.
- **Protozoons:** Trichomoniasis.
- **Ectoparasites:** Pediculosis pubis, and scabies.

BACTERIAL SEXUALLY TRANSMITTED DISEASES

1. GONORRHEA

Gonorrhea is an STD caused by the Gram-negative diplococci **Neisseria gonorrhea**, a bacterium species that involves infection of columnar and transitional epithelium primarily causing gonococcal endocervicitis. The vagina is not involved being lined by squamous epithelium.

CLINICAL PICTURE (Symptoms & Signs)

A. May be **asymptomatic**

B. In **symptomatic cases;** symptoms usually appear 2 to 5 days after exposure. They include;

- Mucopurulent cervical discharge; as occurs in acute cervicitis.
- Lower abdominal pain, anorexia, and fever if PID develops.
- Dysuria may occur in both men and women if there is associated

urethritis and cystitis.
- Symptoms of bartholinitis may occur if infection of Bartholin gland occurs.
- Symptoms of pharyngitis and proctitis may be present.
- Disseminated gonococcal infection; may cause arthritis, endocarditis, and septicemia.
- Other sequelae of infection include; infertility, and chronic pelvic pain.

TREATMENT

Gonococci are fragile organisms that rarely develop resistance to commonly used antibiotics.

Treatment of uncomplicated cases is usually sufficient trough single dose therapy:

CDC regimens for treatment of uncomplicated N. Gonorrhea (2021)

A- Uncomplicated genital, urinary and rectal infections:

Recommended regimen: Ceftriaxone 250- 500 mg IM in a single dose, **PLUS** Doxycycline 100 mg/ 12hours for 7 days if associated chlamydial infection has not been excluded.

OR Azithromycin 1gm orally single dose.

- Sex partners within 60 days of appearance of symptoms should be evaluated and treated for gonorrhea and chlamydia.

B- Disseminated gonococcal infections: Ceftriaxone 1 g IM or IV every 24 hours then cefixime 400 mg orally twice daily to complete 7-14 days of treatment

SEQUELAE OF UNTREATED GONORRHEAL INFECTIONS:

- Ascending infection may lead to salpingitis and tubal damage that may predispose to infertility and ectopic pregnancy.
- In the new born exposure during vaginal delivery, from an infected mother with endocervicitis, may cause ophthalmitis, blindness, or serious sepsis.
- In males, it leads to urethritis, prostatitis, epididymitis and repeated transmission to the female.

2. CHLAMYDIA TRACHOMATIS

Chlamydia trachomatis is the second most commonly reported STD in many countries. It is an obligatory intracellular Gram-negative bacterium that causes chlamydial genital infection.

Chlamydia grows only intracellular as do viruses. They attack only columnar epithelial cells causing mild inflammatory reaction.

CLINICAL PICTURE (Symptoms & Signs):

- **Asymptomatic** in almost 75% of cases
- **Symptomatic cases** may present with:
  - *Mucopurulent endocervicitis:* mucopurulent discharge, with edematous and hyperemic cervix.
  - *Fever and lower abdominal pain* suggest PID: Infection is more mild, insidious and protracted than that seen with gonococcal PID.

- **Co-infection** with N.gonorrhea is common, and susceptibility to HIV is 3 folds increased.

DIAGNOSTIC INVESTIGATIONS:

Laparoscopy: if done in cases of infertility, may show filmy adhesions, specially between the liver and the undersurface of the diaphragm "Fitz-Hugh- Curtis syndrome" indicating occurrence of mild chlamydia infection and silent PID

TREATMENT: ***CDC regimens for treatment of uncomplicated cases of urethritis and cervicitis (2021)***

Recommended Regimen: Doxycycline: 100 mg orally twice daily for 7 days **OR** Azithromycin: 1 g orally once, in case of pregnancy.

3. CHANCROID

Chancroid is a genital ulcerative **painful** lesion, caused by the Gram-negative coccobacillus

Haemophilus ducreyi. The disease is generally rare.

Clinical picture: incubation period of 3-5 days, Lesions begin as small papules that later break into painful genital vulval ulcers in 2-3 days that may be accompanied with inguinal lymphadenopathy.

Treatment (CDC 2021):

- *Azithromycin: 1g orally once* **OR**
- *Ceftriaxone 250mg IM single dose.*

4. GRANULOMA INGUINALE (DONOVANOSIS)

It is a genital ulcerative disease, caused by a gram-negative bacterium Klebsiella granulomatis.

Epidemiology: The disease is endemic in some countries as India and Africa.

Clinical picture: Painless friable ulcerative genital lesions without lymphadenopathy.

Treatment (CDC 2021):

- *Doxycycline* 100 mg orally twice a day for 21 days **OR**
- *Azithromycin* 1 g orally once per week for 21 days in cases of pregnancy.

5. LYMPHOGRANULOMA VENERUM

Lymphogranuloma venereum (LGV) is caused by ***C. trachomatis serotypes L1, L2, and L3***, producing local ulcerations and destruction of genital tissues.

Epidemiology: LGV is a rare disease that is endemic in some parts of Africa and India.

Clinical picture: Primary genital granulomatous lesions (appearing 3-5 weeks after exposure) that may ulcerate, and is associated with tender inguinal lymphadenopathy.

Treatment (CDC 2021):

- Doxycycline 100 mg orally twice a day for 21 days **OR**
- *Azithromycin* 1 g orally once per week for 21 days in cases of pregnancy.

VIRAL SEXUALLY TRANSMITTED DISEASES

1. HERPES SIMPLEX VIRUS

Herpes simplex virus (HSV) is a double strand DNA virus of two main types:

- ***HSV 1:*** primarily transmitted by non-venereal routes particularly following contact with infected saliva. It is responsible for facial and oropharyngeal infections like herpetic gingivostomatitis and the common cold sore or fever blister.
- ***HSV 2:*** is usually transmitted by venereal route to the sexual partner, or maternally to the new born during vaginal delivery. It is responsible for genital herpes and neonatal infections, and has been linked epidemiologically with carcinoma of the cervix.

CLINICAL PICTURE (Symptoms & Signs)

- **The first attack:** is the most severe with pain, vulvitis, and discharge. It occurs after an incubation period of around 21 days, and is manifested by multiple painful anogenital vesicles or painful ulcers with erythematous base that usually resolve after 3-4 weeks.
- **Subsequent attacks:** are normally shorter and less severe.

TREATMENT

- Treatment is mainly supportive with analgesics
- Acyclovir 400 mg orally three times a day for 7–10 days. Treatment can be given during pregnancy.

2. HUMAN PAPILLOMA VIRUS (HPV)

The genital virus in this double stranded DNA family is responsible for a variety of mucocutaneous genital lesions, commonly resulting in genital warts (condyloma accuminata). HPV is also associated with female LGT cancers especially cervical intraepithelial neoplasia (CIN).

EPIDEMIOLOGY:

- **Types of HPV:** HPV types 6 and 11 are the usual causes for visible external warts. While, HPV types16, 18, 31, 33, and 35 have all been strongly associated with CIN and with external genital squamous intraepithelial neoplasia.
- **Mode of transmission:**
  - *Contact with infected male partners:* yield a risk of 60-80% for contracting genital warts, with an incubation period (IP) ranging from 6 weeks to 18 months (mean 3 months).
  - *Maternal to fetal transmission;* very rare, through transplacental,

intrapartum, or postnatal routes. Fetal affection may cause neonatal and respiratory papillomatosis.

CLINICAL PICTURE (Symptoms & Signs):

- **Asymptomatic:** genital HPV are frequently asymptomatic.
- **Skin lesions** (genital warts or condylomata acuminata): appear on the vulva and perianal region as warty like, scattered and multifocal **painless** lesions.

PREVENTION:

1. **Use of barrier contraception** and **avoid multiple sexual partners** to avoid HPV infection.
2. **HPV vaccination**: children aged 9-14 should be routinely vaccinated to prevent anogenital warts and related cancer.

TREATMENT OF EXTERNAL CONDYLOMAS (CDC 2021)

- **Imiquimod or Podofilox solution or gel** is applied by the patient.
- **Podophyllin resin 10- 25%** solution on the external warts, contraindicated in pregnancy.
- **Destruction:** using cryotherapy, Trichloroacetic acid (TCA) or Bichloroacetic acid (BCA) or surgical removal (are applied by the physician).

3. HUMAN IMMUNODEFICIENCY VIRUS (HIV)

HIV is a retrovirus that causes acquired immunodeficiency syndrome (AIDS). The syndrome is responsible for severe deficiencies in cell mediated immunity, leading to unusual opportunistic infections, malignancy, and eventually death.

HIV has a variable and long asymptomatic latent period that ranges from 2 months to 17 years in which patients are asymptomatic carriers.

EPIDEMIOLOGY

- Men are affected more than women, more common in homosexuals.
- Modes of transmission:
  - **Horizontal transmission:** from one sexual partner to another through micro-abrasions of the mucous membranes or through skin penetration by needles.
  - **Vertical transmission:** from an infected mother to the fetus by trans-placental route.

CLINICAL PICTURE (Symptoms & Signs):

1. **Asymptomatic:** 80% of HIV infected cases remain asymptomatic for around 10 years.
2. **Initial HIV exposure:** fever, sweats, headache, pharyngitis, myalgia, and arthralgia.
3. **AIDS is the final stage**: with associated generalized lymphadenopathy,

Kaposi sarcoma, opportunistic infections, malaise, diarrhea, and weight loss.

TREATMENT (CDC 2021):

1. Immediate medical care and referral to specialized centers.
2. Prophylaxis for opportunistic infections by vaccination against pneumococcal infection, Hepatitis B, measles, and hemophiles influenza B.
3. There is now a great development in antiretroviral therapy that help in reducing the disease prevalence and improved prognosis.
4. In pregnant ladies, CS is indicated in cases of high viral load.

SYPHILIS

Syphilis is a systemic STD caused by the spirochete Treponema pallidum.

EPIDEMIOLOGY:

- Mode of transmission: Syphilis can be transmitted either by: Direct contact with an infected lesion or intrauterine vertical transmission.
- Once the organism reaches circulation, there is a rapid systemic spread via the blood and lymphatics to reach different organs.

CLINICAL PICTURE (Symptoms & Signs):

1. **Primary syphilis:** the initial lesion is a **painless**, ulcerated, **chancre,** usually on the external genitalia, although vaginal and cervical lesions may be detected. The IP ranges from 10-90 days. The primary lesion usually resolves in 2-6 weeks.
2. **Secondary syphilis:** in untreated patients this chancre is followed in 6 weeks to 6 months by a secondary or bacteremia stage with infection of the skin and mucous membranes that usually resolve in 2-6 weeks.
   - Mucous patches; maculo-papular rash of the palms, soles, and mucous membranes.
   - Condyloma latum (moist flat confluent plaques), and generalized lymphadenopathy.
3. **Tertiary syphilis:** with internal organ affection.
4. **Congenital syphilis:** due to materno-fetal transmission may cause still birth, nonimmune hydrops, jaundice, infant hepatosplenomegaly, and skin rash. Identification of and treatment of pregnant women with diagnosed syphilis is the mainstay of prevention for congenital syphilis.

TREATMENT (CDC 2021)

- **Primary, secondary and tertiary cases:**
  - *Benzathine penicillin G* 2.4 million units IM in a single dose.
  - *Doxycycline* 100 mg orally twice daily for 2 weeks in non-pregnant patients with penicillin allergy.

- **Neurosyphilis:**
  - *Aqueous crystalline penicillin* G 18-24 million units/day IV infusion for 10-14 days.
  - *In penicillin sensitive cases:* Penicillin desensitization.
- **Congenital syphilis:**
  - Aqueous crystalline penicillin G 50.000 unit/kg IV single every 12 hours during the first 7 days of life and then every 8 hours thereafter for a total of 10 days. Or *In penicillin sensitive cases:* Penicillin desensitization

NB:

- Pregnant patients are treated by the same penicillin regimen according to the disease stage. And cases sensitive to penicillin need penicillin desensitization.

7 FEMALE PELVIC MEDICINE

PELVIC ORGAN PROLPASE (POP)

- **Pelvic organ prolapse (POP)** refers to descent of one or more of the pelvic organs below their normal anatomical positions.
- **POP** occurs in nearly **15-30%** of women; however, the incidence significantly increases by both age and repeated vaginal deliveries, due to progressive weakness of pelvic supports.
- **The AVF (or RVF) position** of the uterus helps in preventing it's downwards descent during increased intra-abdominal pressure (as during straining, chronic cough, and constipation, etc…).

SUPPORTS OF THE PELVIC ORGANS:

The uterus and other pelvic organs are kept in normal anatomical positions by their surrounding ligaments, muscles and fascia. The main supports of pelvic organs include:

A) **Cervical Ligaments:** form the **main support** of the uterus and pelvic organs.
   - *The Mackenrodt's ligaments:* the most important support of the uterus.
   - *The uterosacral ligaments.*
   - *The pubocervical ligament.*

B) **Pelvic floor muscles:** most important are **levator ani** with its 3 parts; (see anatomy)
   - *The pubo-coccygeus:* the most important part of the muscle support.
   - *The ischio*-coccygeus.
   - *The Ilio*-coccygeus.

C) **Endopelvic Fascia:** including vaginal, cervical, perivesical, and rectal fascia.

D) **Additional support of pelvic organs** is also provided by the perineal body, perineal muscles and their innervations.

TYPES OF PROLPASE

A) VAGINAL PROLAPSE:

1. **Anterior vaginal wall prolapse:**
   - Urethrocele: descent of urethra + lower 2-3 cm of attached vagina.
   - Cystocele: descent of the urinary bladder + upper 2/3 of anterior vaginal

wall.

- Cysto-urethrocele (urethra and urinary bladder descent): with full vaginal length.

2. **Posterior vaginal wall prolapse:**
   - Deficient perineum: defective in the perineal body and wide vaginal introitus.
   - Rectocele: descent of middle 1/3 of posterior vaginal wall + rectum.
   - Enterocele: descent of upper 1/3 of posterior vaginal wall + small bowel loops.
3. **Apical prolapse:**
   - Vault prolapse: descent of the blind ended vaginal apex after hysterectomy

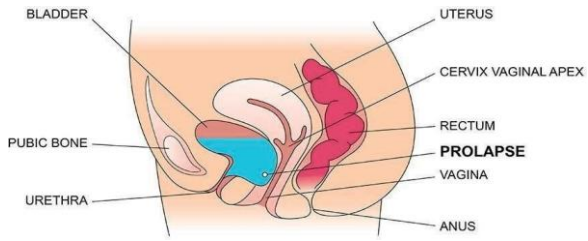


Fig 7- 1: Cystocele



Fig 7- 2: Rectocele

B) UTERINE PROLAPSE

In uterine prolapse descent of the uterus is judged by the level of the cervix, where the external cervical os normally lies at the level of the ischial spines.

- **1st Degree:** The cervix descends downwards through the vagina, but the external os does not reaching the introitus (hymenal ring).
- **2nd Degree:** The external os reaches or slightly protrudes outside the introitus.
- **3rd Degree:** The whole uterus except the fundus descends outside the introitus.
- **Procidentia:** The whole uterus including the fundus descends outside the introitus.

NB: sometimes complete procidentia and 3rd degree uterine prolapse are considered as same entity.

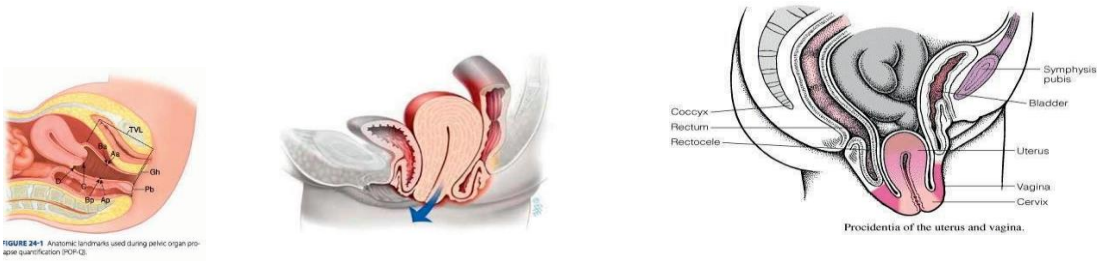


Fig 7-3: Degrees of uterine prolapse

RISK FACTORS FOR POP:

Weakness of the pelvic organs support (fascia, ligaments and muscles), is the main predisposing factor for POP. This occur through;

A) REPEATED CHILDBIRTH TRAUMA:

Damage to the pelvic supports during vaginal birth trauma is the **most frequently encountered** risk factor for POP. It is aggravated by the following conditions;

- *Difficult labor* with oversized fetus and/or prolonged 2nd stage.
- *Short interval successive deliveries* without appropriate involution periods.
- *Direct pelvic floor injury* as in perineal lacerations without prompt repair.

B) MENOPAUSAL ATROPHY:

Older menopausal women are at higher risk for POP due to degenerative changes in pelvic supports secondary to prolonged estrogen deficiency. Risk increases by increased age.

C) CONGENITAL WEAKNESS:

POP may occur in some women in a young age and in absence of childbirth trauma. This is mostly attributed to congenital weakness in pelvic fascia, ligaments, and muscles. It may be associated with poor innervations (as in spina bifida).

D) IATROGENIC WEAKNESS:

This might be induced by improper technique for supporting vaginal vault during hysterectomy, hence increasing the risk for vaginal vault prolapse.

N.B: Precipitating Factors: chronically increased intra-abdominal pressure as occurring in chronic cough and constipation.

ANATOMICAL CHANGES ASSOCIATED WITH LONG STANDING PROLAPSE:

1. **Vaginal skin Keratinization:** everted vaginal skin in cases of prolonged vaginal prolapse becomes thickened, and white with keratin, being exposed to air and trauma.
2. **Vaginal and cervical ulcerations:** may occur at the most dependent part of the vagina or cervix as a result of congestion and circulatory changes rather than friction with the thighs (decubitus ulcers).

3. **Cervical hypertrophy:** due to chronic congestion and associated cervicitis.
4. **Supravaginal elongation of the cervix:** due to stretch on the Mackenrodt's ligament.
5. **Descent of the base of the bladder with stretching of the urethral sphincter**: may lead to frequent desire to micturate and associated stress incontinence.
6. **Kinking and compression on the ureters:** in severe degrees of cystocele, may cause back pressure changes in the form of hydroureter and even hydronephrosis.

CLINICAL PICTURE OF GENITAL PROLAPSE:

A) COMMON SYMPTOMS

In general, **mild cases** with **minimal degrees** of utero-vaginal prolapse are **asymptomatic.** The **more severe degrees** of POP may be associated with one or more of the following symptoms:

1. **Sensation of pelvic heaviness;** especially towards the end of the day that improves or disappears by rest.
2. **A mass filling the vagina or protruding from the introitus:** that may be felt by the patient on straining or squatting, and disappears by lying down on the back, or by reduction.
3. **Low backache:** is common in cases of *uterine prolapse*. Pain is dull aching, or dragging, that radiates to the thighs and legs. Pain is explained by marked traction on uterosacral ligaments by the prolonged uterine descent.
4. **Urinary symptoms:** are common in the presence of a *cystocele*, as:
   - *Frequency of micturition by day;* due to mechanical irritation of the trigone.
   - *Frequency may develop by night (nocturia);* due to residual urine cystitis.
   - *Stress urinary incontinence SUI: due to descent and stretching of the bladder neck and urethral sphincter.*
   - *Inability to complete micturition* unless the anterior vaginal wall is reduced upwards and supported by the patient's fingers, to empty residual urine in a bladder pouch.
5. **Rectal symptoms:** as heaviness and difficulty when trying to defecate in cases of *recto-cele*.
6. **Sexual dissatisfaction:** in cases of deficient perineum and wide introitus.
7. **Pelvic congestion symptoms:** in the form of *Dysmenorrhea* and *Leucorrhea:*

B) SIGNS ON CLINICAL EXAMINATION:

1.GENERAL EXAMINATION

Chest and Abdominal examination are essential to rule out causes for chronic increase in intra-abdominal pressure, as chronic cough, hepatosplenomegaly, ascites, pelvic masses, etc…

2.LOCAL EXAMINATION:

Local examination of the prolapsed mass is performed with the patient in the lithotomy position while maximum straining:

A) **Inspection:** to detect
- Type of prolapse (vaginal, or uterine) and its degree.
- Presence of skin changes or decubitus ulcers.
- Associated patulous introitus.

B) **Digital Palpation:**
- PR examination to test for extent of rectal involvement in rectocele.
- Test for levator ani function and strength by pressing index and middle fingers on perineal body muscles.
- Test for complete procidentia by being able to approximate the index and thumb fingers above the fundus of prolapsed uterus.

C) **Diagnosis of associated conditions;** as
- *Supravaginal elongation of the cervix:* using a uterine sound measure.
- *Stress Urinary Incontinence (SUI):* by asking the patient to cough during examination to detect involuntary escape of urine through urethra (see Incontinence)
- *Test for enterocele:* via eliciting impulse on cough, and or gurgle sensation at the vaginal vault. Combined PV & PR examination may add to accuracy of diagnosis.

PREVENTION OF POP:

- Adequate spacing of deliveries to allow proper involution.
- Avoid bearing down in the 1st stage of labor before full cervical dilatation.
- Avoid difficult and prolonged labor especially in the 2nd stage.
- Avoid damage to perineal muscles and pelvic fascia by proper episiotomy whenever indicated, and proper technique of repair.
- Proper treatment of chronic cough and constipation, with prompt management of pelviabdominal swellings.
- Avoid marked obesity and excessive rapid weight gain.

MANAGEMENT OF POP:

The choice of treatment for genital prolapse depends on several factors including;
- Type and degree of prolapse.
- Desire for future pregnancies.

- Need for preservation of coital function.
- Acceptance and readiness for surgicaltreatment.



Fig 7-4: pessary



Fig 7-5: pessary placement

NON-SURGICAL MEASURES:

- *Pelvic floor exercise (Kegel exercises):* may delay the need for surgery in mild cases. And done also in the first 3-6months after delivery.
- *Pessary treatment:* is only a temporary option aiming at reduction of the prolapsed organs until the patient is ready for surgery, or in cases unfit for surgery.

SURGICAL TREATMENT FOR POP:

Surgical treatment is the only curative approach for symptomatic fit cases.

1. ANTERIOR COLPORRHAPHY:

For repair of the anterior vaginal wall and associated **cystocele or urethrocele**.

- Associated stress urinary incontinence is managed by either;
  - Periurethral fascia plication using Kelly's sutures, or
  - Tension Obturator Tape (TOT).

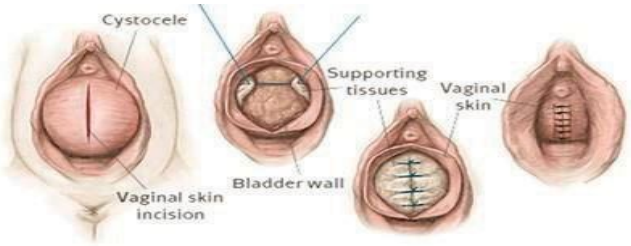


Fig 7-6: Anterior colporrhaphy for cystocele

2) POSTERIOR COLPOPERINEORRHAPHY:

- For repair of posterior vaginal wall prolapse and associated **rectocele or deficient perineum**
- Vagina is packed with gauze for compression hemostasis for 24 hours post-operatively

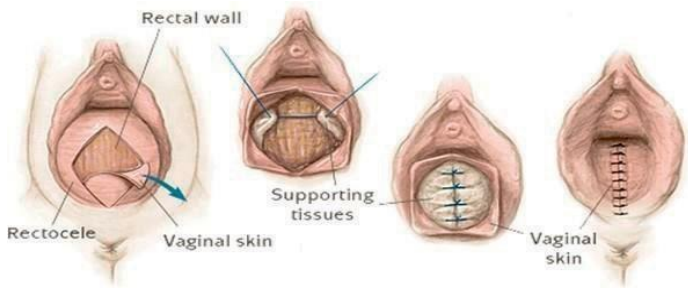


Fig 7-7: Posterior colpoperineorrhaphy

3. CLASSICAL REPAIR:

- Cases of **cysto-rectocele** are best managed by anterior colporrhaphy + posterior col-poperineorrhaphy.
- Associated 1st degree uterine prolapse is managed by **shortening of Mackenrodt's ligament** and their plication in front of the cervix.

4. MANCHESTER (FOTHERGILL'S) OPERATION:

- Cases of 2nd degree uterine prolapse with marked **supravaginal elongation of the cervix** are best managed by **amputation of the vaginal portion of the cervix** and **shortening of the Mackenrodt's ligament**.
- Associated cystorectocele is managed by classical repair.

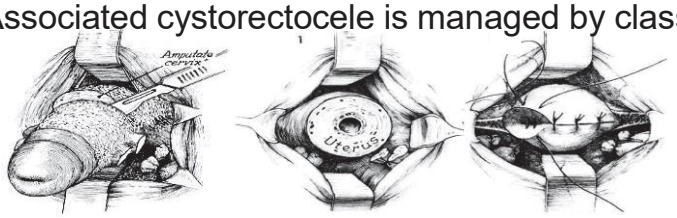


Fig 7-8: Manchester operation

5. VAGINAL HYSTERCTOMY AND PERINEAL REPAIR:

1. Indicated in perimenopausal and postmenopausal patients with marked uterovaginal prolapse.
2. The vaginal vault is supported by plication to the cardinal and uterosacral ligaments.
3. **Anterior colporrhaphy** and **posterior colpo-perineorrhaphy** are then completed.

SURGICAL PROCEDURES FOR REPAIR OF VAGINAL VAULT PROLAPSE:

1. **Abdominal sacro-colpopexy:** Abdominal procedure, done in cases of 2nd or 3rd degree uterine prolapse in young patients desiring fertility.
2. **Vaginal sacrospinous ligament fixation:** Vaginal procedure done in cases of 2nd or 3rd degree uterine prolapse in young patients desiring fertility.
3. **Le Fort's operations:** The procedure aims at obliteration of the vagina over the prolapsed pelvic organs. The procedure is reserved for the very old, flail, non- sexually active patient, unfit for lengthy surgical procedures.

NB: to improve the success rates of surgical repair for POP, the following should be taken into consideration:

- Proper preoperative preparation and treat any condition that increases the

intraabdominal pressure.

- Abstinence of intercourse for 1-2 months after surgery to allow proper healing.
- Delay pregnancy for 1 year after surgery to avoid recurrence.
- Episiotomy should be considered in subsequent vaginal deliveries to avoid recurrence.

URINARY INCONTINENCE

DEFINITION:

- **Continence of urine;** is the ability to hold urine at all times except during voiding**.**
- **Incontinence of urine**: is the involuntary leakage of urine that is objectively demonstrable.

CLASSIFICATION AND TYPES OF FEMALE URINARY INCONTINENCE:

A) **Urethral incontinence:**

1. Stress urinary incontinence **(SUI).**
2. Urge incontinence (detrusor over activity **DO**).
3. Retention with overflow.
4. Nocturnal enuresis.
5. Urethro-vaginal fistula (double stream, not incontinence).

B) **Extra-urethral incontinence:**

- Genito-urinary fistulas **(GUF),** apart from urethro-vaginal type.

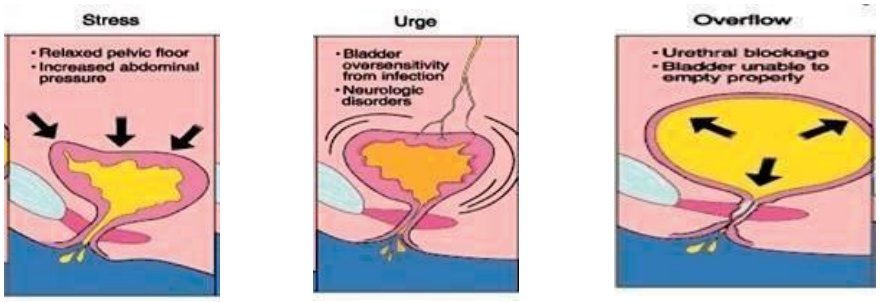


Fig 7-9: Types of urinary incontinence

STRESS URINARY INCONTINENCE (SUI)OR GENUINE INCONTINENCE

DEFINITION AND INCIDENCE:

- **SUI** describes the **involuntary** escape of urine through the **urethra** during **suddenly in-creased** intra-abdominal pressure, as during coughing, sneezing, or laughing.
- It may be encountered in around **5%** of females, and increases with increase age. It is seen in pregnant females due to increase relaxin and progesterone hormones.
- **Urodynamic incontinence (UDI):** is the term used when SUI is confirmed with objective urodynamic testing.

ETIOLOGY AND RISK FACTORS:

a) **Weakness of pelvic floor muscles and supports:**

- *Childbirth trauma:* due to overstretching of the pelvic floor muscles and the endopelvic fascia, with damage to its nerve supply, especially after prolonged and difficult deliveries.
- *Postmenopausal:* due to atrophic changes affecting pelvic fascia 2ry to ***estrogen deficiency***.
- *Congenital (rare):* occurring in young and nulliparous women with no risk factors.

b) **Anterior vaginal wall prolapse:** due to descent of the bladder neck and proximal urethra.

c) **Hypermobile urethra:** not properly fixed to the surrounding tissue or fascia.

d) **Chronic increase in IAPR:** marked obesity, chronic lung disease, or chronic constipation, may precipitate the condition in women with weak pelvic floor musculature and or vaginal prolapse.

SYMPTOMS:

1. Involuntary leakage of urine through the urethra during coughing or sneezing.
2. Symptoms of associated prolapse as heaviness and mass filling vagina may be present.

SIGNS AND TESTS ON CLINICAL EXAMINATION:

1) **Cough stress test:** eliciting involuntary escape of urine through the urethra during cough while the bladder is partially filled and the patient in an erect or better lithotomy position.

2) **Bonney's test: (done in cases of cystocele, to elicit hidden SUI, if present)**

- With the patient in lithotomy position, the bladder neck is elevated gently by the index and middle fingers, placed in the vagina on each side of the urethra, (as if trying to correct mild vaginal and bladder neck descent).
- The patient is asked to cough or strain again;
  - If **no urine escapes** during cough; then no associated SUI is present with the cystocele.
  - If **urine escapes** during cough, then there is associated weakness of bladder neck, and should be corrected surgically with anterior colporrhaphy.

3) **The Q-tip test:** to detect mobility of the urethro-vesical junction on straining. More than 30-degree mobility during straining indicates a hypermobile urethra.

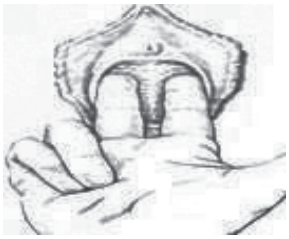


Fig 7-10: Bonney's test

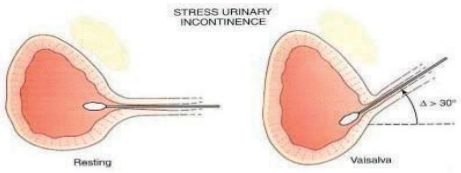


Fig 7-11: Q-tip test

URODYNAMIC TESTS FOR DIAGNOSIS OF SUI:

A special catheter with a sensor probe is introduced in bladder through the urethra to allow performing **cystometry, uroflowmetry, and urethral pressure profile**.

Parameters of Normal Bladder Function on Cystometry and Uroflowmetry:
– *Detrusor pressure filling: < 15 cm H20.* – *Absence of detrusor contractions.* – *First desire to void: 150-200 ml.* – *Capacity: strong desire 400-600 ml.* – *No leakage during cough.* – *Residual urine after voiding: < 50 ml.*

Urodynamic tests in SUI will reveal:

- involuntary leakage of urine through the urethra during increased IAPR.
- Absence of detrusor contractions during filling (differentiate SUI fromDO).

PREVENTION OF SUI:

a. Avoid prolonged second stage of labor and minimize child birth trauma.
b. Pelvic floor exercises especially in puerperium, or after pelvic surgery.
c. Avoid marked obesity and overweight.
d. Treatment of chronic cough/constipation especially in postmenopausal.

MANAGEMENT OF SUI:

A. CONSERVATIVE MEASURES for Mild cases:

Mild SUI may show some improvement via;

1. ***Pelvic floor muscles strengthening exercise:*** as Kegel exercises.
2. ***Estrogen therapy:*** vaginal cream in cases of menopausal atrophy may improve urethral blood flow and increase alpha adrenergic receptor sensitivity.
3. ***Pessary treatment:*** in some cases of vaginal prolapse may improve associated SUI.

B. SURGICAL TREATMENT for Moderate and Severe Cases:

Surgery is the **gold standard** in treatment of **USI.**

1. KELLY'S PERIURETHRAL FASCIAL PLICATION SUTURES:

- It is a **vaginal procedure,** It is a simple procedure associated with a 70% success rate but unfortunately will fall to only 30% after 5 years.

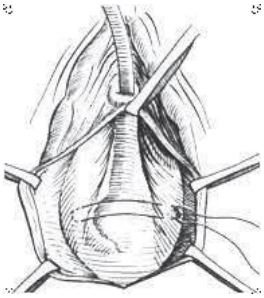


Fig 7-12: Kelly's

2. BURCH COLPOSUSPENSION OPERATION:

- It is an **abdominal procedure,** is considered the **gold standard** procedure with success rates of **95%** after **1 year**, and nearly 75% after 15 years follow up.

sutures

- Recently the procedure has been performed laparoscopically.

3. THE SLING PROCEDURES.

a) Tension-free Vaginal Tape (TVT):

- TVT is performed using synthetic mesh like tape introduced vaginally, on each side of the proximal urethra, then each end of the tape is pulled out through the abdominal wall to which it is finally fixed.

b) Trans- Obturator Tape (TOT):

- TOT is performed using a synthetic mesh-like tape (similar to TVT).
- TOT is simpler than TVT with less complications, performed with regional anesthesia as an outpatient procedure, with success rates approaching TVT.

Fig 7-13: TVT

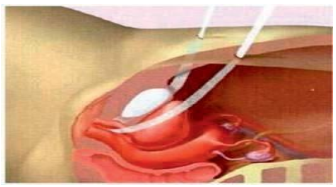
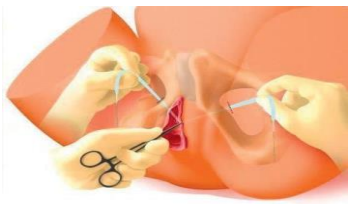


Fig 7-14: TOT



URGE INCONTINENCE (DETRUSOR OVERACTIVITY)

DEFINITION AND INCIDENCE:

Urge incontinence describes **involuntary** leakage of urine through the urethra ***before starting to void*** when the bladder is partially full, posing difficulty in postponing urination urges.

Urge incontinence is usually associated with detrusor overactivity **(DO),** which is on urodynamic studies, is characterized by involuntary detrusor contractions during the bladder filling.

Urge incontinence is the second most common cause of female urethral incontinence.

ETIOLOGY:

- Idiopathic (most common).
- Local bladder irritation (e.g. infection, stone, ulcer, polyps).
- May be an association with a neuropathy.

SYMPTOMS:

- **Urgency:** sudden desire to void.
- **Frequency:** urination 7 or more times a day.
- **Nocturia:** awaking from bed more than once at night to void.

INVESTIGATIONS:

1. **Cystometry: DO** will be diagnosed whenever studies reveal *detrusor pressure filling > 15 cm H20* due to increased systolic contractions with failure to inhibit detrusor contractions during cystometry.
2. **Midstream Urine Examination:** to exclude cystitis and to perform culture and sensitivity if pus cells are high.
3. **Cystoscopy and IVP:** may be needed to exclude intravesical pathology (stone, polyp, etc.)

TREATMENT:

a. **Bladder training exercises;** tends to increase the interval between voids. It is time consuming and requires cooperative patients.
b. **Anticholinergic drugs:** to reduce the vesical pressure and increase the bladder volume (e.g.; detrusitol 2mg twice daily), Oxybutynin (antimuscarinic).

GENITO-URINARY FISTULAS

Genitourinary fistulas are **abnormal communications** between the **urinary organs** (bladder, ureter, and urethra), and the **genital organs** (uterus, cervix, and vagina). They are usually associated with continuous leakage of urine through the vagina (true incontinence) (except for urethra-vaginal fistula).

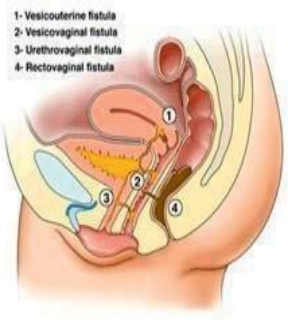


Fig 7-15: Common types of GUF

VESICO-VAGINAL FISTULA (VVF)

OVERVIEW AND ETIOLOGY

VVF is considered as the **commonest type of GUF** encountered in gynecologic practice.

Previously, it occurred due to **ischemic necrosis (necrotic VVF)** 2ry to **prolonged compression** of the fetal head against the bony pelvis, in cases with Obstructed labor.

VVF may still occur due to **direct traumatic injury** during CS and hysterectomy operations, as well as the faulty use of the obstetric forceps procedures, however, this also have been declined due to better surgical techniques and practice.

SYMPTOMS and CLINICAL PRESENTATION:

1. **Complete Urinary Incontinence:**
   - *Continuous leakage of urine from the vagina* in a patient with a history of a recent delivery, pelvic surgery is the main presenting symptom.
   - *Loss of the desire to void urine,* due to absence of bladder filling, is the rule especially when the fistula is large and in a low position.
   - *In necrotic fistulas;* Incontinence may present days or weeks after a difficult delivery, or months after pelvic irradiation.
2. **Partial Urinary Incontinence:**
   - *Interrupted leakage of urine* with *occasional desire to micturate;* may occur in some cases with a small VVF situated high in the bladder, enabling the patient to retain some of the urine in her bladder. This type should be differentiated from UVF.
3. **Pruritus and vulvitis:**

These may occur secondary to prolonged irritation of vulval and vaginal skin with the continuously leaking urine.

4. **Cystitis & UTI:**

UTI may complicate prolonged or neglected cases, due to ascending infection from the vulva. Pyelonephritis is a serious complication which may be the major cause for morbidity in such cases.

SIGNS ON CLINICAL EXAMINATION:

a. **With the patient in Sims' position:**

Inspection of the anterior vaginal wall is performed using Sims' vaginal speculum, to visualize fistulous opening and detect urine leakage from VVF orifice.



Fig 7-16: Sims' position for exami-nation of GUF



Fig 7-17: Sims' speculum

b. With the patient in lithotomy position:

Digital palpation of the anterior vaginal wall is performed to detect the site, size, of VVF and surrounding scarring and fibrosis.

SPECIAL INVESTIGATIONS

1. Retrograde colored dye injection into the bladder (or gauze test):

Retrograde injection of 200 ml. of colored dye (methylene blue or indigo carmine) into the bladder is performed via a urethral catheter. Colored dye will be seen dribbling from the fistulous opening at the anterior vaginal wall using Sim's speculum, or coloring gauzes which were put into the vagina before dye injection.

2. Intravenous pyelography (IVP): done if kidney functions are normal

IVP is helpful to evaluate the amount of residual urine the bladder, delineate the course of the ureters, and exclude ureteric fistulae. Cysto-urethrography helps to exclude urethral fistulas and delineate the fistulous track to the vagina.

3. Cystoscopy & Urethroscopy:

- *Cystoscopy:* visualizes VVF and determines their relation to the ureteric openings in the bladder. It can also exclude multiple VVF, ureteric fistula, and can reveal any associated bladder pathology.

PREVENTION OF VVF:

a. Avoid prolonged and obstructed labor by prompt diagnosis of CPD.
b. Avoid difficult forceps procedures.
c. Proper surgical technique in pelvic surgery to avoid bladder or ureteric injuries.
d. Immediate intraoperative repair of bladder or ureteric injuries once diagnosed.
e. Early diagnosis and proper management of pelvic malignancy (with proper techniques of radiotherapy).

MANAGEMENT OF VESICO-VAGINAL FISTULA:

A. CONSERVATIVE MANAGEMENT

- In case of a **necrotic fistula**, detected few days after a difficult labor, an indwelling silicone catheter is introduced through the urethra and left for a period of 4-6 weeks, to divert the flow of urine away from the fistula, prevent

bladder distension, and allow wound healing.

- Surgical repair is usually postponed for 3-6 months until edema subsides and tissue involution is complete, when surgery becomes easier and safer, allowing for better dissection and healing.
- Spontaneous closure of a small V V F may be achieved in some cases with complete healing of the tear, or failing this, it will be left much smaller in size.

B. SURGICAL TREATMENT

- Surgical anatomical repair is the **gold standard** in management of VVF.
- The **first attempt** for fistula repair carries the **best prognosis**. Repeated trials for repair are less successful due to the extensive fibrosis left after each attempt.
- Proper postoperative bladder drainage is crucial to success of surgery.

OPERATIONS FOR VESICO-VAGINAL FISTULA:

1. ***VAGINAL Flap-splitting operation (dedoublement):***

- This approach is suitable for *low fistulas accessible vaginally.*

2. ***ABDOMINAL Repair of VVF:***

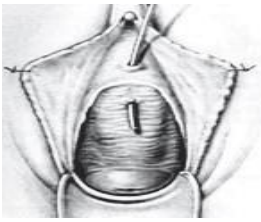


Fig 7-18: Small VVF

- *High and recurrent fistulas* may be better approached by **abdominal repair** through a transverse suprapubic incision, and a retroperitoneal approach.

Post-operative Care:

1. The bladder should be kept constantly empty to avoid any tension on the sutures, via a catheter (for 10- 14 days).
2. In cases with successful repair of VVF, pregnancy should be avoided for a period of 1 year after operation, and at full term delivery by USCS is mandatory to avoid complications of vaginal delivery on pelvic floor and recurrence of fistula.

URETERO-VAGINAL FISTULA (UVF)

This variety of fistula **nearly always** occurs as a result of **injury to the ureter** during gynecologic operations as hysterectomy but may also develop following a difficult delivery by CS.

Symptoms:

- **Incomplete urinary incontinence;** the bladder fills normally from the intact ureter, and the patient voids normally with a **preserved** sensation of bladder filling and **desire to micturate**, while the urine continuously **dribbles through the vagina** from the affected ureter at the site of UVF.

Inspection by Sims' speculum:

- **The fistula** is always **small** and situated **high** up in the vagina lateral to the cervix.

Investigations:

- **UVF** can be **differentiated** from **VVF** by the **methylene blue test (gauze is uncolored in cases of UVF)**.
- **Cystoscopy** shows a normal bladder with presence of the **ureteric efflux on only one side**.

Prevention:

- Intra-operative proper **anatomical identification** of the ureter during pelvic surgery.
- Preoperative identification of ureteric course through an **IVP** or placement of **ureteric catheters** in cases with abnormal ureteric course or large pelvic tumors

Surgical treatment:

- **Abdominal re-implantation** of the ureter into the bladder.

8 BENIGN GYNECOLOGIC CONDITIONS

ABNORMAL UTERINE BLEEDING

The **normal** menstrual cycle is characterized by **cyclic** shedding of the endometrium that occurs every **21-35** days (mean **28** days), with a **duration** of **3-5** days and an average **blood loss** of **30-50** ml/cycle.

Abnormal uterine bleeding **(AUB)** describes abnormal patterns of uterine bleeding other than the above mentioned normal menstrual blood flow.

PATTERNS OF AUB:

- **Heavy Menstrual Bleeding** (**HMB)(formerly called menorrhagia):** excessive bleeding during menstruation.
- **Intermenstrual Bleeding** (**IMB)(formerly called Metrorrhagia):** bleeding in between cycles.
- **Meno-metrorrhagia:** bleeding includes both **HMB** and **IMB** (acyclic irregular bleeding).
- **Postmenopausal Bleeding (PMB):** vaginal bleeding after menopause.
- **Contact bleeding (CB):** bleeding on touching the cervix (e.g.: post-coital bleeding).
- **Other abnormal bleeding patterns** of the menstrual cycle:
  - ***Hypomenorrhea:*** scanty menstrual flow usually of short duration.
  - ***Oligomenorrhea:*** infrequent menstrual cycles (duration > 35 days between cycles).
  - ***Oligohypomenorrhea:*** infrequent cycles with scanty flow or short duration.
  - ***Polymenorrhea:*** frequent menstrual cycles recurring every < 21 days (with normal amount of flow).
  - ***Polymenorrhagia:*** frequent HMB recurring every < 21 days.

CAUSES OF AUB (FIGO 2011):

a) **PREGNANCY**: Normal pregnancy implantation (Hartman sign) or bleeding in early pregnancy complications as: abortion, ectopic, v. mole.
b) **CONTRACEPTIVE methods:** Hormonal or IUCD complications.
c) **STRUCTURAL CAUSES:** referred to as **PALM**

a. **P**olyps: endometrial and endocervical polyps

b. **A**denomyosis: myometrial hypertrophy due to intramyometrial endometriosis.
c. **L**eiomyomas: benign myometrial tumors.
d. **M**alignancies: endometrial, cervical carcinomas, and functioning ovarian tumors.

d) **NON-STRUCTURAL CAUSES:** referred to as **COEIN**

a. **C**oagulation disorders: as ITTP and Von-Willebrand disease.
b. **O**vulatory dysfunction: as chronic anovulation as PCOS.
c. **E**ndometrial inflammatory reaction.
d. **I**atrogenic causes; as anticoagulant therapy.
e. **N**on-classified etiology: as systemic hypertension, capillary fragility, severe thyroid disorders, chronic liver disease and renal failure.

CLASSIFICATION OF AUB ACCORDING TO AGE GROUPS:

1. **Childhood and Adolescence bleeding**: as precocious puberty, FB, rarely malignancy as sarcoma Botryoids (grape like fleshy mass) of the vagina.
2. **Childbearing period bleeding**: in the form of HMB, IMB, or CB (see below).
3. **Perimenopausal bleeding**: mostly due to anovulation (see below).
4. **Postmenopausal bleeding**: mostly atrophic, but malignancy should be excluded (see below).

Heavy menstrual bleeding (HMB)

Heavy menstrual bleeding formerly called Menorrhagia**,** is the term used to describe excessive blood loss **during** menstruation that exceeds > **80** ml in **amount** and/or > **7** days in **duration**, or both. **HMB** occurs more frequently in association with **benign myometrial lesion** rather than malignant conditions.

COMMON CAUSES OF MENORRHAGIA (HMB):

- *Leiomyomas:* especially larger intramural and submucous leiomyomas.
- *Adenomyosis:* in association of symmetrical uterine enlargement and myohypertrophy.
- *Intrauterine contraceptive device (IUD):* non-hormonal and Cu IUDs.
- *Systemic disease* and *coagulation disorders:* as hypertension & ITTP.

EVALUATION OF THE AMOUNT OF MENSTRUAL BLEEDING:

1. Number of blood-soaked diapers changed daily.
2. Duration of menstrual bleeding.
3. Presence of large blood clots and associated anemia.
4. Affection of the patient's general condition and her life style.

The pictorial blood assessment chart (PBAC) using a scoring sheet, patients are asked to record daily number of sanitary products (pads or tampons) that are lightly, moderately, or completely saturated. Point totals > 100 per menses indicate heavy menstrual bleeding.

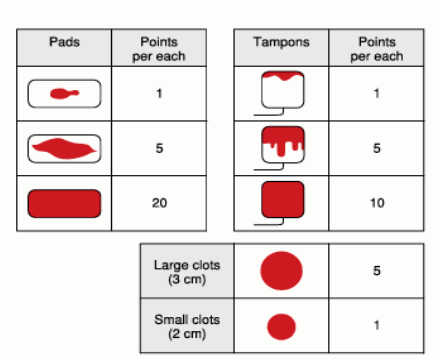


Fig 8-1: PBAC

Intermenstrual bleeding (IMB)

Intermenstrual bleeding (IMB) formerly called metrorrhagia occurring spontaneously not in relation to the time of menstruation.

IMB may be associated with various **benign endometrial** conditions, but is more also commonly associated with **hormonal** disturbances, **premalignant** lesions, and **malignancies** of the uterine body and cervix.

COMMON CAUSES OF METRORRHAGIA (IMB):

- *Combined OCPs* (COCPs), and *Gestagen only pills* (POPs): may cause breakthrough bleeding and intermittent spotting.
- *Endometrial polyps:* due to irregular shedding from the surface of the endometrium.
- *Leiomyomas:* ulcerated infected submucous fibroid polyp's tip.
- *Endometrial hyperplasia (EH), Endometrial* and *endocervical carcinomas* and estrogen producing *ovarian tumors.*

PERIMENOPAUSAL BLEEDING

Perimenopausal bleeding describes **AUB** occurring in **elderly** women, commonly > 40 years of age till the age of menopause. It may occur in the form of Menorrhagia, Metrorrhagia, or Meno metrorrhagia.

Although the underlying etiology is usually functional or benign, yet malignant neoplasms should be thoroughly excluded, before treatment is initiated.

COMMON CAUSES OF PERIMENOPAUSAL BLEEDING:

- *Anovulatory DUB* is the **most common cause** (see later).
- *IUCD or OCPs induced bleeding.*
- *Leiomyomas and Adenomyosis.*
- *Endometrial and endocervical polyps.*

- *Cervical carcinoma*, and less commonly endometrial cancer and functioning ovarian tumors.

POSTMENOPAUSAL BLEEDING

Postmenopausal bleeding (PMB) describes vaginal bleeding that occurs in menopausal women (ie. bleeding occurring 1 year after complete cessation of menstruation).

PMB is always **abnormal** even if mild, and although in most cases the underlying etiology is **benign (atrophic endometritis or atrophic vaginitis)**, yet **premalignant** and **malignant** lesions should be thoroughly looked for and **excluded** in every case.

COMMON CAUSES OF PMB:

- *Benign conditions:* atrophic vaginitis, atrophic endometritis (**most common causes**).
- *Iatrogenic:* as with hormone replacement therapy in menopause (HRT).
- *Systemic disease:* as uncontrolled hypertension and ITTP.

-*Premalignant lesions:* vulvar dystrophies, endometrial and endocervical polyps, EH, and CIN lesions.

- *Gynecologic malignancies:* endometrial, cervical, vaginal, and vulvar cancers, and estrogen producing ovarian tumors (**most serious causes**).

CONTACT BLEEDING

Contact bleeding (CB) describes bleeding that occurs on touching the cervix. It usually presents as post coital bleeding that can occur also following PV or speculum examination.

In **most** cases the underlying cause is a **benign**, **inflammatory**, or **traumatic cervical lesions,**
or less commonly vaginal lesions, however CB in perimenopausal and postmenopausal women is largely related to **cervical cancer,** and should be thoroughly looked for and excluded in such age groups (see cancer cervix).

COMMON CAUSES OF CB:

- Benign cervical lesions: mucous polyps and cervical erosion (most common).
- Premalignant cervical conditions: High grade CIN lesions (common in high-risk cases).
- Vaginal causes: severe Trichomonas vaginitis, or menopausal atrophic vaginitis.
- Vaginal trauma: as friction and defloration injuries.

DYSFUNCTIONAL UTERINE BLEEDING (DUB)

Dysfunctional uterine bleeding **(DUB)** describes **AUB** that occurs in absence of recognizable pelvic pathology, systemic disease, blood dyscrasias, or pregnancy. This diagnosis is made by exclusion; it more commonly occurs in late childbearing period (perimenopausal women).

DUB occurs also in association with chronic **anovulatory disturbances,** that may cause later on endometrial hyperplasia **(EH)** in response to continuous estrogen **(E2) (proliferative)** effect on the endometrium unopposed by progesterone **(PRG)**.

MANAGEMENT OF A CASE OF AUB

- ***Speculum examination:*** to exclude cervical polyp, erosion, ulcer, mass, or associated vaginal pathology.
- ***PAP smear test:*** to exclude high grade intracervical lesions (CIN).
- ***Pelvic US:*** to exclude structural lesions: uterine, endometrial and adnexal pathology and determine endometrial thickness (normal cut-off value in childbearing period is 12mm and in postmenopausal period is 5mm).
- ***Laboratory tests and hormonal assays:*** for exclusion of pregnancy, suspected liver disease, endocrine disorders, and blood dyscrasias (as ITP, & von Willebrand disease).
- ***Office hysteroscopy:*** is beneficial for diagnosis of small endocervical and endometrial lesions.
- ***D&C biopsy:*** is the **gold standard** in diagnosis of endometrial hyperplasia (EH) in cases with DUB.
- ***MRI & CT scan:*** are beneficial in presence of an adnexal mass or suspected malignancy.

TREATMENT OF A CASE OF AUB

Treatment of AUB is a **treatment of the cause**.

Treatment will largely depend on the **severity** of AUB, previous treatment if present, together with the **patient's age** her **desire for further fertility**, and the final diagnosis of the primary etiology.

Treatment of anemia, systemic diseases, endocrine disorders, and blood dyscrasias, should be initiated once diagnosis of any of these conditions is established.

A) **MEDICAL TREATMENT:** HMB can be treated via;

- ***Antifibrinolytics*****:** as Tranexamic acid, 500 mg tds orally, to inhibit fibrinolysis.
- ***Non-Steroidal Anti-inflammatory Drugs (NSAIDs)*****:** as Mefenamic acid 500 mg two to three times daily orally, decrease amount of menstrual flow by up to 50% through decreased PGL synthesis.

B) HORMONAL TREATMENT: IMB or HMB not controlled by the above measures can be treated via;

1. ***Cyclic Combined estrogen / progesterone:*** e.g.; OCP for 21 days /month. These regulate endometrial ripening and shedding inducing regular cycles with reduced menstrual flow. Treatment is usually extended for 3 successive cycles.
2. ***Cyclic Gestagens:*** oral tablets 10 mg daily for 14- 21 days /month (for 3 successive cycles).
3. ***Progesterone impregnated IUD (LNG-IUD):*** continuous slow release of low dose gestagen from an IUD is an effective method for inducing hypomenorrhea in severe HMB.

C) SURGICAL TREATMENT:

- **D&C** for control of acute attack, and obtaining biopsy to exclude EH and EC. It is indicated in cases with severe bleeding especially in presence of abnormal endometrial thickness or suspected polyp by TVS.
- **Endometrial ablation:** in some cases, with recurrent DUB and failed hormone therapy, especially if the patient is risky for major surgery. Endometrial ablation is a procedure that aims at destroying the basal layer of the endometrium. **Myomectomy: see fibroid.**
- **Hysterectomy*:*** abdominal, vaginal, laparoscopic or LAVH (Laparoscopically Assisted Vaginal hysterectomy) as last resort.

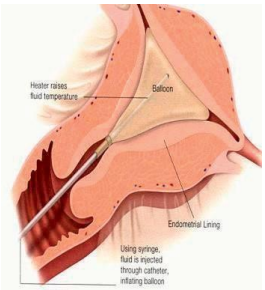


Fig 8-2: Endometrial ablation

UTERINE LEIOMYOMAS

Uterine **leiomyomas** are the commonest **benign** tumors of the female genital organs. They arise from a single muscle cell within the **myometrium,** and are popularly known as *myomas* or *fibroids*.

Myomas are detectable in almost **20%** of women in the *child bearing period*, and in up to **70%** of uteri examined at *autopsy.*

Tumors usually **grow slowly** in size over many years, and have a very **low** malignant potential. They will mostly pass **unnoticed,** due to their slowly progressive course, and will be diagnosed only when they reach a **large size** or

produce significant **symptoms**, or discovered **accidentally** during routine gynecologic examination.

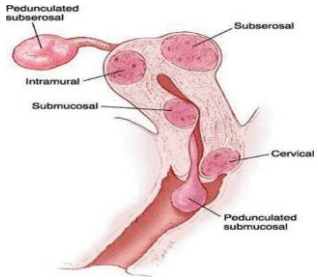


Fig 8-3: Sites of Leiomyomas

ETIOLOGIC FACTORS: (exact etiology is unknown)

A. **Abundant estrogen Environment:** Leiomyomas are estrogen dependent as evidenced by;
   - Leiomyomata appear only in the childbearing period.
   - More common in nulliparous and low parity women.
   - No new tumors develop before puberty or after menopause.
   - Increase in size occurs during pregnancy as well as during estrogen therapy.
   - More common in women with higher body mass index (BMI).

B. **Genetic Factors:**
   - More common in dark races than in white and yellow races.

PATHOLOGY OF LEIOMYOMAS

1. Site Of Myomas:

a. **Corporeal Leiomyomas:** up to **95%** of leiomyomas develop within the myometrium of the uterine body. They are usually **multiple**, and of **variable size**. They are classified according to their relation to the uterine mucosa (endometrium) into:
   - *Interstitial Myomas **(ISM):*** within the center of the myometrium.
   - *Subserosal Myomas **(SSM):*** raising the peritoneal covering of the uterus, the serosa, externally. It may acquire a pedicle forming a *pedunculated **SSM***, or burrow within the leaves of the broad ligament forming a *2ry broad ligamentary myoma **(BLM).***
   - *Submucosal Myomas **(SMM):*** indenting the endometrial lining, and encroaching on the endometrial cavity. It may acquire a pedicle forming a ***SMM polyp.***

b. **Cervical Leiomyomas:** represent **< 5%** of uterine myomas, and are usually solitary.
   - *From the portio-vaginalis:* growing downward to project into the vagina.

- *From the supravaginal cervix:* growing upwards in the true pelvis.

c. Broad-ligament Myomas (BLM): rare, representing < 1% of leiomyomas.

- *1ry BLM:* or true BLM, arising from muscle fibers within the broad ligament and not connected to the uterus (lateral to the ureter).

2ry BLM: SSM that grow externally from the side of the uterus to burrow within the anterior and posterior leaves of the broad ligament. The myoma is thus connected to the uterus either directly or through a pedicle (medial to the ureter, displacing it laterally).

FIGO Classification for the Site of the myomas

These tumors are classified using the FIGO system into:

Type 0: Pedunculated submucous (completely intracavity)
Type 1: Submucous <50% intramural
Type 2: Submucous ≥50% intramural
Type 3: 100% intramural but contacts the endometrium
Type 4: Intramural
Type 5: Subserous ≥50% intramural
Type 6: subserous <50% intramural
Type 7: Subserous pedunculated
Type 8: Other (e.g., cervical, broad ligament, parasitic)

2. ***Size:*** variable size, ranging from size of a peanut to that of fetal head.
3. ***Shape:*** rounded or spherical in shape.

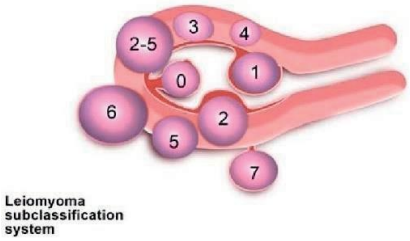


Fig 8-4: FIGO Classification

4. ***Number:*** corporeal myomas are usually multiple, whereas cervical and BLM are usually single.
5. ***Consistency:*** **firm** except of degenerated where it may become soft, cystic, or hard (see later).
6. ***Cut Section:*** Leiomyomas are well circumscribed tumors characterized by;
   - ***Whorled appearance****:* due to interlacing bundles of muscle cells with fibrous tissue
   - *Paler than rest of myometrium:* due to poorer vascularity.
   - *Pseudo-capsule:* myomas are surrounded by a false capsule formed of compressed normal myometrium (capsule does not belong to the tumor).
   - *The blood vessels:* lie in the capsule and send radial branches towards the

center. This explains why calcifications occur in the periphery, while degeneration occurs at the center (poorer blood supply).

- *Fibroid polyp:* when a submucous fibroid attains a pedicle and protrudes into the cavity forming a SMF polyp, the capsule usually ruptures and retracts. The fibroid polyp receives blood supply from vessels in the pedicle.



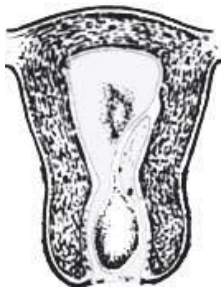


Fig 8-5: Cut section in hysterectomy specimen

PATHOLOGIC CHANGES IN MYOMAS:

1. **Atrophy:** It is the *commonest* change occurring in the *menopause* due to *estrogen deficiency*.
2. **Hyaline degeneration:** This is the *commonest* change in the *child bearing period*, mostly occurs in the *center* of the tumor due to *diminished vascularity*.
3. **Red degeneration: (Necro-biosis):** It is the commonest change during pregnancy, possibly as the tumor may outgrow its blood supply, so *incomplete necrosis* -from which the tumor is capable of *recovery*- occurs
   - *Clinically;* myomas become enlarged, tender, with *acute pelvic pain* and mild *pyrexia*.
   - *Management:* is essentially *conservative* by rest and analgesics as NSAIDs.
4. **Malignant transformation into Leiomyosarcoma:** Leiomyosarcoma is very rare occurring in less than 0.1% of myomas.

- *Malignancy is suspected whenever:*
  - Tumors that elicit a rapid growth without evidence of degeneration.
  - Tumors that first appear, or increase in size, in the menopause.
  - Tumors that present with postmenopausal bleeding (PMB).
  - Tumors that rapidly recur after surgical excision.

CLINICAL PICTURE

- **Age:** Leiomyomas are more prevalent among age groups ranging **35-45** years.
- **Parity:** more prevalent among infertile patients, nulliparous and low parity population.
- **NO SYMPTOMS:**
  - The majority of *solitary and small fibroids are asymptomatic*,

especially SSM and ISM.

- The nearer the fibroid to the endometrial cavity, the more will be the chances of being symptomatic, even if they are relatively small (as in small SMM and SMF polyp).

SYMPTOMS OF LEIOMYOMAS:

1) Abnormal Uterine Bleeding (AUB)

a. Heavy menstrual bleeding (HMB) or Menorrhagia: this is the **commonest symptom** for myomas. It occurs mostly with SMM. Excessive bleeding may be due to:

- Increased surface area of the endometrium.
- Mechanical interference with uterine contraction.
- Increased myometrial vascularity; including venous congestion and engorgement.

b. **Metrorrhagia:** may be due to:

- Sloughing in the tip of a SMF polyp.
- Associated EH, and/or hormonal imbalance due to chronic anovulation.
- Sarcomatous changes in a myoma or coincidental uterine carcinoma.

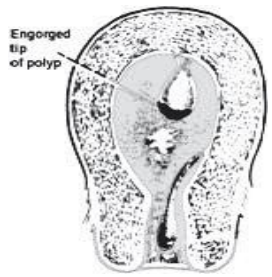


Fig 8-6: Engorged tip of SMF polyp

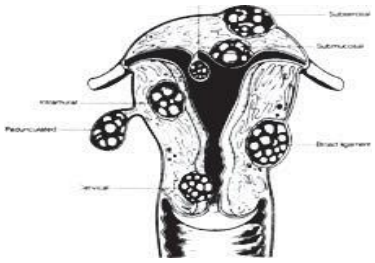


Fig 8-7: Different sites of uterine myomas

2) PELVIC PAIN:

Fibroids are **painless unless complicated,** types of pain include**:**

- *Colicky pelvic pain:* as with extrusion of a SMF polyp through the cervix.
- *Dysmenorrhea;* congestive or spasmodic dysmenorrhea: as with SMF and SMF polyp.
- *Acute abdominal pain:* in red degeneration, and torsion of a pedunculated SSM.

3) PRESSURE SYMPTOMS (specially with SSM or Cervical myomas):

- *Frequency of micturition and dysuria* due to pressure on the urinary bladder.
- *Retention of urine* due to stretching and pressure on the urethra by large cervical myo-ma.
- *Referred back and thigh pain,* 2ry to pressure of myomas on pelvic nerves.

4) Recurrent pregnancy loss (RPL) or INFERTILITY:

- Interference with implantation due to distortion of uterine cavity; as in SMM.
- Tubal obstruction may be caused by multiple ISM and/or bilateral cornual fibroids.

5) OTHER SYMPTOMS:

- ***Pregnancy Complications:*** Preterm labor (PTL), dysfunctional labor, malpresentations, and post-partum hemorrhage.
- ***Vaginal discharge:*** Leucorrhea due to pelvic congestion, or infected fibroid polyp (foul discharge).
- ***Progressive Abdominal Enlargement***

SIGNS ON CLINICAL EXAMINATION:

No specific signs on general examination except for those of pallor if anemia is present.

A) ABDOMINAL AND LOCAL EXAMINATION:

- Large myomas will be felt abdominally, as a central firm mass, with bossy irregular contour, that is usually non tender, of limited mobility, and dull on percussion.
- ***PV examination;*** may detect cervical myomas and SMF polyp protruding through the cervix.
- ***Bimanual examination;*** will detect uterine enlargement, which may be firm with bossy con-tour, non-tender, and with restricted mobility.

INVESTIGATIONS:

1. **Ultrasonography (US: TAS or TVS depending on the size of the myoma)**:

U.S. is considered the ***gold standard*** in the diagnosis of ***uterine leiomyomata***.

- Assessment of the site, size, number, and relation of myomas to the endometrial cavity.
- It is also valuable in detecting adnexal pathology and excluding pregnancy complications.

2. **Saline infusion sonography (SIS**), is sensitive in diagnosis of SMM and SMF polyps.
3. **Hysterosalpingography (HSG):** HSG is valuable in diagnosis of *SMM* that appears as a filling defect. It is also important for testing tubal patency in patients complaining of ***infertility.***
4. **Hysteroscopy**: to diagnose small *endometrial polyps, SMF polyps, and SMM*.
5. **Laparoscopy:**

Laparoscopy may be used to diagnose SSM, and complex *adnexal masses*

MANAGEMENT OF UTERINE LEIOMYOMAS

1. CONSERVATIVE APPROACH: in small asymptomatic myoma

- ***Periodic follow up:*** every 6 – 12 months is appropriate for asymptomatic small leiomyomas.
  - **N.B.:** Conservative approach is abandoned whenever:
    - leiomyoma reaches a large size (≥ 14 weeks).
    - broad ligament myoma (as ureteric compression and affection of renal function may occur).
    - SMM in infertile patients or RPL (as this myoma may be the cause of failure of implantation).

2. MEDICAL TREATMENT:

To control moderate **bleeding** and **pain** through oral or IM drugs as:

a. ***Nonsteroidal Anti-inflammatory drugs (NSAIDs):*** these drugs act by inhibiting PGL synthesis and restoration of PGL / Prostacyclin ratio. They are used to minimize menstrual pain and bleeding.
b. ***Antifibrinolytic drugs*** as Tranexamic acid to control AUB.
c. ***Venotonic drugs*** as Daflon 500mg: that acts by improving vascular tone and decrease pelvic venous congestion.

3. HORMONAL TREATMENT:

To **minimize** menstrual blood flow and **regulate** withdrawal bleeding.

a. ***Gestagens:*** (see AUB).
b. ***GnRH Agonists:*** Long-acting GnRH agonists IM on **monthly** intervals for 3-6 months, results in pseudo-menopausal state, leading to reduction in the size and vascularity of leiomyomas. This is only used **temporary** as it is expensive, induces unpleasant menopausal like symptoms. GnRH therapy may help preparing the patient before myomectomy to decrease size and vascularity of the myoma.

4. MINIMALLY INVASIVE PROCEDURES

Several modalities have emerged in dealing with small to **intermediate size symptomatizing leiomyoma**. These techniques require **proper selection of cases** suitable.

A. Uterine Artery Embolization (UAE):

- *Advantages:* The procedure is relatively safe. It results in 60% reduction in size of myomas, with control of menorrhagia in up to 90% of cases.
- *Disadvantages*: The main disadvantage is significant postoperative pain (like red degeneration). UAE may not be also suitable for patients seeking future fertility.



Fig 8-8: Uterine Artery Embolization

B. Laparoscopic Myolysis:

- *The procedure:* Laparoscopic myolysis using laser or coagulation current, or cryomyolysis.

C. MRI-guided High frequency Ultrasound (HIFU):

- *The procedure:* This uses ultrasound generated heat to cause cell death. Fibroids are localized by MRI and heat from a phased array ultrasound transducer is directed towards the myoma to produce protein denaturation.

5. DEFINITIVE SURGICAL TREATMENT

Surgical removal of leiomyomas is the only definitive treatment for these benign tumors.

- **Myomectomy**: Myomectomy includes enucleation (shelling) of the myoma from its "false" capsule, and closing the dead space formed, leaving the uterus for further fertility. It can be done open abdominal, vaginal, laparoscopic, hysteroscopic depending on site and size of the myoma.
- **Hysterectomy:** Hysterectomy is performed in presence of multiple or large tumors in an elderly patient that has completed her family and not desirous for further childbearing, or whenever malignancy is suspected or patient is menopausal

ENDOMETRIOSIS

Endometriosis is the presence of **endometrial glands** and stroma **outside** the endometrial cavity.

It is estimated to be present in almost **10%** in women in their childbearing period, in around **20%** of women with chronic pelvic pain, and in almost **30%** of women with infertility.

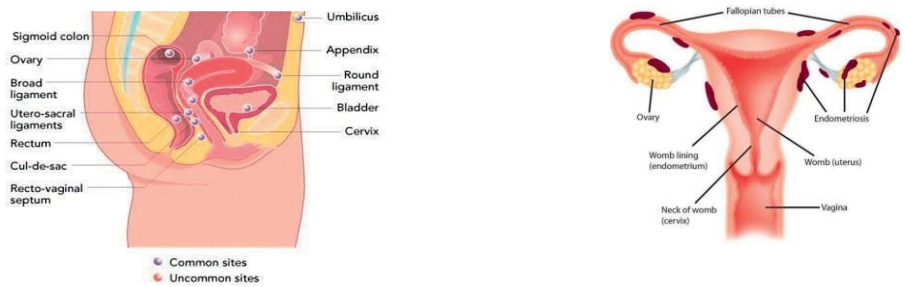


Fig 8-9: Sites of Pelvic Endometriosis

This endometrium **bleeds at menstruation** producing pelvic **pain**, and in some cases pelvic adhesions causing **subfertility**.

THEORIES FOR ETIOLOGY AND PATHOGENESIS:

Several theories have been proposed about the etiology of endometriosis, but no one theory can explain its occurrence in multiple different sites.

1. ***Sampson's theory of Retrograde Menstruation:*** suggests that viable endometrial tissue is transported through the fallopian tubes during menstruation, resulting in intra-abdominal pelvic implants. This theory is the most acceptable, but it *explains only pelvic endometriosis.*
2. ***Halban's theory of Lymphatic Spread:*** proposes that endometrial tissue is transported via lymphatic system to various intraperitoneal sites. It *does not explain distant sites of endometriosis* as the diaphragm, breast, lungs and brain.
3. ***Meyer's theory of Coelomic Metaplasia:*** proposes that multipotential cells in peritoneal tissue undergo metaplasia into functional endometrial tissue.
4. ***Theory of Altered Immune Response:*** women with endometriosis may have an altered immune response that makes them less likely to clear the viable endometrial cells from the pelvic cavity (reduced natural killer cells and macrophage activity).

RISK FACTORS:

- Women with white races and higher socioeconomic standard.
- Women in their childbearing period (specially 25- 35 years).
- Women with Infertility, nulliparity, and low parity.
- Women with family history of endometriosis in the mother or sister.

SITES OF ENDOMETRIOSIS

b. ***Pelvic (Common sites):*** Douglas pouch, uterosacral ligament, the ovaries, pelvic peritoneum.

c. ***Extra pelvic (Rare sites):*** anywhere in the body as far as the lungs, brain, breast, and umbilicus.

PATHOLOGY

A. **Pelvic Endometriosis:** appears as multiple small dark blue or brown small lesions of altered
blood, and lined by endometrial tissue. They give a naked eye appearance of burned match-head spots.

B. **Ovarian Endometriosis:** is present in one of two forms:

1. Superficial Lesions:

- Tiny dark red or brown lesions with adhesion formation *(powder burns).*

2. ***Deep lesions:*** *(ovarian endometrioma or chocolate cysts):*

- They are usually of moderate size, filled with dark altered blood (hemosiderin laden cells) and surrounded by adhesions.

CLINICAL PICTURE:

Many cases may remain **asymptomatic** for a long period of time, until accidentally discovered during **laparoscopy or laparotomy** performed for different indications not related to endometriosis.

SYMPTOMS:

- Symptoms when present are only **suggestive** as they are not specific to endometriosis.
- Symptoms do not necessarily correlate with the **severity or extent** of the disease:
- The 2 major clinical presentations of endometriosis are **pain** and/or **infertility.**

A) PAIN:

- **Chronic pelvic Pain:**
- Diffuse chronic pelvic pain > 6 months, is strongly suggestive of endometriosis.
- **Dysmenorrhea:**
- 2ry dysmenorrhea is one of the commonest clinical presentations.
- Pain starts during menstruation, and increases rapidly with the flow, due to distension of ectopic endometrial glands with blood.
- Pain decreases gradually after the end of the cycle due to absorption of blood within the endometriotic implants.
- **Dyspareunia:**
- Deep dyspareunia due to the presence of tender endometriotic nodules in pouch of Douglas, presence of ovarian endometriomas, or due to associated fixed RVF uterus.

B) INFERTILITY:

Endometriosis may cause infertility by causing peritoneal or peri tubal adhesions.

C) OTHER SYMPTOMS:

- **GIT symptoms:** pain during defecation, intestinal cramps, and rarely cyclic rectal bleeding.
- **Urinary symptoms:** dysuria, frequency, and rarely cyclic hematuria.
- **Distant deposits:** symptoms according to organ involved (brain, lungs, and umbilicus)

SIGNS:

- **General Examination:** no general specific signs for endometriosis.
- **PV and Bimanual Examination:**

- The uterus may be fixed in RVF position.
- Tender nodules on uterosacral ligaments and pouch of Douglas.
- Endometriomas (chocolate cysts); may be palpated as tender tense cystic fixed adnexal masses in non-obese patients.

INVESTIGATIONS:

A) Laparoscopy:

- Diagnostic Laparoscopy (with or without biopsy) is the **gold standard** in **diagnosis** and **staging** of endometriosis.
- Laparoscopy is indicated in cases of chronic pelvic **pain**, **infertility**, and unresolved adnexal **masses**.
- **Typical lesion:** endometriotic implants appear as brown or black pigmentations (powder burn lesions) on peritoneal surfaces, surrounded by adhesions.

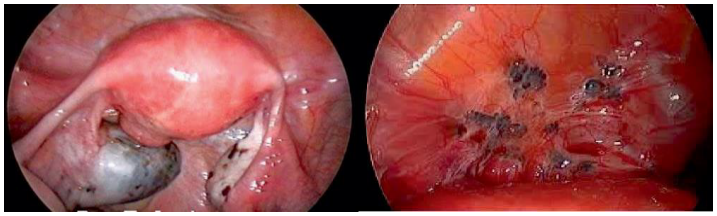


Fig 8-9: Pelvic endometriosis by Laparoscopy

B) Ultrasonography:

- US shows ovarian endometrioma with "ground glass appearance"
- Small endometriotic lesions may not be visualized by US.

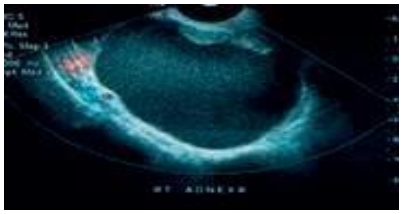


Fig 8-10: Endometrioma by US



Fig 8-11: Endometrioma by laparoscopy

C) CA-125:

- CA-125 is a cell surface antigen found on coelomic epithelium (Normal 5-35 u/ml).
- It shows slight to moderate increase in levels in many cases of endometriosis, but is not a specific marker.
- It is a useful marker for response to treatment in cases with initially elevated levels.

MANAGEMENT OF ENDOMETRIOSIS:

Management of endometriosis is planned according to several factors including; age of the patient, severity of symptoms, future reproductive plans,

and extent of the disease at laparoscopy. ***EXPECTANT MANAGEMENT:***

Expectant management is appropriate for young patients, with **minimal symptoms** and **minimal signs**. The goal of expectant management is:

- Effective control of pain, using **NSAIDs**.
- Follow up for growth of lesions (endometriotic implants or endometriomas).

MEDICAL TREATMENT:

Medical treatment is indicated in cases with **marked symptoms**. Symptoms of endometriosis improve during pregnancy (predominance of progesterone hormone) and menopause.

1. **Continuous Combined Oral Contraceptive Pills (COCs):**
   - *Aim:* induces amenorrhea with a pseudo pregnancy state.
   - *Mechanism:* pseudo-decidualization, followed by atrophy of ectopic endometrium.
   - *Dose:* continuous COCs once daily for 3-6 months.
2. **Continuous Synthetic Progesterone (Gestagens):**
   - *Aim:* induces amenorrhea with a pseudo pregnancy state.
   - *Mechanism:* pseudo-decidualization, followed by atrophy of ectopic endometrium.
   - *Doses:* continuous gestagen (oral or IM injections) treatment for a 3-6 months course
3. **Long acting Gn RH agonists**
   - *Aim:* induces amenorrhea with a pseudo menopause state.
   - *Mechanism:* menopause like hyopo-estrogenic state, with atrophy of ectopic endometrium.
   - *Doses:* given SC monthly or every 3 months for 3-6 months.
   - *Disadvantages:* hot flushes, vaginal dryness, and accelerated bone loss.
   - Adverse effects may be minimized by the use of low dose estrogen (add back therapy).

B) SURGICAL TREATMENT OF ENDOMETRIOSIS:

Surgery is the best choice in cases with pelvic adhesions and / or larger endometriomas (more than 4cm in diameter).

1. **CONSERVATIVE SURGERY:**

- Excision, fulguration, or laser ablation of all visible lesions for pain and fertility treatment.
- Lysis of pelvic peritubal and periovarian adhesions.
- Ovarian cystectomy of sizable endometriomas (> 4.0 cm).

Conservative surgery can be achieved via either;

a. ***Laparoscopy:*** is the gold standard for conservative surgical approach. It is suitable for the infertile patients aiming to restore normal anatomy with creating minimal postoperative adhesions.

b. ***Laparotomy:*** is reserved for the severe cases with extensive adhesions

or large endometriomas.

2. **DEFINITIVE SURGERY:**
   - ***TAH & BSO*** is the procedure usually performed in elderly patients, with extensive disease, and no desire for fertility. It has limited place and limited indications.
   - Estrogen replacement therapy (ERT) postoperatively is recommended in the relatively young patient to relieve associated menopausal symptoms.

- ***Summary for Treatment Options for a case of Endometriosis:***
  - Minimal lesion + desire for fertility: conservative surgery + assisted reproduction.
  - Endometrioma > 4.0 cm: Ovarian cystectomy and laser ablation via laparoscopy, or laparotomy
  - Extensive lesion + no desire for fertility+ elderly: TAH-BSO + excision and ablation of im-plants.

ADENOMYOSIS

Adenomyosis is the presence of endometrial glands and stroma deep within the myometrium. It has been formerly considered as internal endometriosis, affecting primarily **multiparous** women in their late thirties and early forties.

ETIOLOGY AND THEORIES: The etiology of adenomyosis is **unknown**
- Theories postulate that it may be due to growth of endometrial glands deeply between muscle fibers of the myometrium during uterine involution in puerperium.
- Adenomyosis is associated with smooth muscle hypertrophy around the ectopic glands.

PATHOLOGY:

A. **Diffuse type:** the uterus is symmetrically enlarged, globular in shape, with a firm consistency reflecting myometrial thickening.

B. **Localized type:** adenomyosis may be localized at an area (fundus, posterior wall, anterior wall or part of them) very similar to uterine leiomyomata. Unlike the myoma localized adenomyosis is <u>not encapsulated.</u>

CLINICAL PICTURE:

SYMPTOMS: Heavy Menstrual Bleeding (HMB) (Menorrhagia) and 2ry dysmenorrhea.
SIGNS on BIMANUAL EXAMINATION: Symmetrically enlarged **tender** uterus (**Halban's sign).**
INVESTIGATIONS: Ultrasonography (TAS/TVS/3DUS): will reveal thickened myometrium, sometimes associated with localized areas of different

echogenicity somewhat similar to leiomyomas.

TREATMENT:

- ***HMB or Menorrhagia*** can be managed by:
  - *Antifibrinolytics* as tranexamic acid and *venotonics* as Diosmin.
  - *Gestagens* as norethisterone acetate 5 mg t.d.s. for 2 weeks / month.
  - COCPs 21 days/month.
  - *LNG IUD* (Mirena) effective in control of bleeding within 3 months of insertion (see AUB)
- ***Hysterectomy*** is the treatment of choice in cases of severe perimenopausal bleeding not responding to medical or hormonal treatment. Preservation of the ovaries can be done in patients less than 45 years to avoid menopausal symptoms.

BENIGN CONDITIONS OF THE VULVA

Pruritus vulvae, pain, dermatoses, swellings and vulvodynia are common manifestations of vulvar diseases. Proper diagnosis of many vulvar conditions is often delayed because of the delay from the patient's side to reach medical advice.

PRURITUS VULVAE

Pruritus means a *sensation of itching, usually arousing the desire to scratch.*

ETIOLOGY:

A. It is mostly associated with ***irritant vaginal discharge*** as in Trichomonas vaginalis and Candida albicans.

B. Or uncommonly occurring ***without underlying associated vaginal discharge as in cases of:***

- **Dermatoses** as psoriasis, eczema, or dermatitis at other body sites.
- New medication, or an irritant as urine, condoms, toilet paper, moist wipes, scented body soaps.
- Skin drying and mechanical trauma and aggravates the itch/ scratch cycle.
- Patients frequently use nonprescription remedies for relief of vulvovaginal itching or perceived odor. These products commonly contain multiple known contact allergens, and their use is discouraged.

Biopsy is recommended if the cause of symptoms is not obvious; if focal, hyper-/hypopigmented, or exophytic lesions are present.

SOLID VULVAR SWELLINGS

Most solid vulvar tumors are **benign** such as: condylomata, leiyomyoma, fibroma, lipoma.

Less commonly, malignant lesions arise on the vulva and are typically of squamous

cell carcinoma or less commonly metastatic lesions.
If the diagnosis is not obvious by examination, biopsy is mandatory.

CYSTIC VULVAR SWELLINGS

1. Bartholin Gland Duct Cyst and Abscess:

Etiology: Obstruction of Bartholin gland's duct *is* common and may follow infection or trauma by repeated cycling. In some cases, cyst contents may become infected and lead to abscess formation.

Symptoms: Most Bartholin gland cysts are small and asymptomatic. Larger or infected cysts may cause vulvar pain.

Signs: Cysts are typically unilateral, ovoid, and fluctuant or tense. If infected, there is surrounding erythema and tenderness. The mass is located in the inferior labia majora. Bartholin abscesses will exhibit an area of softening, where rupture will most likely occur.

Treatment:

If symptomatic, cyst is managed by marsupialization (deroofing of the duct, to allow direct discharge to the lower vagina and preserve the gland function). Infected cyst is managed by antibiotic therapy, while abscess is managed by incision and drainage.



Fig. 8-12:Bartholin duct cyst cyst

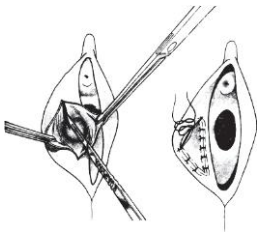


Fig. 8-13: Marsupialization of the

2. Urethral Diverticulum:

Ductal occlusion of the Skene gland or paraurethral glands may lead to paraurethral cystic enlargement and possible abscess formation. They may present with postvoiding dribbling, dysuria and dyspareunia. Finger compression elicits discharge from the diverticulum into the urethra.

3. Epidermal Inclusion Cysts:

These may be sebaceous cysts or clitoridal cyst (specially following FGM) due to traumatic implantation of epidermal cells into deeper tissues and become filled with keratin.
If symptomatic or secondarily infected, excision is recommended.

4. Vulvar Hematoma:

This may develop following injury or trauma. These are more frequent in children. If

small and not enlarging, conservative management with cool packs, pain control is enough.

In large or rapidly expanding hematomas, surgical incision and ligation of the bleeding vessel is needed.

NON-NEOPLASTIC OVARIAN SWELLINGS

Non neoplastic ovarian swellings, also described as **functional cysts** of the ovary, represent almost 24% of all adnexal masses encountered in gynecologic practice.

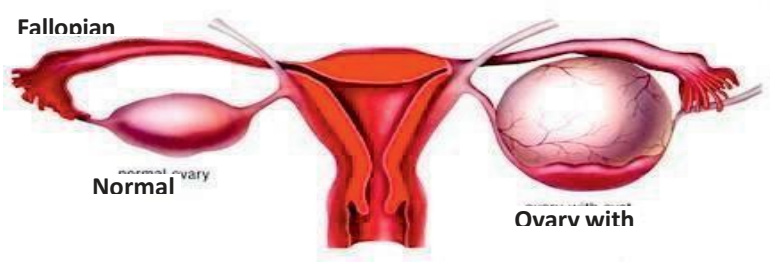


Fig8-14: Normal ovary & Cystic ovarian swelling

CLASSIFICATION OF NON-NEOPLASTIC OVARIAN SWELLINGS, according to their origin:

1. Follicular cysts.
2. Corpus luteum cysts.
3. Theca lutein cysts.
4. Endometriotic cysts (see endometriosis).
5. Inflammatory cysts (ovarian and tubo-ovarian) (see upper genital tract infections).
6. Germinal inclusion cyst.

GENERAL FEATURES OF NON-NEOPLASTIC OVARIAN SWELLINGS:

- **Age incidence:** They occur in the child bearing period mostly at extremes of age.
- **Bilaterality:** usually unilateral, except for theca lutein.
- **Size:** small in size (3-6 cm in diameter, rarely exceed 7 cm).
- **Fate:** spontaneous regression in few weeks is the rule.
- **Complications:** functional ovarian cysts rarely undergo torsion, hemorrhage, or rupture.
- **Management:** functional cysts usually regress spontaneously, so conservative management is the rule. If they don't regress, COCPs may be used. If they reach large size, or become complicated, surgical treatment, usually via laparoscopy, becomes mandatory.

1. FOLLICULAR CYSTS

Follicular cysts are the **commonest non neoplastic** cysts of the ovary. They usually arise from cystic over-distension of an atretic follicle, or due to unruptured dominant ovarian follicle.

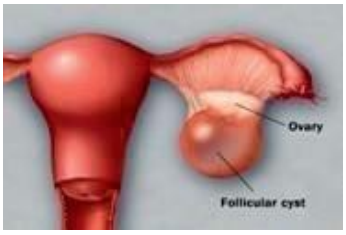


Fig 8-15: Follicular cyst of the ovary



Fig 8-16: Follicular cyst by US

CLINICAL PICTURE AND DIAGNOSIS

A) Symptoms:

1. *Asymptomatic:* most follicular cysts are only accidentally discovered.
2. *Menstrual disorders:* in the form of *delayed cycles* or *intermenstrual spotting* may occur due to persistent estrogen production by granulosa cells
3. *Pain:* mild pelvic pain may occur. Rarely acute severe pain in cases of complicated cyst by hemorrhage or rupture.

B) Signs on bimanual pelvic examination:

- Tenderness may be elicited at the ovarian point (above the mid inguinal ligament).
- Fullness at one of the vaginal fornices, associated with mild tenderness.
- A small cystic mass may be felt in thin patients.

C) Special Investigation:

1. Pelvic TAS or TVS is the ***gold standard in diagnosis***, showing a unilateral, unilocular, thin-walled cyst, < 7.0 cm size, filled with clear fluid, with no internal echoes**.**
2. Laparoscopy is diagnostic whenever diagnosis is in doubt, or cyst complicated.

2. CORPUS LUTEUM CYSTS

Corpus luteum (CL) cysts are the **second most common non neoplastic cysts** of the ovary. They arise from excessive hemorrhage inside the CL during the stage of vascularization. They mostly occur during the child bearing period, and may occur in association with early pregnancy.



Fig 8-17: Corpus luteum cyst



Fig 8-18: CL cyst by US

CLINICAL PICTURE AND DIAGNOSIS:

a) Symptoms:

1. *Asymptomatic* in most cases.
2. *Menstrual disturbance:* in the form of short delay in cycles or intermenstrual spotting may occur due to persistent progesterone (PRG) production.
3. *Pain:* acute lower abdominal pain may be present at one iliac fossa if the cyst is complicated with hemorrhage or rupture, and may be misdiagnosed as appendicitis.

b) Signs on bimanual pelvic examination:

- Tenderness is commonly elicited at the mid-ovarian point at one iliac fossa.
- Tenderness and fullness at one of the vaginal fornices.
- A tender cystic adnexal mass may be felt in thin patients.

c) Special Investigations:

1. Pelvic TAS or TVS is the **gold standard** in diagnosis showing a unilateral, unilocular, cyst < 7 cm, filled with blood that appears as fine particles within clear fluid.
2. laparoscopy is conclusive whenever diagnosis is in doubt or cyst complicated

DIFFERENTIAL DIAGNOSIS:

- From follicular cyst and simple serous cyst.
- Other causes of pain in right iliac fossa (as appendicitis, renal colic, & ectopic pregnancy)

3. THECA LUTEIN CYSTS

Theca lutein cysts of the ovary usually develop in response to abnormally high levels of gonadotrophins (FSH & LH) as in OHSS, or chorionic gonadotrophins (hCG) as in GTD.



Fig 8-19: Theca lutein cysts

CLINICAL PICTURE AND DIAGNOSIS:

a. **History:** drugs for induction of ovulation, multifetal pregnancy, or GTD.

b. **Symptoms:** Pelvic pain which may be severe resembling acute abdomen.

c. **Signs on abdominal and bimanual examination:**

- Large theca lutein cysts may be felt suprapubically. They are usually tender with restricted mobility.
- Abdominal distension and ascites may be present in Grade 3 and 4 OHSS.

d. **Special investigations:**

- US is the gold standard for diagnosis.
- Serum B-hCG shows very high levels in GTD.

4. GERMINAL INCLUSION CYSTS

They are microscopic cysts, that result from invagination of the germinal epithelium into the substance of the ovary at the site of ovulation. Previously they were considered of no clinical importance, but now they are regarded as forerunners for ovarian epithelial cancers.

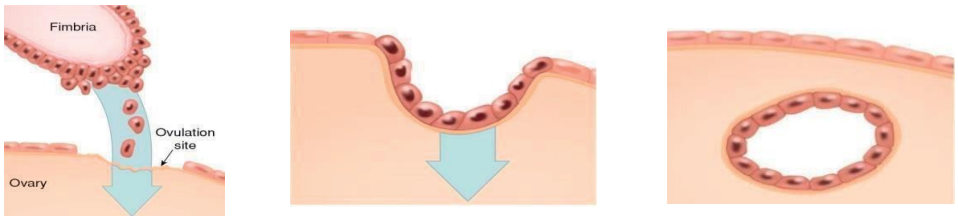


Fig 8-20: Epithelial cells are implanted on the denuded surface of the ovary at the site of ovulation. Subsequently, an inclusion cyst is formed

MANAGEMENT OF NON-NEOPLASTIC OVARIAN SWELLINGS:

A) CONSERVATIVE MANAGEMENT:

- **Small non complicated** functional cysts are managed conservatively since spontaneous regression is the rule in most of these cysts.
- Follow up **US** is used to monitor complete cyst resolution.
- COCPs may accelerate the regression of these functional cysts.

B) SURGICAL TREATMENT:

Large cysts > 7.0 cm, **complicated** cysts (by hemorrhage or torsion), and persistent ones are managed surgically via **Laparoscopy.** Operation includes:

1. Ovarian cystectomy for cysts that reached large size or starting complication.
2. In cases of torsion: Undoing torsion of a cyst before it becomes ischemic and gangrenous.
3. Ovariotomy (removal of a diseased ovary): only in cases with severe hemorrhage or gangrene in which ovarian preservation becomes impossible.

9 GYNECOLOGIC ONCOLOGY

ENDOMETRIAL HYPERPLASIA AND ENDOMETRIAL CARCINOMA

OVERVIEW

Endometrial carcinoma (EC) is now considered the **most common** gynecologic malignancy. It is also the **most curable** owing to the fact that most cases are diagnosed at an **early stage**.

EC occurs on top of atypical endometrial hyperplasia with an excess-estrogen environment.

RISK FACTORS FOR EC/ EH:

- ***Nulliparity or low parity*** (due to absent progesterone effect that occurs during pregnancy and lactation)
- ***Early menarche and Late menopause*** age > 55 years
- ***PCOS*** (chronic anovulation results long periods of E2 unopposed by PRG)***Marked Obesity*** (increased peripheral conversion of androgen to E1 in adipose tissue)
- ***Unopposed estrogen therapy*** (as estrogen re-placement therapy in menopause ERT)
- ***Long term Tamoxifen therapy*** (in treatment of estrogen receptor positive breast cancer)



Fig 9-1: Endometrial carcinoma

- ***Estrogen producing ovarian neoplasm*** (e.g.; granulosa theca cell tumor)
- ***Family history:*** Endometrial cancer is the most common extracolonic manifestation of **Lynch syndrome (hereditary nonpolyposis colorectal cancer [HNPCC]**, which is an autosomal-dominant syndrome and these cases develop more than 10 years earlier than sporadic endometrial cancer.

N.B.: Long term use of **OCPs** is usually associated with a **decreased risk** of EC by nearly **50%**, due to the effect of gestagens in protection from endometrial hyperplasia (EH).

ENDOMETRIAL HYPERPLASIA (EH)

Endometrial hyperplasia (EH) is a pathologic entity that describes an **increase** in **size** and **number** of irregularly proliferating **endometrial glands**, leading to abnormal endometrial **thickness** and increased tendency for bleeding**.**

Hyperplasia of the endometrium occurs in response to prolonged **estrogenic** effect on the endometrium in **absence of** an adequate counteracting progestational effect. It occurs more frequently in the above predisposing risk factors.

PATHOLOGIC TYPES OF EH

A. ***Simple EH:*** cystic endometrial glandular dilatation with **disparity** in **size** (estrogen effect) and no evidence of secretory activity (PRG effect), giving it a *Swiss cheese* appearance microscopically.

B. ***Simple EH with atypia:*** presence of nuclear atypia (abnormal mitosis, hyperchromatosis, and increased nuclear cytoplasmic ratio) in the above type.

C. ***Complex EH:*** new endometrial glands will be formed with resulting **increase in number** and **crowding** with minimal intervening stroma *(back-to-back arrangement* microscopically*).*

D. ***Complex EH with atypia:*** presence of nuclear atypia (as described previously) in the above type

NB: the presence of atypia will raise the malignant potential of EH.

MALIGNANT POTENTIAL FOR EH: ***(WHO classification)***

Type of EH	Progression to EC
• *Simple EH*	*1 %*
• *Complex EH*	*3 %*
• *Simple EH with atypia*	*8 %*
• *Complex EH with atypia*	*29 %*

Endometrial intraepithelial neoplasia (EIN) classification (nomenclature similar to CIN, VaIN, and VIN) is recognized by the ACOG. However, it has not been universally implemented.

PATHOLOGY OF ENDOMETRIAL CARCINOMA:

A. **GROSS:**

- ***Localized type:*** localized area of endometrial affection in the form of **nodule** or **polyp**.
- ***Diffuse type:*** diffuse endometrial thickening or multiple endometrial polyps.

B. **MICROSCOPY:**

- ***Adenocarcinoma:*** is the commonest pathologic type of EC, and carries

best prognosis.

C. GRADING:

Tumor grade is determined by both the degree of abnormality of the glandular architecture, and the degree of nuclear atypia.

- ***Well differentiated tumor (Grade I):*** normally looking glandular epithelium with **< 5%** solid structure. They are slowly invasive and carry the **best prognosis**.
- ***Moderately differentiated tumor (Grade II):*** glandular structure admixed with papillary and occasional solid areas **< 50**%. It is more aggressive and carries **intermediate prognosis**.
- ***Poorly differentiated tumor (Grade III):*** the structure becomes predominantly solid, with minimally identified glandular epithelium. The tumor is highly aggressive with early deep myometrial invasion and generally **poor prognosis**.

D. SPREAD:

1. **Direct Spread: commonest** mode of spread
   - *Myometrial invasion:* depth of invasion correlates directly with prognosis of EC.
   - *Spread* to the cervix.
   - *Spread to ovaries,* fallopian tubes, omentum, and peritoneal surfaces.
2. **Lymphatic Spread:** Occurs later than direct spread
   - *Para-aortic nodes:* draining the fundus.
   - *Inguinal nodes:* draining the cornu.
   - *Para-cervical nodes:* draining the isthmus and body.
3. **Vascular Spread:** occurs late usually with higher grade tumors.
   - Metastases to nearby intra-pelvic organs.
   - Distant extra-pelvic metastases rarely occur (liver, lung, brain & bones).

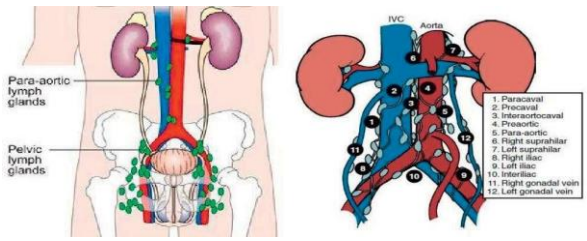


Fig 9-2: Lymphatic spread of EC

CLINICAL PICTURE:

EC usually presents as **AUB** in **postmenopausal** women, with added risk to those who are nulliparous, obese, diabetic and hypertensive (corpus cancer syndrome).

SYMPTOMS:

1. ***Post-menopausal bleeding;*** is the **commonest presenting** symptom in

cases with EC.

2. ***Metrorrhagia:*** perimenopausal women present with **AUB**, which is recurrent and profuse even after medical or hormonal treatment.
3. ***Other symptoms:*** offensive vaginal discharge, pyometra.

SIGNS:

a. *Bimanual Examination:*
   - The uterus may be slightly symmetrically enlarged in size.
   - Consistency is usually softer than expected in a menopausal uterus.
   - The adnexae may be free, or may show a solid ovarian neoplasm, or metastases.

b. *Speculum examination:* is important to exclude;
   - Cervical mass or ulcer (in cases of cervical involvement).
   - Foul odor cervical discharge (in cases of pyometra).

SPECIAL INVESTIGATIONS

1. ***Transvaginal Sonography (TVS): used for screening to measure the endometrial thickness***

- *Increased endometrial thickness* in postmenopausal women above the cut-off value of 5.0 mm is strongly suspicious of EH. Thicker endometrium > 10 mm correlates with EC.
  - *Associated uterine lesion* as polypi, or pyometra.
  - *Associated adnexal masses:* as ovarian neoplasms.

2. ***Fractional Curettage (FC):***
   - **FC** the **gold standard** method for diagnosis of **EC**.
   - An endocervical biopsy is first obtained before cervical dilatation.
   - Full curettage of the endometrial cavity (isthmus, anterior and posterior walls, fundus, and uterine cornu). This is important to confirm the diagnosis of EC, its histologic type and grade, and for detection of spread to the cervix.

3. ***Other Investigations:***
   - Preoperativework Up including;
     - Chest X-ray to exclude lung metastases.
     - Abdominal US to exclude liver metastases, abdominal masses, and ascites.
     - CT scan for pelvis and abdomen, may point to LN involvement.
     - MRI is sensitive in assessment of depth of myometrial invasion if present.

Simplified SURGICAL FIGO STAGING OF ENDOMETRIAL CANCER 2018:

Stage	Stage Description
I	**The cancer is confined to the Corpus uteri.**

IA	No or less than half myometrial invasion
IB	More than half myometrial invasion
II	**The cancer has spread to the cervical stroma (NB: involvement of cervical glands is still considered stage I).**
III	**Local spread outside the uterus.**
IIIA	Spread to serosa of corpus uteri and/ or adnexa.
IIIB	Spread to the vagina or the parametrium.
IIIC	Spread to pelvic lymph nodes.
IV	**The cancer has spread to distant organs.**
IVA	The cancer has spread to the mucosa of the rectum or urinary bladder.
IVB	The cancer has spread to the lungs, liver, bones or brain.

MANAGEMENT OF ENDOMETRIAL CARCINOMA:

Prevention:

Weight loss, chronic progestin supplementation in PCO cases. At the onset of menopause, women are strongly required to report unexpected bleeding or spotting to their provider ***"as any case of postmenopausal bleeding should be considered malignant until proved otherwise".***

Only for women with **Lynch syndrome**, EMB (endometrial biopsy) is recommended every 1 to 2 years beginning at age 30 to 35 years. Prophylactic hysterectomy is recommended once women have completed childbearing.

Active Management:

Following **surgical staging** the treatment will depend upon several factors including; the **stage** of the disease, which is the most important factor, the **histologic type** and **grade** of the tumor, and the patient's **age** and **general health**.

A. *Surgery:*

- An exploratory **laparotomy** with total abdominal hysterectomy and bilateral salpingo-oophorectomy (**TAH-BSO)** is the gold standard treatment, performed in all fit patients with Stage I. while Wertheim's operation is reserved to fit patients with Stage II.
- Peritoneal **washings** are taken with normal saline for cytological examination.
- Pelvic **lymphadenectomy** is performed on **high-risk** cases including those with serous, clear cell types. **Sentinel node biopsy (SLNB)** has

been used as an acceptable replacement for lymphadenectomy.

B. ***Radiation Therapy:***

- Cases of stage I can be followed **without** adjuvant radiotherapy.
- Other stages, and High-risk cases, with **serous** or **clear cell** carcinoma, should receive **external beam pelvic radiation**.
- Patients medically unfit for surgery may receive radiation therapy alone.

C. **Chemotherapy:**

In practice, chemotherapy is frequently combined, with radiotherapy in patients with advanced endometrial cancer to reduce toxicity,

D. ***Hormone Therapy:***

One of the unique characteristics of endometrial cancer is its **hormonal responsiveness**. Thus, for Stage I young women who desire fertility, continuous progestin treatment or a levonorgestrel-releasing IUD can be a therapy that helps to reverse pathology.

RECURRENT DISEASE:

- More than 80% of recurrences occur within the first two years of treatment.
- The vaginal vault is the primary site for recurrent disease
- The Treatment of recurrent disease is individualized depending on the site and extent of recurrence.

CIN AND CANCER OF THE CERVIX

OVERVIEW

Cervical cancer is the second most common cancer of the female genital tract. It has higher incidence in **developing countries**.

Widespread use of **Pap test** for screening has reduced the incidence and mortality from cervical cancer by more than 80%.

Racial, ethnic, geographic and socioeconomic factors play important roles in the **prevalence** of this disease. **Sexual** and **hygienic** habits are also important factors. The slow progression of cervical cancers from precancerous lesions **(CIN III)**, to invasive cancer provided the rationale for the preventive and early treatment strategies to control the disease.



Fig 9-3: Cervical cancer

CERVICAL INTRAEPITHELIAL NEOPLASIA (CIN)

BENIGN SQUAMOUS METAPLASIA:

Both the **squamous cells** (covering the ectocervix) and the **columnar cells** (lining the endocervix) come from **immature reserve cells** present at the **squamo-columnar junction SCJ** (the junction between the ectocervical squamous epithelium and endocervical glandular epithelium).

The **immature reserve cells** at the **SCJ** continuously generate both squamous cells and columnar cells by **metaplasia** throughout the female's life.

The **SCJ** changes its place by age, moving slowly continuously towards the internal cervical os:

- *During childhood;* it is at the outer edge of the external os.
- *During the reproductive life;* it moves slightly inwards few millimeters within the external os.
- In postmenopausal women; it moves further inwards to be placed inside the cervical canal.

The Transformation Zone (TZ): is the **area where squamous metaplasia occurs. It lies be-tween original SCJ and the new SCJ.**



Fig 9-4: Transformation zone TZ

ABNORMAL METAPLASIA AND CIN:

Continuous exposure of **reserve cells** at the **TZ** to **carcinogenic factors**, as **viral DNA** may lead to the production of **atypical cells** (through atypical metaplasia) which might progress by time to **preinvasive** or **invasive** cervical **cancer**.

Cervical Intraepithelial Neoplasia (CIN): describes the presence of **atypical cells** in the cervical epithelium **without invasion** of the basement membrane.

- **CIN 1**: indicates the lower third epithelial involvement.
- **CIN 2:** indicates the lower 2/3 epithelial involvement.
- ***CIN 3 or CIS:*** *indicates* full-thickness involvement epithelial involvement but without invasion of the basement membrane.

CIN 1, is now recognized as evidence of HPV infection, most of which is transient and unlikely to progress. CIN 2 and CIN 3 are considered to be true precancerous lesions, especially CIN 3.

In contrast, the cervical columnar epithelium is only one cell layer thick, so its histologic abnormalities are just *adenocarcinoma in situ (**AIS**).*

In **1989, the Bethesda System** cytologic nomenclature replaced CIN with *squamous intraepithelial lesion (SIL) as such:*

- CIN I as *low-grade squamous intraepithelial lesions (**LSIL**).*
- CIN 2, CIN 3, and CIS as *high-grade squamous intraepithelial lesion (**HSIL**).*

These 2 nomenclatures are now recommended for histology as well as guidelines for the management of these lesions.

RISK FACTORS FOR CIN AND CANCER CERVIX:

1. **Abnormal sexual activity:**
   - Early age at first sexual intercourse (the transformation zone being near to external os).
   - Multiple sexual partners (subject cervical epithelium to variable foreign proteins).
2. **Viral sexually transmitted diseases:**

A. ***Human Papilloma Virus (HPV):***

- High-risk HPV types include 16, 18, 31, 33, 35, 45, & 58, and account for nearly 95% of precancerous lesions worldwide.
- If the virus remains present for several years, in patients with low immunity and repeated exposures, cervical intraepithelial neoplasia (CIN) may develop.

B. ***Herpes Simplex Virus type II (HSV-2):***

- HSV-2 DNA and messenger RNA sequences have been found in cervical cancer cells.
- They may also increase the likelihood of HPV and HIV infection.

3. **Low socioeconomic standards** and **poor hygiene (chronic irritation).**
4. **Cigarette smoking.** (By-products in cigarettes may concentrate in cervical epithelium, with depletion of Langerhan's cells that assist in cell mediated immunity).
5. **Immuno-suppression** (HIV infection, or immunosuppressant drugs used in organ transplant).

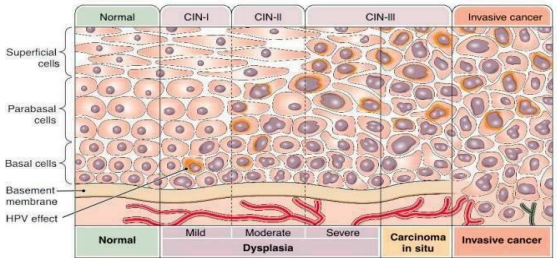


Fig 9-5: Histologic grading of CIN

DIAGNOSIS OF CIN:

- **CIN** usually affects women in **younger age** groups (25-45 years).

- It is usually **asymptomatic or may present by contact bleeding.**
- **Diagnosed** through routine screening (every 3 years for low-risk population and yearly in high-risk population) by the **Pap smear** test.
- Abnormally suspicious Pap smears are followed by **colposcopy** and **biopsies** from abnormal areas.

1. CERVICAL PAP SMEAR Test:

- The Pap smear test is based on cytologic examination of cells shed from both the ectocervix and endocervix.
- It is performed using a **cytobrush** to wipe cells from the surface of the TZ of the ectocervix and from the endocervical canal.
- Cells obtained are spread on a glass slide fixed by ethyl alcohol and stained by **Papanicolaou stain**.
- The test is done as an office procedure with an accuracy rate up to 95%.

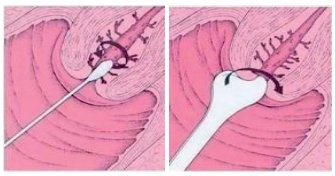
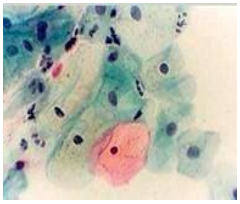
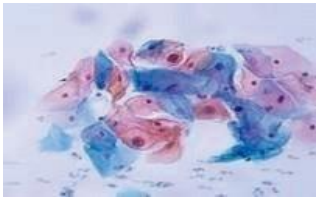


Fig 9-6: Pap smear test

- Currently, most cervical cytology specimens are collected into liquid-based media that is FDA approved.

- **Interpretation of a Pap smear:**
  1. ***Normal:*** *absence of any abnormal cells*.
  2. ***ASCUS*** *(Atypical squamous cells of undetermined significance):* Cells suggestive, but not fulfilling the criteria for squamous intra-epithelial lesions **(SIL)**. For its evaluation, testing for HPV is required.
  3. ***LSILs*** *(Low grade squamous intra-epithelial lesions)* concomitant with **CIN 1**.
  4. ***HSILs*** *(High grade squamous intra-epithelial lesions)* concomitant with **CIN 2 & CIN 3**.
  5. Squamous cell carcinoma.
  6. ***Atypical*** *glandular cells (**AIS**).*
  7. *Endocervical **Adenocarcinoma**.*





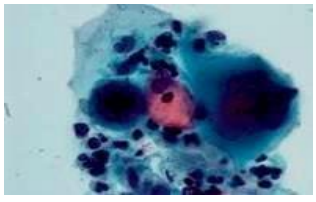


Fig 9-7: Normal Pap Smear CIN 1 *CIN3*

NB: In case of a gross lesion by naked eye, **direct biopsy** will replace the Pap test.



2. COLPOSCOPY:

- Colposcopy allows **inspection** of the **TZ** with a magnification up to 20 times after applying 3%-5% acetic acid solution.
- Colposcopic directed **biopsies** are performed from areas of abnormal epithelium (aceto-white areas, punctuation,
- mosaicism, leukoplakia, and Schiller's iodine negative areas).

 Endocervical curettage is performed to rule out dysplasia when the TZ in the cervical canal is not properly visualized in presence of an abnormal pap smear.

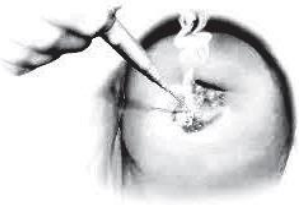


Fig 9-9: Diathermy destruction of CIN

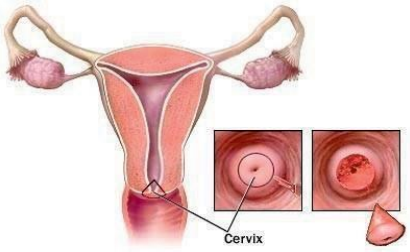


Fig 9-10: Cold knife excision of TZ

3. HPV TEST:

has become an important screening tool either paired with cytology *(co-testing) as in cases of ASCUS* or, recently, as a stand-alone screening test.

CIN MANAGEMENT:

- ***LSIL: CIN 1***

Follow up by Pap smear after 3 months without treatment, or antibiotic treatment. Persistent CIN 1 can be treated by ablation (Laser, diathermy or cryocautery)

- ***HSIL: CIN 2 and CIN 3***

Loop Electrosurgical Excision Procedure (LEEP) or LLTZ (Long Loop excision of the transformation zone) or conization (remove a cone from the cervix, with the base towards the ectocervix and the apex towards the cervical internal os), all these procedures aim to remove the entire TZ (transformation zone) or hysterectomy if the patient is not desiring fertility.

- ***Adenocarcinoma in Situ***

Cold-knife conization is favored over loop electrosurgical excision procedure (LEEP) to allow for healthy edges to be examined for the presence of malignant cells.

INVASIVE CARCINOMA OF THE CERVIX

INCIDENCE AND ETIOLOGY:

Invasive cervical cancer has significantly declined in the last 3 decades, owing

to the effective screening programs using the Pap smear test and the use of HPV vaccine. However, this decreased the ectocervical cancer incidence and raising the incidence of endocervical cancer.

The age incidence for cancer cervix is commonly between 45-55 years which is almost 10 years younger than that for endometrial carcinoma, and 10 years older than that for CIN lesions.

PATHOLOGY:

A. Squamous cell carcinoma of the Ectocervix (70%):

- ***Gross:***
  - Friable necrotic mass easily bleeding on touch, or
  - Deep ulcer with everted and indurated edges, or
  - Indurated nodule that may break forming an ulcer.
- ***Microscopically:***
  - The tumor is formed of cell nests with or without keratin.
  - The cells are commonly large with malignant criteria but may be small undifferentiated.

B. Adenocarcinoma of the Endocervix (30%):

- ***Gross:***

They may distend the cervix giving it a barrel-shaped appearance. By the time it appears at the external os, the carcinoma has often spread extensively outside the cervix.

- ***Microscopically:***

The cells are malignant columnar cells arranged in glandular pattern with malignant criteria. In poorly differentiated tumors, the glandular pattern is replaced with unarranged cell nests.

- ***GRADING:***

a. ***Well differentiated (grade I):*** the majority of cells (more than 95%) resemble the normal squamous epithelium of the cervix. They grow slowly and are less likely for early spread.

b. ***Moderate and poorly differentiated (grade 2 & 3);*** cervical carcinoma is predominantly of the moderate or poorly differentiated type. Abnormal cells predominate; they are more likely for rapid growth and early spread.

- ***SPREAD:***

1. **Direct spread:** To adjacent tissues including upper and lower vagina, parametrium, uterosacral ligaments. Later spread may involve the bladder and rectum.
   Spread to the parametrium may lead to ureteric obstruction with subsequent development of renal failure (**most common cause of death).**
2. **Lymphatic spread: Most common route** of spread. It involves;
   a. *Primary groups:* include spread from the cervical lymphatics along to the

paracervical LNs, to the obturator and external iliac nodes, and occasionally backwards to the internal iliac group.

b. *Secondary groups:* involve drainage into the common iliac and lateral sacral groups. Ultimately the common iliac group drains into the para-aortic lymph nodes.

3. **Blood spread:** Least common, usually in late or advanced cases. Metastases may occur to distant organs as to the liver, lungs, brain and bone.

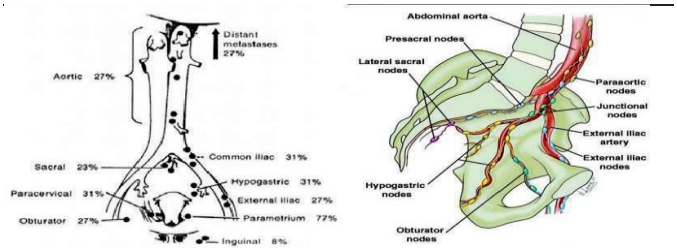


Fig 9-11: Lymphatic drainage and spread of cancer cervix

CLINICAL PICTURE:

A) SYMPTOMS:

- *Contact bleeding;* is the commonest presentation (postcoital bleeding or bleeding on touch). uncontrolled hemorrhage from a tumor bed may occur. In such cases, bleeding can often be controlled with a combination of Monsel paste (ferric sulfate) and vaginal packing.
- *Vaginal discharge* which is excessive bloody or malodorous.
- *Pain:* deep pelvic pain and loin pain may be associated with advanced disease.

B) SIGNS:

1. ***General examination:***
   - Early stages do not affect the patient's general condition.
   - Advanced stages are associated with; chronic blood loss, urinary manifestations and ureteric obstruction which may lead to severe anemia, uremia (leading cause of death) and cachexia.
2. ***Palpation by digital vaginal examination (PV):***
   - Early stages: mass or ulcer that bleeds profusely when touched by the finger due to friability.
   - Later stages: as the disease progresses, the cervix loses its mobility, becomes fixed, and the surrounding parametrium becomes tender and indurated (**detected by PR examination**).
3. ***Bimanual pelvic examination:***
   - The uterus is usually normal in size.
   - Symmetrical uterine enlargement may occur if pyometra or hematometra may develop.
4. ***Inspection via a speculum vaginal examination:***

- Early stages: small friable warty mass, nodule, or an ulcer that bleeds on touch.
- Later stages: mass or ulcer extending to the vaginal walls obliterating the vaginal fornices.

5. ***Rectal examination (PR):***
   - For detecting parametrial extension and uterosacral involvement

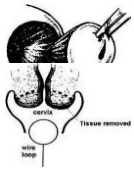


Fig 9-12

DIAGNOSIS:

Histopathologic examination of **cervical tissue biopsies (either direct biopsy, punch biopsy, conization, or endocervical biopsy in cases of endocervical carcinoma)** containing the abnormal epithelium is the gold standard for diagnosis of cervical cancer.

CLINICAL STAGING AND PREOPERATIVE PREPARATION:

The clinical staging system incorporates: cold-knife conization, pelvic examination under anesthesia (EUA), cystoscopy, proctoscopy, chest radiograph, intravenous pyelogram (IVP), computed tomography [CT] or positron emission tomography (PET) scan.

FIGO classification is used for gynecologic cancers. In contrast, the American *j*oint Committee on Cancer (AJCC) developed the *TNM Staging System,* which is based on the extent of the tumor (T), the extent of spread to the lymph nodes (N), and the presence of metastasis (M).

Simplified Clinical FIGO Staging for CARCINOMA OF THE CERVIX 2018

Stage	Stage Description
0	**CIS (Carcinoma in Situ): with no invasion of the basement membrane.**
I	**Carcinoma Is strictly confined to cervix (extension to corpus is disregarded**
IA	Microscopic lesion, with stromal invasion with a maximum depth <5 mm
IB	Clinical lesions confined to the cervix greater than IA
II	**Carcinoma causes local spread beyond cervix**
IIA	Involves vagina, but not the lower third.
IIB	it involves the parametrium, but not reaching the lateral pelvic wall.

III	**Carcinoma has extended more beyond the cervix**
IIIA	involvement of lower third of vagina, but No extension to pelvic wall.
IIIB	carcinoma extended to the lateral pelvic wall, on rectal examination there is no cancer-free space between tumor and pelvic wall (all cases with hydronephrosis or nonfunctioning kidney should be Included).
IIIC	involves pelvic and/or para-aortic lymph nodes.
IV	**Distant Spread**
IVA	Carcinoma has clinically involved mucosa of bladder or rectum.
IVB	Spread to distant organs.

PREVENTION OF CANCER CERVIX:

Invasive cancer cervix is the most preventable female genital cancer. It is achieved via;

A. **PRIMARY PREVENTION:**

- **An HPV vaccine** has been approved for prevention of HPV infection especially types 6, 11, 16 & 18.
- The vaccine is over **95% effective** in preventing HPV 16, 18, related cervical cancers.
- The vaccine is effective when given in age groups from 9-26 years, in non-previously infected population.

B. **SECONDARY PREVENTION:** early detection and proper management of CIN lesions

- *Regular screening* (Pap smears): Regular screening resulted in a significant decrease in the incidence of invasive cancer cervix.
- *Treatment of CIN lesions* as previously mentioned.

TREATMENT OF INVASIVE CANCER CERVIX

In general, early-stage disease (Stage I & IIA) is effectively eradicated surgically. For advanced disease (stage IIB & higher), chemoradiation is primarily selected.

- **Stage IA:**
  - For fertility preservation, Therapeutic **conization.**
  - Old patient/family completed = Simple extrafascial hysterectomy.
- **Stage IB:**
  - For fertility preservation, may be treated with **radical trachelectomy** (with non-absorbable cerclage placement) and lymphadenectomy. Although this technique is promising, further studies to validate its efficacy are needed.
  - Old patient/family completed = Simple extrafascial hysterectomy ie TAH.

- **Stage IIA:**

Radical/Wertheim hysterectomy (total abdominal hysterectomy, bilateral salpingo-oophorectomy, removal of upper cuff of the vagina and lymph node sampling) + Concurrent chemoradiation: Surgery is appropriate for those who are physically able to tolerate an aggressive surgical procedure, to avoid the long-term effects of radiation therapy, as young women who desire ovarian preservation and retention of a functional, nonirradiated vagina.

- **Stage IIB – IV:**

Chemoradiation and palliative treatment: Current evidence indicates that chemotherapy given concurrently with radiation therapy significantly improves survival rates. More recently, adding monoclonal antibody, to combination chemotherapy increases the median overall survival.

RADIOTHERAPY IN CANCER CERVIX

- All stages of cervical cancer are radiosensitive.
- In early stages, cure rates are comparable to those for surgery.
- In more advanced stages, cancer will not be curable by surgery alone, and radiotherapy becomes the preferred first line of treatment. It may be given alone, or in combination with surgery or chemotherapy.

- It involves external beam radiotherapy (EBRT) to the pelvis followed by intracavitary treatment or brachytherapy.

Pelvic Exenteration for Primary Disease:

This ultraradical surgery comprises removal of the uterus, fallopian tubes, ovaries, vagina, bladder, rectum, and surrounding tissues. Exenteration is rarely performed nowadays due to its high morbidity.

CARCINOMA OF THE CERVICAL STUMP (after subtotal hysterectomy)

Cancer cervix may develop in the stump of the cervix left after subtotal hysterectomy performed for benign indications as leiomyomata, DUB, atonic postpartum hemorrhage.

Surgery (radical cervicectomy and lymphadenectomy) is complicated by presence of adhesions and abnormal anatomic relations induced by the previous hysterectomy.

Radiotherapy: EBRT may be the best option as internal irradiation is compromised by the absence of the uterus as a container for intrauterine applicators, while vaginal irradiation may not deliver the sufficient dose without risk of damage to the bladder and rectum.

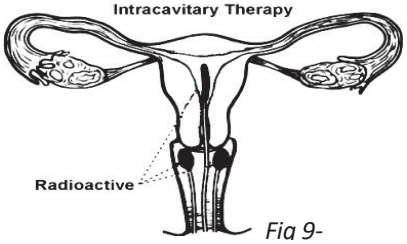


Fig 9-

CARCINOMA OF CERVIX DURING PREGNANCY

Survival rates between pregnant and nonpregnant women with cervical cancer do not differ when matched by age, stage, and year of diagnosis.

• Stages I- IIA Cancer in Pregnancy

Women with early stages may wait till fetal maturity then deliver and be reevaluated.

• Stages IIB- IV Cervical Cancer in Pregnancy

For patients with late stages, pregnancy may impair prognosis. A woman who elects to delay treatment, to provide quantifiable benefit to her fetus, will need to accept an undefined risk of disease progression.

VIN AND VULVAL CANCER VULVAR INTRAEPITHELIAL NEOPLASIA (VIN)

Etiology and Risk factors of VIN/ Cancer vulva:

ACOG summarized VIN to have three categories of squamous intraepithelial lesions (SIL):

- **Vulvar LSIL** (flat condyloma, or condyloma accuminata HPV 6, 11).
- **Vulvar HSIL** (VIN usual type): associated with oncogenic HPV infection, particularly HPV 16, occurs in younger age specially with tobacco smoking.
- **Differentiated VIN**: associated with chronic vulval skin irritation as in lichen sclerosis and occurs in older women.

Other rare intraepithelial vulvar neoplasms include melanoma and Paget disease.

Clinical Picture:

- Asymptomatic and discovered with routine gynecologic examinations.
- Ulcer, warty papule, white hyperkeratotic plaque, hyperpigmented lesions; or erythematous areas.
- Lesions may be raised or flat.

Investigations:

Most suspicious focal vulvar lesions are biopsied. Vulvoscopy may be needed to take the biopsy

Vulvar changes are enhanced by applying a 5% acetic-acid. Biopsies can be obtained using a Keyes punch after provision of a local anesthetic injection.

Prognosis and Prevention:

HPV vaccination lowers the risk for both vulvar LSIL and vulvar HSIL. **Smoking cessation** and improving compromised immune status are also important.
As differentiated VIN may *be* found with **vulvar dermatoses, treating** *these* conditions may reduce cancer risk.

Management:

- **Vulvar LSIL:** Is self-limited and not precancerous and is not treated unless symptomatic.
- **Vulvar HSIL:** Vulvar HSIL is a premalignant condition. Thus, all vulvar HSIL is treated by **local excision** with laser ablation or with use of **Topical 5-FU** (off label, as it is not FDA approved), which is potentially caustic and teratogenic.

Cavitational ultrasonic surgical aspiration ***(CUSA)*** may be used to treat vulvar HSIL confined solely to non-hairy skin, however; The tissue specimen is fragmented and lacks the diagnostic accuracy.

INVASIVE CANCER OF THE VULVA

Overview:

Vulvar cancer is rare. It constitutes approximately **6% of all gynecologic cancers**.

Vulvar cancer typically arises within the covering **squamous epithelium**. Vulvar squamous neoplasia arises predominantly along the **Hart line** (which marks the border between the vulvar keratinized squamous epithelium, and the nonkeratinized squamous mucosa).

Pathology:

Grossly: Lesions may be raised, ulcerated, pigmented, or warty.
Microscopically: approximately **90% are squamous cell carcinoma**. Malignant melanoma is the second most common, but rare histologic subtypes also may be encountered.
Spread: *Direct extension* to adjacent structures, as the vagina, urethra, and anus.

- *Lymphatic permeation* to regional ***lymph nodes*** **(common).**
- *Hematogenous spread* to distant sites, including the lungs, liver, brain and bones (rare).

Clinical picture:

Women with VIN and vulvar cancer commonly present with **pruritus** and a **visible lesion**. However, pain, bleeding, ulceration, and inguinal *mass* may present.

Diagnosis:

Biopsy is needed (as mentioned in VIN) for histological diagnosis.
Radiological investigations (CT, MRI, PET Scan) are needed to determine any metastasis and the extent of the lesion.

MANAGEMENT OF INVASIVE CANCER VULVA:

- ***Surgery:*** Surgery is often an integral part of vulvar cancer treatment. Potential procedures include, **wide local excision**, **simple vulvectomy** (with 1cm safety margin) and sentinel lymph node biopsy, and **radical total vulvectomy** (which includes also inguino-femoral lymphadenectomy).
- ***Sentinel Lymph Node Biopsy only:*** (without the need for inguinofemoral LN sampling) is used in lesions less than 4cm. This dramatically reduces surgical morbidity yet adequately assesses nodal involvement.
- ***In advanced vulvar cancer*** involving the proximal urethra, anus, or rectovaginal septum many centers have been using pre or post operative radiation or chemo radiation in addition to conservative surgical excision. Also, it may be used alone in surgically unfit patients.

BENIGN OVARIAN TUMORS

CLASSIFICATIONS OF OVARIAN NEOPLASMS:

The ovary is a common site for various types of neoplasms that are usually described according to different classifications which include the following:

A) **PATHOLOGIC CLASSIFICATION:** Ovarian neoplasms are classified pathologically into ***benign, Low malignant potential (LMP) or border line*** and ***malignant*** tumors. Benign neoplasms are much more common than malignant ones.

B) **CLINICAL CLASSIFICATION:** Ovarian neoplasms are clinically divided into ***cystic*** and ***solid*** tumors. Cystic tumors are more common and are mostly benign, while solid tumors are much less common and are mostly malignant.

C) **HISTOGENETIC CLASSIFICATION:** The most internationally accepted classification is the WHO histogenetic classification based on the cell of origin of the tumor, either from:

- ***Surface epithelium*** (epithelial tumors): **commonest,** around **90%** of ovarian tumors**,** occurs in **postmenopausal** women.
- ***Ovarian germ cells*** (germ cell tumors): around **10%** of ovarian tumors, **commonest in childbearing period**.
- ***Ovarian stroma*** (sex cord and stromal cell tumors): least encountered **≤ 1%** of ovarian tumors, usually at the extreme of ages especially **postmenopausal.**

1. EPITHELIAL TUMORS OF THE OVARY:

These tumors originate from the ***surface epithelium*** which embryologically develops into Mullerian and Wolffian epithelium, and can thus differentiate into different types of epithelia. They are classified into;

a. **Serous cystadenoma:** lined by cells resembling tubal epithelium.
b. **Mucinous cystadenoma:** lined by cells resembling cervical epithelium.
c. **Endometrioid tumors:** lined by cells resembling endometrial lining.
d. **Brenner tumor**: lined by transitional cells resembling urinary tract epithelium.

2. Germ cell tumors:

These ***germ cells*** that may show either:

a. ***Differentiation into Embryonic tissue:*** this will give various forms of **teratoma**, which contains elements of the three germ layers (ectoderm, endoderm, and mesoderm). This is the most common germ cell tumor.
b. ***Differentiation into Extraembryonic tissue:*** this will give either:
   - Trophoblastic tissue; with development of **choriocarcinoma** (malignant from the start).
   - Yolk sac tissue: with development of **yolk sac tumor** (Endodermal sinus tumor) (malignant from the start).
c. ***No differentiation of germ cells:*** this will result in the development of

gonadoblastoma.

3. SEX CORD STROMAL TUMORS:

These are the least commonly encountered tumors, originating from:

a. *The primitive sex cord;* which developmentally gives origin to granulosa cells in the ovary, or Sertoli cells in the testicle. It may give rise to malignant tumors either **Granulosa cell tumor or Androblastoma** (also known as Sertoli-Leydig cell tumor).

b. *The stroma of the gonad;* which differentiates into theca cells in the ovary or fibrous tissue stroma. It may give rise to either **Fibroma** or **Thecoma.**

PATHOLOGY OF BENIGN OVARIAN NEOPLASMS:

I. EPITHELIAL OVARIAN TUMORS:

They are essentially benign but could be either border line or malignant.

A. SEROUS CYSTADENOMA

This is the ***commonest benign ovarian neoplasm***. They are bilateral in up to 50% of cases, usually of moderate sizes, it may present in one of the following three types:

1. **Simple serous cysts:** The commonest type of serous cysts. They are unilocular, thin walled, and filled with thin clear serous fluid. Simple serous cysts have the lowest malignant potential in all epithelial tumors.
2. **Multilocular serous cyst.**
3. **Papillary serous cysts.**



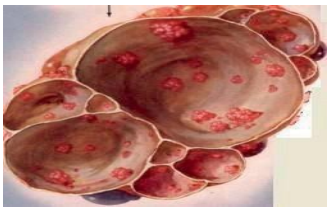


Fig 9-14: Simple serous cyst *Fig 9-15: Papillary serous cyst*

B. MUCINOUS CYSTADENOMA

These are the ***second most common benign*** ovarian neoplasms. They are usually unilateral, having a bluish or yellowish transparent color. Essentially benign with very low malignant potential (<5%)

- **Cut section:** Mucinous cystadenomas are characterized by being **multilocular** with plenty of locules containing thick gelatinous like mucin material. They are known to reach huge size sometimes filling the entire abdominal cavity.

- **Pseudomyxoma peritonii:**



Fig9-16: Mucinous cyst

This is a rare condition, resulting from rupture of a mucinous cyst, in which, gelatinous mucin material is found free in the peritoneal cavity.

Diagnosis is seldom made before laparotomy, and prognosis is poor even after surgery, as reaccumulation of gelatinous material after removal is the rule.

Treatment: Chemotherapy (radioactive intraperitoneal colloid installation), after surgical removal of the tumor and contents.

c. BRENNER TUMOUR:

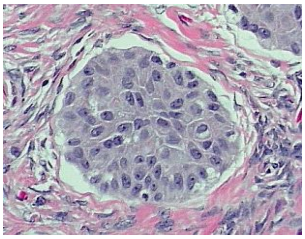
These are **rare** tumors arising from Wolffian metaplasia of the surface ovarian epithelium, they are bilateral in 10% of cases and mostly **benign**.

- Brenner tumors are **solid** in consistency, usually of **small** (< 2 cm) **to moderate size**, and are incidental discovered during pathological examination of the specimen.
- **Microscopically:** the tumor is characterized by ***epithelial cell nests*** with characteristic **coffee bean** nuclei.
- **Hormonal function:** occasionally some of Brenner tumors may secrete estrogen and may even present by vaginal bleeding (i.e.; functioning Brenner tumor).

Fig 9-17: Brenner tumor



Fig 9-18: Coffee bean appearance by microcopy



I. GERM CELL TUMORS:

Germ cell tumors constitute almost **10%** of all ovarian neoplasms, occurring mostly in **childbearing age** women. They are generally benign tumors.





Fig 9-19: Bilateral dermoid cysts



Fig 9-20: Cut section in dermoid cyst with hair and cartilage

A. BENIGN CYSTIC TERATOMA (BCT) (Dermoid cyst):

BCT, also known as dermoid cysts. They represent the commonest ovarian neoplasm during the ***childbearing*** *period* and especially ***during pregnancy***.

BCT are bilateral in 10% of cases. They are usually of moderate size. Most cysts have a long pedicle that makes them more liable to complications as torsion.

- **Cut section:** Cysts are greyish in color**,** mostly multilocular, with variable tissue contents including hair, teeth, bone, cartilage, mixed in a sebaceous material.
- BCT have **very low** malignant potential <1% (squamous cell carcinoma).

N.B.: Most tumors are asymptomatic, incidentally discovered, unless complicated by torsion or rupture causing chemical peritonitis.

B. STRUMA OVARII:

Struma ovarii is an example of a monodermal teratoma (composed of a single tissue) that is composed of hormonally active ***thyroid tissue***, capable of producing thyroid hormone.

C. GONADOBLASTOMA:

A **benign solid** tumor composed of germ cells. Although gonadoblastoma is initially benign, 50% of these tumors may predispose to development of malignant Dysgerminoma. Almost all patients with a gonadoblastoma have an abnormal gonad, with a Y chromosome in 90% of cases.

N.B.: Prophylactic gonadectomy should be performed in patients with dysgenetic gonad, particularly with a Y chromosome (as in AIS) for fear of future development of malignant neoplasms.

II. SEX CORD STROMAL TUMORS:

They are almost always **benign**.

A. FIBROMA:

Fibromas are **rare** tumors derived from stromal cells, seen mostly in women around age of 50 years. They are **benign**, **solid**, tumors, **unilateral** in >90% of cases, usually mobile with a long pedicle and a lobulated glistening white surface.

Ascites and Right pleural effusion may occur with many of the fibromas

giving the picture of **Meig's syndrome**. (Meig's syndrome is the association of ovarian Fibroma with ascites and right-side pleural effusion that disappears on removal of the tumor).

B. THECOMA (THECA CELL TUMOR):

Theca cell tumors are **rare**, with the majority occurring in post-menopausal women. Almost all are **benign**, **solid**, and **unilateral**. **Thecomas** usually produce excess **estrogen.** *They* May be occasionally luteinized and these luteinized thecomas are hormonally inactive or are androgen producing.

COMPLICATIONS OF BENIGN OVARIAN NEOPLASMS:

Benign ovarian tumors are liable to several complications that may form the main presenting clinical picture in many cases. These include; **torsion, hemorrhage, rupture, infection, incarceration,** and **malignant transformation.**

TORSION:

It is the commonest complication. Torsion may be gradual or sudden, incomplete or complete and it may be initiated by a sudden movement of the patient.

- *Gradual incomplete torsion;* leads to congestion, enlargement of the tumor and thrombosis.
- *Sudden complete torsion;* causes sudden occlusion of the arterial blood supply leading to necrosis and gangrene of the cyst.

- **Factors predisposing to torsion:**

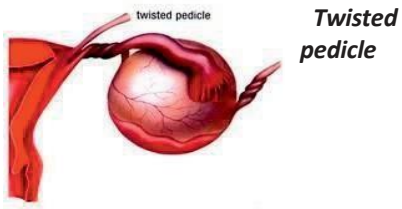


Fig 9-21: Torsion ovarian cyst

1. *Moderate size tumors with long pedicle of the cyst:* BCT is the commonest tumor to undergo torsion due to its long pedicle. Large tumors cannot usually undergo rotation.
2. *Free mobility with absence of surrounding adhesions;* facilitate torsion.
3. *Pregnancy and puerperium:* Due to displacement of the tumor by pregnancy and laxity of abdominal wall after delivery.

- **Clinical picture:** is that of ***acute abdomen***, where the tumor is tense and tender on palpation.
- **Treatment:**
  a. Ovarian cystectomy: untwisting of the pedicle and removal of the cyst with preservation of the ovary has a place if there is still adequate

healthy ovarian tissue present.

b. Ovariotomy (removal of the diseased ovary), usually via laparotomy, is the treatment of choice in a gangrenous cyst, in absence of viable healthy ovarian tissue.

2.HEMORRHAGE:

Hemorrhage in a cyst may occur as a result of torsion, trauma to the abdomen or may occur spontaneously especially during pregnancy.

- **Clinical picture**: is that of ***acute abdomen***, where the tumor is tense and tender on palpation.
- **Treatment**: Ovariotomy usually via laparotomy.

3.RUPTURE:

Rupture of an ovarian cyst may occur as a result of torsion or hemorrhage, or may be traumatic, during labor or due to rough vaginal examination.

- **Clinical picture:**

a. Severe pain with picture of acute abdomen (abdominal wall rigidity, tenderness, and rebound tenderness)

b. Intraperitoneal hemorrhage may occur. Palpation reveals dullness in the flanks with shifting dullness.

c. Rupture of a dermoid cyst may lead to severe chemical peritonitis, while rupture of a mucinous cystadenoma may lead to pseudomyxoma peritonii.

- **Treatment:**

a. Sedatives to alleviate pain.

4.INFECTION:

Ovariotomy done as an emergency procedure usually via laparotomy, and peritoneal lavage performed to remove contents of the cyst.

Ovarian cyst may be infected either;

- In the **puerperium** (following abortion or labor), or
- From a nearby **infected organ** (as appendicitis or pyosalpinx).

- **Clinical picture:**
  - General picture of infection as general malaise, fever, tachycardia, tachypnea, etc…
  - Bimanual examination shows a fixed, tender tumor with rapid increase in size.
- **Treatment:**
  - Antibiotics to control infection (broad spectrum + anaerobic types).

5.INCARCERATION:

Ovarian cyst may be impacted in Douglas pouch, particularly in large size tumors.

- **Clinically:** it may cause pressure symptoms on bladder and rectum. During pregnancy it may cause obstructed labor.
- **Treatment** is by surgical removal of the tumor by cystectomy or ovariotomy.

6.MALIGNANT TRANSFORMATION:

- Papillary serous cystadenoma has malignant potential up to 50%.
- Mucinous cystadenoma has a 5% incidence of malignant transformation.
- Benign cystic teratoma carries less than 1% risk of malignancy.
- Gonadoblastoma in 50% of cases transform into dysgerminoma.

CLINICAL PICTURE OF BENIGN OVARIAN NEOPLASMS:

A) SYMPTOMS:

1. **Asymptomatic:** many of the benign ovarian tumors are discovered during ultrasonography.
2. **Abdominal swelling:** may be felt by the patient only if the tumor is large.
3. **Lower abdominal pain:**
   - Acute pain; if the tumor is complicated by torsion, rupture, hemorrhage, or infection.
   - Chronic dull aching pain; in large-sized swellings as in mucinous cystadenoma.
4. **Pressure symptoms:** if the tumor is huge or incarcerated.
   - Abdominal pressure symptoms; Epigastric pain, dyspnea and palpitation.
   - Pelvic pressure symptoms: Frequency of micturition, retention of urine.
5. **Menstrual disorders:** ovarian neoplasms are usually not associated with menstrual disturbances, except in rare functioning tumors (thecoma & functioning Brenner).

B. SIGNS: These will depend largely on the size of the tumor.

1.Small tumors: are only detected on ***bimanual pelvic examination***, felt on one side of the uterus, separate from the uterus (the movement of the mass is not transmitted to the cervix).

2.*Large tumors:** can be detected by ***abdominal examination:

- ***Inspection:*** abdominal enlargement.
- ***Palpation:*** abdominal mass to one side of the midline, with a well-defined borders, smooth or lobulated surface, usually tense, commonly mobile from above downwards.
- ***Percussion:*** central dullness is elicited on the mass, with resonant flanks (except if associated with marked ascites, where flanks become dull with shifting dullness).

- **N.B.:** in excessively large cyst, as mucinous cysts, *ovarian cachexia* may occur.

INVESTIGATIONS:

Ovarian swellings can be suspected clinically but always need further confirmation by:

1. **Ultrasonography:** This is the **gold standard** in diagnosis of ovarian swellings.
   - US is important to diagnose the ovarian origin of the swelling.
   - Smaller tumors are best detected by TVS, while larger ones by TAS.
   - Both TAS and TVS can differentiate between unilateral and bilateral tumors, cystic and solid ones, unilocular or multilocular cysts, and suspicions for malignancy (heterogeneous echogenicity of the tumor, high vascularity and low resistance Doppler indices, presence ascites, etc.…).
2. **Tumor markers:** to exclude malignancy
   - CA-125 is significantly elevated in epithelial ovarian cancer; however, it may be also elevated in benign conditions as endometriosis and endometriomas.
   - CEA and CA 19-19 are peculiarly elevated with mucinous carcinoma.
3. **Laparoscopy:**

May be indicated if ultrasonography was inconclusive in evaluation of an adnexal swelling as differentiating between

- Ovarian cyst and tubal or tubo-ovarian cystic masses, or
- Solid ovarian fibroma and pedunculated subserous myoma SSM.

TREATMENT OF BENIGN OVARIAN NEOPLASMS:

The **aim of treatment** in benign ovarian neoplasms is **surgical removal** of the tumor, with or without the affected ovary, for fear of complications. The type of operation depends upon the age of the patient, her desire for future pregnancies, together with the size and type of the tumor, and whether complicated or not.

1. OVARIAN CYSTECTOMY:

It consists of ***shelling out*** or ***enucleation*** of the cyst with ***preservation of the ovary***. It is indicated in young patients and particularly with bilateral cysts of small size as in dermoid cysts. Ovarian cystectomy may be done either by ***laparotomy*** or by ***laparoscopy***.

2. Ovariotomy or ovariectomy:

It consists of ***removal of an enlarged or diseased ovary***. However; oophorectomy means removal of a healthy ovary.

3. PAN HYSTERECTOMY (TAH+ BSO):

If the patient is premenopausal, and has completed her family, total abdominal hysterectomy with bilateral salpingo- oophorectomy **(TAH +BSO)** is usually indicated. This is considered as a preventive step against the future development of ovarian cancer by advance of age.

MALIGNANT OVARIAN TUMORS

Ovarian cancer is the **third most common malignancy** of female genital tract, after cancer of endometrium and cervix, but is the most lethal of them all. More women die from ovarian cancer than from carcinoma of the cervix and body of the uterus combined.

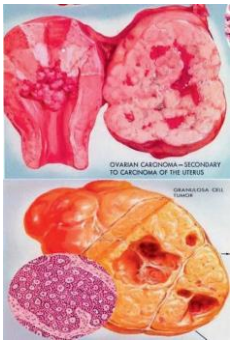
2/3 of patients have advanced disease when diagnosed. Ovarian cancer remains the fifth leading cause of cancer-related death. Ovarian cancer is named as a **"silent killer**" that lacks early signs or symptoms.

HISTOGENETIC CLASSIFICATION OF MALIGNANT OVARIAN TUMOURS:

1. **Epithelial ovarian cancer:**
    a. *Serous cystadenocarcinoma.*
    b. *Mucinous cystadenocarcinoma.*
    c. *Endometrioid adenocarcinoma.*
    d. *Undifferentiated adenocarcinoma.*
2. **Germ cell tumors:**
    a. *Choriocarcinoma.*
    b. *Endodermal sinus tumor.*
    c. *Dysgerminoma.*
    d. *Malignant teratoma.*
3. **Sex cord stromal tumors:**
    a. *Granulosa cell tumor.*
    b. *Androblastoma: Sertoli-Leydig cell tumor.*
    c. *Gynandroblastoma.*
4. **Metastatic tumors:**

mostly krukenberg tumors.

Fig 9-22: malignant ovarian tumor



EPITHELIAL OVARIAN CANCER

Numerous **reproductive, environmental, and genetic risk factors** have been associated with ovarian cancer. The most important is a family history of breast or

ovarian cancer, and up to 25% of patients have an inherited genetic predisposition.

ETIOLOGY

The etiology of epithelial ovarian cancer is unknown. Possible etiologic factors include:

1. **Reproductive Factor:**

75% have no identifiable genetic link for their ovarian cancer. Repeated stimulation of the ovarian surface epithelium is hypothesized to lead to malignant transformation.

a. ***Prolonged suppression of ovulation:*** as that occurring during pregnancy, lactation, and prolonged use of OCP is associated with **decreased incidence** of epithelial ovarian cancer.

b. ***Prolonged and repeated use of drugs for induction of ovulation:*** as in treatment of anovulatory infertility or ovarian super-ovulation protocols in ICSI & IVF procedures, has been associated with a slightly **increased risk** of ovarian cancer after several years of cessation of treatment.

2. **Hereditary 'Genetic' factor:**

25% of epithelial ovarian cancer will occur in women with ***hereditary predisposition*** i.e. two or more relatives had ovarian or breast cancer. Familial ovarian cancer is identified:

a. ***Hereditary breast / ovarian cancer syndrome***, ***with BRCA1 or BRCA2 gene positive*** Prophylactic bilateral salpingo-oophorectomy (BSO) should be offered to *BRCA1 and BRCA2* carriers at age 35- 45 years.

b. ***Hereditary non polyposis colorectal cancer syndrome with endometrial, breast, or ovarian cancer (Lynch type II):*** In women with HNPCC syndrome, hysterectomy is mandatory when performing prophylactic BSO because of coexisting endometrial cancer risks.

RISK FACTORS FOR EPITHELIAL CANCER:

1. **Age:** Increasing age is the strongest patient-related risk factor. The mean age at diagnosis is 60 years and the risk of malignancy increases with advancing age.
2. **Nulliparity & infertility**; are associated with increased risk, compared to multipara.
3. **White race:** have more prevalence compared to black and African women.
4. **Prior history of endometrial or breast cancer** in the same patient.
5. **Family history of ovarian cancer** in first degree relatives.

N.B: Multiparity, prolonged lactation, and prolonged use of OCP seem to have a relatively protective effect against epithelial ovarian cancer.

PATHOLOGY OF EPITHELIAL OVARIAN CANCER:

A. Macroscopic appearance: :

Most epithelial ovarian cancers have both cystic and solid elements (with more solid components), with marked variation in the tumor size and consistency. The tumor shows areas of extensive hemorrhage and necrosis.

In advanced stages, the tumor may be bilateral, the capsule may be ruptured, and the tumor may be fixed to adjacent structures as the uterus, bladder, colon, etc…

A. **Microscopic picture: adenocarcinomas** *of the Serous, Mucinous, or Endometrioid, types.*

B. **Grading:** according to cell differentiation into:

1. ***Well differentiated tumors (Grade I):*** showing malignant cells with a preserved glandular pattern
2. ***Moderately differentiated tumors (Grade II):*** showing sheets of highly malignant cells with presence of limited glandular pattern.
3. ***Undifferentiated tumors (Grade III):*** showing only sheets of highly malignant cells with absence of a well-defined glandular pattern.

MALIGNANT GERM CELL TUMORS

Germ cell tumors are rarely **malignant**, they include *dysgerminoma, endodermal sinus tumor (yolk sac tumor), choriocarcinoma,* and *malignant teratoma.*

SPECIAL FEATURES OF MALIGNANT GERM CELL TUMOURS:

Malignant germ cell tumors differ from epithelial ovarian cancer in the following;

a. **Age incidence:** they occur in **younger age** group of women. They should be suspected whenever a unilateral solid tumor is diagnosed by US in a young female.
b. **Associated tumor markers:** Most germ cell tumors produce **specific markers**, including lactic dehydrogenase (LDH) in dysgerminoma, human chorionic gonadotrophins (hCG) in choriocarcinoma, and alpha feto protein (AFP) in endodermal sinus tumor (yolk sac tumor).
c. **Association with abnormal gonads:** dysgerminoma is the commonest tumor occurring in patients with abnormal dysgenetic gonads.
d. **Sensitivity to irradiation:** In contrast to epithelial ovarian cancers, dysgerminoma is a **radiosensitive** tumor due to its lymphocytic infiltration.
e. **The place for conservative surgery:** Unilateral salpingo-oophorectomy is an appropriate treatment option in stage **1A** tumors when the patient is young and desirous for further fertility preservation.

A) DYSGERMINOMA:

Dysgerminoma is the **commonest** malignant germ cell tumor. It mostly occurs in patients with **abnormal gonads**, as in gonadal dysgenesis, or testicular feminization syndrome. In such cases dysgerminoma may arise in a **gonadoblastoma.** A dysgerminoma may secrete ***LDH*****,** which may serve as specific **tumor marker**.

B) ENDODERMAL SINUS TUMOUR "EST" (Yolk Sac Tumor):

EST or yolk sac tumor is the **second most common** malignant germ cell tumor of the ovary.

Most EST are associated with elevated **alpha *-fetoproteins (AFP)*** that are used as a **tumor marker**. ***Shiller-Duval bodies*** are a characteristic of this tumor on microscopic specimen examination.

C) CHORIOCARCINOMA:

These tumors are *very rare*. They are formed of sheets of malignant cytotrophoblasts and syncitiotrophoblasts. They **secrete hCG** which is used as a tumor marker.

D) MALIGNAT TERATOMA:

This is a very rare malignant germ cell tumor may occur in the form of **Immature solid teratoma or more rarely as Malignant transformation in a benign cystic teratoma (dermoid cyst).**

MALIGNANT SEX-CORD STROMAL TUMORS

Malignant sex cord stromal tumors may arise from functioning or nonfunctioning stroma

1. **Functioning sex–cord tumors** may give rise to:
   - ***Granulosa cell tumor;*** with estrogenic activity.
   - ***Sertoli-leydig cell tumor;*** with androgenic activity
   - ***Gynandroblastoma;*** a rare tumor with both estrogenic and androgenic effect.

2. **Non- functioning stroma** may very rarely give rise to ***fibrosarcoma*** of the ovary.

A) GRANULOSA CELL TUMORS:

Granulosa cell tumors mostly secrete hormones namely ***estrogen***, while others appear to secrete ***inhibin***.

Estrogen secreting tumors may lead to pseudo-precocious puberty, IMB, or postmenopausal bleeding. In **25%-50%** of cases, granulosa cell tumors are associated with endometrial hyperplasia or endometrial carcinoma in postmenopausal patients.

- **Macroscopically:** Tumors are usually unilateral, solid, yellow or yellow-grey in color.
- **Microscopically:** The tumor is formed of granulosa cells arranged in rosette-like shape. ***Call-Exner bodies are*** pathognomonic.

B) SERTOLI-LEYDIG CELL TUMORS:

These are among the **rarest** of all ovarian tumors comprising **<** 0.2% of such tumors.

The majority of Sertoli-Leydig tumors are **androgenic**, causing a state of ***defeminisation*** followed by ***virilizing effects*** in the form of acne, hirsutism, clitoromegaly, deepening of voice, and increased muscle bulk.

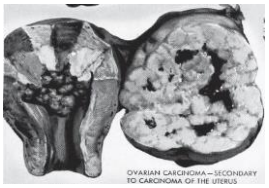


Fig 9-23

METASTATIC OVARIAN CANCER

Metastatic ovarian cancer forms about 5% of all ovarian tumors. The primary may arise from **genital tumors**, or **extra genital tumors;** from cancers of the breast, GIT (as stomach, colon, & biliary tract), or the thyroid gland.

Pathologically it is classified into **2** types;

A. **Typical metastatic cancer;** which resembles the primary cancer
B. **Atypical metastatic cancer;** e.g. Krükenberg tumor, The primary is usually in the **pylorus of the stomach,** less commonly in the colon, breast or biliary tract.

- **Macroscopically:** Krukenberg tumors are **bilateral** solid ovarian tumor mostly presenting as nodules in the Douglas pouch.
- **Microscopically:** ***signet ring cells*** in which the nucleus is pushed aside by abundant cytoplasm.

C) SPREAD OF OVARIAN CANCER:

1. **Direct extension** to adjacent organs such as tube, uterus, colon and bladder.
2. **Trans-coelomic spread**, by exfoliation of surface cells into the **peritoneal cavity**, giving deposits in the pouch of Douglas, on the surface of intestine or omentum forming an omental cake. This is **the commonest way of spread**.
3. **Lymphatic spread** mainly to para-aortic lymph nodes.
4. **Hematogenous spread** is always late and uncommon. This would be to vital organs such as the liver, lung, brain and bones.

STAGING OF OVARIAN CANCER

Surgical staging for ovarian cancer **via** an **exploratory laparotomy**, via a **mid-line subumbilical suprapubic incision** in which the following is performed:

1. *Aspiration* of ascitic fluid (or saline peritoneal washings if no ascites present) for cytologic examination.
2. *Performing* a TAH-BSO, together with Infracolic omentectomy, and pelvic and para-aortic lymph node sampling.
3. *Exploration* of the pelvic and peritoneal cavity to assess the tumor extension and *Resect* any visible enlarged masses >1.0 cm (debulking or cytoreductive surgery).

Simplified SURGICAL FIGO STAGING OF OVARIAN CANCER 2018:

Stage	Stage Description
I	**Tumor confined to the ovaries**
IA	Tumor limited to 1 ovary, capsule intact, no tumor on surface, negative washings.
IB	Tumor involves both ovaries, otherwise like IA.
IC	Tumor limited to 1 or both ovaries, with surgical spill/ with capsule rupture before surgery or tumor on ovarian surface/ with malignant cells in ascites or peritoneal washings.
II	**Tumor involves 1 or both ovaries with pelvic extension (below the pelvic brim) or primary peritoneal cancer.**
IIA	Extension and/ or implants on uterus and/ or fallopian tubes
IIB	Extension to other pelvic intraperitoneal tissues.
III	**Tumor involves 1 or both ovaries with cytologically or histologically confirmed spread to the peritoneum outside the pelvis and/ or metastasis to retroperitoneal lymph nodes.**
IIIA	+ve retroperitoneal lymph nodes and/ or microscopic extrapelvic (above the brim) peritoneal involvement
IIIB	Macroscopic extrapelvic, peritoneal metastasis ≤2cm and/ or +ve retroperitoneal nodes.
IIIC	Macroscopic extrapelvic, peritoneal metastasis ≥2cm and/ or +ve retroperitoneal nodes.
IV	**Distant metastasis**
IVA	Pleural effusion with +ve cytology
IVB	Hepatic and/ or splenic parenchymal metastasis, metastasis to extraabdominal organs as brain or bone.

DIAGNOSIS OF OVARIAN CANCER: This is a disease of 60s years of age in the majority of cases, with a very insidious onset and misleading nonspecific symptoms in the majority of cases.

SYMPTOMS:

A. **Early-stage disease:** is usually **asymptomatic**.

B. **Later stages** may be associated with:

- Nonspecific GIT symptoms; in the form of dyspepsia, indigestion, and

anorexia.

- Abnormal uterine bleeding, occurring only in estrogen producing ovarian tumors.

NB: Because symptoms are commonly nonspecific and mild, most tumors will be diagnosed late.

SIGNS:

The most important physical sign is the ***palpation of a pelvic mass***. In early stages of the disease this may be accidentally discovered during routine gynecologic examination for checkup. In late cases a large ***pelvic or pelviabdominal mass*** may be palpated with features suggestive of malignancy as being bilateral, solid, fixed, associated with ascites, unilateral LL oedema, and rapid growth.

FEATURES SUGGESTING MALIGNANCY in ovarian tumors

1. History and symptoms:

- Tumors discovered at older age (postmenopausal) are highly suspicious for malignancy.
- Tumors discovered in women with family history of breast, colonic, or ovarian cancer.
- Tumors associated with rapid weight loss, rapid progressive abdominal enlargement, and persistent GIT symptoms.
- Functioning tumors with feminizing or virilizing effects.

2. General examination:

- Malignant cachexia (with marked and rapid weight loss).
- Palpable supraclavicular lymph nodes especially on the left side, (Virchow's glands).
- Pleural effusion, however it may be present in Meig's syndrome.
- Presence of an associated breast mass on breast examination.
- Unilateral lower limb oedema (unilateral pressure by tumor with venous and lymphatic obstruction).

3. Abdominal Examination:

- ***Inspection:*** Abdominal enlargement, overlying skin showing peau d'orange.
- ***Palpation:*** Tumor which is solid (or partially solid), fixed especially if bilateral.
- ***Percussion:*** Presence of ascites.

4. Pelvic examination:

1. Non-tender solid nodules in Douglas pouch (Krukenberg tumour).
2. Bilateral, especially if solid adnexal masses are very presumptive.
3. Fixed pelvic masses especially if amalgamated with pelvic organs (frozen pelvis).

5. At Laparotomy:

- Presence of ascites, especially if altered blood-stained ascites.
- Bilaterality, fixation, and invasion of the capsule.
- Extracystic papillae and adhesions to surrounding structures.
- Peritoneal nodules or secondary deposits in omentum, intestine, liver, or lymph nodes.
- Variable consistency with a cut section of the tumor shows hemorrhage and necrosis.

SPECIAL INVESTIGATIONS in ovarian cancer:

1. **Tumor markers:**

- **CA 125**; Alone, CA125 is a poor marker for ovarian cancer detection. However, a more sensitive Risk of Ovarian Cancer Algorithm (ROCA). It calculates a risk score for each woman using a combination of their CA 125 results with their age (1 in premenopausal, and 3 if menopausal). Patients at risk for ovarian cancer will then undergo TVS to determine whether additional intervention is needed.
- **Other markers** include; serum B-hCG (choriocarcinoma), serum alpha fetoprotein (EST), and serum lactic acid dehydrogenase (dysgerminoma), CA 19-9 (mucinous cyst), and carcin-embryonic antigen (CEA).

2. **Pelvic ultrasound:**

- *TAS & TVS* are the gold standard diagnostic tool that can accurately detect pelvic masses of variable size and suggest their ovarian origin.
- *Sonographic features suggesting malignancy include;* heterogenous echopattern, intracystic and extracystic papillae, bilaterality, presence of ascites, and low resistance Doppler flow of the tumor vessels (indicating high vascularity) (1 if only 1 malignant feature, and 3 if more than one malignant feature).
- Coupling sonographic features x value of CA125 x menopausal status will indicate the *Risk of Malignancy Index (RMI) for this patient,* a RMI more than 200, indicates a high risk of being malignant tumor.

3. **Genetic screening;** testing for *BRCA1 or BRCA2* mutation carriers in women with strong family history of breast and ovarian cancer.
4. **Chest X-Ray:** for detection of pleural effusion and/or secondaries in the lungs.
5. **Plain X-Ray abdomen:** can detect calcification in dermoid cysts and Psammoma bodies.
6. **CT & MRI:** useful in detection of spread to liver and or lymph nodes.
7. **Barium meal/enema:** to exclude primary cancer in the stomach or colon (or spread).
8. **Upper/lower G.I. Endoscopy:** to detect a primary in the stomach or colon (or

spread)

9. **IVP:** to evaluate the course of the ureters and to exclude back pressure on the kidneys.
10. **Paracentesis:** needle aspiration of ascetic fluid for cytologic examination.
11. **Endometrial curettage:** In cases of abnormal uterine bleeding to exclude a primary or spread to the uterus.

EXPLORATORY LAPAROTOMY IN OVARIAN CANCER:

The final diagnosis of ovarian cancer can only be made at exploratory laparotomy, which will serve not only for **diagnosis** but also for **surgical staging** and **primary surgical treatment**.

SURGICAL TREATMENT OF OVARIAN CANCER

A) EARLY STAGE OVARIAN CANCER:

If no gross evidence for extension of the disease beyond the ovary;

1. **TAH-BSO and Infracolic omentectomy** is the standard treatment for patients with disease limited to the ovary (stages **I-IIA**). Surgical staging is completed via peritoneal wash and lymph node sampling for microscopic assessment.
2. **Fertility- sparing unilateral salpingo-oophorectomy (USO)** is performed in all reproductive aged women desiring fertility and diagnosed with malignant ovarian **germ cell tumors stage IA,** as this conservative approach does not adversely affect survival. Following removal of the affected ovary, *surgical staging* by laparotomy or laparoscopy proceeds (lymphadenectomy is most important for dysgerminomas)

B) ADVANCED STAGE OVARIAN CANCER:

In cases with tumor extension beyond the ovaries:

1. **Initial Debulking (*Primary Cytoreductive surgery*):**

   The aim of initial surgery is to remove *all cancer* tissue leaving only residual small tumor deposits no more than 1.0 cm in diameter.

   - Debulking surgery includes a **TAH-BSO** + omentectomy + excision of pelvic masses and peritoneal deposits >1cm + segmental bowel resection if involved within the tumor mass.
   - Advantages of initial debulking surgery:
     a. It Improves *survival* in patients with advanced disease.
     b. It Improves *response to chemotherapy*, when residual tumor implants left are less than 1cm in size.

2. **Chemotherapy whether single or multiple agents**: Chemotherapy, most commonly used: carboplatin and paclitaxel (Taxol), is indicated in ***all cases*** of stage **IIB-IV** disease.
   A. As adjuvant therapy in all cases after primary cytoreductive debulking surgery
   B. Palliative therapy in patients with irresectable tumors, or with recurrent disease.

Toxicity from chemotherapy:

Chemotherapeutic agents are highly toxic at therapeutic doses, and therefore need close monitoring during treatment cycles. Toxicity includes; nausea, vomiting, myalgia and arthralgia. In severe cases renal damage, peripheral neuropathy, hearing loss, dehydration and electrolyte imbalance may occur.

10 IMAGING TECHNIQUES IN GYNECOLOGY

X-RAY RADIOGRAPHY:

1. Plain X-ray films of the pelvis to diagnose large tumors, calcifications in pelvic tumors, dermoid cyst and renal stones.
2. X-ray chest for:
   A. Preoperative assessment
   B. Diagnose of metastases
   C. Suspected T.B.
3. I.V.P. (intravenous pyelography) for:
   A. Assessment of renal function.
   B. Diagnosis of urinary stones
   C. Diagnosis of genito-urinary fistula

ULTRASOUND:

POSSIBLE DIAGNOSTIC APPLICATIONS OF US IN GYNECOLOGY:

1. Evaluation of genital anomalies.
2. Evaluation of pelvic masses.
3. Evaluation of lower abdominal pain & inflammatory processes especially in patients with muscular guarding or rigidity.
4. Monitoring of follicular growth as part of infertility workup.
5. Localization of I.U.C.D. (either post insertion or in occasions of missed device).
6. Diagnosis of abnormal uterine bleeding
7. Diagnosis and evaluation of uterine fibroids (number, size, site, type, relation to the cavity)
8. Measuring endometrial thickness.
9. Diagnosis of ovarian cysts and neoplasms.
10. Evaluation of urinary incontinence and residual urine volume.
11. Sonohysterography for tubal patency.
12. Can differentiate between benign and malignant conditions with a reasonable degree of efficacy.

POSSIBLE THERAPEUTIC APPLICATIONS OF US IN GYNAECOLOGY:

1. US-guided follicular aspiration as a part of IVF programs.
2. US-guided aspiration of simple cysts.
3. US- guided drainage of ascites.

POSSIBLE USES OF US IN OBSTETRICS:

1.During the first trimester:

a. Diagnosis of pregnancy.

b. Exclusion of extrauterine pregnancy.
c. Diagnosis of unhealthy pregnancy.
d. Determination of number of embryos.
e. Determination of gestational age.
f. Early diagnosis of some congenital anomalies and some chromosomal abnormalities e.g. Nuchal thickness in Down syndrome.
g. Cervical measurements to exclude cervical incompetence.
h. Guide for chorionic villous sampling.
i. Helpful in conservative management of ectopic pregnancy.



Fig 10-1: Obstetric US

2. Uses of US after the first trimester:

a. Diagnosis of lie, presentation & position.
b. To exclude IUFD.
c. Estimation of fetal weight.
d. Diagnosis of congenital anomalies.
e. Assessment of fetal well-being through biophysical profile (gross fetal movements, breathing movements, tone, and amount of liquor) & through Doppler study of placental, umbilical and fetal BVs.
f. Placental location and exclude placental separation (accidental hemorrhage).
g. Amount of liquor.

POSSIBLE US GUIDED PROCEDURES OF US IN OBSTETRICS:

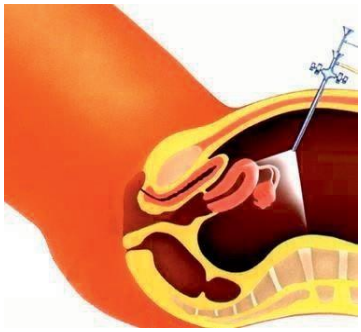
1. Chorionic villous sampling.
2. Amniocentesis: either diagnostic (chromosomal study - bilirubin level - immunoglobulins in infections) Or therapeutic.
3. Cordocentesis.
4. Intrauterine blood transfusion in Rh isoimmunization.

11 ENDOSCOPY IN GYNECOLOGY

LAPAROSCOPY

Laparoscopy is a procedure which allows direct inspection of the peritoneal cavity and pelvic organs by introduction of an optic lens through the umbilicus after properly insufflating the peritoneum with CO2 gas. The procedure also allows the performing of variable types of pelvic surgery.

INDICATIONS:



A. Diagnostic laparoscopy:

1. Infertility whether primary or secondary.
2. Chronic pelvic pain.
3. Diagnosis of pelvic endometriosis.
4. Diagnosis of some undiagnosed pelvic masses.
5. Assessment of congenital anomalies of uterus & tubes.
6. Follow up after radical surgery for malignancy.

B. Operative laparoscopy:

1. **Tubal surgery:**
   a. Management of adhesion by adhesolysis.
   b. Management of ectopic pregnancy.
   c. Removal of hydrosalpinx or pyosalpinx or tubal disconnection.
   d. Tubal sterilization.
1. **Ovarian surgery:**
   a. Selective removal of ovarian masses by cystectomy, ovariotomy or oophorectomy.
   b. Drilling of polycystic ovarian disease in cases resistant to medical treatment.
2. **Uterine surgery:**
   a. Myomectomy for myomas of small number and moderate size.
   b. Hysterectomy whether total or subtotal is feasible in selective cases.
3. **Endometriosis:** laser ablation, fulguration and diathermy cauterization of endometriotic foci, ovarian cystectomy for endometrioma.

Complications:

1. Anesthetic and cardiopulmonary complications, more especially CO2 embolism, arrhythmias, gastric reflux, hypotension.
2. Extraperitoneal insufflation due to malposition of insufflation needle will result in

cutaneous surgical emphysema.

3. Electrosurgical complication: These can result in diathermy burns to important structures such as the bowel, nerves and may end in fatalities if they are not recognized during surgery.
4. Hemorrhagic complications:
   a) The **most serious complication** of these are injuries the aorta and IVC. This can happen during introduction of insufflation needle or the trocar specially in very thin patients.
   b) Injury to the cutaneous abdominal wall vessels specially the inferior epigastric.
   c) Injury to an intraperitoneal vessel by any type of surgery.
5. Gastrointestinal complications: Injury by an insufflations needle or a trocar or during dissection especially in patients with previous abdominal surgery.
6. Urologic injury: Injury to the bladder or ureter may happen secondary to mechanical dissection or thermal trauma.
7. Neurological injury: Peripheral nerve injury is usually related to poor positioning of the patient. Nerve injury may also occur as a result of surgical dissection or diathermy injury.
8. Incisional hernia and wound dehiscence
9. Infection: wound infection, crepitus, pelvic cellulites and pelvic abscess are rare happening.

Advantage of operative laparoscopy:

- Minimal hospital stays and early return to work.
- Minimal patient discomfort.
- Minimal patient adhesion and therefore minimal iatrogenic infertility.
- Better cosmetic results.
- Rare wound complications.
- Allow proper inspection and excludes the need for what is called exploratory laparotomy.

HYSTEROSCOPY

The procedure of hysteroscopy entails the introduction of lens inside the uterine cavity through the cervix after distending the uterine cavity with suitable medium (CO_2-saline-glycine 1.5%) to visualize the endometrial cavity and perform variable intrauterine operative procedures.

INDICATIONS:

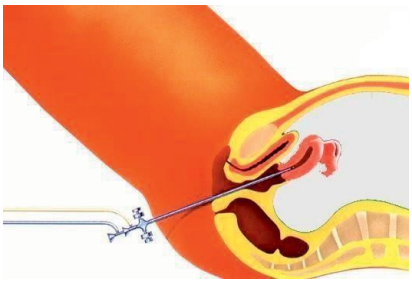
A. Diagnostic:

1. Infertility. Variable findings may be found such as polyps, synechiae,

submucous myoma, septate uterus.

2. Habitual and recurrent abortions.
3. Irregular uterine bleeding.

B. Operative:



1. Polypectomy.
2. Resection of uterine septae.
3. Resection of a submucous myoma.
4. Division of intrauterine synechiae.
5. Removal of a missed IUD.
6. Resection or coagulation of the endometrium.
7. Intrauterine tubal catheterization for cornual obstruction.
8. Sterilization by injecting a sclerosing agent or a plug to the cervix.

Complications:

1. Anesthesia complications.
2. Perforation of the uterus.
3. Bleeding due to myometrial vascular trauma.
4. Complications related to distending media:
    - Fluid overload, especially in lengthy procedures.
    - CO_2 can result in embolization.
    - Low viscosity fluids (e.g. glycine 1.5%) can result in **serious electrolyte** may be fatal (most serious complication).

12 OPERATIVE GYNECOLOGY

SOUNDING OF THE UTERUS

The uterine sound is made of malleable metal and its tip is blunt. It is graduated in inches or centimeters. It is slightly curved near the tip so that it can adapt to the angle of uterine flexion.

- Before introducing the uterine sound, bimanual examination is done to determine whether the uterus is AVF, or RVF, to introduce the sound in the proper direction and avoid uterine perforation
- Cervix is exposed by a speculum and the anterior lip of the cervix is grasped by a volsellum.
- Sound is introduced until resistance is felt at the internal os to measure the length of the cervical canal, then it is pushed until it reaches the fundus to measure the length of the uterine cavity.

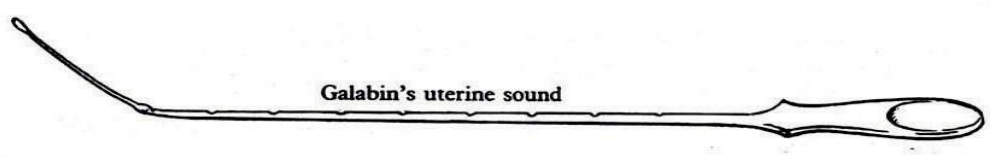


Fig 12-1: Sound

Uses of the uterine sound:

1. Degree of supravaginal elongation of the cervix in cases of prolapse.
2. Diagnosis of uterine hypoplasia.
3. Measurement of the uterine length before introduction of I.U.C.D.
4. Diagnosis of cervical stenosis (inability to introduce the sound).
5. Abnormality inside the uterus as polyp or septum.
6. Differentiation between chronic inversion and fibroid polyp bulging through the cervix. In the latter case, the uterine length is normal.
7. Detection of retained I.U.C.D.

Complications of the uterine sound:

1. Perforation.
2. Infection.

3. Abortion if done for pregnant uterus.

DILATATION OF THE CERVIX

Types of dilators:

1. Hegar: uniform thickness.
2. Fenton: tapers gradually towards the tip.

N.B.: The number of the dilator indicates the diameter in millimeters.

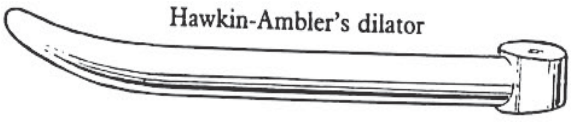


Fig 12-2: Dilator

Uses of dilators:

A) Dilatation alone:

1. Spasmodic dysmenorrhea.
2. Cervical stenosis.
3. Drainage of pyometra or hematometra.

B) Dilatation preliminary to another operation:

1. Operations on the cervix: as trachelorrhaphy, amputation, cautherization.
2. Operations on the uterus: as curettage, evacuation, polypectomy.
3. Operations on the tubes: as Rubin test if a large cannula is used.

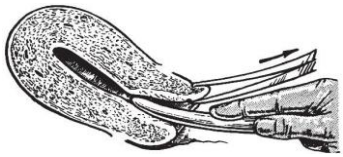


Fig 12-3: Dilatation

Complications of dilatation:

1. *Anesthetic complications*.
2. *Perforation of the uterus* (It is detected when the sound or dilator can be introduced more than the expected length of the uterine cavity). If perforation occurs, the patient is kept under observation and antibiotics are given, but if there is shock, signs of internal hemorrhage or injury to abdominal organs, laparotomy is necessary.
3. *Lacerations of the cervix* are more liable to occur if the cervix is dilated very rapidly. Lacerations may become infected leading to chronic cervicitis. Incompetent isthmus and RPL later on.
4. *Infection.*
5. *Shock* is liable to occur if dilatation is carried out without anesthesia or with improper anesthesia

.

CURETTAGE

Types of curettes:

1. The loop curette: either sharp or blunt or combined. Blunt curette is used in cases of recent pregnancy or suspected malignancy as the uterus is easily perforated.
2. Biopsy curette (as Novak biopsy curette): very small diameter (3mm), so used without anesthesia. It removes only a small strip of endometrium, so not used to diagnose malignancy of the endometrium.

Indications of curettage:



Fig 12-4: Curette

(A) Diagnostic (endometrial biopsy):

1. Detection of the endometrial pattern in hormonal disorders.
2. To determine the cause of uterine bleeding and to differentiate between dysfunctional bleeding and bleeding due to organic cause.
3. Diagnosis of malignant diseases of the endometrium or endocervix.
4. Diagnosis of diseases of the endometrium as tuberculous endometritis.

(B) Therapeutic:

1. Post-abortive and puerperal bleeding due to retained products.
2. Endometrial and cervical polypi.
3. Dysfunctional uterine bleeding.

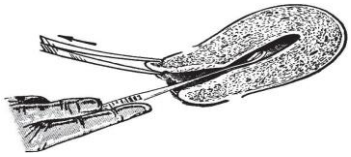


Fig 12-5: Curetting

Complications of curettage:

- In addition to complications of dilatation (see before),
- Over curettage (specially for a pregnant uterus or in cases of infection) leading to Asherman' syndrome due to removal of the basal parts of the endometrial glands and formation of intrauterine synechiae (adhesions).

- Rupture of the scar in subsequent pregnancy.

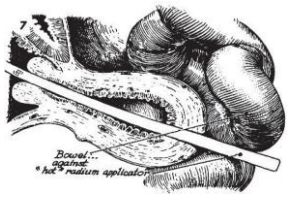


Fig 12-6: Perforation of the uterus

HYSTERECTOMY

Hysterectomy may be done abdominally, vaginally, laparoscopically, laparoscopically assisted vaginal hysterectomy (LAVH) or using robotic surgery.

Indications of hysterectomy:

(A) Obstetrical:

1. Rupture of the uterus.
2. Uncontrollable postpartum hemorrhage.
3. Couvelaire's uterus.
4. Placenta accreta.

(B) Gynecological:

1. Inflammatory as some cases of genital tuberculosis.
2. Neoplastic:
   a. Benign: some cases of fibroids and benign ovarian tumors in old patients.
   b. Malignant tumors of the cervix, body and ovaries.
3. Displacements: some cases of uterine prolapse or chronic inversion.
4. Some cases of endometriosis.
5. Some cases of abnormal uterine bleeding.

Types of Abdominal hysterectomy (AH):

1. **Subtotal AH:** We remove the body and leave the cervix. It is quicker, usually done in some cases of post-partum hemorrhage and rupture of the uterus, or if there is extensive adhesions around the cervix.
2. **Total AH:** We remove the body and cervix and it is better than the subtotal (see below).
3. **Pan hysterectomy:** Total hysterectomy and bilateral salpingo-oophorectomy.
4. **Radical hysterectomy:** Wertheim's operation (it is a panhysterectomy with removal of upper cuff of the vagina, and sampling of pelvic lymph nodes)

Complications of abdominal hysterectomy (or any abdominal surgery):

1. Anesthesia complications as postoperative cyanosis, postoperative vomiting, pulmonary complications (collapse or infection after inhalation of vomitus).
2. Hemorrhage (primary, reactionary or secondary).
3. Shock.

4. Gastrointestinal complications as acute gastric dilatation, paralytic ileus or adhesions (those lead to vomiting and distension), Injuries to abdominal organs, fecal and rectovaginal fistula.
5. Pulmonary complications: bronchopneumonia or lobar pneumonia, pulmonary collapse.
6. Urinary complications: incontinence of urine (which may be due to bladder or ureteric injury), anuria (due to bilateral ligation of ureters or renal failure).
7. Venous thrombosis: that may lead to pulmonary embolism.
8. Wound infection and burst abdomen.
9. Remote complications:
   a. Complications of the abdominal scar as keloid formation and incisional hernia.
   b. Vaginal discharge due to infection at the vaginal vault.
   c. Vaginal vault prolapse after hysterectomy.
   d. Dyspareunia after hysterectomy (due to a tender scar in the vaginal vault).
   e. Menopausal symptoms following bilateral oophorectomy in a young patient.

Vaginal Hysterectomy (VH)

Indications of VH:

1. Some cases of uterine prolapse or chronic inversion.
2. Some cases of DUB.
3. Some cases of small fibroids.

Advantages of VH:

1. Absence of an abdominal scar.
2. Lower incidence of intestinal complications and peritonitis.
3. An associated genital prolapse can be treated at the same time.

Disadvantages of VH:

1. It is difficult in the presence of pelvic adhesions.
2. The ovaries cannot be removed in some cases.
3. It cannot be done if the size of the uterus is larger than a 14 weeks pregnant uterus (unless the uterus is bisected before removal).
4. Vault prolapse is more likely to occur than after abdominal hysterectomy.

13 GYNECOLOGIC HISTORY AND EXAMINATION

GYNECOLOGICAL HISTORY

The most important evidence is always provided by history given by the patient. In many cases a provisional diagnosis can be reached before starting detailed physical examination.

1. PERSONAL HISTORY:

- Patient's name, age, marital status, gravidity and parity, address, occupation.
- Special habits of medical importance

NB: Take the husband's personal history in Infertility cases.

2. COMPLAINT:

Complaint is written in the ***patient's own words*** as; excessive vaginal bleeding, pelvic pain, painful intercourse, progressive abdominal enlargement, mass filling or protruding from the vagina, etc...

3. PRESENT HISTORY:

- Analysis of the **onset, course,** and **duration,** of the **present complaint**.
- Ask about **symptoms associated** with the present complaint.
- Ask about symptoms related to **other body system affection** as: urinary, GIT, cardiac, pulmonary, neurologic, etc...
- Ask about **investigations and any treatment** given to the patient.

NB: **SEXUAL HISTORY is taken** in cases of infertility.

- Frequency of sexual intercourse (times /week).
- Associated symptoms; as presence of dyspareunia (painful intercourse).

4. PAST HISTORY:

A) Medical Disorders:

- *Systemic disease:* hypertension, diabetes mellitus, cardiac, renal, or liver disease.
- *Endocrine disorders:* especially thyroid and adrenal disorders.
- *Blood dyscrasias:* as ITP & von Willebrand disease (VWD).

B) Surgical Procedure:

- *Gynecologic & Obstetric procedures:* D&C, CS, Myomectomy, Hysterectomy, Laparoscopic and hysteroscopic procedures.
- *Surgical procedures:* as pelvic surgery (appendicitis, peritonitis, pelvic masses), bariatric procedures (gastric banding, and sleeve operations), thyroidectomy, breast lumps, cholecystectomy, etc…

5. FAMILY HISTORY:

Of similar condition in the family, genetic disorders, hereditary malignancies, twins and consanguinity (in obstetric sheet).

6. MENSTRUAL HISTORY:

- ***Age at menarche:*** normally 11-13 years.
- ***Cyclic pattern:*** normally every 25-32 days (mean 28 days).
- ***Duration of bleeding:*** normally 3-5 days.
- ***Menstrual Index:*** duration of bleeding/ duration of the cycle, eg 5/ 30 (menses coming every 30 days and lasting for 5 days).
- ***Amount:*** normally changing 3-4 pads/day, not associated with blood clots.
- ***Dysmenorrhea:*** Menstrual pain and its relation to the onset of bleeding.
- ***Last menstrual period (LMP):*** calculated from 1st day of onset of menses.

7. OBSTETRIC HISTORY:

- ***Previous abortions, pregnancies and deliveries:*** in sequence of their occurrence.
  - *Duration of each gestation* (term, preterm, or abortion).
  - *Route of each delivery:* vaginal delivery (VD), caesarean section (CS), surgical evacuation (SE).
  - *Place of the delivery:* home, private clinic, governmental hospital.
  - *The gender and condition of the outcome:* living, still born (SB), neonatal death (NND), abortus.
  - *Complications postpartum or post abortive:* as hemorrhage, sepsis or lactation problems.
- ***The Obstetric Code:*** The obstetric history can be described for simplicity by:
  - ***The 4 digit code (FPAL)*** (Fullterm, Preterm, Abortion, Living):
    - *First 3 digits:* stand for gravidity (fullterm, preterm, & abortions).
    - *First 2 digits:* stand for the parity (fullterm and preterm).
    - *The 4th digit:* stands for number of living offspring now.
  - ***The Gravidity / Parity code*** (GPL):
    - *Gravidity* describes the *number of pregnancies*, including the current pregnancy if present (term + preterm + abortions).

- *Parity* stands for *number of previous deliveries* (term + preterm).
- *Living* stands for the number of living offsprings today.

- ***Contraceptive history:*** *All the methods of contraception used in-between pregnancies.*

GYNECOLOGIC EXAMINATION

1. GENERAL EXAMINATION:

- Vital signs: measurement of the BP, Pulse, and temp.
- Gait, height, weight, and body mass index (BMI) = weight kg /height in meter square.
- Hair distribution and development of the secondary sexual characters.
- Breast examination is an important part of any gynecological examination.
- Inspection of the back: for spina bifida, hair tuft, severe kyphosis, or scoliosis.

2. ABDOMINAL EXAMINATION: *(**bladder should be empty).***

- ***Inspection:*** abdomen exposed from costal margin till symphysis pubis.
  - Abdominal masses, scars of previous operations, pigmentation, etc…
  - Hernia orifices, and divarication of the rectii muscles (during coughing).
- ***Palpation of all abdominal quadrants***
  - ***9 quadrants:*** (right and left hypochondruim, right and left lumbar, right and left iliac fossa hypogastrium, umbilical, and suprapubic regions).
  - ***4 quadrants:*** (right and left upper quadrant and right and left lower quadrants).
  - Palpation for enlarged liver, spleen, renal, abdominal, and pelviabdominal masses.
- ***Percussion of the abdomen:***
  - Normally there is resonance in the center and in the flanks due to air in intestinal loops.
  - Tumors show uniform dullness all over the tumor area.
  - Moderate ascites shows central resonance (floating bowel) with dull flanks (fluid). Shifting dullness will occur when patient is turned on her side with upper flanks becoming resonant (due to shift of ascites to more dependent parts and intestines floating over fluid).
  - Tense ascites gives generalized dullness on percussion (fluid everywhere).

3. BREAST EXAMINATION:

Breast examination is an integral part of a comprehensive gynecologic examination. It is an important method for both **screening** for benign and malignant breast tumors, and **diagnosis** of breast conditions as a breast lump, fibrocystic breast disease, mastitis, breast abscess, and galactorrhea.

A) **Inspection:** with patient in upright position, for contour, symmetry, skin changes or nipple retraction.

B) **Palpation:** with patient in upright then supine position, palpate:

- The 4 breast quadrants, areola, and nipple with the flat of the hand.

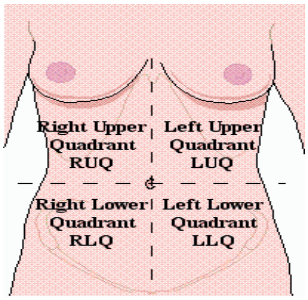


Fig 13-1: 4 quadrants of the abdomen



Fig 13-2: Breast examination

- Each axilla is examined for lymph node groups (medial, lateral and apical) while the patient's arm is supported.
- The supraclavicular fossae are also palpated for lymphadenopathy.

4. LOCAL EXAMINATION:

- Patient's attendance is by itself considered consent for a pelvic examination. However, presence of a third party (a female relative, a chaperone or a nurse), is ethically mandatory.
- The patient's bladder must be empty, and only ***disposable gloves*** should be used.
- Instruments used are either ***disposable*** or should be ***sterilized by autoclaving***, to avoid the risk of transferring infecting vaginal organisms from one woman to another.
- The patient is examined locally while lying down in the lithotomy position.



Fig 13-3: Lithotomy position

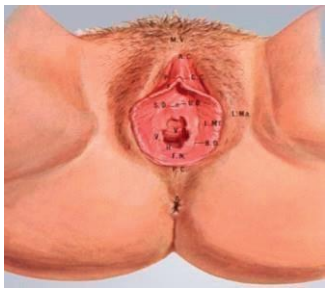


Fig 13-4: Inspection of the vulva

A) *Inspection of the Vulva and Perineum:*

- *The vulva:* is inspected for anatomical abnormalities, swellings, tumors, warts, or ulcers, together with presence of abnormal vaginal discharge or

bleeding.

- *The external urethral meatus:* for any redness, discharge, or presence of urethral caruncle.
- *The perineum:* for old lacerations, and the presence of piles around the anal orifice.
- *Inspection while straining:* for presence of genital prolapse and stress incontinence.

B) *Digital Vaginal Palpation (PV):*

The index finger is inserted into the vagina first, and as the patient relaxes, it is followed by the middle finger, and together slope gradually deeper in the vagina following its direction (upwards and backwards). As the fingers are withdrawn, the following structures are palpated:

- *The cervix:* noting its size, consistency, mobility, direction, and level. Presence of lacerations, hypertrophy, dilatation, and any descent on straining, or any contact bleeding.
- *The vaginal walls:* palpated for any mass, ulcer, or fistulous opening.
- *The bladder and urethra:* can be palpated through the anterior vaginal wall.
- *The rectum and pouch of Douglas:* can be palpated through the posterior vaginal wall.
- *Bartholin's glands:* are palpated to detect any swelling.
- *The condition of the levator ani muscles:* can be determined by two fingers in the vagina, and the thumb on the perineum. Ask the patient to cough and feel the muscle as it contracts.



Fig 13-5: PV Examination



Fig 13-6: Bimanual Examination

C) *Bimanual Pelvic Examination:*

The index and middle fingers of the right hand are placed in the posterior vaginal fornix, while the left hand is placed flat on the abdominal wall.

1. **Examination of the Uterus:**

   The vaginal fingers of the right hand should push the cervix upwards. The abdominal hand is placed below the umbilicus and gradually moved lower suprapubically until the fundus is caught and pressed against the fingers of both hands.

2. **Examination of the adnexae:**

The fingers of the vaginal hand move towards one lateral fornix with the abdominal hand following, trying to gently squeeze the adnexa to palpate any masses, swellings, or elicit tenderness. The same procedure is then repeated in the other lateral fornix.

3. **Examination of the Pouch of Douglas:**

The vaginal hand is moved to the posterior fornix, while the abdominal hand follows trying to reach the vaginal fingers behind the uterus suprapubically.

- **Points to be Determined During Bimanual Pelvic Examination:**

  - **Uterine Size, Shape, Consistency, and Contour:** the uterus is a pear-shaped organ with firm consistency, and smooth outer contour. It normally measures nearly 7.5 X 2.5 X 5cm in Long/AP/Transverse diameters. It may be felt with difficulty in women with marked trunk obesity.
  - **Uterine Position:** The uterus is commonly Anteverted flexed (**AVF**) in position, with the external os pointing towards the posterior fornix. In nearly 15% of cases the uterus is retroverted flexed (**RVF**), with the external os pointing towards the anterior fornix.
  - **Uterine Tenderness and Mobility:** The uterus is slightly tender when squeezed between the two hands. It is normally mobile, but might be fixed with adhesions or pelvic masses.
  - **Presence of Adnexal Tenderness, Fullness, or Masses:**
    - The normal tubes are never palpable even in the very thin patient, while the normal ovaries can be felt in the thin patient, in absence of pelvic tenderness.
    - Presence of adnexal tenderness, fullness, or masses (cystic or solid), is always abnormal, and should raise suspicion for presence of pelvic pathology as; PID, ectopic pregnancy, and tubo-ovarian masses.
  - **Evaluation of vaginal discharge:**
    - The amount and character of vaginal discharge, if present, are noted on the examining fingers while being withdrawn outside the vagina at the end of vaginal examination.

D) Speculum Examination:

Inspection of the cervix and vaginal walls is achieved by the aid of a vaginal speculum. The one commonly used is the **Cusco's speculum**, which is inserted vertically then rotated to be horizontally placed inside the vagina.

The cervix is inspected to detect ectopy, laceration, polyp or ulceration.



1. At the introitus *2. Inside vagina* *3. Opening to view cervix*

Fig 13-7: Steps for Speculum examination

E) Rectal Examination (PR):

Rectal examination is indicated in:

1. Examination of virgins.
2. In cancer cervix: to detect extension to parametrium along the uterosacral ligament.
3. In cases of prolapse: to detect presence of rectocele and exclude presence of enterocele.
4. Cases with complete perineal tears if diagnosis is not sure by inspection alone.
5. Cases with mass in Douglas pouch.
6. Cases with postmenopausal bleeding (piles, cancer rectum).
7. Any patient complaining of rectal symptoms.



Fig 13-8: PR examination

